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15. April 1983

Projective bundles on a complex torus

By *G. Elencwajg* at Nice and *M. S. Narasimhan* at Bombay

Let X be a complex manifold. If P is a holomorphic projective bundle on X, we have an obstruction, which is a torsion element in $H^2(X, \mathcal{O}^*)$, for P to be of the form $\mathbb{P}(E)$, where E is a holomorphic vector bundle on X. This obstruction class is obtained by means of the connecting homomorphism

$$H^1(X, PGL(r, \mathcal{O})) \to H^2(X, \mathcal{O}^*)$$

associated with the exact sequence of sheaves of groups on X:

$$1 \to \mathcal{O}^* \to GL(r, \mathcal{O}) \to PGL(r, \mathcal{O}) \to 1.$$

A. Grothendieck has asked, in the algebraic case, whether every torsion element in $H^2_{et}(X, \mathcal{O}^*)$ is the obstruction class of a projective bundle ([4], p. 76). We deal with the analogue of this question in the case of a complex torus.

Theorem 1. *Let X be a complex torus. Then every torsion element in $H^2(X, \mathcal{O}^*)$ is the obstruction class of a holomorphic projective bundle on X. In fact, the projective bundle may be chosen to be a flat bundle.*

For results of this type on abelian varieties we refer to [1] and [5].

It follows from the theorem that there exists, on any complex torus of complex dimension $\geqq 2$, a holomorphic projective bundle which is not associated to any holomorphic vector bundle (see Proposition 4. 1).

The proof of the theorem is purely algebraic. Let Γ be the fundamental group of X. Using the fact that X is a $K(\Gamma, 1)$ space we reduce the problem to one of showing that every element of $H^2(\Gamma, \mu_r)$ (where μ_r denotes the group of rth roots of unity) is the obstruction for lifting a projective representation of Γ into a representation of Γ into the special linear group. The latter problem is solved by explicit construction.

2. Projective representations of Γ

We shall denote by $SL(r)$ (resp. $PGL(r)$) the group $SL(r, \mathbb{C})$ (resp. $PGL(r, \mathbb{C})$) and by g the complex dimension of X. Consider the exact sequence

$$1 \to \mu_r \to SL(r) \to PGL(r) \to 1.$$

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Let $\rho: \Gamma \to PGL(r)$ be a homomorphism. The obstruction for lifting ρ into a homomorphism of Γ into $SL(r)$ is an element $\delta(\rho)$ in $H^2(\Gamma, \mu_r)$.

Note that there is a natural homomorphism $PGL(r) \times PGL(r') \to PGL(rr')$ induced by the homomorphism $SL(r) \times SL(r') \to SL(rr')$ given by

$$(A, B) \to A \otimes B, \quad A \in SL(r), \quad B \in SL(r').$$

If

$$\rho : \Gamma \to PGL(r) \quad \text{and} \quad \rho' : \Gamma \to PGL(r')$$

are two homomorphisms, we thus have a natural homomorphism denoted by $\rho \otimes \rho'$, of Γ into $PGL(rr')$. We have

$$\delta(\rho \otimes \rho') = \Phi_1(\delta(\rho)) \cdot \Phi_2(\delta(\rho'))$$

where $\Phi_1 : H^2(\Gamma, \mu_r) \to H^2(\Gamma, \mu_{rr'})$ and $\Phi_2 : H^2(\Gamma, \mu_{r'}) \to H^2(\Gamma, \mu_{rr'})$ are induced respectively by the inclusions of μ_r and $\mu_{r'}$ in $\mu_{rr'}$.

Remark. For several reasons it is better to think of the obstruction as an element of $H^2(\Gamma, \mu_\infty)$ where μ_∞ denotes the group of all roots of unity. But we shall not do so here.

Proposition 2. 1. *Every element ξ in $H^2(\Gamma, \mu_r)$ occurs as the obstruction class of a projective representation $\rho : \Gamma \to PGL(r^g)$. (More precisely the image of ξ in $H^2(\Gamma, \mu_{r^g})$ is the obstruction class of ρ.)*

Remark. We do not claim that ξ is the obstruction class of a homomorphism of Γ into $PGL(r)$.

For the proof of the proposition we need some lemmas.

Lemma 2. 2. *Let $\xi \in H^2(\Gamma, \mu_r)$. Then we can choose a basis $(e_1, \ldots, e_{2g})$ of Γ with the following property: If Γ_i denotes the subgroup $\mathbb{Z}e_{2i-1} + \mathbb{Z}e_{2i}$, there exists an element $\xi_i \in H^2(\Gamma_i, \mu_r)$, $1 \leqq i \leqq g$, such that*

$$\xi = \sum_{i=1}^{g} \tilde{\xi}_i$$

where $\tilde{\xi}_i \in H^2(\Gamma, \mu_r)$ is the image of ξ_i under the homomorphism $H^2(\Gamma_i, \mu_r) \to H^2(\Gamma, \mu_r)$ induced by the projection $\Gamma \to \Gamma_i$ (with respect to the direct sum decomposition $\Gamma = \bigoplus_i \Gamma_i$).

Proof. Note first that, since X is a $K(\Gamma, 1)$ space, we have $H^i(\Gamma, \mathbb{Z}) \xrightarrow{\sim} H^i(X, \mathbb{Z})$, $i \geqq 0$. [3], Ch. XVI, § 9. In particular we see that $H^3(\Gamma, \mathbb{Z})$ is torsion free. Consider the exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \xrightarrow{e^{\frac{2\pi i}{(r)}(\cdot)}} \mu_r \longrightarrow 1$$

and the corresponding exact cohomology sequence

$$H^2(\Gamma, \mathbb{Z}) \to H^2(\Gamma, \mu_r) \to H^3(\Gamma, \mathbb{Z}) \to H^3(\Gamma, \mathbb{Z}).$$

Since $H^3(\Gamma, \mathbb{Z})$ is torsion-free, the map $H^2(\Gamma, \mathbb{Z}) \to H^2(\Gamma, \mu_r)$ is surjective. Let η be an element of $H^2(\Gamma, \mathbb{Z})$ mapping into ξ.

Let $Z^2(\Gamma, \mathbb{Z})$ be the space of 2-cocycles of Γ with values in $\mathbb{Z}$ and $\mathrm{Alt}^2(\Gamma, \mathbb{Z})$ the space of alternating 2-forms on Γ with values in $\mathbb{Z}$. One knows that the mapping $Z^2(\Gamma, \mathbb{Z}) \to \mathrm{Alt}^2(\Gamma, \mathbb{Z})$ mapping a cocycle f into the alternating 2-form

$$\varphi(x, y) = f(x, y) - f(y, x)$$

induces an isomorphism of $H^2(\Gamma, \mathbb{Z})$ onto $\mathrm{Alt}^2(\Gamma, \mathbb{Z})$ ([6], Ch. I, § 2, Lemma on page 16). Let Φ be the alternating 2-form corresponding to η. By a theorem of Frobenius ([2], § 5, Th. 1) we can choose a basis $(e_1, \ldots, e_{2g})$ of Γ and integers $d_1, \ldots, d_g \geqq 0$ such that

1) $\Phi(e_{2i-1}, e_{2i}) = d_i \quad (i = 1, \ldots, g)$,

2) $\Phi(e_i, e_j) = 0$ for all other (i, j) with $i \leqq j$.

Let $\eta_i \in H^2(\Gamma_i, \mathbb{Z})$ be the element which maps into the alternating form Φ_i on Γ_i given by $\Phi_i(e_{2i-1}, e_{2i}) = d_i$. If $\xi_i \in H^2(\Gamma_i, \mu_r)$ is the image of η_i under the canonical map

$$H^2(\Gamma_i, \mathbb{Z}) \to H^2(\Gamma_i, \mu_r),$$

then $\{\xi_i\}$ satisfy the required conditions.

Lemma 2. 3. *Let Γ_0 be a free abelian group on two generators and ψ be an element of $H^2(\Gamma_0, \mu_r)$. Then there exists a projective representation $\rho_0 : \Gamma_0 \to PGL(r)$ whose obstruction class is ψ.*

Proof. Let (e_1, e_2) be a basis of Γ_0 and let ψ correspond to the (multiplicative) alternating form $\varphi(e_1, e_2) = \lambda \in \mu_r$. Let A and B be two elements of $SL(r)$ such that $ABA^{-1}B^{-1} = \lambda I$.

[For example, if $(f_1, \ldots, f_r)$ is the canonical basis in $\mathbb{C}^r$ let $\tilde{A}$ be the linear transformation in $\mathbb{C}^r$ defined by $\tilde{A}(f_i) = \lambda^{i-1} f_i$ $(i = 1, \ldots, r)$ and let $\tilde{B}$ be defined by $\tilde{B} f_i = f_{i+1}$ for $i < r$ and $\tilde{B} f_r = f_1$. Take

$$A = \frac{\tilde{A}}{\sqrt[r]{\det \tilde{A}}} \quad \text{and} \quad B = \frac{\tilde{B}}{\sqrt[r]{\det \tilde{B}}}\Bigg].$$

If $\underline{A}$ and $\underline{B}$ be the images of A and B in $PGL(r)$, then $\underline{A}$ and $\underline{B}$ commute in $PGL(r)$ and let $\rho_0 : \Gamma \to PGL(r)$ be the homomorphism such that $\rho_0(e_1) = \underline{A}$ and $\rho_0(e_2) = \underline{B}$. We verify easily that the obstruction class of ρ_0 is ψ.

Proof of Proposition 2. 1. Let $\xi \in H^2(\Gamma, \mu_r)$. Choose $\xi_i \in H^2(\Gamma_i, \mu_r)$ as in Lemma 2. 2. By Lemma 2. 3, we can find a homomorphism $\rho_i : \Gamma_i \to PGL(r)$ with $\delta(\rho_i) = \xi_i$. Let $\tilde{\rho}_i = \rho_i \circ p_i$ where $p_i : \Gamma \to \Gamma_i$ is the projection with respect to the decomposition $\Gamma = \oplus \Gamma_i$. Define $\rho : \Gamma \to PGL(r^g)$ to be $\tilde{\rho}_1 \otimes \cdots \otimes \tilde{\rho}_g$. Then ρ has the required property.

3. Proof of Theorem 1

Consider the Kummer exact sequence

$$1 \longrightarrow \mu_r \longrightarrow \mathcal{O}^* \xrightarrow{(.)^r} \mathcal{O}^* \longrightarrow 1$$

and the corresponding exact cohomology sequence

$$H^2(X, \mu_r) \longrightarrow H^2(X, \mathcal{O}^*) \xrightarrow{(.)^r} H^2(X, \mathcal{O}^*).$$

From this we see that every r-torsion element of $H^2(X, \mathcal{O}^*)$ is the image of an element in $H^2(X, \mu_r)$. Now X being a $K(\Gamma, 1)$ space, the natural homomorphism

$$H^2(\Gamma, \mu_r) \longrightarrow H^2(X, \mu_r)$$

is an isomorphism.

If χ is an r-torsion element in $H^2(X, \mathcal{O}^*)$, choose $\xi \in H^2(\Gamma, \mu_r)$ mapping into χ. Let ξ' be the image of ξ in $H^2(\Gamma, \mu_{r^g})$ and $\rho: \Gamma \to PGL(r^g)$ a homomorphism whose obstruction class is ξ' (such a ρ exists by Proposition 2. 2). Let P_ρ be the flat holomorphic projective bundle associated to ρ. Using the commutativity of the diagram

$$\begin{array}{ccc} H^2(\Gamma, \mu_r) & \longrightarrow & H^2(X, \mathcal{O}^*) \\ & \searrow \quad \nearrow & \\ & H^2(\Gamma, \mu_{r^g}) & \end{array}$$

we see that the obstruction class of P_ρ is χ.

4. An application

Proposition 4. 1. *On any complex torus of complex dimension $\geqq 2$, there exists a holomorphic projective bundle which does not arise from a holomorphic vector bundle.*

Proof. Applying Theorem 1, it suffices to show that $H^2(X, \mathcal{O}^*)_{tors} \neq 1$. We see, using the Kummer sequence, that we have an exact sequence

$$1 \to \frac{NS(X)}{r \cdot NS(X)} \to H^2(X, \mu_r) \to H^2(X, \mathcal{O}^*)_{r\text{-}tors} \to 1,$$

where $NS(X)$ is the Neron-Severi group of X. On the other hand we have

1) $$\# \frac{NS(X)}{r \cdot NS(X)} = r^s,$$

where s is the rank of the Neron-Severi group and

2) $$\# H^2(X, \mu_r) = r^{\binom{2g}{2}}.$$

Since $s \leqq h^{1,1}(X) = g^2$ we have $\binom{2g}{2} > g^2 \geqq s$ für $g \geqq 2$.

Hence $H^2(X, \mathcal{O}^*)_{r\text{-}tors} \neq 1$ for $r > 1, g \geqq 2$.

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Konvergenz formaler komplexer Räume mit konvexem oder konkavem Normalenbündel

Von *Siegmund Kosarew* in Regensburg

0. Einleitung

Diese Arbeit stellt einen Beitrag zu der allgemeinen Frage dar, wann ein formaler komplexer Raum $\mathfrak{X}$ konvergent ist. Dies bedeutet, daß es eine abgeschlossene Einbettung komplexer Räume $Y \hookrightarrow X$ gibt, so daß $\mathfrak{X}$ isomorph zu $\hat{X}$ ist, wobei $\hat{X}$ die formale Komplettierung von X bzgl. Y bezeichne. Die unterliegende Menge von $\mathfrak{X}$ wird hier stets als kompakt vorausgesetzt. Die bisher bekannten Beispiele formaler komplexer Mannigfaltigkeiten, die nicht konvergent sind, besitzen ein triviales Normalenbündel. Somit ist es naheliegend, nach Bedingungen an das Normalenbündel bzw. den Normalenraum zu suchen, unter denen dann die Konvergenz impliziert würde. Im Falle der Kodimension 1 wurde die Konvergenz in der algebraischen Situation von Hironaka [15] bei amplen Normalenbündel bewiesen und von Artin in [6] bei amplem Konormalenbündel. Die Konstruktion von Hironaka liefert einen kanonischen Repräsentanten für die Algebraisierung. Seine Methoden lassen sich leider auf den Fall der Kodimension >1 nicht ohne weiteres übertragen. Bei Artin folgt die Konvergenz durch Anwendung seiner Algebraisierungssätze für isolierte Singularitäten und formale Modifikationen. Diese Theorie ist nun im analytischen Fall von Ancona/Tomassini in [2] und Bingener in [9] entwickelt worden. Mit Hilfe ihrer Ergebnisse folgt dann der entsprechende Analytisierungssatz bei schwach negativem Normalenraum unter einer gewissen Glattheitsvoraussetzung, auf die allerdings nicht verzichtet werden kann. Dies ist in Paragraph 5 enthalten.

Der Fall, wo das Normalenbündel in einem geeigneten Sinne positiv oder konkav ist, erfordert dagegen eine völlig andere Vorgehensweise. Die hier verwendeten Hilfsmittel sind Methoden der Deformationstheorie. Als entscheidend erweist sich, daß man eine abgeschlossene Einbettung $Y \hookrightarrow X$ formaler komplexer Räume unter gewissen Voraussetzungen stets in die Nullschnitteinbettung $Y \hookrightarrow N_{Y|X}$ des Normalenbündels deformieren kann.

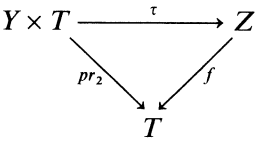
Ferner wird noch das Hauptresultat von [18] benutzt, sowie Aussagen über einige Kohomologiegruppen auf Vektorraumbündel, die sich in Paragraph 3 finden. Man erhält schließlich eine Konvergenzaussage im glatten Fall, aber bei beliebiger Kodimension (4. 9). Die zum Beweis benutzten Hilfsmittel sind zum Teil etwas allgemeiner formuliert bzw. bewiesen und sind manchmal unabhängig von Interesse.

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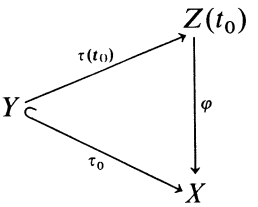
Meinen Dank möchte ich noch Herrn Prof. Dr. J. Bingener und Herrn Dipl.-math. H. Stieber aussprechen, die durch vielerlei Bemerkungen und Ratschläge zur endgültigen Form dieser Arbeit mit beigetragen haben.

§ 1. Deformation holomorpher und formaler Einbettungen

1. 1. Definition. Sei $\tau_0 : Y \hookrightarrow X$ eine abgeschlossene holomorphe Einbettung komplexer Räume, T ein komplexer Raumkeim mit ausgezeichnetem Punkt t_0. Unter einer Deformation von τ_0 über T versteht man ein kommutatives Diagramm von Keimen komplexer Räume



wobei f flach und $\tau(t_0) : Y \hookrightarrow Z(t_0)$ eine abgeschlossene Einbettung ist, welche zu τ_0 isomorph ist, d.h. es gibt ein kommutatives Diagramm



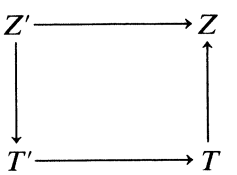
mit einem Isomorphismus φ.

1. 2. Bezeichnung. Sei $\tau_0 : Y \hookrightarrow \mathfrak{X}$ eine abgeschlossene Einbettung des komplexen Raumes Y in den formalen komplexen Raum $\mathfrak{X}$. Ist τ_0 durch ein Definitionsideal von $\mathfrak{X}$ gegeben, so nennen wir τ_0 kurz eine *formale* Einbettung.

1. 3. Bemerkungen. i) Sei X parakompakt — was im folgenden immer vorausgesetzt wird. Es gibt dann stets einen Repräsentanten von τ, der eine abgeschlossene Einbettung ist (siehe [18] I. 1. 2).

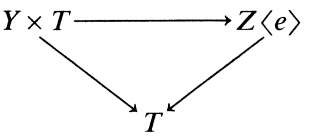
ii) Der entsprechende Deformationsbegriff kann für jede abgeschlossene Einbettung $\tau_0 : Y \hookrightarrow X$ von formalen komplexen Räumen — insbesondere für formale Einbettungen — formuliert werden.

iii) Bezeichnet man für zwei obige Diagramme $\mathscr{V} = (\tau, f, Z)$, $\mathscr{V}' = (\tau', f', Z')$ mit Hom $(\mathscr{V}, \mathscr{V}')$ die Menge aller mit τ, τ' und φ, φ' verträglichen kartesischen Diagramme



so erhält man auf natürliche Weise ein Gruppoid F über der Kategorie der komplexen Raumkeime.

iv) Jede lokal triviale Deformation (τ, f, Z) von $\tau_0\colon Y \hookrightarrow X$ induziert für jedes $e \in \mathbb{N} \cup \{\infty\}$ auf natürliche Weise eine lokal triviale Deformation von $(\tau_0)_e\colon Y \hookrightarrow X_e$ der Einbettung in die e-te infinitesimale Umgebung von Y in X:



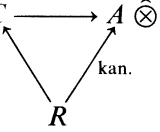
Diese Konstruktion ist mit Basiswechsel verträglich. X_∞ ist dabei mit $\hat{X}$ zu identifizieren.

v) Sei X eine komplexe Mannigfaltigkeit. Dann ist jede Deformation von $\tau_0\colon Y \hookrightarrow X$ lokal trivial.

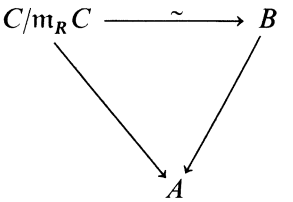
Die Aussagen i bis iv verifiziert man sofort. Wir beweisen hier nur v. Dies ergibt sich aus dem folgenden Lemma.

1. 4. Lemma. *Seien A, B, C, R analytische Algebren. Es gebe exakte Sequenzen*

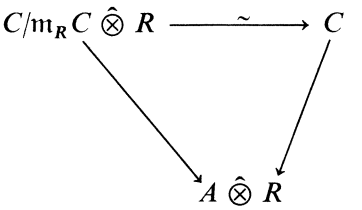
$$0 \to \mathfrak{a} \to B \to A \to 0, \qquad C \to A \hat{\otimes} R \to 0$$



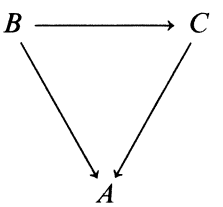
wobei $R \to C$ flach sei. Ferner existiere ein kommutatives Diagramm



Behauptung. *Ist B eine glatte analytische Algebra, so gibt es ein kommutatives Diagramm von R-Algebrenhomomorphismen*



Beweis. Da B glatt ist, kann man eine Liftung $B \to C$ von $B \xrightarrow{\sim} C/\mathfrak{m}_R C$ finden. Das Diagramm



kommutiert dann automatisch. Der induzierte R-Algebrenhomomorphismus $B \hat{\otimes} R \to C$ ist dann ein Isomorphismus, da dies nach Komplettieren bzgl. $\mathfrak{m}_R$ der Fall ist. Ferner ist offenbar nach Konstruktion das Diagramm in der Behauptung des Lemmas kommutativ.

Sei $\tau: Y \hookrightarrow X$ eine abgeschlossene Einbettung formaler komplexer Räume. Mit $\mathrm{Def}(\tau, T)$ bezeichnen wir die Menge der Isomorphieklassen von lokal trivialen Deformationen von τ über dem komplexen Raumkeim T. $\mathfrak{G}_T$ sei die Garbe von Gruppen auf X, deren Schnitte über $U \subset X$ offen, gerade aus denjenigen T-Automorphismen $U \times T \to U \times T$ bestehen, die auf $(U \cap Y) \times T$ und auf $U \times \{t_0\}$ die Identität induzieren.

In Analogie zu [11] 5. 5 hat man den folgenden Satz:

1. 5. Satz. *Mit obigen Bezeichnungen gilt: Es gibt eine kanonische Bijektion*

$$\mathrm{Def}(\tau, T) \to H^1(X, \mathfrak{G}_T)$$

von Mengen mit ausgezeichnetem Punkt.

Diese Bijektion ist in T funktoriell.

Wir wenden uns nun der Beschreibung von $\mathfrak{G}_{p_1}$ zu, wobei $p_1 = (0, \mathbb{C}[t]/(t^2))$ den Doppelpunkt bezeichne. Es gilt die Aussage:

1. 6. Lemma. *Sei $\tau: Y \hookrightarrow X$ eine abgeschlossene Einbettung formaler komplexer Räume, $\mathscr{I} := \mathscr{K}er\,(\mathscr{O}_X \to \tau_* \mathscr{O}_Y)$. Dann ist*

$$\mathfrak{G}_{p_1} \cong \mathscr{H}om_{\mathscr{O}_X}(\Omega_X, \mathscr{I}).$$

Ersetzt man τ durch $\tau_e: Y \hookrightarrow X_e$, $e \in \mathbb{N}$, und bezeichnet man die entsprechende Garbe von Gruppen auf X_e mit $\mathfrak{G}_{p_1}\langle e\rangle$, so ist

$$\mathfrak{G}_{p_1}\langle e\rangle \cong \mathscr{H}om_{\mathscr{O}_X}(\Omega_X, \mathscr{I}/\mathscr{I}^{e+1}).$$

Beweis. Es gilt:

$$\mathfrak{G}_{p_1} \subset \{g \in \mathscr{H}om_{\mathbb{C}\text{-Alg.}}(\mathscr{O}_X, \mathscr{O}_X[\varepsilon]) : g(a) = a + d(a)\varepsilon\} \cong \mathscr{H}om_{\mathscr{O}_X}(\Omega_X, \mathscr{O}_X).$$

Bei dieser Inklusion gehen die Elemente von $\mathfrak{G}_{p_1}$ genau auf den Untermodul $\mathscr{H}om_{\mathscr{O}_X}(\Omega_X, \mathscr{I})$. Der zweite Teil der Behauptung folgt mit Hilfe des ersten unter Benutzung der Tatsache, daß $\mathscr{I}/\mathscr{I}^{e+1}$ von $\mathscr{I}^e$ annuliert wird.

1. 7. Lemma. *Sei $\tau: Y \hookrightarrow X$ eine abgeschlossene Einbettung des komplexen Raumes Y in die komplexe Mannigfaltigkeit X und T ein komplexer Raumkeim. Für $e \in \mathbb{N}$ bezeichne $\mathfrak{H}_T\langle e\rangle \subset \mathfrak{G}_T$ den Normalteiler derjenigen Automorphismen in $\mathfrak{G}_T$, welche auf $X_e \times T$ die Identität induzieren. $i_e: X_e \hookrightarrow X$ sei die Inklusion, $\mathscr{I} := \mathscr{K}er\,(\mathscr{O}_X \to \tau_* \mathscr{O}_Y)$.*

Behauptung. *Die kanonische Sequenz*

$$1 \to \mathfrak{H}_T\langle e\rangle \to \mathfrak{G}_T \to (i_e)_* \mathfrak{G}_T\langle e\rangle \to 1$$

ist exakt.

Beweis. Es ist nur die Exaktheit an der vorletzten Stelle einzusehen. Dazu kann man annehmen, daß X eine offene Steinsche Menge im $\mathbb{C}^n$ ist.

Ein $g \in \Gamma(X_e, \mathfrak{G}_T\langle e\rangle)$ wird dann auf eindeutige Weise durch ein Element

$$G \in \Gamma\left(X_e \times T, (\mathscr{I}/\mathscr{I}^{e+1}) \hat{\otimes}_{\mathbb{C}} \mathfrak{m}_T\right)^n$$

gegeben. Zu G existiert aber nun ein Urbild in $\Gamma(X \times T, \mathscr{I} \hat{\otimes}_{\mathbb{C}} \mathfrak{m}_T)^n$, welches umgekehrt ein $\tilde{g} \in \Gamma(X, \mathfrak{G}_T)$ definiert mit Bild g in $\Gamma(X_e, \mathfrak{G}_T\langle e\rangle)$.

1. 8. Bemerkung. Die Aussage des Lemmas bleibt gültig, falls $\tau\colon Y \hookrightarrow \mathfrak{X}$ eine formale Einbettung des komplexen Raumes Y in die formale komplexe Mannigfaltigkeit $\mathfrak{X}$ ist.

1. 9. Proposition. *Sei $\tau\colon Y \hookrightarrow X$ eine abgeschlossene Einbettung komplexer Räume, $\mathscr{I} := \mathscr{K}er(\mathscr{O}_X \to \tau_* \mathscr{O}_Y)$, $T \hookrightarrow T'$ eine kleine Erweiterung komplexer Raumkeime, d.h. für $K := \operatorname{Ker}(\mathscr{O}_{T'} \to \mathscr{O}_T)$ gilt $\mathfrak{m}_{T'} K = 0$.*

Behauptung. *Ist X glatt, so hat man für jedes $e \in \mathbb{N}$ ein kommutatives Diagramm mit exakten Zeilen*

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \mathscr{I}\Theta_X \otimes_{\mathbb{C}} K & \longrightarrow & \mathfrak{G}_{T'} & \longrightarrow & \mathfrak{G}_T & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \mathscr{I}\hat{\Theta}_X \otimes_{\mathbb{C}} K & \longrightarrow & \mathfrak{G}_{T'}\langle\infty\rangle & \longrightarrow & \mathfrak{G}_T\langle\infty\rangle & \longrightarrow & 1 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & (\mathscr{I}\Theta_X/\mathscr{I}^{e+1}\Theta_X) \otimes_{\mathbb{C}} K & \longrightarrow & \mathfrak{G}_{T'}\langle e\rangle & \longrightarrow & \mathfrak{G}_T\langle e\rangle & \longrightarrow & 1,
\end{array}$$

wobei die Kerne jeweils in den Zentren enthalten sind.

Der *Beweis* kann entsprechend wie in [11] 5. 7, 5. 9 geführt werden.

1. 10. Proposition. *Sei $\tau\colon Y \hookrightarrow X$ eine abgeschlossene Einbettung des komplexen Raumes Y in die komplexe Mannigfaltigkeit X, $\mathscr{I} := \mathscr{K}er(\mathscr{O}_X \to \tau_* \mathscr{O}_Y)$.*

$$\rho_i\colon H^i(X, \mathscr{I}\Theta_X) \to H^i(\hat{X}, \mathscr{I}\hat{\Theta}_X)$$

bezeichne den natürlichen Morphismus. $\mathscr{C}$ sei die Kategorie der lokalen artinschen $\mathbb{C}$-Algebren.

Behauptung. i) *Sei ρ_1 surjektiv, ρ_2 injektiv. Dann ist der Morphismus von Funktoren*

$$\operatorname{Def}(\tau, -) \to \operatorname{Def}(\hat{\tau}, -)$$

glatt.

ii) *Ist ρ_1 zusätzlich bijektiv, so ist obiger Funktormorphismus glatt und unverzweigt.*

Dabei werden $\operatorname{Def}(\tau, -)$ *bzw.* $\operatorname{Def}(\hat{\tau}, -)$ *als Funktoren von $\mathscr{C}$ nach der Kategorie der Mengen aufgefaßt.*

Der *Beweis* ergibt sich unmittelbar aus der Kohomologiesequenz für das Diagramm in 1. 9 unter Benutzung von 1. 5.

Die Proposition 1. 10 ist ein Spezialfall einer allgemeineren Aussage, zu deren Formulierung wir die Bezeichnungen von [8] § 1 und [20] 2. 1 übernehmen:

1. 11. Proposition. *Sei* (S, s_0) *ein komplexer Raumkeim,* $(An/(S, s_0))$ *die Kategorie der komplexen Raumkeime über dem Raumkeim* (S, s_0). *Gegeben sei folgendes Diagramm von Gruppoiden über* $(An/(S, s_0))$

$$\begin{array}{ccc} F & \xrightarrow{\quad f \quad} & G \\ & \searrow \quad \swarrow & \\ & (An/(S, s_0)) & \end{array}$$

Für F und G sei die Bedingung (S1) *erfüllt und es gebe Obstruktionstheorien* O^F, O^G *für F bzw. G, die mit dem Funktor f verträglich sind.*

Behauptung. *Sind für jede kleine Erweiterung artinscher Raumkeime* $T \hookrightarrow T'$ *in* $(An/(S, s_0))$ *mit* $K := \mathscr{K}er\,(\mathcal{O}_{T'} \to \mathcal{O}_T)$ *die Bedingungen erfüllt:*

$$D^F_{a_0}(K) \longrightarrow D^G_{f(a_0)}(K) \text{ surjektiv},$$

$$O^F_a(K) \longrightarrow O^G_{f(a)}(K) \text{ injektiv}$$

für jedes $a_0 \in F(p_0)$ *und jedes* $a \in F(T)$ *mit* $a|_{p_0} = a_0$, *so ist f formal glatt.*

Ist darüberhinaus $D^F_{a_0}(K) \to D^G_{f(a_0)}(K)$ *bijektiv, so ist f minimal formal glatt.*

Der *Beweis* dieser Behauptung ist eine einfache Konsequenz der Definition einer Obstruktionstheorie und der Hindernismoduln.

1. 12. Bemerkung. Besitzt in der Situation von Proposition 1. 10 τ eine formal semiuniverselle Deformation $\mathscr{V} \to S$, so ist im Falle i die unduzierte Deformation $\mathscr{V}\langle\infty\rangle \to S$ von $\hat{\tau}$ eine formal verselle Deformation, im Fall ii eine formal semiuniverselle Deformation von $\hat{\tau}$.

§ 2. Deformationen von Einbettungen in ihr Normalenbündel

2. 0. Wir betrachten zunächst folgende allgemeine Situation: Sei k ein algebraisch abgeschlossener Körper, A eine kommutative k-Algebra und $\mathfrak{a} \subset A$ ein Ideal. Dann heißt $R_{\mathfrak{a}}(A) := A[\mathfrak{a}u^{-1}, u]$ die Rees-Algebra von A bzgl. des Ideals $\mathfrak{a}$. Hierbei ist u eine Unbestimmte. Sei $r \in \mathbb{N}$.

Wir setzen

$$(\mathfrak{a}^r)^* := \{\sum_{i \in \mathbb{Z}} a_i u^i \in R_{\mathfrak{a}}(A) : a_i \in \mathfrak{a}^r, \forall\, i \in \mathbb{Z}\}.$$

$(\mathfrak{a}^r)^* \subset R_{\mathfrak{a}}(A)$ ist ein Ideal und es gilt insbesondere

$$\mathfrak{a}^* = \mathfrak{a} R_{\mathfrak{a}}(A) + \mathfrak{a}u^{-1} R_{\mathfrak{a}}(A)$$

sowie $(\mathfrak{a}^*)^r = (\mathfrak{a}^r)^*$ für jedes $r \in \mathbb{N}$.

Im folgenden Lemma stellen wir die für uns wichtigen Eigenschaften von $R_{\mathfrak{a}}(A)$ dar.

2. 1. Lemma. *Mit den obigen Bezeichnungen gilt*:

i) $R_{\mathfrak{a}}(A)$ *und* $R_{\mathfrak{a}}(A)/(\mathfrak{a}^r)^*$ *sind flache* $k[u]$*-Moduln.*

ii) *Es gibt eine natürliche Surjektion vom Grad null*

$$R_{\mathfrak{a}}(A) \to A/\mathfrak{a}[u].$$

iii) $(\mathfrak{a}^r)^*/u(\mathfrak{a}^r)^* \cong \coprod_{k \geqq r} (\mathfrak{a}^k/\mathfrak{a}^{k+1}) u^{-k}$.

iv) *Sei* $a \in A^*$, *dann folgt*:

$$(\mathfrak{a}^r)^*/(u-a)\,(\mathfrak{a}^r)^* \cong \mathfrak{a}^r.$$

v) $(\mathfrak{a}^r)^*_u = \mathfrak{a}^r[u^{-1}, u] = \mathfrak{a}^r \otimes_k k[u]_u$.

vi) *Ist* A *noethersch und* $\hat{A} := \hat{A}^{\mathfrak{a}}$, *so gilt* $R_{\mathfrak{a}}(\hat{A}) = R_{\mathfrak{a}}(A) \otimes_A \hat{A}$.

Beweis. Zu i. Für jedes $a \in A$ ist $u-a$ ein Nichtnullteiler in $R_{\mathfrak{a}}(A)$ und $R_{\mathfrak{a}}(A)/(\mathfrak{a}^r)^*$. Da $k[u]$ ein Dedekindring ist und jedes maximale Ideal von $k[u]$ von einem Element der Form $u-c$, $c \in k$, erzeugt wird, folgt aus dem Lokal-Global-Prinzip die Aussage i.

Zu ii. Die Abbildung

$$R_{\mathfrak{a}}(A) \to A/\mathfrak{a}[u], \quad \sum_{i \in \mathbb{Z}} a_i u^i \mapsto \sum_{i \in \mathbb{N}} \bar{a}_i u^i$$

ist ein Epimorphismus von $A[u]$-Algebren, wie man leicht nachrechnet.

Ebenso bestätigt man sofort iii.

Zu iv betrachte man die Komposition der natürlichen Abbildungen

$$\mathfrak{a}^r \xrightarrow{\alpha} (\mathfrak{a}^r)^*/(u-a)\,(\mathfrak{a}^r)^* \xrightarrow{\beta} \mathfrak{a}^r[u^{-1}, u]/(u-a)\,\mathfrak{a}^r[u^{-1}, u] \xrightarrow{\gamma} \mathfrak{a}^r,$$

wobei α homogen vom Grad null ist, β durch die Inklusion $(\mathfrak{a}^r)^* \subset \mathfrak{a}^r[u^{-1}, u]$ und γ durch die Zuordnung $u \mapsto a$ gegeben ist. Offenbar ist γ bijektiv und $\gamma \circ \beta \circ \alpha = \mathrm{id}$. Aus der Injektivität von β, die man sofort verifiziert, folgt iv.

Die Aussage v ist klar, da $\mathfrak{a}^r[u] \subset (\mathfrak{a}^r)^* \subset \mathfrak{a}^r[u^{-1}, u]$.

Der Teil vi ist trivial.

2. 2. Sei nun X ein komplexer Raum, $\mathscr{E}$ ein kohärenter $\mathcal{O}_X$-Modul,

$$\mathbb{V}(\mathscr{E}) = \mathrm{Specan}\,(S^{\cdot}_{\mathcal{O}_X}(\mathscr{E}))$$

der zu $\mathscr{E}$ assoziierte lineare Faserraum über X. $\pi: \mathbb{V}(\mathscr{E}) \to X$ bezeichne die kanonische Projektion, $j: X \hookrightarrow \mathbb{V}(\mathscr{E})$ den Nullschnitt und $\mathscr{J} \subset \mathcal{O}_{\mathbb{V}(\mathscr{E})}$ dessen definierendes Ideal. Jedem $S^{\cdot}(\mathscr{E})$-Modul $\mathscr{M}$, der lokal von endlicher Präsentation ist, kann man auf natürliche Weise einen kohärenten $\mathcal{O}_{\mathbb{V}(\mathscr{E})}$-Modul $\mathscr{M}^{\sim}$ zuordnen (vgl. [7] § 1). Diese Zuordnung ist ein exakter Funktor, der mit der Bildung von Tensorprodukten verträglich ist.

Sei $S^+(\mathscr{E}) := \coprod_{k \geqq 1} S^k(\mathscr{E})$. Dann ist $S^+(\mathscr{E})$ ein kohärentes Ideal in $S^\bullet(\mathscr{E})$. Man verifiziert jetzt leicht die folgende Aussage:

2. 3. Lemma. *Sei* $\mathscr{G} \in \mathrm{Coh}(X)$. *Dann gilt*:

i) $(S^+(\mathscr{E})^k)^\sim = \mathscr{J}^k$, *für alle* $k \in \mathbb{N}$.

ii) $(S^\bullet(\mathscr{E})/S^+(\mathscr{E})^k)^\sim = \mathscr{O}_{V(\mathscr{E})}/\mathscr{J}^k \cong \coprod_{i=0}^{k-1} \mathscr{J}^i/\mathscr{J}^{i+1}$.

iii) $\mathscr{J}^k/\mathscr{J}^{k+1} \cong S^k(\mathscr{E})^\sim$.

iv) $\left(\coprod_{i=0}^{k-1} S^i(\mathscr{E}) \otimes_{\mathscr{O}_X} \mathscr{G}\right)^\sim = \pi^*\mathscr{G}/\mathscr{J}^k\pi^*\mathscr{G}$.

Man vergleiche auch in diesem Zusammenhang [1].

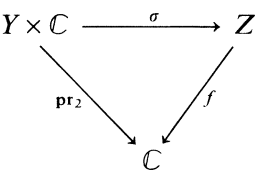
Wir kommen jetzt zu einer für uns später sehr wichtigen Aussage. Dazu sei $\tau\colon Y \hookrightarrow X$ eine abgeschlossene Einbettung formaler komplexer Räume und

$$N_{Y|X} = \mathrm{Specan}\,(S^\bullet_{\mathscr{O}_Y}(\mathscr{J}/\mathscr{J}^2))$$

der Normalenraum von $Y \hookrightarrow X$, wobei $\mathscr{J} := \mathscr{K}er\,(\mathscr{O}_X \to \tau_*\mathscr{O}_Y)$ sei. $j\colon Y \hookrightarrow N_{Y|X}$ bezeichne den Nullschnitt.

2. 4. Satz. *Es sei* $Y \subset X$ *lokal vollständiger Durchschnitt. Dann existiert ein kommutatives Diagramm formaler komplexer Räume*

(*)

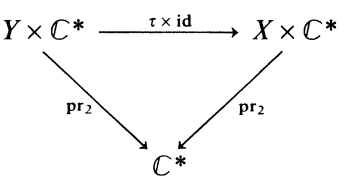


mit folgenden Eigenschaften:

σ ist eine abgeschlossene Einbettung und f ist flach.

$\sigma(0)\colon Y \to Z(0)$ ist isomorph zu j.

Über $\mathbb{C}^*$ ist das Diagramm (*) *isomorph zu*



Ist Y eine komplexe Mannigfaltigkeit, X eine formale komplexe Mannigfaltigkeit und $\tau\colon Y \hookrightarrow X$ eine formale Einbettung, so induziert das Diagramm (*) *eine lokal triviale Deformation von $\hat{j}\colon Y \hookrightarrow \hat{N}_{Y|X}$ mit allgemeiner Faser $Y \hookrightarrow X$.*

Beweis. Wir setzen $Z := \mathrm{Specan}\,(R_{\mathscr{I}}(\mathscr{O}_X))$. Dann folgt die Behauptung des Satzes aus dem Lemma 2. 1. Falls Y und X glatt sind und $Y = |X|$, so ist *lokal* $Y \hookrightarrow X$ isomorph zu $Y \hookrightarrow \hat{N}_{Y|X}$ und man wende jetzt 2. 1 vi sowie 1. 3 v an. Daraus ergibt sich dann die lokale Trivialität (nach Komplettieren von $Z(0)$ bzgl. des Nullschnitts).

§ 3. Kohomologie in Umgebung des Nullschnitts von linearen Faserräumen

In diesem Abschnitt wollen wir einen Verschwindungs- bzw. Vergleichssatz für gewisse kohärente Modulgarben auf linearen Faserräumen beweisen. Spezialfälle hierfür finden sich in [4].

3. 0. Sei X ein komplexer Raum und $\pi: E \to X$ ein linearer Faserraum über X. $j: X \hookrightarrow E$ bezeichne den Nullschnitt, der durch das kohärente Ideal $\mathscr{J} \subset \mathscr{O}_E$ gegeben sei. Ferner sei $\mathscr{G}$ ein $\mathscr{O}_X$-Modul. Abkürzend setzen wir

$$\mathscr{F} := \pi^* \mathscr{G}, \quad \mathscr{F}_k := \mathscr{F}/\mathscr{J}^{k+1}\mathscr{F} \quad \text{für} \quad k \in \mathbb{N}.$$

3. 1. Lemma. *Die exakten Sequenzen von $j^{-1}\mathscr{O}_E$-Moduln*

$$0 \longrightarrow j^{-1}(\mathscr{J}^k \mathscr{F}) \longrightarrow j^{-1}(\mathscr{F}) \longrightarrow j^{-1}(\mathscr{F}_{k-1}) \longrightarrow 0,$$

$$0 \longrightarrow j^{-1}(\mathscr{J}^k \mathscr{F}/\mathscr{J}^{k+1}\mathscr{F}) \longrightarrow j^{-1}(\mathscr{F}_k) \longrightarrow j^{-1}(\mathscr{F}_{k-1}) \longrightarrow 0$$

spalten auf natürliche Weise als Sequenzen von $\mathscr{O}_X$-Moduln für jedes $k \geqq 1$. Ferner gilt

$$\mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{J}^k) \cong j^{-1}(\mathscr{J}^k \mathscr{F}),$$
$$\mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{O}_{E_k}) \cong j^{-1}(\mathscr{F}_k).$$

Dabei sei E_k die k-te infinitesimale Umgebung von X in E.

Beweis. Für $\mathscr{G} = \mathscr{O}_X$ findet sich obige Aussage in [16] I. 5. 1. 1. Es ist nun

$$j^{-1}(\mathscr{F}) = j^{-1}(\pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_X} \mathscr{O}_E) = \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}\mathscr{O}_E \text{ und ebenso } j^{-1}(\mathscr{F}_k) = \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{O}_{E_k}).$$

Die Behauptung ergibt sich dann durch Tensorieren der entsprechenden Sequenzen für $\mathscr{F} = \mathscr{O}_E$ mit $\mathscr{G}$ über $\mathscr{O}_X$.

3. 2. Corollar. *Für $k \in \mathbb{N}$ gilt:*

$$j^{-1}(\mathscr{J}^k \hat{\mathscr{F}}) \cong \prod_{i \geqq k} \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{J}^i/\mathscr{J}^{i+1}),$$
$$j^{-1}(\mathscr{F}_k) \cong \coprod_{i=0}^{k} \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{J}^i/\mathscr{J}^{i+1}).$$

Für jedes $r \in \mathbb{N}$ sind die Sequenzen

$$0 \to H^r(X, j^{-1}(\mathscr{J}^k \mathscr{F})) \to H^r(X, j^{-1}(\mathscr{F})) \to H^r(X, \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{O}_{E_{k-1}})) \to 0$$

und

$$0 \to H^r(X, \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{J}^k/\mathscr{J}^{k+1})) \to H^r(X, \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{O}_{E_k})) \to H^r(X, \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{O}_{E_{k-1}})) \to 0$$

exakt und spalten auf.

3. 3. Sei X ein komplexer Raum. Dann kann man auf natürliche Weise zu X einen differenzierbaren Raum $(X, \mathscr{E}_X)$ und einen stetigen Raum $(X, \mathscr{C}_X)$ konstruieren, sowie einen Morphismus $\mathbb{C}$-geringter Räume $(X, \mathscr{C}_X) \to (X, \mathscr{E}_X)$. Die Garben $\mathscr{E}_X$ und $\mathscr{C}_X$ auf X sind fein. Ist $X \subset U$ abgeschlossen, $U \subset \mathbb{C}^n$ offen und $\mathscr{O}_X = \mathscr{O}_U/\mathscr{I}|_X$, so gilt $\mathscr{E}_X = (\mathscr{E}_U/(\operatorname{Re}\mathscr{I}, \operatorname{Im}\mathscr{I})\,\mathscr{E}_U)|_X$ und $\mathscr{C}_X = (\mathscr{C}_U/(\operatorname{Re}\mathscr{I}, \operatorname{Im}\mathscr{I})\,\mathscr{C}_U)|_X$. Dabei sei $\mathscr{E}$ bzw. $\mathscr{C}$ die Garbe der differenzierbaren bzw. stetigen *reellwertigen* Funktionen auf U.

Zu jedem $\varphi \in \Gamma(X, \mathscr{E}_X)$ und $x \in X$ hat man auf dem holomorphen Tangentialraum die Leviform $\operatorname{Lev}_x(\varphi)\colon T_{X,x} \times T_{X,x} \to \mathbb{C}$ zu φ im Punkt x.

3. 4. Definition. Sei X ein komplexer Raum, $\varphi \in \Gamma(X, \mathscr{C}_X)$ ein Schnitt und $p, q \in \mathbb{N} \cup \{\infty\}$.

φ heißt vom Typ $\{p, q\}$ auf X, falls es ein $\tilde{\varphi} \in \Gamma(X, \mathscr{E}_X)$ gibt, dessen Bild unter der kanonischen Abbildung $\Gamma(X, \mathscr{E}_X) \to \Gamma(X, \mathscr{C}_X)$ φ ist, und $\operatorname{Lev}_x(\tilde{\varphi})$ mindestens $N_x - p + 1$ positive sowie mindestens $N_x - q + 1$ negative Eigenwerte besitzt für jedes $x \in X$. Dabei sei $N_x := \dim_{\mathbb{C}} T_{X,x}$.

Im Falle $q = \infty$ (bzw. $p = \infty$) sagt man, daß φ p-konvex (bzw. q-konkav) ist.

3. 5. Definition. Sei $E \to X$ ein linearer Faserraum über dem kompakten komplexen Raum X und $p, q \in \mathbb{N} \cup \{\infty\}$. E heißt als linearer Faserraum vom Typ $\{p, q\}$, falls es eine offene Umgebung W des Nullschnitts in E gibt und eine Ausschöpfungsfunktion $\varphi \in \Gamma(W, \mathscr{C}_E)$ von W mit den Eigenschaften:

i) $\varphi(w) \in \mathbb{R}_+$ für alle $w \in W$.

ii) $X = \varphi^{-1}(0)$.

iii) φ ist auf $W \setminus X$ vom Typ $\{p, q\}$.

Wir wollen nun den folgenden Satz beweisen.

3. 6. Satz. *Sei X ein kompakter komplexer Raum, $\pi\colon E \to X$ ein linearer Faserraum vom Typ $\{p, q\}$ mit ausgezeichneter Umgebung W des Nullschnitts und Ausschöpfungsfunktion $\varphi \in \Gamma(W, \mathscr{C}_E)$. Ferner sei $\mathscr{G} \in \operatorname{Coh}(X)$, $\mathscr{F} := \pi^*\mathscr{G}$ und $\mathscr{J} \subset \mathscr{O}_E$ das Ideal des Nullschnitts von E.*

Behauptung. i) $H^r\big(X, \mathscr{G} \otimes_{\mathscr{O}_X} j^{-1}(\mathscr{J}^k/\mathscr{J}^{k+1})\big) = 0$, *für alle* $k \gg 0$ *und* $r \geqq p$ *oder* $r < \operatorname{prof}_{W \setminus X} \mathscr{F} - q$.

ii) *Sei $k \in \mathbb{N}$. Für jedes $\alpha \in \mathbb{N}$ ist die natürliche Abbildung*

$$H^r(W, \mathscr{J}^{k+\alpha}\mathscr{F}) \to H^r(W, \mathscr{J}^k\mathscr{F})$$

injektiv für $r \geqq p$ oder $r \leqq \operatorname{prof}_{W \setminus X} \mathscr{F} - q$.

iii) $H^r(W, \mathscr{J}^k\mathscr{F}) = 0$ *für alle* $k \gg 0$ *und* $r \geqq p$ *oder* $r < \operatorname{prof}_{W \setminus X} \mathscr{F} - q$; *im Falle* $r = \operatorname{prof}_{W \setminus X} \mathscr{F} - q$ *gilt noch*

$$\bigcap_{\alpha \in \mathbb{N}} \operatorname{Im}\big(H^r(W, \mathscr{J}^{k+\alpha}\mathscr{F}) \to H^r(W, \mathscr{J}^k\mathscr{F})\big) = 0, \text{ für jedes } k \in \mathbb{N}.$$

iv) *Sei $k \in \mathbb{N}$. Die natürliche Abbildung*

$$H^r(W, \mathscr{J}^k\mathscr{F}) \to H^r(\hat{W}, \mathscr{J}^k\hat{\mathscr{F}})$$

ist bijektiv für $r \geqq p$ oder $r < \operatorname{prof}_{W \setminus X} \mathscr{F} - q$ und injektiv für $r = \operatorname{prof}_{W \setminus X} \mathscr{F} - q$.

Zum *Beweis* des obigen Satzes setzen wir:

$$W_c := \{w \in W : \varphi(w) < c\} \quad \text{für} \quad c \in \mathbb{R}_+ \cup \{\infty\},\ c > 0.$$

(W_c) ist ein Fundamentalsystem offener Umgebungen von $X \subset E$. Seien

$$c', c \in \mathbb{R}_+ \cup \{\infty\},\ 0 < c' < c \quad \text{und} \quad k \in \mathbb{N}.$$

Die Beschränkungsabbildung

$$H^r(W_c, \mathcal{J}^k\mathcal{F}) \to H^r(W_{c'}, \mathcal{J}^k\mathcal{F})$$

ist bijektiv für $r \geqq p$ oder $r < \operatorname{prof}_{W \setminus X}\mathcal{F} - q$ und injektiv für $r = \operatorname{prof}_{W \setminus X}\mathcal{F} - q$. Ferner ist $H^r(W, \mathcal{J}^k\mathcal{F})$ endlich dimensional für $r \geqq p$ oder $r < \operatorname{prof}_{W \setminus X}\mathcal{F} - q$ und noch separiert für $r = \operatorname{prof}_{W \setminus X}\mathcal{F} - q$ (vgl. [19]).

Die Aussagen i und ii folgen leicht mit Hilfe des Corollars 3. 2. Nehmen wir an, wir hätten schon gezeigt, daß für $k \in \mathbb{N}$ die Abbildung $\rho_r : H^r(W, \mathcal{J}^k\mathcal{F}) \to H^r(\hat{W}, \mathcal{J}^k\hat{\mathcal{F}})$ injektiv ist, falls $r \geqq p$ oder $r \leqq \operatorname{prof}_{W \setminus X}\mathcal{F} - q$.

Der Endlichkeitssatz zusammen mit ii liefert dann die Aussage iii, wie man leicht sieht.

Die Surjektivität der Abbildung ρ_r für $r \geqq p$ oder $r < \operatorname{prof}_{W \setminus X}\mathcal{F} - q$ ist eine einfache Konsequenz von ii und der exakten Kohomologiesequenz. Die Behauptung des Satzes ergibt sich nun aus folgender Proposition.

3. 7. Proposition. *Sei X ein kompakter komplexer Raum, $\mathcal{G} \in \operatorname{Coh}(X)$. E, $\mathcal{F}$ und $\mathcal{J}$ seien wie in* 3. 0. *Es gebe ein Fundamentalsystem offener Umgebungen $(W_c)_{c \in R}$ des Nullschnitts $X \subset E$, so daß jeweils $H^r(W_c, \mathcal{J}^k\mathcal{F})$ separiert ist für festes $r, k \in \mathbb{N}$.*

Behauptung. *Zu jeder offenen Umgebung W von X in E gibt es ein $c \in R$, so daß für das kommutative Diagramm*

$$\begin{array}{ccc} H^r(W, \mathcal{J}^k\mathcal{F}) & \xrightarrow{\alpha} & H^r(\hat{W}, \mathcal{J}^k\hat{\mathcal{F}}) \\ \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} \\ H^r(W_c, \mathcal{J}^k\mathcal{F}) & \longrightarrow & H^r(\hat{W}_c, \mathcal{J}^k\hat{\mathcal{F}}) \end{array}$$

die Inklusion $\operatorname{Ker}\alpha \subset \operatorname{Ker}\beta$ *erfüllt ist.*

Zur Durchführung des *Beweises* stellen wir zuerst die folgende lokale Betrachtung an: Sei $U \Subset \mathbb{C}^n$ offen, Steinsch, $X \subset U$ ein abgeschlossener komplexer Unterraum und $\mathcal{G} \in \operatorname{Coh}(X)$ mit einem Epimorphismus $\mathcal{O}_U^l \to \mathcal{G}$ über U. Ferner sei $E \subset X \times \mathbb{C}^m$ ein linearer Unterfaserraum, der durch $At = 0$ definiert sei mit $A \in M(m' \times m, \Gamma(U, \mathcal{O}_U))$ und $t = (t_1, \ldots, t_m)$. $\pi : E \to X$ bezeichne die kanonische Projektion und $\mathcal{J} \subset \mathcal{O}_E$ sei das Ideal des Nullschnitts j. Man hat dann durch $\mathcal{O}_U^l \to \mathcal{G}$ induzierte Epimorphismen

$$\varphi : \quad \mathfrak{m}^k \mathcal{O}^l_{U \times \mathbb{C}^m} \to i_*(\mathcal{J}^k\mathcal{F})$$

wobei $\mathcal{F} := \pi^*\mathcal{G}$, $i : E \hookrightarrow U \times \mathbb{C}^m$ die Inklusion sei und $\mathfrak{m} \subset \mathcal{O}_{\mathbb{C}^m}$ die Idealgarbe des Nullpunktes im $\mathbb{C}^m$. Die homogene Entwicklung in $j^{-1}(\mathcal{F})$ bzw. $j^{-1}(\mathcal{J}^k\mathcal{F})$ längs den Fasern von E (siehe 3. 1) ist durch die homogene Entwicklung in $\mathcal{O}^l_{U \times B}$ nach t induziert mit $B = (\mathbb{C}^m, 0)$. Sei $W \subset E$ offen. W heiße „standard" offene Menge von E, falls es eine

offene Steinsche Teilmenge $X' \Subset X$ und einen Polyzylinder $P \Subset \mathbb{C}^m$ mit dem Ursprung als Mittelpunkt gibt, so daß $W = E \cap (X' \times P)$. Für jedes solche W und $f \in \Gamma(W, \mathscr{J}^k\mathscr{F})$ hat man dann eine homogene Entwicklung $f = \sum_{e \geq k} f_e$ längs P. Sei $\tilde{V} \subset U \times \mathbb{C}^m$ offen und Steinsch. Für $f \in \Gamma(E \cap \tilde{V}, \mathscr{F})$ sei

$$|f|_{\tilde{V}} := \inf\{|F|_{\tilde{V}} : F \in \Gamma(\tilde{V}, \mathcal{O}^l_{U \times \mathbb{C}^m}),\ \varphi(F) = i_*(f)\}.$$

Hierbei bezeichne $|\ldots|_{\tilde{V}}$ die Supremumspseudonorm der Komponenten von $\Gamma(\tilde{V}, \mathcal{O}^l_{U \times \mathbb{C}^m})$.

Ist z.B. $U' \subset U$ offen und Steinsch und sind $P' \Subset P \Subset \mathbb{C}^m$ konzentrische Polyzylinder mit Multiradien $r' = \varepsilon r$, $0 < \varepsilon < 1$ und dem Mittelpunkt 0 und sei

$$Q := U' \times P,\ Q' := U' \times P',$$

so gilt für jedes $f = \sum_{e \in \mathbb{N}} f_e$ aus $\Gamma(E \cap Q, \mathscr{F})$

$$|f_e|_{Q'} \leqq \frac{1}{(2\pi)^m} \binom{m+e-1}{m-1} \varepsilon^e |f|_Q.$$

Sei umgekehrt $\hat{f} = \sum_{e \in \mathbb{N}} f_e$ aus $\Gamma(E \cap Q, \hat{\mathscr{F}})$ mit $|f_e|_Q < M$, so gilt: $\hat{f}$ ist auf Q' konvergent und

$$|\hat{f}|_{Q'} \leqq \frac{M}{(2\pi(1-\varepsilon))^m}.$$

Wir wählen nun eine feste endliche Kartenüberdeckung $u_i : X_i \hookrightarrow U_i \Subset \mathbb{C}^n$, $i \in I$, von X, so daß für E_{X_i} jeweils die obige Situation hergestellt sei. l, n und m können dabei simultan für alle $i \in I$ gewählt werden. Sämtliche im folgenden vorkommenden Normen und Abschätzungen beziehen sich auf diesen Atlas von X bzw. E.

Sei $\xi \in H^r(W, \mathscr{J}^k\mathscr{F})$ mit $\xi \in \operatorname{Ker}\alpha$ vorgegeben. Nach eventueller Verkleinerung von W als Umgebung von $X \subset E$ kann man annehmen, daß ξ durch einen beschränkten Cozyklus z auf einer Überdeckung von W repräsentiert ist, die nur aus standard offenen Mengen besteht. Auf W besitzt dann z die homogene Entwicklung $z = \sum_{e \geq k} z_e$ längs den Fasern von E.

Nach Schrumpfen $W' \Subset W$, $X \subset W'$, von W gibt es eine Konstante $K > 0$ mit $|z_e|_{W'} \leqq K$ für alle $e \in \mathbb{N}$. Da nach Voraussetzung der Cozyklus z formal zerfällt, d.h. $z = \delta\hat{c}$, $\hat{c} = \sum_{e \geq k} \hat{c}_e$, $\hat{c}_e$ homogen vom Grad e, zerfallen auch sämtliche homogene Komponenten $z_e = \delta\hat{c}_e$, da die homogene Entwicklung mit Beschränkung verträglich ist. Wegen der Separiertheitsvoraussetzung und dem Satz von Banach für offene Abbildungen von Fréchet-Räumen gibt es nach weiterem Schrumpfen von W', $X \subset W'' \Subset W'$ eine Konstante $L > 0$ und $c(e) \in C^{r-1}(W'', \mathscr{F})$ mit $z_e = \delta c(e)$ und $|c(e)|_{W''} \leqq L|z_e|_{W'}$.

Man kann o.B.d.A. annehmen, daß $c(e)$ bereits homogen vom Grade e ist. Setzen wir jetzt $c := \sum_{e \geq k} c(e)$, so ist c in einer kleinen Umgebung von X konvergent und dort gilt $z = \delta c$. Also ist $z \in \operatorname{Ker}\beta$ für ein geeignetes $c \in R$. Die Behauptung der Proposition 3. 7 ist damit bewiesen und somit auch der Satz 3. 6.

3. 8. Proposition. *Sei X eine kompakte Mannigfaltigkeit, $\pi : E \to X$ ein Vektorraumbündel vom Typ $\{p, q\}$ auf X. Dann bleiben die Aussagen* i, iii, iv *der Behauptung von* 3. 6 *richtig, falls man dort $\mathscr{F}$ durch Θ_E und $\mathscr{G}$ durch $j^*\Theta_E$ ersetzt, wobei wieder $j : X \hookrightarrow E$ der Nullschnitt sei.*

Beweis. Wir haben die exakte Sequenz von Vektorraumbündel auf E

$$0 \to \pi^* E \to T_E \to \pi^* T_X \to 0,$$

welche jedoch im allgemeinen nicht spaltet. Die Behauptung folgt jetzt leicht durch Anwenden der langen exakten Kohomologiesequenz und des Fünferlemmas auf die exakte Sequenz

$$0 \to \mathcal{J}^k \pi^* \mathcal{E} \to \mathcal{J}^k \Theta_E \to \mathcal{J}^k \pi^* \Theta_X \to 0$$

bzw. deren Komplettierung, mit Hilfe von 3. 6. Dabei sei $\mathcal{J} := \mathcal{K}er(\mathcal{O}_E \to j_* \mathcal{O}_X)$ und $\mathcal{E}$ die Garbe der holomorphen Schnitte von E.

§ 4. Konvergenz formaler Einbettungen mit konkavem Normalenbündel

4. 1. Proposition. *Seien $\tau: X \hookrightarrow \mathfrak{X}$ und $\sigma: X \hookrightarrow \mathfrak{Y}$ formale Einbettungen des komplexen Raumes X in die formalen komplexen Mannigfaltigkeiten $\mathfrak{X}$ und $\mathfrak{Y}$, $\mathcal{J} := \mathcal{K}er(\mathcal{O}_{\mathfrak{X}} \to \tau_* \mathcal{O}_X)$. $\mathfrak{X}_e$ bzw. $\mathfrak{Y}_e$ bezeichne die e-te infinitesimale Umgebung von X in $\mathfrak{X}$ bzw. $\mathfrak{Y}$. Sei $e \geqq 1$ und $f_e: \mathfrak{X}_e \to \mathfrak{Y}_e$ ein Isomorphismus zwischen den Einbettungen*

$$\tau_e: X \hookrightarrow \mathfrak{X}_e \quad \text{und} \quad \sigma_e: X \hookrightarrow \mathfrak{Y}_e,$$

d. h. $f_e \circ \tau_e = \sigma_e$.

Behauptung. *Die Garbe auf X, deren Schnitte über $V \subset X$ offen gerade aus den Isomorphismen $f_{e+1}: \mathfrak{X}_{e+1}|_V \to \mathfrak{Y}_{e+1}|_V$ besteht welche f_e auf $\mathfrak{X}_e|_V$ induzieren, ist ein prinzipalhomogener Raum unter der kohärenten $\mathcal{O}_X$-Modulgarbe*

$$\mathcal{G} := \mathcal{H}om_{\mathcal{O}_X}(\tau^* \Omega_{\mathfrak{X}}, \tau^{-1}(\mathcal{J}^{e+1}/\mathcal{J}^{e+2})).$$

Es folgt hieraus sofort die Aussage:

4. 2. Corollar. *Unter den Voraussetzungen von* 4. 1 *gibt es eine exakte Sequenz*

$$\mathrm{Isom}(\tau_{e+1}, \sigma_{e+1}) \xrightarrow{\beta} \mathrm{Isom}(\tau_e, \sigma_e) \xrightarrow{o} H^1(X, \mathcal{G}),$$

wobei β die natürliche Einschränkung bezeichne. Ferner sind sämtliche Fasern von β formal prinzipalhomogene Räume unter $H^0(X, \mathcal{G})$.

Der *Beweis* beider Aussagen kann wie in [13] Exp. III, Abschnitt 5 geführt werden.

4. 3. Bemerkung. Da $\mathfrak{X}$ eine formale komplexe Mannigfaltigkeit ist, identifiziert sich die Garbe $\mathcal{G}$ auf kanonische Weise mit

$$\tau^* \Theta_{\mathfrak{X}} \otimes_{\mathcal{O}_X} \tau^{-1}(\mathcal{J}^{e+1}/\mathcal{J}^{e+2}), \quad \text{wobei} \quad \Theta_{\mathfrak{X}} = \Omega_{\mathfrak{X}}{}^{\vee}.$$

4. 4. Satz. *Sei $\tau: X \hookrightarrow \mathfrak{X}$ eine formale Einbettung des kompakten komplexen Raumes X in die formale komplexe Mannigfaltigkeit $\mathfrak{X}$, $\mathcal{J} := \mathcal{K}er(\mathcal{O}_{\mathfrak{X}} \to \tau_* \mathcal{O}_X)$. X sei in $\mathfrak{X}$ lokal vollständiger Durchschnitt.*

Behauptung. *Ist das Normalenbündel $N_{X|\mathfrak{X}}$ ein Bündel vom Typ $\{p, q\}$, wobei $p=1$ oder $q < \operatorname{rg} N_{X|\mathfrak{X}} + \operatorname{prof} \mathcal{O}_X - 1$ ist, so gilt*

$$H^1\left(X, \tau^* \Theta_{\mathfrak{X}} \otimes_{\mathcal{O}_X} S^k_{\mathcal{O}_X}(\tau^{-1}(\mathcal{I}/\mathcal{I}^2))\right) = 0$$

für alle $k \gg 0$.

Diese Aussage ist eine Folgerung von 3. 6 i.

Wir wollen noch den Fall betrachten, wo das Normalenbündel ampel ist.

4. 5. Satz. *Sei X eine kompakte komplexe Mannigfaltigkeit der Dimension n, N ein amples Vektorraumbündel auf X. $\mathcal{N}$ bezeichne die Garbe der holomorphen Schnitte von N.*

Behauptung. *Für jede lokal freie Garbe $\mathcal{F} \in \operatorname{Coh}(X)$ gilt*

$$H^r\left(X, \mathcal{F} \otimes_{\mathcal{O}_X} S^k_{\mathcal{O}_X}(\mathcal{N}^{\vee})\right) = 0$$

für alle $k \gg 0$ und für $r \leqq n-1$.

Der *Beweis* folgt unter Benutzung des Serre'schen Dualitätssatzes, wenn man noch beachtet, daß in Charakteristik null $S^{\cdot}(\mathcal{F}^{\vee}) = S^{\cdot}(\mathcal{F})^{\vee}$ erfüllt ist.

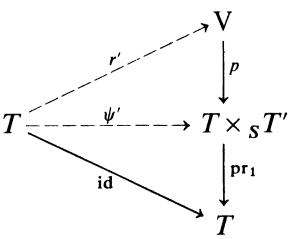
Für einen komplexen Raumkeim T mit ausgezeichnetem Punkt t_0 bezeichne T_k die k-te infinitesimale Umgebung von t_0 in T und $i_k: T_k \hookrightarrow T$ die Inklusion.

4. 6. Satz. *Sei (S, s_0) ein komplexer Raumkeim, $F \to (An/(S, s_0))$ ein Gruppoid über der Kategorie der analytischen Raumkeime über S. Für $T \in (An/(S, s_0))$ und je zwei Objekte $a, b \in F(T)$ sei der Funktor* **Isom** *$(a, b) : (An/T) \to$ (Mengen) darstellbar.*

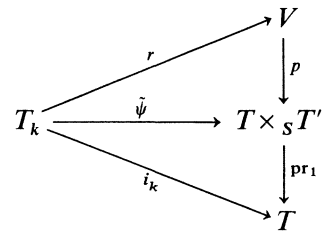
Behauptung. *Seien $a \in F(T)$, $b \in F(T')$. Zu jedem $k_0 \in \mathbb{N}$ existiert dann ein $k_1 \in \mathbb{N}$ mit folgender Eigenschaft: Gibt es einen S-Morphismus $\varphi: T_k \to T'_k$, $k \geqq k_1$, mit $a_k \cong \varphi^* b_k$, so existiert ein S-Morphismus $\psi: T \to T'$ mit $\psi|T_{k_0} = \varphi|T_{k_0}$ und $a \cong \varphi^* b$. Dabei sei $a_k := i_k^* a$ und entsprechend b_k.*

Beweis. Es bezeichne pr_i, $i=1, 2$ die jeweiligen kanonischen Projektionen von $T \times_S T'$ auf T bzw. T', $\tilde{a} := \mathrm{pr}_1^* a$, $\tilde{b} := \mathrm{pr}_2^* b$. Der Funktor **Isom** $(\tilde{a}, \tilde{b})$ ist nach Voraussetzung durch einen Raumkeim $p: V \to T \times_S T'$ darstellbar, $\bar{a} := p^* \tilde{a}$, $\bar{b} := p^* \tilde{b}$, $\rho: \bar{a} \to \bar{b}$ sei der universelle Isomorphismus. Wir betrachten das folgende Diagramm in $(An/S, s_0))$

(*)



Aus dem Artinschen Approximationssatz in der Formulierung von Wavrik [23] Theorem I_n folgt dann: Zu jedem $k_0 \in \mathbb{N}$ gibt es ein $k_1 \in \mathbb{N}$ mit der Eigenschaft: Existiert ein kommutatives Diagramm (*) auf den k_1-ten infinitesimalen Umgebungen der ausgezeichneten Punkte, so existieren r', ψ', die (*) kommutativ machen und mit den vorgegebenen Morphismen bis zur Ordnung k_0 übereinstimmen. Sei also $k \geqq k_1$ und $\varphi: T_k \to T'_k$ ein Morphismus mit $a_k \cong \varphi^* b_k$. Wir setzen $\tilde{\psi} := (i_k \times_S (i'_k \circ \varphi)) \circ \Delta_{T_k \times_S T_k}$, wobei $i_k: T_k \hookrightarrow T$ und $i'_k: T'_k \hookrightarrow T'$ die Inklusionen sind. Wegen der universellen Eigenschaften von V existiert folgendes kommutative Diagramm



Der Isomorphismus $a_k \xrightarrow{\sim} \varphi^* b_k$ ergibt sich dabei als Rückzug des universellen Isomorphismus ρ vermöge r. Somit existieren r' und ψ' wie im Diagramm (*) und approximieren r bzw. $\tilde{\psi}$ bis zur Ordnung k_0. Sei nun $\psi := \mathrm{pr}_2 \circ \psi'$. Dann erfüllt ψ die gewünschten Eigenschaften in der Behauptung.

4. 7. Satz. *Seien $p: X \to S$, $q: Y \to S$ eigentliche und flache Abbildungen komplexer Räume, $X' \subset X$, $Y' \subset Y$ S-flache abgeschlossene Unterräume und $f_0: X' \to Y'$ ein S-Isomorphismus.*

Behauptung. *Der Funktor*

$$\mathbf{Isom}_S(X, Y; f_0): (An/S) \to (Mengen)$$

derjenigen Isomorphismen, die mit f_0 verträglich sind, ist durch einen abgeschlossenen Unterraum des komplexen Raumes $\mathrm{Isom}_S(X, Y)$ über S darstellbar.

Beweis. Seien $i: X' \hookrightarrow X, j: Y' \hookrightarrow Y$ die jeweiligen S-Inklusionen. Wir betrachten die holomorphe Abbildung komplexer Räume

$$\alpha: \mathrm{Isom}_S(X, Y) \to \mathrm{Hom}_S(X', Y), \quad \varphi \mapsto \varphi \circ i.$$

Sei $Z := \alpha^*(j \circ f_0)$. Z stellt dann offensichtlich den Funktor $\mathbf{Isom}_S(X, Y; f_0)$ dar.

Wir benötigen noch das Hauptresultat von [18].

4. 8. Satz. *Sei $\tau: X \hookrightarrow W$ eine abgeschlossene Einbettung des kompakten komplexen Raumes X in die q-konkave komplexe Mannigfaltigkeit W, wobei $q < \dim W - 1$ ist.*

Dann besitzt τ eine semiuniverselle Deformation.

Wir sind nun in der Lage, ein Hauptergebnis dieser Arbeit zu beweisen.

4. 9. Theorem. *Sei $\tau: X \hookrightarrow \mathfrak{X}$ eine formale Einbettung der kompakten komplexen Mannigfaltigkeit X in die formale komplexe Mannigfaltigkeit $\mathfrak{X}$. Ist das Normalenbündel $N_{X|\mathfrak{X}}$ ein q-konkaves Bündel mit $q < \dim \mathfrak{X} - 1$, so ist τ bereits konvergent.*

4. 10. Bemerkung. Unter leichten Zusatzvoraussetzungen an die Ausschöpfungsfunktion φ von $W, X \subset W \subset N_{X|\mathfrak{X}}$, folgt sofort, daß $N_{X|\mathfrak{X}}$ insbesondere 1-positiv im Sinne von Commichau-Grauert ist (vgl. [10]). Nach dem dortigen Hauptsatz ist die Analytisierung $\sigma: X \hookrightarrow Y$ von τ eindeutig bestimmt, da für jeden weiteren Einbettungskeim $\sigma': X \hookrightarrow Y'$ dann jeder formale Isomorphismus $\mathfrak{f}: \hat{Y} \to \hat{Y}'$ schon konvergent ist.

4. 11. Corollar. *Sei* $\tau: X \hookrightarrow \mathfrak{X}$ *wie in* 4. 9 *und* $\dim X \geqq 2$. *Ist das Normalenbündel* $N_{X|\mathfrak{X}}$ *ampel und von globalen Schnitten erzeugt, so folgt die Konvergenz von* τ.

Wir beweisen zunächst das Theorem. Sei $W \subset N_{X|\mathfrak{X}}$ eine q-konkave Umgebung des Nullschnitts gemäß 3. 5. Die Einbettung $X \hookrightarrow W$ besitzt, da $q < \dim W - 1$, eine semiuniverselle Deformation $\mathcal{V} \to S$ (vgl. 4. 8). $\mathcal{V}\langle\infty\rangle \to S$ bezeichne die induzierte Deformation von $X \hookrightarrow \hat{W}$ über S (vgl. 1. 3 iv). Wegen 1. 12 und 3. 8 ist $\mathcal{V}\langle\infty\rangle \to S$ eine *formal* semiuniverselle Deformation von $X \hookrightarrow \hat{W}$. Es bezeichne $\mathfrak{A} \to (\mathbb{C}, 0)$ die nach 2. 4 existierende Deformation von $X \hookrightarrow \hat{W}$ in $X \hookrightarrow \mathfrak{X}$ mit allgemeiner Faser $X \hookrightarrow \mathfrak{X}$. Es existiert nun ein kartesisches Diagramm der entsprechenden formalen Deformationen

(I)
$$\begin{array}{ccc} \mathfrak{A} & \longrightarrow & \hat{\mathcal{V}}\langle\infty\rangle \\ \downarrow & & \downarrow \\ \hat{\mathbb{C}} & \xrightarrow{\varphi} & \hat{S} \end{array} .$$

Sei $e \in \mathbb{N}$ vorgegeben. Aus dem Diagramm I erhält man mit 1. 3 iv das kartesische Diagramm

(II)
$$\begin{array}{ccc} \mathfrak{A}\langle e\rangle & \longrightarrow & \hat{\mathcal{V}}\langle e\rangle \\ \downarrow & & \downarrow \\ \hat{\mathbb{C}} & \xrightarrow{\varphi} & \hat{S} \end{array} .$$

Wendet man jetzt auf diese Situation 4. 6 und 4. 7 an, so existiert ein kartesisches Diagramm

(III)
$$\begin{array}{ccc} \mathfrak{A}\langle e\rangle & \longrightarrow & \mathcal{V}\langle e\rangle \\ \downarrow & & \downarrow \\ (\mathbb{C}, 0) & \xrightarrow{\psi} & S \end{array} .$$

Sei $p \in \mathbb{C}^*$ ein Punkt, der im Definitionsbereich eines geeigneten Repräsentanten von ψ liegt und $s := \psi(p)$. $\sigma: X \hookrightarrow Y$ bezeichne die über s liegende abgeschlossene Einbettung $\mathcal{V}(s)$. Wegen III gilt, daß $\sigma_e: X \hookrightarrow Y_e$ isomorph zu $\tau_e: X \hookrightarrow \mathfrak{X}_e$ ist. Wählt man e genügend groß, so ist nach 4. 2 und 4. 4 die formale Einbettung $\hat{\sigma}: X \hookrightarrow \hat{Y}$ isomorph zu $\tau: X \hookrightarrow \mathfrak{X}$, womit das Theorem bewiesen ist.

Nun zum *Beweis* des Corollars 4. 11. Wir führen dazu einen etwas stärkeren Konkavitätsbegriff für Bündel ein.

4. 12. Definition. Sei X eine n-dimensionale kompakte komplexe Mannigfaltigkeit, N ein holomorphes Vektorraumbündel vom Rang m auf X. N heiße q-konkav im Sinne von Sommese [22] 1. (1. 1), wenn es eine stetige Funktion $\varphi: N \to \mathbb{R}_+$ gibt mit den Eigenschaften:

i) $\varphi(ze) = |z|^2 \varphi(e)$ für alle $e \in N$ und $z \in \mathbb{C}$.

ii) φ ist auf $N \setminus X$ unendlich oft differenzierbar und die Leviform von φ hat dort mindestens $n+m-q+1$ negative Eigenwerte.

Sei N q-konkav in obigem Sinne und $\varphi: N \to \mathbb{R}_+$ eine Ausschöpfungsfunktion. φ induziert auf dem tautologischen Geradenbündel $L(N)$ auf $\mathbb{P}(N)$ eine hermitesche Metrik, deren Leviform außerhalb des Nullschnitts ebenfalls mindestens

$$n+m-q+1 = \dim \mathbb{P}(N) - (q-1) + 1$$

negative Eigenwerte besitzt. $L(N)$ ist damit *metrisch* $(q-1)$-konkav [21].

Sei $\mathcal{N}$ die Garbe der holomorphen Schnitte von N und $\bar{N} := \operatorname{Projan}(S^{\bullet}_{\mathcal{O}_X}(\mathcal{N}^{\vee} \otimes \mathcal{O}_X))$ die natürliche Kompaktifizierung von N. Es ist dann $X \subset N \subset \bar{N}$.

Wir betrachten die monoidale Transformation von $\bar{N}$ längs X

$$\begin{array}{ccc} \tilde{X} & \longrightarrow & \tilde{\bar{N}} \\ \downarrow & & \downarrow f \\ X & \longrightarrow & \bar{N} \end{array}$$

Es gilt $\tilde{X} = \mathbb{P}(N)$, $N_{X|\bar{N}} = N$, $N_{\tilde{X}|\tilde{\bar{N}}} = N_{\tilde{X}|\tilde{N}} = L(N)$. Nach Schneider [21] Zusatz zu Satz 1 gibt es eine relativ kompakte offene Umgebung $\tilde{W}$ von $\tilde{X} \subset \tilde{N}$ und eine differenzierbare Ausschöpfungsfunktion $\tilde{\varphi}: \tilde{W} \to \mathbb{R}_+$ mit $\tilde{\varphi}^{-1}(0) = \tilde{X}$, welche außerhalb von $\tilde{X}$ streng $(q-1)$-konkav ist. Sei $W := f(\tilde{W})$. Dann ist W offen in N und es folgt, daß N ein $(q-1)$-konkaves Bündel ist (3. 5). Die Aussage des Corollars kann dann auf folgende Weise auf die des Theorems zurückgeführt werden:

Da $N_{X|\mathfrak{X}}$ ampel und von den globalen Schnitten erzeugt ist, folgt nach einem Satz von Sommese [22] (1. 3) (vgl. auch [12]), daß $N_{X|\mathfrak{X}}$ $(m+1)$-konkav im Sinne von Sommese ist. $N_{X|\mathfrak{X}}$ ist somit ein m-konkaves Bündel. Wegen

$$n \geqq 2 \quad \text{gilt} \quad m < n+m-1 = \dim \mathfrak{X} - 1$$

und das Theorem ist somit anwendbar.

§ 5. Konvergenz formaler Einbettungen mit schwach negativem Normalenraum

5. 1. Sei $f\colon \mathfrak{X} \to \mathfrak{Y}$ ein Morphismus formaler komplexer Räume, $K \subset |\mathfrak{X}|$, $L \subset |\mathfrak{Y}|$ Steinsche Kompakta mit $f(K) \subset L$. Wir setzen

$$X_K := \operatorname{Spec}\left(\Gamma(K, \mathcal{O}_{\mathfrak{X}})\right), \qquad Y_L := \operatorname{Spec}\left(\Gamma(L, \mathcal{O}_{\mathfrak{Y}})\right), \qquad \text{und} \qquad f_{K,L}\colon X_K \to Y_L$$

bezeichne den durch f induzierten Morphismus. Nach [14] O_{IV} 17. 3. 3 gelten die Inklusionen

$$f_{K,L}^{-1}\left(\operatorname{Reg}(Y_L)\right) \cap \operatorname{Reg}(f_{K,L}) \subset \operatorname{Reg} X_K,$$
$$\operatorname{Reg}(X_K) \cap \operatorname{Fla}(f_{K,L}) \subset f_{K,L}^{-1}\left(\operatorname{Reg}(Y_L)\right).$$

Hierbei haben wir mit $\operatorname{Reg}(-)$ bzw. $\operatorname{Fla}(-)$ den regulären bzw. flachen Ort bezeichnet. Insbesondere ist $\operatorname{Reg}(f_{K,L}) \subset \operatorname{Fla}(f_{K,L})$.

Bekanntlich gibt es zu jedem formalen komplexen Raum $\mathfrak{X}$ ein kohärentes Ideal $\mathfrak{s}_{\mathfrak{X}} \subset \mathcal{O}_{\mathfrak{X}}$, welches folgende Eigenschaft besitzt: Zu jedem Steinschen Kompaktum $K \subset |\mathfrak{X}|$ definiert $\Gamma(K, \mathfrak{s}_{\mathfrak{X}})$ gerade den singulären Ort von $\operatorname{Spec}\left(\Gamma(K, \mathcal{O}_{\mathfrak{X}})\right)$. Man kann annehmen, daß $r(\mathfrak{s}_{\mathfrak{X}}) = \mathfrak{s}_{\mathfrak{X}}$ ist.

Einen formalen komplexen Raum $\mathfrak{X}$ nennen wir dementsprechend *regulär* außerhalb eines durch ein kohärentes Ideal $\mathcal{J} \subset \mathcal{O}_{\mathfrak{X}}$ definierten Unterraumes $\mathfrak{X}' \subset \mathfrak{X}$, falls die Inklusion $\mathfrak{s}_{\mathfrak{X}} \supset \mathcal{J}$ erfüllt ist.

Zum Begriff der formalen Modifikation formaler komplexer Räume, der im folgenden benutzt wird, sei auf [9] § 5 verwiesen.

5. 2. Lemma. *Sei $f\colon \mathfrak{X} \to \mathfrak{Y}$ eine relativ algebraische formale Modifikation formaler komplexer Räume und $x \in \mathfrak{X}$ ein Punkt. Zu jedem Steinschen Kompaktum $L \subset |\mathfrak{Y}|$ mit $f(x) \in L$ gibt es ein Steinsches Kompaktum $K \subset |\mathfrak{X}|$ mit $x \in K$ und $f(K) \subset L$, so daß die Inklusion*

$$\operatorname{Reg}(f_{K,L}) \supset X_K \setminus V(I_K)$$

erfüllt ist. Hierbei sei $I_K = \Gamma(K, \mathcal{I}_{\mathfrak{X}})$ und $\mathcal{I}_{\mathfrak{X}}$ das maximale Definitionsideal von $\mathfrak{X}$.

Beweis. Wir können annehmen (vgl. [9] (5. 3)), daß $\mathfrak{Y}$ Steinsch ist und $f|L$ durch Anwendung des Funktors „an" auf eine Modifikation $\tilde{f}\colon Z \to Y_L$ längs $V(I_L)$ entsteht, wobei Z ein Schema ist, lokal von endlichem Typ über Y_L. Es sei $i\colon \mathfrak{X}_{/L} \to Z$ der kanonische Morphismus. Dann gibt es eine affine offene Menge $U \subset Z$ mit $x \in i^{-1}(U)$ und so daß $\Gamma(U, \mathcal{O}_Z)$ eine $\Gamma(L, \mathcal{O}_{\mathfrak{Y}})$-Algebra von endlichem Typ ist. Sei $K \subset i^{-1}(U)$ ein Steinsches Kompaktum mit $x \in K$. Da nun die kanonische Abbildung $\Gamma(U, \mathcal{O}_Z) \to \Gamma(K, \mathcal{O}_{\mathfrak{X}})$ regulär ist und $\tilde{f}$ eine Modifikation von Y_L längs $V(I_L)$, folgt hieraus mit [14] IV 6. 8. 3 die Behauptung.

Als Konsequenz bemerken wir:

5. 3. Proposition. *Sei $f\colon \mathfrak{X} \to \mathfrak{Y}$ eine Modifikation formaler komplexer Räume. Dann ist $\mathfrak{X}$ regulär außerhalb der unterliegenden Menge $|\mathfrak{X}|$ genau dann, wenn $\mathfrak{Y}$ regulär außerhalb von $|\mathfrak{Y}|$ ist.*

Beweis. Zunächst reduziert man die Aussage mit Hilfe des formalen Chow-Lemmas von [9] (7. 1) auf den relativ algebraischen Fall, wo dann das letzte Lemma anwendbar ist. Die Behauptung ergibt sich nun leicht aus den beiden Inklusionen in 5. 1 unter Verwendung von [9] (5. 4).

Wir kommen zu einem weiteren Hauptresultat dieser Arbeit.

5. 4. Theorem. *Sei* $\tau: X \hookrightarrow \mathfrak{X}$ *eine formale Einbettung des kompakten komplexen Raumes* X *in den formalen komplexen Raum* $\mathfrak{X}$. *Es sei* $\mathfrak{X}$ *regulär außerhalb von* X.

Ist die Konormalengarbe $\mathscr{I}/\mathscr{I}^2$ *von* τ *ampel, so folgt die Konvergenz von* τ. *Ferner ist die Analytisierung von* $\mathfrak{X}$ *bis auf Isomorphie eindeutig bestimmt.*

5. 5. Bemerkung. Falls der Normalenraum $N_{X|\mathfrak{X}}$ von τ schwach negativ ist, folgt insbesondere, daß $\mathscr{I}/\mathscr{I}^2$ ampel ist (vgl. 3. 6 i, 2. 3 iii).

Beweis des Theorems. Wir zeigen zunächst die Existenz einer Analytisierung von $\mathfrak{X}$. Die Voraussetzung über $\mathscr{I}/\mathscr{I}^2$ impliziert nach [9] (6. 1), (6. 11), daß es eine formale Modifikation $f: \mathfrak{X} \to \mathfrak{Y}$ gibt, wobei $\mathfrak{Y}$ ein einpunktiger formaler komplexer Raum ist, d.h. $\mathfrak{Y} = (0, A)$ mit $A = \mathbb{C}[[x_1, \ldots, x_n]]/\mathfrak{a}$. Aus der Regularitätsvoraussetzung und 5. 3 ergibt sich, daß $\operatorname{Spec}(\mathscr{O}_{\mathfrak{Y},0})$ regulär außerhalb des abgeschlossenen Punktes ist. Es folgt nach Artin [5] (3. 8), daß A isomorph zur Komplettierung eines algebraischen $\mathbb{C}$-Schemas in einem Punkt ist. Also ist $\mathfrak{Y}$ konvergent und damit nach [2] Théorème D, [9] (9. 1) auch $\mathfrak{X}$.

Die Eindeutigkeit der so erhaltenen Analytisierung von $\mathfrak{X}$ erhält man aus [17]. Es sei hier darauf hingewiesen, daß die in [17] geforderte Bedingung der analytischen Dünnheit der Zentren überflüssig ist. Der dortige Beweis kann z.B. mit Hilfe des Chow-Lemmas [9] (7. 2) und des algebraischen Chow-Lemmas entsprechend modifiziert werden.

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Arithmetic specializations in polynomials

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90 years ago David Hilbert published in this journal a work on the irreducibility of polynomials [4]. His principal result was as follows: Given an irreducible polynomial $F(x, y)$ in $\mathbb{Q}[x, y]$, where $\mathbb{Q}$ is the field of rational numbers, there exists an infinite sequence of rational integers x_ν $(\nu = 1, 2, \ldots)$ such that $F(x_\nu, y)$ is irreducible in $\mathbb{Q}[y]$. Later on Dörge [2], Emmy Noether [6] and Siegel [9] made first contributions to better understanding and the development of the phenomena touched upon by the Hilbert's Theorem. It is recognized now as one of his famous theorems which influenced the solutions of many deep problems of Galois Theory and Algebraic Geometry.

The topic of this article is influenced by Hilbert's work too and concerns the connection between the multiplicative structure of the number x_0 in an algebraic number field $\mathbb{K}$ (of finite degree over $\mathbb{Q}$) and the multiplicative structure of the specialized polynomial $F(x_0, y)$ in $\mathbb{K}[y]$. The case of $\mathbb{K} = \mathbb{Q}$ was considered previously [10], [11], and as an application of the results obtained on this way the following statement was proved [12]: The numbers

$$x_\nu = [\exp \sqrt{\log \log \nu}] + \nu!\, 2^{\nu^2} \qquad (\nu > e^e)$$

satisfy the conclusion of the Hilbert's Theorem for all $F = F(x, y)$ if ν exceeds a bound $\nu_0(F)$ which may be explicitly expressed through the coefficients and the degree of F.

The Theorem which is proved below contains all previous basic results as special cases and gives a new ground for further brakethrough into the arithmetic structure of polynomials.

1. Introduction

Let $\mathbb{K}$ be an algebraic number field of degree k over $\mathbb{Q}$, and let r_1 and r_2 be the numbers of real and pairs of conjugate complex embeddings of $\mathbb{K}$ into the fields of real numbers $\mathbb{R}$ and complex numbers $\mathbb{C}$ respectively, $k = r_1 + 2r_2$. We denote by $\mathbb{K}^{(i)}$ $(i = 1, 2, \ldots, k)$ the corresponding images of $\mathbb{K}$ in $\mathbb{R}$ and $\mathbb{C}$; similar, $\varkappa$ and $\varkappa^{(i)}$ are the corresponding elements of $\mathbb{K}$ and $\mathbb{K}^{(i)}$. We define the "size" of $\varkappa$ as

$$[\varkappa] = \max_{(i)} |\varkappa^{(i)}| \qquad (i = 1, 2, \ldots, k).$$

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Hence for any two numbers $\varkappa_1$ and $\varkappa_2$ of $\mathbb{K}$ we have

$$\overline{|\varkappa_1+\varkappa_2|}\leqq\overline{|\varkappa_1|}+\overline{|\varkappa_2|},\quad \overline{|\varkappa_1\varkappa_2|}\leqq\overline{|\varkappa_1|}\,\overline{|\varkappa_2|}. \tag{1.1}$$

The size of a polynomial $P(x)$ in $\mathbb{K}[x]$ is defined as

$$\overline{|P(x)|}=\max\overline{|\text{the coefficients of } P(x)|},$$

and similar in case of the polynomials in many variables.

It is easy to see that given two polynomials $P_1(x)$ and $P_2(x)$ in $\mathbb{K}[x]$ one has

$$\overline{|P_1(x)+P_2(x)|}\leqq\overline{|P_1(x)|}+\overline{|P_2(x)|}, \tag{1.2}$$

$$\overline{|P_1(x)\,P_2(x)|}\leqq\min\left(1+\deg P_1(x),\,1+\deg P_2(x)\right)\overline{|P_1(x)|}\,\overline{|P_2(x)|} \tag{1.3}$$

which is a consequence of (1.1).

We denote $|\cdots|_v$ (also $|\cdots|_w$ in the next sections) the valuations (metrics) on $\mathbb{K}$, both infinite and finite, and $\mathbb{K}_v$ the completion of $\mathbb{K}$ in respect to $|\cdots|_v$. Thus, $|\cdots|_v$ on $\mathbb{K}$, in case of being infinite, is a continuation of the ordinary absolute value $|\cdots|_\infty$ from $\mathbb{Q}$ onto $\mathbb{K}$ (we signify it as $v|\infty$); hence, $\mathbb{K}_v$ is $\mathbb{R}$ or $\mathbb{C}$ and $k_v=[\mathbb{K}_v:\mathbb{Q}_v]$ is 1 or 2. In case of finite $|\cdots|_v$ it is a continuation on $\mathbb{K}$ of the p-adic metrics $|\cdots|_p$ of $\mathbb{Q}$ with a prime p, $|p|_p=p^{-1}$ (we signify it as $v|p$); $\mathbb{K}_v$ is a finite extension of the field $\mathbb{Q}_p$ of p-adic numbers (in fact it is $\mathbb{K}_\mathfrak{p}$ for the corresponding prime divisor $\mathfrak{p}$ over p), $k_v=[\mathbb{K}_v:\mathbb{Q}_v]\leqq k$. In both cases we have

$$\sum_{v|\infty}k_v=\sum_{v|p}k_v=k, \tag{1.4}$$

where the sums are taken over all valuations $|\cdots|_v$ on $\mathbb{K}$ continuing $|\cdots|_\infty$ and $|\cdots|_p$ from $\mathbb{Q}$ respectively.

We denote by $\mathbb{K}^v$ the image of $\mathbb{K}$ in $\mathbb{K}_v$ obtained by the embedding of $\mathbb{K}$ into $\mathbb{K}_v$. We do not identify $\mathbb{K}$ and $\mathbb{K}^v$ and sometime distinguish also between the numbers $\varkappa\in\mathbb{K}$ and $\varkappa^v\in\mathbb{K}^v$; in case of infinite $|\cdots|_v$ it is habitual, since $\mathbb{K}^v$ is $\mathbb{K}^{(i)}$ for some i, but we do so in case of finite $|\cdots|_v$ too, hence, we have to think on $\mathbb{K}^v$ or $\mathbb{K}^\mathfrak{p}$ as an non-archimedean analogy of $\mathbb{K}^{(i)}$ (and distinguish between the "p-adic conjugates" $\varkappa^\mathfrak{p}$ for $\mathfrak{p}|p$ (this agrees with (1.4))). Only in those cases where any misunderstanding is excluded we write $\varkappa$ instead of $\varkappa^v$. Thus, we write the well known "product formula" as

$$\prod_v|\varkappa|_v^{k_v}=1\qquad(\varkappa\in\mathbb{K},\ \varkappa\neq0). \tag{1.5}$$

We denote S_∞ and S_0 the sets of all infinite and all finite valuations on $\mathbb{K}$ respectively.

The height $H_\mathbb{K}(\varkappa)$ of a non-zero $\varkappa\in\mathbb{K}$ (or the $\mathbb{K}$-height of $\varkappa$) is defined by the product

$$H_\mathbb{K}(\varkappa)=\prod_v\max\left(1,\,|\varkappa|_v^{k_v}\right),$$

and the v-component $(\varkappa)_v$ of $\varkappa$ is defined as

$$(\varkappa)_v=\max\left(1,\,|\varkappa|_v^{-k_v}\right).$$

It follows from (1. 5) that

$$\prod_v (\varkappa)_v = H_K(\varkappa^{-1}) = H_K(\varkappa). \tag{1. 6}$$

Let now I_K be the ring of integers of K and let $F(x, y)$ be a polynomial in $I_K[x, y]$ which will be supposed absolutely irreducible and satisfying

$$\deg_y F(x, y) = n \geqq 2, \qquad F(0, 0) = 0, \qquad \frac{\partial}{\partial y} F(0, 0) \neq 0. \tag{1. 7}$$

One can define then a power series

$$f(x) = \sum_{\nu=1}^{\infty} f_\nu x^\nu \qquad (f_\nu \in K) \tag{1. 8}$$

which satisfies the equation

$$F(x, f(x)) = 0. \tag{1. 9}$$

Due to the well known theorem of Eisenstein (Lemma 3. 3 below) there exists a natural number $E = E_f$ such that all of the numbers $E^\nu f_\nu$ $(\nu = 1, 2, \ldots)$ lie in I_K. Taken a number $x_0 \in K$ denote $f(x_0)_v$ the sum of the series (1. 8) in the metrics of $|\cdots|_v$ when the series converges in $x = x_0$ (in particular, it is convergent for those finite v, for which $|x_0|_v < |E_f|_v$).

Suppose $F(x_0, y)$ splits into irreducible factors in $K[y]$ as follows

$$F(x_0, y) = F_1(y) \cdots F_r(y). \tag{1. 10}$$

Then we have the "conjugate" decompositions

$$F^v(x_0, y) = F_1^v(y) \cdots F_r^v(y)$$

obtained from (1. 10) by the embedding of K (and the coefficients of the polynomials) into K^v. The equality (1. 9) shows that $f(x_0)_v$ is a root of $F^v(x_0, y)$, hence a root of some $F_j^v = F_j^v(y)$, what we shall signify as

$$f(x_0)_v \in F_j^v \qquad (j = 1, 2, \ldots, r).$$

The following Theorem shed the light upon the law of the distribution of the sums $f(x_0)_v$ among the factors of $F(x_0, y)$.

Theorem. *Let $F(x, y)$ be a polynomial in $I_K[x, y]$, absolutely irreducible and satisfying* (1. 7). *Let $x_0 \neq 0$ be a number in K and let $F(x_0, y)$ split into irreducible polynomials in $K[y]$ as* (1. 10),

$$\deg F_j(y) = d_j \qquad (j = 1, 2, \ldots, r).$$

Then it follows for all j:

$$\sum_{f(x_0)_v \in F_j^v} \frac{\log (x_0)_v}{\log H_K(x_0)} = \frac{d_j}{n} + O\left(\left(1 + \frac{\log \Delta_K}{\log H_K(x_0)}\right)\sqrt{\frac{\log |F|}{\log H_K(x_0)}}\right), \tag{1. 11}$$

where

$$\Delta_K = \prod_{q | D_K} q^{\frac{1}{q-1}},$$

the sum is taken over all v for which the series (1. 8) *with* $x = x_0$ *converges in the metric* $|\cdots|_v$ *to the root of* $F_j^v(y)$, $(x_0)_v$ *is the* v*-component of* x_0, $H_{\mathbb{K}}(x_0)$ *is the height of* x_0 *in* $\mathbb{K}$, $\overline{|F|}$ *is the size of* $F(x, y)$, $D_{\mathbb{K}}$ *is the discriminant of* $\mathbb{K}$ *and the symbol* O *involves a value which can be effectively determined through the degrees of* $F(x, y)$ *and* $\mathbb{K}$. *If* $\mathbb{K}$ *is the field generated by the coefficients of* $F(x, y)$ *the remainder term in* (1. 11) *is of the form*

$$O\left(\sqrt{\frac{\log \overline{|F|}}{\log H_{\mathbb{K}}(x_0)}}\right). \tag{1. 12}$$

We do not specify the dependence of the remainder terms in (1. 11) and (1. 12) on the degrees of $F(x, y)$ and $\mathbb{K}$ since nothing is done in this work to get the best possible values, though the explicit form of the remainder term may be expressed easy from the estimates obtained and involved below.

In fact to get formulas similar to (1. 11) we need not to assume that $F(0, 0) = 0$ and $\frac{\partial}{\partial y} F(0, 0) \neq 0$, but the existence of the Laurent series satisfying (1. 9). We prefer to use stronger conditions since then it is guaranteed that we get the series over the minimal possible field (generated by the coefficients of $F(x, y)$) and a good estimate for the Eisenstein number is provided.

We postpone the discussion of some simple corollaries from the Theorem until the last section of the work.

2. Preliminaries

Write $F(x, y)$ in the form

$$F(x, y) = a_0(x)\, y^n - a_1(x)\, y^{n-1} - \cdots - a_n(x), \qquad a_i(x) \in I_{\mathbb{K}}[x].$$

We want to have $a_0(x)$ identically 1, and we achieve it by multiplying $F(x, y)$ by $a_0(x)^{n-1}$ and setting $Y = a_0(x)\, y$. Since we suppose $F(x, y)$ irreducible over any algebraic number field, the resulting polynomial $F^*(x, Y)$ will possess the same property of irreducibility, as it follows from Gauss lemma. The power series $f^*(x) = a_0(x) f(x)$ will have the coefficients in $\mathbb{K}$ and satisfy $F^*(x, f^*(x)) = 0$. If $x_0 \in \mathbb{K}$ is such that $a_0(x_0) \neq 0$, there is a direct correspondence between the decompositions of $F(x_0, y)$ and $F^*(x_0, Y)$ in $\mathbb{K}[y]$ and $\mathbb{K}[Y]$ respectively. The formulae (1. 11) change is obvious, if one takes into account that

$$\overline{|F^*(x, Y)|} \leqq (1 + \deg_x F(x, y))^{n-1}\, \overline{|F(x, y)|}.$$

Thus without the change of notations we shall assume that $F(x, y)$ is of the form

$$F(x, y) = y^n - a_1(x)\, y^{n-1} - \cdots - a_n(x), \qquad a_i(x) \in I_{\mathbb{K}}[x]. \tag{2. 1}$$

Before proceeding with the main arguments we have yet to exclude a set of exceptional numbers x_0, like the roots of $a_0(x) = 0$ above. To do so it would be enough to assume that

$$H_{\mathbb{K}}(x_0) > (\overline{|F|} + 1)^{C_1}, \tag{2. 2}$$

where $C_1 = C_1(n, n_1, k)$ may be expressed in explicit form through $n = \deg_y F(x, y)$, $n_1 = \deg_x F(x, y)$ and k (all the "constants" below are similar values).

Suppose $x_0 \in \mathbb{K}$ is such that $F(x_0, y)$ has a multiple root. Then the resultant

$$R(x) = R_y\left(F, \frac{\partial}{\partial y} F\right)$$

of the polynomials $F(x, y)$ and $\frac{\partial}{\partial y} F(x, y)$ as the polynomials in y over $\mathbb{K}[x]$ will have a root x_0. Since $F(x, y)$ is irreducible and $\frac{\partial}{\partial y} F(x, y)$ has a lower degree in y, the resultant $R(x)$ is not identically zero. Representing the resultant as the determinant formed by the coefficients of the polynomials, one finds that $R(x)$ has integer coefficients in $\mathbb{K}$ and

$$\deg R(x) \leqq (2n-1)\, n_1\,, \qquad \overline{|R(x)|} \leqq n!\, n^n\, \overline{|F|}^{(2n-1)n_1}\,.$$

The Lemma 3. 1 below shows that all the roots $x_0 \in \mathbb{K}$ of the equation $R(x) = 0$ have the $\mathbb{K}$-heights less than

$$(2n_1\, n!\, n^{n+1}\, \overline{|F|}^{2nn_1})^k\,.$$

Since the asymptotic nature of the formulae (1. 11) implies that the remainder terms are less than 1, these formulae will be consistent only in case of x_0 satisfying an estimate of the form (2. 2). Hence, we may assume (2. 2) which would imply that $F(x_0, y)$ has no multiple root.

3. General lemmata

We collect in this section several general results concerning mostly analytic and arithmetic properties of algebraic power series and systems of homogeneous linear diophantine equations.

Lemma 3. 1. *Let $P(x)$ be a polynomial in $I_{\mathbb{K}}[x]$ of degree m and let $\varkappa$ be a root of $P(x)$ in $\mathbb{K}$. Then*

$$H_{\mathbb{K}}(\varkappa) \leqq (m\, \overline{|P|})^k\,.$$

Proof. Let

$$P(x) = \alpha_0 x^m + \alpha_1 x^{m-1} + \cdots + \alpha_m\,, \qquad \alpha_i \in I_{\mathbb{K}}\,.$$

Then

$$\varkappa = \frac{1}{\alpha_0}(-\alpha_1 - \alpha_2 \varkappa^{-1} - \cdots - \alpha_m \varkappa^{-m+1})\,. \tag{3. 1}$$

We have for $v \in S_\infty$ in case of $|\varkappa|_v > 1$:

$$|\varkappa|_v \leqq \frac{1}{|\alpha_0|_v}(|\alpha_1|_v + \cdots + |\alpha_m|_v) \leqq m\, \overline{|P|}\, |\alpha_0|_v^{-1}\,.$$

Hence,

$$|\varkappa|_v \leqq \max\left(1, m\, \frac{\overline{|P|}}{|\alpha_0|_v}\right) = m\, \overline{|P|}\, |\alpha_0|_v^{-1}\,.$$

We have again from (3. 1) in case of $v \in S_0$: if $|\varkappa|_v > 1$, then

$$|\varkappa|_v \leqq \frac{1}{|\alpha_0|_v} \max(|\alpha_1|_v, \ldots, |\alpha_m|_v) = |\alpha_0|_v^{-1}.$$

Now we find from the definition of $H_K(\varkappa)$ and the product formula (1. 5):

$$H_K(\varkappa) = \prod_v \max(1, |\varkappa|_v^{n_v}) \leqq \prod_{v \in S_\infty} (m\overline{|P|}\,|\alpha_0|_v^{-1})^{n_v} \prod_{v \in S_0} |\alpha_0|_v^{-n_v}$$
$$= (m\overline{|P|})^k \prod_v |\alpha_0|_v^{-n_v} = (m\overline{|P|})^k.$$

Lemma 3. 2. *The coefficients of the series* (1. 8) *satisfy*

$$\overline{|f_\nu|} \leqq A^\nu \qquad (\nu = 1, 2, \ldots), \tag{3. 2}$$

where $1 < A < C_2\overline{|F|}^{8k(nn_1)^2}$, C_2 *is determined through* n, n_1, k.

The *proof* utilizes trivial changes in the proof of the Lemma 3. 1 of [12].

Lemma 3. 3. *The number* E_f *defined for the series* (1. 8) *is a divisor of the absolute norm* $\mathrm{Nm}\left(\frac{\partial}{\partial y} F(0, 0)\right)^2$, *hence*

$$E_f \leqq \overline{|F|}^{2k}.$$

For the *proof* see [12], Lemma 3. 2.

Lemma 3. 4. *If the polynomial* $F(x, y) \in I_K[x, y]$ *is absolutely irreducible,* $\mathfrak{q}$ *is a prime ideal of* I_K *and the norm of* $\mathfrak{q}$ *exceeds* C_3,

$$C_3 = (4\overline{|F|})^{ck}, \qquad c = s^{2^s}, \qquad s = \frac{(n+n_1)(n+n_1-1)}{2},$$

then the reduced polynomial

$$\bar{F}(x, y) = F(x, y) \pmod{\mathfrak{q}}$$

is irreducible in $\mathbb{F}_\mathfrak{q}[x, y]$, *where* $\mathbb{F}_\mathfrak{q}$ *is the field of residue classes* $I_K \pmod{\mathfrak{q}}$.

This is a generalization of the well known Theorem due to Ostrowski [7]. The *proof* may be obtained by a slight change in the proof of Corollary 2B in [8], p. 193.

Lemma 3. 5. *Let* $\mathfrak{q}$ *be a prime ideal of* I_K *satisfying the conditions of Lemma* 3. 4 *and not occurring in* E_f. *Then the power series over* $\mathbb{F}_\mathfrak{q}$

$$\bar{f}(x) = \sum_{\nu=1}^{\infty} \bar{f}_\nu x^\nu$$

which is obtained from (1. 8) *by means of the mapping*

$$f_\nu \to \bar{f}_\nu = f_\nu \pmod{\mathfrak{q}} \qquad (\nu = 1, 2, \ldots)$$

is algebraic over $\mathbb{F}_\mathfrak{q}[x]$ *of degree* n.

The *proof* follows from the remark that $\bar{f}(x)$ satisfies $\bar{F}(x, \bar{f}(x)) = 0$ and the irreducibility of $\bar{F}(x, y)$ in $\mathbb{F}_\mathfrak{q}[x, y]$ by Lemma 3. 4.

Now we use Minkowski's Theorem on successive minima to demonstrate the existence of a "small" basis in $I_{\mathbb{K}}$.

Lemma 3. 6. *There exists a basis* $\beta_1, \ldots, \beta_k$ *of* $I_{\mathbb{K}}$ *with*

$$\boxed{\beta_1} \cdots \boxed{\beta_k} \leqq 2^{\frac{k(k+1)}{2}} \sqrt{|D_{\mathbb{K}}|}, \tag{3. 3}$$

where $D_{\mathbb{K}}$ *is the discriminant of* $\mathbb{K}$.

Proof. Let $\omega_1, \ldots, \omega_k$ be a basis of $I_{\mathbb{K}}$. Set

$$l_i(\bar{x}) = \omega_1^{(i)} x_1 + \cdots + \omega_k^{(i)} x_k \qquad (i = 1, 2, \ldots, k)$$

and consider the body

$$T_\lambda = \{\bar{x} \in \mathbb{R}^k : \max_{(i)} |l_i(\bar{x})| \leqq \lambda\}, \qquad \lambda > 0.$$

It is obvious that $T_\lambda = \lambda T_1$ and the volume of the body T_1 is equal to

$$V(T_1) = \frac{2^{r_1 + r_2} \pi^{r_2}}{\sqrt{|D_{\mathbb{K}}|}}.$$

Signifying $\lambda_1, \lambda_2, \ldots, \lambda_k$ the successive minima of the body T_1, by Minkowski's Theorem we find

$$\lambda_1 \lambda_2 \cdots \lambda_k \leqq 2^k V(T_1)^{-1} = \left(\frac{2}{\pi}\right)^{r_2} \sqrt{|D_{\mathbb{K}}|}.$$

Hence, there exists an integral basis $\alpha_1, \ldots, \alpha_k$ of $\mathbb{K}$ with

$$\boxed{\alpha_1} \leqq \boxed{\alpha_2} \leqq \cdots \leqq \boxed{\alpha_k}, \tag{3. 4}$$

$$\boxed{\alpha_1}\boxed{\alpha_2} \cdots \boxed{\alpha_k} \leqq \left(\frac{2}{\pi}\right)^{r_2} \sqrt{|D_{\mathbb{K}}|}. \tag{3. 5}$$

Now one can find such a basis $\beta_1, \ldots, \beta_k$ of $I_{\mathbb{K}}$ that

$$\begin{aligned} \alpha_1 &= u_{11} \beta_1, \\ \alpha_2 &= u_{21} \beta_1 + u_{22} \beta_2, \\ &\vdots \\ \alpha_k &= u_{k1} \beta_1 + \cdots + u_{kk} \beta_k \end{aligned}$$

with rational integers u_{ij} satisfying $0 \leqq u_{ij} < u_{ii}$ $(1 \leqq j < i \leqq k)$. It follows from (3. 4) that

$$\boxed{\beta_i} \leqq 2^i \boxed{\alpha_i} \qquad (i = 1, 2, \ldots, k),$$

hence, (3. 5) gives (3. 3).

Next is a refined form of a well known Siegel's lemma.

Lemma 3. 7. *Let*

$$L_i(\bar{x}) = \sum_{j=1}^{N} \beta_{ij} x_j \qquad (i = 1, 2, \ldots, M)$$

be a set of $M < N$ *linear forms in* N *variables with coefficients in* $I_{\mathbb{K}}$, *and let*

$$B = \max_{(i,j)} \boxed{\beta_{ij}} > 0.$$

Then there exist integers $x_1, \ldots, x_N$ *in* I_K *with*

$$L_i(\bar{x}) = 0 \qquad (i = 1, 2, \ldots, M),$$

$$0 < \max_{(j)} |x_j| \leqq C_4 (C_4 k N B)^{\frac{M}{N-M}},$$

where C_4 *is given by*

$$C_4 = (k+1)!\, 2^{\frac{k(k+1)}{2}} \sqrt{|D_K|}.$$

The *proof* follows Mahler's arguments in [5], pp. 91—94, and utilizes Lemma 3. 6 to estimate constants involved therein.

4. Auxiliary power series

In this section we start the proof of the Theorem and first of all construct special power series which play the fundamental role in further arguments. What we are doing here may be considered as the "diophantine approximations" in the field of power series, and also as the construction of some rational functions on the curve $F(x, y) = 0$. The topic is quite well known, and a competent reader will trace the influence of Siegel's E and G-functions method [9], but he will also find a "tricky" Lemma 4. 5, which in fact gave birth to the method we describe here.

Lemma 4. 1. *Given a natural number* $h \geqq 4n$ *and a real number* ε *with* $0 < \varepsilon < 1$ *there exist polynomials*

$$A_j(x) \in I_K[x] \qquad (j = 0, 1, \ldots, n-1)$$

of degrees no greater than h *and sizes no greater than*

$$H_0 = C_5 (2 E_f A)^{\frac{n(h+1)}{\varepsilon}}, \qquad C_5 = C_4 (C_4 k)^{\frac{1}{\varepsilon}},$$

and not all equal to zero, such that the power series

$$V(x) = \sum_{j=0}^{n-1} A_j(x) f^j(x) \tag{4. 1}$$

with respect to x *has the order*

$$\operatorname{ord} V(x) \geqq (1 - \varepsilon)\, n(h+1).$$

Proof. Put

$$A_j(x) = \sum_{i=0}^{h} a_{ij} x^i \qquad (j = 0, 1, \ldots, n-1), \tag{4. 2}$$

where a_{ij} are unknown integers in I_K. We find

$$f^j(x) = \sum_{i=j}^{\infty} d_{tj} x^t,$$

$$d_{tj} = \sum_{v_1 + \cdots + v_j = t} f_{v_1} \cdots f_{v_j}, \qquad E_f^t d_{tj} \in I_K.$$

Then in force of (23) we find

$$E_f^t \overline{|d_{tj}|} \leqq t^n (E_f A)^t \qquad (1 \leqq j \leqq n-1;\ t = j, j+1, \ldots).$$

Further, $V(x)$ for the polynomials (4. 2) takes the form

$$V(x) = \sum_{l=0}^{\infty} b_l x^l,$$

$$b_l = a'_{l0} + \sum_{j=1}^{n-1} \sum_{\substack{i+t=l \\ 0 \leqq i \leqq h,\, t \geqq j}} a_{ij} d_{tj},$$

where $a'_{l0} = a_{l0}$ for $l = 0, 1, \ldots, h$ and is zero for $l \geqq h+1$.

Now put $l_0 = [(1-\varepsilon)(h+1)n]$ and consider the system of in $a_{ij} \in I_K$ linear equations

(4. 3) $$b_l = 0 \qquad (l = 0, 1, \ldots, l_0 - 1).$$

The number M of these equations is l_0 and the number N of unknowns is $(h+1)n > l_0$. Multiplying each of the equations (4. 3) by $E_f^{l_0}$ we shall have the equations with the coefficients in I_K of size no greater than

$$B = l_0^n (E_f A)^{l_0}.$$

By the Lemma 3. 7 we find that there exists a nontrivial solution with

$$\max_{(i,j)} \overline{|a_{ij}|} \leqq C_4 \left(C_4 k (h+1)\, n l_0^n (E_f A)^{l_0}\right)^{\frac{l_0}{(h+1)n - l_0}}.$$

Since

$$\frac{l_0}{(h+1)n - l_0} \leqq \frac{1-\varepsilon}{\varepsilon}, \qquad (h+1)\, n l_0^n \leqq 2^{(h+1)n} \qquad (h \geqq 4n)$$

it follows that

$$\max_{(i,j)} \overline{|a_{ij}|} \leqq C_4 \left(C_4 k (2E_f A)^{(h+1)n}\right)^{\frac{1-\varepsilon}{\varepsilon}} < H_0.$$

Lemma 4. 2. *Starting with the equality*

(4. 4) $$f^n(x) = a_1(x) f^{n-1}(x) + \cdots + a_n(x), \qquad a_i(x) \in I_K[x],$$

which corresponds to (1. 9) *and* (2. 1), *put for* $r = n, n+1, \ldots$

$$f^r(x) = a_{1r}(x) f^{n-1}(x) + \cdots + a_{nr}(x), \qquad a_{ir}(x) \in I_K[x].$$

Then

$$\max_{(i)} \deg a_{ir}(x) \leqq n_1 (r - n + 1),$$

$$\max_{(i)} \overline{|a_{ir}(x)|} < C_6^r \overline{|F|}^{r-n+1} \qquad (r = n, n+1, \ldots)$$

where C_6 *is determined through* $n_1 = \deg_x F(x, y)$.

Lemma 4. 3. *Let*

$$S(x)=s_0(x)+s_1(x)f(x)+\cdots+s_{n-1}(x)f^{n-1}(x), \qquad s_i(x)\in I_K[x],$$
$$T(x)=t_0(x)+t_1(x)f(x)+\cdots+t_{n-1}(x)f^{n-1}(x), \qquad t_i(x)\in I_K[x],$$

and suppose that degrees and sizes of the polynomials $s_i(x)$ do not exceed s and S respectively, and that degrees and sizes of polynomials $t_i(x)$ do not exceed t and T respectively. Then

$$S(x)\,T(x)=u_0(x)+u_1(x)f(x)+\cdots+u_{n-1}(x)f^{n-1}(x), \qquad u_i(x)\in I_K[x],$$

and degrees and sizes of the polynomials $u_i(x)$ satisfy

$$\max_{(i)} \deg u_i(x)\leqq s+t+n_1(n-1),$$

$$\max_{(i)} \overline{|u_i(x)|}<C_7 \min(s,t)\, ST\overline{|F|}^{n-1},$$

where C_7 is determined through n and n_1.

The *proofs* of both Lemma 4. 2 and Lemma 4. 3 follow from the properties of sizes of polynomials given by (1. 2) and (1. 3).

Lemma 4. 4. *Suppose that the power series $\mathfrak{B}$ is determined by*

$$\mathfrak{B}=\mathfrak{B}(x,f(x))=\frac{\partial}{\partial y}F(x,y)|_{y=f(x)},$$

$D=\dfrac{d}{dx}$ *is the operator of differentiation with respect to x, and $V=V(x)$ is determined by Lemma* 4. 1. *Then for any integer $s\geqq 1$*

$$D^s(\mathfrak{B}^{2s}V)=A_{0s}(x)+A_{1s}(x)f(x)+\cdots+A_{n-1\,s}(x)f(x)^{n-1} \tag{4. 5}$$

where

$$A_{is}(x)\in I_K[x], \qquad \deg A_{is}(x)\leqq h+2nn_1 s,$$
$$\overline{|A_{is}(x)|}<C_8^s\overline{|F|}^{2ns}H_0\prod_{r=0}^{s-1}(s+h+2nn_1 r)$$

and C_8 is determined through n and n_1.

Proof. Let

$$P(x,y)=\sum_{i=0}^{n-1}A_i(x)\,y^i,$$

where $A_i(x)$ are determined in Lemma 4. 1, and suppose $0\leqq l\leqq s$. Then

$$D^l\big(\mathfrak{B}^{2s}P(x,f(x))\big)=\mathfrak{B}^{2s-2l}P_l(x,f(x)), \tag{4. 6}$$

where

$$P_l(x,y)=\sum_{i=0}^{n-1}A_{il}(x)\,y^i, \qquad A_{il}(x)\in I_K[x].$$

We prove (4. 6) by induction on l. For $l=0$ the assertion is obvious with $P_0(x,y)=P(x,y)$. Suppose that it is true for a given $l<s$ and prove it for $l+1$.

Writing $\mathfrak{B}(x, y)$ instead of $\frac{\partial}{\partial y} F(x, y)$ we have (everywhere substitute $y=f(x)$ after partial differentiation):

$$D^{l+1}\left(\mathfrak{B}^{2s}P(x,f(x))\right)=D\left(D^{l}\mathfrak{B}^{2s}P(x,f(x))\right)=D\left(\mathfrak{B}^{2s-2l}P_l(x,f(x))\right)$$
$$=2(s-l)\,\mathfrak{B}^{2s-2l-1}\left(\frac{\partial}{\partial y}\mathfrak{B}+\frac{\partial}{\partial y}\mathfrak{B}\cdot f'(x)\right)P_l(x,f(x))$$
$$+\mathfrak{B}^{2s-2l}\left(\frac{\partial}{\partial x}P_l+\frac{\partial}{\partial y}P_l\cdot f^1(x)\right).$$

Since

$$f'(x)=-\frac{\frac{\partial}{\partial x}F(x,f(x))}{\frac{\partial}{\partial y}F(x,f(x))}=\frac{\mathfrak{A}}{\mathfrak{B}},$$

where

$$\mathfrak{A}=a_1'(x)f^{n-1}(x)+\cdots+a_n'(x),$$
$$\mathfrak{B}=nf^{n-1}(x)-(n-1)\,a_1(x)f^{n-2}(x)-\cdots-a_{n-1}(x),$$

we obtain

$$D^{l+1}(\mathfrak{B}^{2s}P)=\mathfrak{B}^{2s-2(l+1)}P_{l+1}(x,f(x)),$$
$$P_{l+1}(x,f(x))=2(s-l)\left(\mathfrak{B}\frac{\partial}{\partial x}\mathfrak{B}+\mathfrak{A}\frac{\partial}{\partial y}\mathfrak{B}\right)P_l(x,f(x))+\mathfrak{B}\left(\mathfrak{B}\frac{\partial}{\partial x}P_l+\mathfrak{A}\frac{\partial}{\partial y}P_l\right).$$

Since $f(x)$ satisfies the equation (4. 4), we see that $P_{l+1}(x,f(x))$ can be expressed linearly over $I_K[x]$ in terms of $1, f(x),\ldots,f^{n-1}(x)$, hence, we have (4. 6). Taking $l=s$ we find (4. 5). Let now

$$h_l=\max_{(i)}\deg A_{il}(x),\qquad H_l=\max_{(i)}\overline{|A_{il}(x)|}\qquad(l=1,2,\ldots)$$

while $h=h_0$ and H_0 are determined in Lemma 4. 1. Writing $\mathfrak{A}$, $\mathfrak{B}$, $\frac{\partial}{\partial x}\mathfrak{B}(x,f(x))$, $\frac{\partial}{\partial y}\mathfrak{B}(x,f(x))$, $P_l(x,f(x))$, $\frac{\partial}{\partial x}P_l(x,f(x))$, $\frac{\partial}{\partial y}P_l(x,f(x))$ in terms of $1, f(x),\ldots,f^{n-1}(x)$ over $I_K[x]$ and using Lemma 4. 2 and Lemma 4.3 to estimate degrees and sizes of resulting polynomial coefficients, we find

$$h_{l+1}\leqq h_l+2nn_1,$$
$$H_{l+1}<C_8\{(s-l)\overline{|F|}^2H_l\overline{|F|}^{2(n-1)}+\overline{|F|}^2h_lH_l\overline{|F|}^{2(n-1)}\}$$
$$=C_8(s-l+h_l)\overline{|F|}^{2n}H_l,$$

where C_8 is determined through n and n_1. Hence we find $h_l\leqq h+2nn_1l$ and

$$H_l<(C_8\overline{|F|}^{2n})^lH_0\prod_{r=0}^{l-1}(s+h+2nn_1r)$$

which completes the proof.

Lemma 4. 5. *Suppose that the hypotheses of Lemma* 4. 4 *hold, and let* π_s *be the maximal divisor of* $s!$ *composed of the powers of rational primes greater than* C_3 *and not occurring in* $E_f D_{\mathbb{K}}$:

$$\pi_s = \prod_{\substack{C_3 < q \leqq s \\ q \nmid E_f D_{\mathbb{K}}}} q^{\operatorname{ord}_q s!} \qquad (s > C_3),$$

and $\pi_s = 1$ *if there are no primes specified above. Then all the coefficients of all the polynomials* $A_{is}(x)$ *determined by* (4. 5) *are divisible by* π_s.

Proof. All the coefficients of the power series for $D^s(\mathfrak{B}^{2s} V(x))$ are divisible by π_s, since the series $\mathfrak{B}^{2s} V(E_f x)$ has integer coefficients in $\mathbb{K}$ and the operator D^s gives each term of the series a factor of the form $r(r-1)\cdots(r-s+1)$ which is divisible by $s!$.

Let q be a rational prime divisor of π_s and $\mathfrak{q}$ any prime ideal in $I_{\mathbb{K}}$ with $\mathfrak{q} \mid q$. We obtain from (4. 5) the following equation over $\mathbb{F}_{\mathfrak{q}}$:

$$\bar{A}_{0s}(x) + \bar{A}_{1s}(x)\bar{f}(x) + \cdots + \bar{A}_{n-1\,s}(x)\bar{f}^{n-1}(x) = 0,$$

where the $\bar{A}_{is}(x)$ and $\bar{f}(x)$ are obtained from the $A_{is}(x)$ and $f(x)$ by embedding their coefficients into $\mathbb{F}_{\mathfrak{q}}$. By Lemma 3. 5 the degree of $\bar{f}(x)$ over $\mathbb{F}_{\mathfrak{q}}[x]$ is equal to n, so that all of the polynomials $\bar{A}_{is}(x)$ must be zero. Hence, all the coefficients of all the polynomials $A_{is}(x)$ are divisible by $\mathfrak{q}$. Since q is not a divisor of $D_{\mathbb{K}}$ it is a product of distinct prime ideals $\mathfrak{q}$, and with any of them we can argue, as it is described. Hence, we find that all of the polynomials $\dfrac{A_{is}(x)}{q}$ have integer coefficients (in $I_{\mathbb{K}}$).

If q divides π_s with an exponent greater than 1, then, after dividing all of the coefficients of the polynomials $A_{is}(x)$ and all of the terms of the power series $D^s(\mathfrak{B}^{2s} V(x))$ by q, we repeat the argument. In this way we find that all the coefficients of all the polynomials $A_{is}(x)$ are divisible by $q^{\operatorname{ord}_q \pi_s}$; thus, they are divisible by the product of these factors, i.e., by π_s.

Now we turn our attention to the regular functions $f(\omega) = f(\omega)_v$ defined by the sums of the power series (1. 8) in the metrics of v for $x = \omega \in \mathbb{K}_v$. Let $R_v(f)$ be the corresponding radius of convergence of the series (1. 8). Hence, $f(\omega)_v$ is defined for $\omega \in \mathbb{K}_v$ in the disc $|\omega|_v < R_v(f)$ and we consider, of course, the case of $R_v(f) > 0$.

Lemma 4. 6. *Let* $A_0(x), \ldots, A_{n-1}(x)$ *be polynomials in* $\mathbb{K}[x]$ *of degrees no greater than* h *and not all identically equal to zero. Then the function*

$$V(\omega) = A_0(\omega) + A_1(\omega) f(\omega)_v + \cdots + A_{n-1}(\omega) f(\omega)_v^{n-1} \tag{4. 7}$$

has no more than $nh + (n-1)\, n_1$ *zeros in the disc* $|\omega|_v < R_v(f)$.

Proof. Let $f_0(x), f_1(x), \ldots, f_{n-1}(x)$ be all the power series expansions in a neighbourhood of $x = 0$ of the algebraic function $y(x)$ satisfying the equation

$$y^n = a_1(x)\, y^{n-1} + \cdots + a_n(x).$$

Then one of the series $f_i(x)$, say $f_0(x)$, is $f(x)$ and all other $f_i(x)$ are obtained, in general, by Puiseux expansions.

Note first that any symmetric function of $f_1(x),\dots,f_{n-1}(x)$ is a polynomial in x and $f(x)$ over I_K. Indeed, signifying S_r the elementary symmetric functions of

$$f_1(x),\dots,f_{n-1}(x),$$

we find:

$$S_1=a_1(x)-f(x),\qquad S_r=(-1)^{r-1}a_r(x)-f(x)\,S_{r-1}\quad(1<r<n-1),$$

$$(-1)^{n-1}S_{n-1}=\frac{a_n(x)}{f(x)}=-a_{n-1}(x)-\cdots-a_1(x)f^{n-2}(x)+f^{n-1}(x).$$

Hence, all of S_r lie in $I_K[x,f(x)]$.

Let now

$$G(x,y)=A_0(x)+A_1(x)\,y+\cdots+A_{n-1}(x)\,y^{n-1}.$$

The resultant $R_y(G,F)$ of the polynomials $G(x,y)$ and $F(x,y)$ with respect to y may be represented in the form

$$R_y(G,F)=G\big(x,f(x)\big)\prod_{i=1}^{n-1}G\big(x,f_i(x)\big)=G\big(x,f(x)\big)\,M\big(x,f(x)\big),$$

where $M\big(x,f(x)\big)$ belongs to $I_K[x,f(x)]$ due to the remark made above. Obviously, $R_y(G,F)$ is a polynomial in x of degree no greater than $nh+(n-1)\,n_1$ (take the determinant representation of the resultant). Since $f(x)$ is of degree n over $K[x]$ and $G(x,y)$ is of degree no greater than $n-1$ in y, $R_y(G,F)$ is not identically zero.

Passing to the function $V(\omega)$ we see that

$$R_y(G,F)=V(\omega)\,M\big(\omega,f(\omega)\big)$$

and since $f(\omega)_v$ is regular in the disc $|\omega|_v<R_v(f)$, the same is true for $M\big(\omega,f(\omega)\big)$. Hence, any zero of $V(\omega)$ in the disc $|\omega|_v<R_v(f)$ is a zero of the polynomial $R_y(G,F)$ in $K[\omega]$ and the total number of such zeros does not exceed the degree of the polynomial.

5. Passing to auxiliary polynomials

In the sequel we fix the numbers h and ε provided that $\varepsilon h\geqq n_1$, $0<4e n\varepsilon<1$.

Let $x_0\in K$, $x_0\neq 0$, and suppose $|x_0|_v<R_v(f)$ for some metrics associated with v. Define the functions $f(\omega)=f(\omega)_v$ and $V(\omega)=V(\omega)_v$ as in Lemma 4. 6 with the polynomials $A_0(x),\dots,A_{n-1}(x)$ in (4. 7) introduced by the Lemma 4. 1. Then $V(\omega)$ at the point $\omega=0$ has a zero of order not less than $n(h+1)(1-\varepsilon)$, while the Lemma 4. 6 shows that the total number of zeros of $V(\omega)$ in the disc $|\omega|_v<R_v(f)$ is no more than $nh+(n-1)\,n_1$. Hence, if

$$V^{(i)}(x_0)=0\quad(0\leqq i\leqq s-1),\quad V^{(s)}(x_0)\neq 0,\tag{5. 1}$$

we must have $s\leqq 2\varepsilon nh$ since we admit $\varepsilon h\geqq n_1$.

Return now for a while to the power series (4. 5) and set

$$V_s(x) = D^s\left(\mathfrak{B}^{2s} V(x, f(x))\right).$$

We find

$$\text{ord } V_s(x) \geqq \text{ord } V(x) - s \geqq (1-3\varepsilon)\, n(h+1). \tag{5. 2}$$

Define

$$\begin{aligned} W(x) &= \frac{1}{\pi_s} V_s(x) \\ &= B_0(x) + B_1(x) f(x) + \cdots + B_{n-1}(x) f^{n-1}(x) \end{aligned}$$

where the numbers π_s were introduced in Lemma 4. 5. We know that

$$B_i(x) = \frac{1}{\pi_s} A_{is}(x) \in I_K[x], \tag{5. 3}$$

$$g = \max_{(i)} \deg B_i(x) \leqq h + 2nn_1 s$$

and we have to estimate yet the maximal size of the polynomials $B_i(x)$ $(i = 0, 1, \ldots, n-1)$.

We estimate first the numbers π_s. Since by Lemma 3. 3

$$E_f \leqq |F|^{2k}$$

and C_3 as defined in Lemma 3. 4, exceeds

$$(4|F|)^{k3^8} > |F|^{2k},$$

the inequality $q > C_3$ implies $q \nmid E_f$. Taking $s > C_3$ we find

$$\pi_s = \prod_{\substack{q > C_3 \\ q \nmid D_K}} q^{\left[\frac{s}{q}\right] + \left[\frac{s}{q^2}\right] + \cdots} \geqq s! \left(\prod_{q \leqq C_3} q^{\frac{1}{q-1}}\right)^{-s} \Delta_K^{-s},$$

where

$$\Delta = \prod_{q \mid D_K} q^{\frac{1}{q-1}}$$

and in all the cases q runs through the rational primes. Since

$$\sum_{q \leqq C_3} \frac{\log q}{q-1} = \log C_3 + O(1)$$

we find

$$\pi_s \geqq s! (C_3 C_9 \Delta_K)^{-s} \geqq s! (4 C_9 |F|^{ck} \Delta_K)^{-s} \tag{5. 4}$$

with C_9 the absolute constant.

Now we use the estimates of Lemma 4. 4, (5. 3) and (5. 4) to estimate the sizes of the polynomials $B_i(x)$ (we take $s \geqq 1$, since if $s=0$ the final result is given by Lemma 4. 1). We find

$$\max \overline{|B_i(x)|} < (C_8 \overline{|F|}^{2n})^s H_0 (4 C_9 \overline{|F|}^{ck} \Delta_K)^s \frac{1}{s!} \times \prod_{r=0}^{s-1} (s+h+2nn_1 r)$$

$$< (C_{10} \overline{|F|}^{2n+ck})^s H_0 \left(\frac{h}{s}\right)^s,$$

where C_{10} is determined through n and n_1. Since $0 < \varepsilon < (4en)^{-1}$, we have

$$\left(\frac{h}{s}\right)^s \leqq \left(\frac{h}{2\varepsilon nh}\right)^{2\varepsilon nh} = e^{-2\varepsilon nh \log(2\varepsilon n)} < C_{11}^h,$$

and since by Lemma 4. 1 we have the estimate for H_0, we find from (3. 2):

$$\max \overline{|B_i(x)|} < C_5 \left(C_{12} \overline{|F|}^{(2n+C)2\varepsilon^2 + 9kn^2 n_1^2} \Delta_K^{2\varepsilon^2}\right)^{\frac{n(h+1)}{\varepsilon}} \tag{5. 5}$$

where $C_{12} = C_{12}(n, n_1, k)$ and C_5 is determined in Lemma 4. 1.

We have yet excluded in section 2 such x_0 that $F(x_0, y)$ has a multiple root. Hence

$$\mathfrak{B}(x_0, f(x_0)_v) = \frac{\partial}{\partial y} F(x_0, f(x_0)_v) \neq 0.$$

It follows from (5. 1) that

$$V_s(x_0)_v = \mathfrak{B}^{2s}(x_0, f(x_0)_v)\, V^{(s)}(x_0)_v \neq 0. \tag{5. 6}$$

Let now v correspond to one of the finite metrics of K, $v \in S_0$. Since by the definition of the number E_f we have

$$|f_v E_f^\nu|_v \leqq 1 \qquad (\nu = 1, 2, \ldots),$$

the series $f(\omega)$ is convergent for $\omega \in K_v$ in the disc $|\omega|_v < |E_f|_v$, so that $R_v(f) \geqq |E_f|_v > 0$.

Suppose $|x_0|_v < |E_f|_v$ and write for brevity $\theta_v = f(x_0)_v$. Setting

$$L(y) = B_0 + B_1 y + \cdots + B_{n-1} y^{n-1}, \qquad B_i = B_i(x_0),$$

we see that $B_i \in K$ $(i = 0, 1, \ldots, n-1)$, and due to (5. 6)

$$L(\theta_v) = W(x_0)_v = \frac{1}{\pi_s} V_s(x_0)_v \neq 0.$$

Since we have from (5. 2)

$$\operatorname{ord} W(x) \geqq (1 - 3\varepsilon)\, n(h+1),$$

and the coefficients of the power series $f(E_f x)$ are v-integers, so are the coefficients of $W(E_f x)$ and we find

$$0 \neq |L(\theta_v)|_v \leqq \left|\frac{x_0}{E_f}\right|_v^{\operatorname{ord} W(x)} \leqq \left|\frac{x_0}{E_f}\right|^{(1-3\varepsilon)n(h+1)}. \tag{5. 7}$$

Notice yet that if w corresponds to any other finite metrics of $\mathbb{K}$ and $|x_0|_w < |E_f|_w$ then we shall have

$$|L(\theta_w)|_w \leqq \left| \frac{x_0}{E_f} \right|_w^{(1-3\varepsilon)n(h+1)}. \tag{5. 8}$$

In a similar way we consider the estimates in the infinite metrics. Let v correspond to one of them, $v \in S_\infty$, and we suppose that $|x_0|_v \leqq (4A)^{-1}$. It follows from Lemma 4. 2 that $R_v(f) \geqq A^{-1} > 0$, hence, $f(x_0)_v = \theta_v$ is determined as before. We construct again the polynomial $L(y)$ for which $L(\theta_v) \neq 0$ and we have to estimate $|L(\theta_v)|_v$ from above.

To do so we notice first that the coefficients b_l of the power series $V(x)$, introduced in the proof of Lemma 4. 1, satisfy

$$b_l = 0 \qquad (l = 0, 1, \ldots, l_0 - 1),$$

$$\boxed{b_l} \leqq H_0 + \sum_{j=1}^{n-1} \sum_{i=0}^{h} H_0 l^n A^l < 2nhH_0 l^n A^l \qquad (l = l_0, l_0+1, \ldots).$$

Further we have

$$\frac{1}{s!} |V^{(s)}(x_0)|_v \leqq \sum_{l=l_0}^{\infty} \binom{l}{s} \boxed{b_l} |x_0|_v^{l-s}.$$

Since $s < l_0 \leqq l$ we find $\binom{l}{s} < 2^l$ and using the estimates for the sizes of b_l, we obtain

$$\begin{aligned} \frac{1}{s!} |V^{(s)}(x_0)|_v &\leqq \sum_{l=l_0}^{\infty} 2^{l+1} nhH_0 l^n A^l |x_0|_v^{l-s} \\ &= 2nhH_0 (2A)^{l_0} |x_0|_v^{l_0-s} \sum_{l=0}^{\infty} (2A|x_0|_v)^l (l_0+l)^n. \end{aligned}$$

Since $|x_0|_v < (4A)^{-1}$, the sum of the last series does not exceed

$$\begin{aligned} \sum_{l=0}^{\infty} 2^{-l}(l_0+l)^n &\leqq \sum_{l=0}^{l_0} 2^{-l}(2l_0)^n + \sum_{l=l_0+1}^{\infty} 2^{-l}(2^l)^n \\ &< 3(2l_0)^n. \end{aligned}$$

Hence, we find

$$\frac{1}{s!} |V^{(s)}(x_0)|_v < 6nhH_0 (2l_0)^n (2A)^{l_0} |x_0|_v^{l_0-s}. \tag{5. 9}$$

We shall use (5. 6) to estimate $V_s(x_0)_v$, hence, we need an estimate for $\mathfrak{B}^{2s}(x_0, f(x_0)_v)$. Since $f(x_0)_v = \theta_v$ is a root of the polynomial $F(x_0, y)$, we have

$$\theta_v^n = a_1(x_0)\, \theta_v^{n-1} + \cdots + a_n(x_0), \qquad \theta_v \in \mathbb{K}_v. \tag{5. 10}$$

We claim that

$$|\theta_v|_v \leqq \max\left(1, n(n_1+1)\boxed{F}\right) = n(n_1+1)\boxed{F}.$$

Indeed, if $|\theta_v|_v > 1$, we find from (5. 10)

$$|\theta_v|_v \leqq \left((n_1+1)\overline{|a_1(x)|}+\cdots+(n_1+1)\overline{|a_n(x)|}\right)\max\left(1, |x_0|_v^{n_1}\right)$$
$$\leqq n(n_1+1)\overline{|F|},$$

since $|x_0|_v < 1$. Hence,

$$|\mathfrak{B}(x_0, f(x_0)_v)|_v = \left|\frac{\partial}{\partial y}F(x_0, \theta_v)\right|_v < C_{13}\overline{|F|}^{n-1},$$

where $C_{13} = C_{13}(n, n_1)$.

Thus, we obtain from (5. 6) and (5. 9)

$$\frac{1}{s!}|V_s(x_0)|_v < \left(C_{14}\overline{|F|}\right)^{2ns} h H_0 l_0^n (2A)^{l_0} |x_0|_v^{l_0-s}.$$

For the $W(x_0)_v$ we now obtain

$$|W(x_0)|_v = \frac{1}{\pi_s}|V_s(x_0)|_v \leqq (4C_9\overline{|F|}^{ck}\Delta_K)^s \frac{1}{s!}|V_s(x_0)|_v,$$

and, finally, we obtain for the $L(\theta_v)$:

$$0 \neq |L(\theta_v)|_v < (C_{15}\overline{|F|}^{2n+ck}\Delta_K)^s h H_0 l_0^n (2A)^{l_0} |x_0|_v^{l_0-s}$$
$$< C_5 G_1^{\frac{n(h+1)}{\varepsilon}} |x_0|_v^{(1-3\varepsilon)n(h+1)}, \tag{5. 11}$$

where

$$G_1 = C_{16}\overline{|F|}^{9k(nn_1)^2+8k(nn_1)^2\varepsilon+2ck\varepsilon^2+4n\varepsilon^2}\Delta_K^{2\varepsilon^2},$$

and C_{16} is determined by n, n_1 and k. This is an analogy to (5. 7).

To obtain the analogy to (5. 8) for $w \in S_\infty$ with $|x_0|_w < (4A)^{-1}$, we argue as before, but since we do not have (5. 1) now, we have to use the general formula

$$V_s(x) = D^s\left(\mathfrak{B}^{2s}V(x)\right) = \sum_{r=0}^{s}\binom{s}{r}\left(D^r\mathfrak{B}^{2s}(x, f(x))\right)V^{(s-r)}(x). \tag{5. 12}$$

For the $V^{(s-r)}(x_0)_w$ we shall have an estimate of the type (5. 9) with $s-r$ instead of s. To estimate $D^r\mathfrak{B}^{2s}(x_0, f(x_0)_w)$ we may appeal to complex analysis, since w is infinite and we may consider the corresponding metrics as an ordinary absolute value determined on the field of complex numbers. Setting $\rho = (4A)^{-1}$, we shall have

$$D^r\mathfrak{B}^{2s}(x_0, f(x_0)) = \frac{r!}{2\pi i}\int_{|z-x_0|=\rho}\frac{\mathfrak{B}^{2s}(z, f(z))}{(z-x_0)^{r+1}}dz.$$

For any complex z with $|z| \leqq (2A)^{-1}$ we have

$$|f(z)| \leqq n(n_1+1)\overline{|F|},$$

so that

$$|\mathfrak{B}(z, f(z))| \leqq C_{17}(n, n_1)\overline{|F|}^{n-1}.$$

Hence,

$$|D^r\mathfrak{B}^{2s}(x_0, f(x_0))| \leqq \frac{r!}{2\pi} 2\pi (C_{17}\overline{|F|}^{n-1})^{2s} \rho^{-r}$$
$$= r!(C_{17}\overline{|F|}^{n-1})^{2s}(4A)^r.$$

Now we find from (5. 12)

$$\frac{1}{s!}|V_s(x_0)|_w \leqq \sum_{r=0}^{s} \binom{s}{r} \frac{1}{r!} |D^r\mathfrak{B}^{2s}(x_0, f(x_0)_w)|_w \frac{1}{(s-r)!} |V^{(s-r)}(x_0)|_w$$
$$< (2C_{17}\overline{|F|}^{n-1})^{2s} C_{18} h H_0 l_0^n (2A)^{l_0} |x_0|_w^{l_0 - s},$$

where $C_{18} = C_{18}(n, n_1, k)$, and it follows then

$$|W(x_0)|_w < C_{18}(C_{19}\overline{|F|}^{2n+ck} \Delta_K)^s h H_0 l_0^n (2A)^{l_0} |x_0|_w^{l_0 - s}.$$

Hence we obtain

$$|L(\theta_w)|_w < C_5 G_2^{\frac{n(h+1)}{\varepsilon}} |x_0|_w^{(1-3\varepsilon)n(h+1)} \tag{5. 13}$$

with G_2 of the type G_1, possibly, with another value of C_{16}.

Let us call v (and the metrics of v) "leading" if $|x_0|_v < |E_f|_v$ or $|x_0|_v \leqq (4A)^{-1}$ in accordance with $v \in S_0$ or $v \in S_\infty$. Then we may formulate the results of this section as the following

Lemma 5. 1. *Given numbers h and ε provided that $\varepsilon h \geqq n_1$, $0 < 4en\varepsilon < 1$, and given any leading v there exists polynomial $L(y)$ in $I_K[y]$ of degree no more than $g \leqq (1+\varepsilon_1)h$, $\varepsilon_1 = 4n^2 n_1 \varepsilon$, and of size satisfying* (5. 5) *and such that $L(\theta_v)$ satisfies* (5. 7), *if $v \in S_0$, and* (5. 11), *if $v \in S_\infty$. In addition to that for any leading w we have* (5. 8) *or* (5. 13) *in accordance with $w \in S_0$ or $w \in S_\infty$.*

Notice that $L(\theta_v) \neq 0$, while we do not know whether this is true for θ_w instead of θ_v, and the v-metrics play the fundamental role in the determination of the polynomial $L(y)$.

Let $\bar{\theta}_w$ be a root of $F(x_0, y)$ in $\bar{K}_w$, the algebraic closure of K_w (we write $\bar{\theta}_w$ to distinguish it from $\theta_w = f(x_0)_w$). We need yet the estimates of

$$L(\bar{\theta}_w) = B_0(x_0) + B_1(x_0)\,\bar{\theta}_w + \cdots + B_{n-1}(x_0)\,\bar{\theta}_w^{n-1} \tag{5. 14}$$

in the metrics of w without any suppositions and restrictions on w, and the result is given by the lemma as follows.

Lemma 5. 2. *If w corresponds to any metrics on K and $\bar{\theta}_w$ is a root of $F(x_0, y)$ in $\bar{K}_w$, then*

$$|L(\bar{\theta}_w)|_w \leqq \max(1, |x_0|_w)^{(1+\varepsilon_1)h + (n-1)n_1} \tag{5. 15}$$

in case of $w \in S_0$ and

$$|L(\bar{\theta}_w)|_w < C_5 G_4^{\frac{n(h+1)}{\varepsilon}} \max(1, |x_0|_w)^{(1+\varepsilon_1)h + (n-1)n_1} \tag{5. 16}$$

in case of $w \in S_\infty$, where $\varepsilon_1 = 4n^2 n_1 \varepsilon$ and G_4 is of the type G_1 with another value of $C_{16} = C_{16}(n, n_1, k)$.

Proof. Let us consider first the finite w. Since $\deg B_i(x) \leqq g$ and all of the $B_i(x)$ lie in $I_{\mathbb{K}}[x]$, we have

$$|B_i(x_0)|_w \leqq \max(1, |x_0|_w)^g .$$

Estimating $|\bar{\theta}_w|_w$ from the equation

$$\bar{\theta}_w^n = a_0(x_0)\,\bar{\theta}_w^{n-1} + \cdots + a_n(x_0) \tag{5. 17}$$

we find

$$|\bar{\theta}_w|_w \leqq \max(1, |x_0|_w)^{n_1} .$$

Thus, we get from (5. 14)

$$|L(\bar{\theta}_w)|_w \leqq \max(1, |x_0|_w)^{g+(n-1)n_1},$$

which gives (5. 15), since $g \leqq (1+\varepsilon_1)\,h$.

Let now w be infinite. Using (5. 5) we find

$$\begin{aligned} |B_i(x_0)|_w &\leqq (g+1)\,\overline{|B_i(x)|}\,\max(1, |x_0|_w)^g \\ &< C_5 G_3^{\frac{n(h+1)}{\varepsilon}} \max(1, |x_0|_w)^{(1+\varepsilon_1)h}, \end{aligned}$$

where

$$G_3 = C_{20}\,\overline{|F|}^{9k(nn_1)^2 + 2(2n+ck)\varepsilon^2}\,\Delta_{\mathbb{K}}^{2\varepsilon^2}$$

and C_{20} is determined by n, n_1 and k. From the equation (5. 17) we find

$$|\bar{\theta}_w|_w \leqq n(n_1+1)\,\overline{|F|}\,\max(1, |x_0|_w)^{n_1} .$$

Hence, the equality (5. 14) gives (5. 16).

6. Proof of the Theorem

We signify here, as before, S_0 and S_∞ the sets of all finite and all infinite valuations on $\mathbb{K}$ respectively, and let $S = S(x_0)$ be the set of all leading valuations, as they were introduced in section 5.

Let

$$F(x_0, y) = F_1(y) \cdots F_r(y) \tag{6. 1}$$

be the decomposition into irreducible polynomials in $\mathbb{K}[y]$, each polynomial with the leading coefficient 1, and let $v \in S$. Then $\theta_v = f(x_0)_v$ is a root of $F^v(x_0, y)$, hence, a root of the only polynomial $F_j^v(y)$, say, $F_1^v(y)$. We take the numbers h and ε and find the polynomial $L(y)$ by Lemma 5. 1. Since $L^v(\theta_v) \neq 0$ and $F_1^v(y)$ is irreducible in $\mathbb{K}^v[y]$, it follows that $L^v(y)$ and $F_1^v(y)$ have no common root, the resultant

$$R(L^v, F_1^v) \neq 0 . \tag{6. 2}$$

Let $d = \deg F_1(y)$ and $\theta_{1v}, \ldots, \theta_{dv}$ signify the roots of $F_1^v(y)$ in $\bar{\mathbb{K}}^v$. Since $F_1^v(\theta_v) = 0$, θ_v is contained among θ_{iv}, say $\theta_v = \theta_{1v}$. We have

$$R(L^v, F_1^v) = L^v(\theta_v) \prod_{i=2}^{d} L^v(\theta_{iv}) . \tag{6. 3}$$

Suppose $v \in S_0$. Since $|x_0|_v < |E_f|_v \leqq 1$, the leading coefficient of $F(x, y)$ is 1 and all its coefficients are v-integers, it follows that all of the roots θ_{iv} are v-integers. Hence, from $L(y) \in I_K[y]$ we find

$$|L(\theta_{iv})|_v \leqq 1 \qquad (i = 2, 3, \ldots, d).$$

Now the estimate (5. 7) and the decomposition (6. 3) give

$$|R(L, F_1)|_v \leqq \left| \frac{x_0}{E_f} \right|_v^{(1-3\varepsilon)n(h+1)}. \tag{6. 4}$$

Suppose $v \in S_\infty$. Since then $|x_0|_v \leqq (4A)^{-1} < 1$, it follows that θ_{iv} as the roots of $F(x_0, y)$ satisfy

$$|\theta_{iv}|_v \leqq n(n_1+1)\overline{|F|} \qquad (i = 1, 2, \ldots, d)$$

and we have in accordance with (5. 16)

$$|L(\theta_{iv})|_v < C_5 G_4^{\frac{n(h+1)}{\varepsilon}} \qquad (i = 2, 3, \ldots, d).$$

Hence, we find from (5. 11) and (6. 3)

$$|R(L, F_1)|_v < C_5^d G_5^{\frac{n(h+1)}{\varepsilon}} |x_0|_v^{(1-3\varepsilon)n(h+1)}, \tag{6. 5}$$

where

$$G_5 = G_1 G_4^{d-1}.$$

Notice yet that in force of the second assertion of Lemma 5. 1 we have similar estimates in the metrics of w instead of v, if $w \in S$ and $\theta_w = f(x_0)_w$ is a root of $F_1^w(y)$.

Thus, taking the product of all such estimates with the corresponding exponents k_w, we obtain

$$\prod_{\substack{w \in S \\ \theta_w \in F_1^w}} |R(L, F_1)|_w^{k_w} < G_6 \prod_{\substack{w \in S \\ \theta_w \in F_1^w}} |X_0|_w^{k_w(1-3\varepsilon)n(h+1)}, \tag{6. 6}$$

where G_6 is given by

$$G_6 = C_5^{dk} G_7^{dk\frac{n(h+1)}{\varepsilon}} \prod_{w \in S_0} |E_f|_w^{-k_w} \tag{6. 7}$$

and G_7 by a factor depending only on n, n_1 and k differs from G_1. In view of the product formula (1. 5) and the estimate of Lemma 3. 3 the product in (6. 7) is

$$E_f^k \leqq \overline{|F|}^{2k^2}.$$

Hence, we have

$$G_6 \leqq C_5^{dk}(C_{21} G_1)^{dk\frac{n(h+1)}{\varepsilon}} \overline{|F|}^{2k^2}, \qquad C_{21} = C_{21}(n, n_1, k). \tag{6. 8}$$

Denote T the set of all w not occurring in the set of $w \in S$ for which $\theta_w \in F_1^w$, i.e. T is the complementary set to the set of w in the products (6. 6). We find from Lemma 5. 2: if $w \in T \cap S_0$, then

$$|R(L, F_1)|_w = \prod_{i=1}^{d} |L(\theta_{iw})|_w \leqq \max(1, |x_0|_w)^{(1+\varepsilon_1)hd+(n-1)n_1 d};$$

if $w \in T \cap S_\infty$, then

$$|R(L, F_1)|_w < C_5^d G_4^{d\frac{n(h+1)}{\varepsilon}} \max(1, |x_0|_w)^{(1+\varepsilon_1)hd+(n-1)n_1 d}.$$

Hence, we obtain

$$\prod_{w\in T} |R(L, F_1)|_w^{k_w} < C_5^{dk} G^{dk\frac{n(h+1)}{\varepsilon}} \times \prod_{w\in T} \max(1, |x_0|_w^{k_w})^{(1+\varepsilon_1)hd+(n-1)n_1 d}.$$

Since the set T contains all w with $|x_0|_w \geqq 1$, we have

$$\prod_{w\in T} \max(1, |x_0|_w^{k_w}) = H_K(x_0),$$

and it follows then

$$\prod_{w\in T} |R(L, F_1)|_w^{k_w} < C_5^{dk} G_4^{dk\frac{n(h+1)}{\varepsilon}} \times H_K(x_0)^{(1+\varepsilon_1)hd+(n-1)n_1 d}.$$

The inequality (6. 2) and the product formula (1. 5) show that

$$1 = \prod_w |R(L, F_1)|_w^{k_w},$$

and we find from (6. 6) and (6. 9)

$$1 = \prod_{\substack{w\in S\\ \theta_w\in F_1^w}} |R(L, F_1)|_w^{k_w} \prod_{w\in T} |R(L, F_1)|_w^{k_w} < G_6 \prod_{\substack{w\in S\\ \theta_w\in F_1^w}} |x_0|_w^{k_w(1-3\varepsilon)n(h+1)} \times C_5^{dk} G_4^{dk\frac{n(h+1)}{\varepsilon}} H_K(x_0)^{(1+\varepsilon_1)hd+nn_1 d}.$$

Since h may be taken arbitrary large and G_6 satisfies (6. 8), we obtain

$$1 < (C_{21} G_1)^{\frac{dkn}{\varepsilon}} \prod_{\substack{w\in S\\ \theta_w\in F_1^w}} |x_0|_w^{k_w(1-3\varepsilon)} \times G_4^{\frac{dkn}{\varepsilon}} H_K(x_0)^{(1+\varepsilon_1)d}.$$

Taking into account that G_4 is a value of the type G_1 we find

$$\sum_{\substack{w\in S\\ \theta_w\in F_1^w}} \log|x_0|_w^{-k_w} < (1-3\varepsilon)^{-1}\frac{d}{n}\log H_K(x_0)+\frac{dkn}{\varepsilon}\log(C_{22}G_8),$$

where C_{22} is expressible through n, n_1 and k,

$$G_8=|F|^{9k(nn_1)^2+8k(nn_1)^2\varepsilon+2ck\varepsilon^2+4n\varepsilon^2}\Delta_K^{2\varepsilon^2}.$$

Here we have no restrictions on the choice of ε, except for $0<4en\varepsilon<1$. Hence, if we take

$$\varepsilon=\frac{3}{2}kn^2n_1\sqrt{\frac{\log|F|}{\log H_K(x_0)}}$$

and suppose

$$H(x_0)>(18n^3n_1k)^2|F|,$$

we shall have

$$\sum_{\substack{w\in S\\ \theta_w\in F_1^w}} \frac{\log|x_0|_w^{-k_w}}{\log H(x_0)} < \frac{d}{n}+\rho, \tag{6. 10}$$

$$\rho=C_{23}\sqrt{\frac{\log|F|}{\log H_K(x_0)}}\cdot\left(1+\frac{\log\Delta_K}{\log H_K(x_0)}\right), \tag{6. 11}$$

where $C_{23}=C_{23}(n, n_1, k)$.

This is the principle inequality from which the final result follows in a purely formal way.

Indeed, if there exists a leading w for which θ_w is not the root of $F_1^w(y)$, let $F_2^w(y)$ be the polynomial from the factors in (6. 1) having θ_w as a root. We may repeat the argument and obtain the inequality of the type (6. 10) with F_2^w instead of F_1^w and d the degree of $F_2(y)$ and so on, until all of the leading w will be exhausted. Let $F_1,\ldots,F_s$ be all of the polynomials F_j, occurring in this process, $s\leqq r$. Taking the sum of the inequalities (6. 10) over all F_j with $j=1, 2,\ldots, s$, we find

$$\sum_{w\in S}\frac{\log|x_0|_w^{-k_w}}{\log H_K(x_0)}\leqq\frac{1}{n}\sum_{j=1}^{s}d_j+s\rho,\qquad d_j=\deg F_j(y). \tag{6. 12}$$

Notice now that we have

$$\sum_w \log\max(1,|x_0|_w^{-k_w})=\sum_{w\in S}\log|x_0|_w^{-k_w}+\sum_{\substack{w\in S_0\\ |E_f|_w\leqq|x_0|_w\leqq 1}}\log|x_0|_w^{-k_w}+\sum_{\substack{w\in S_\infty\\ (4A)^{-1}<|x_0|_w\leqq 1}}\log|x_0|_w^{-k_w}. \tag{6. 13}$$

Since in view of (1. 6) the initial sum is equal to $\log H_K(x_0)$ while the third and the fourth sums do not exceed $k\log E_f$ and $k\log(4A)$ respectively, we find

$$\sum_{w\in S}\log|x_0|_w^{-k_w}\geqq\log H_K(x_0)-C_{24}\log|F| \tag{6. 14}$$

with $C_{24}=C_{24}(n, n_1, k)$. Hence, the inequality (6. 12) shows that

$$1\leqq\frac{1}{n}\sum_{j=1}^{s}\deg F_j+n\rho+C_{24}\frac{\log|F|}{\log H_K(x_0)}.$$

If the height $H_{\mathbb{K}}(x_0)$ is so large that

$$n\rho + C_{24}\frac{\log|F|}{\log H_{\mathbb{K}}(x_0)} < \frac{1}{n},$$

we must have the equality

$$\sum_{j=1}^{s} \deg F_j = n$$

and it follows then $s=r$. Hence, the inequalities of the type (6. 10) will be true for all of the polynomials F_j:

$$\sum_{\substack{w\in S\\ \theta_w\in F_1^w}} \frac{\log|x_0|_w^{-k_w}}{\log H_{\mathbb{K}}(x_0)} < \frac{d_j}{n}+\rho \qquad (j=1,2,\ldots,r). \tag{6. 15}$$

Taking any m in the range $1\leqq m\leqq r$ and summing up (6. 15) over all $j\neq m$ we shall find from (6. 14)

$$\sum_{\substack{w\in S\\ \theta_w\in F_m^w}} \frac{\log|x_0|_w^{-k_w}}{\log H_{\mathbb{K}}(x_0)} > \frac{d_m}{n} - n\rho - c_{24}\frac{\log|F|}{\log H_{\mathbb{K}}(x_0)}. \tag{6. 16}$$

We see now that the inequalities (6. 15) and (6. 16) together with the expression of ρ in (6. 11) lead to the formula

$$\sum_{\substack{w\in S\\ \theta_w\in F_m^w}} \frac{\log|x_0|_w^{-k_w}}{\log H_{\mathbb{K}}(x_0)} = \frac{d_m}{n} + O\left(\left(1+\frac{\log \Delta_{\mathbb{K}}}{\log H_{\mathbb{K}}(x_0)}\right)\sqrt{\frac{\log|F|}{\log H_{\mathbb{K}}(x_0)}}\right).$$

We may replace $\log|x_0|_w^{-k_w}$ by $\log\max(1, |x_0|_w^{-k_w})$ and omit here in the summation the condition $w\in S$, since the possible error will not exceed $(\log H_{\mathbb{K}}(x_0))^{-1}$ times the two last sums in the decomposition (6. 13) and will contribute a value $O\left(\frac{\log|F|}{\log H_{\mathbb{K}}(x_0)}\right)$ at most. Thus, we have proven the main assertion (1. 11) of the Theorem.

To prove that the remainder term may be reduced to the form (1. 12) in case of the field $\mathbb{K}$ coinciding with the field generated by the coefficients of the polynomial $F(x, y)$, we argue as follows.

Let α be a coefficient of $F(x, y)$, an algebraic integer. The discriminant of the field $\mathbb{Q}(\alpha)$ is a divisor of the discriminant of the number α, hence, its absolute value does not exceed

$$(2|\alpha|)^{(\deg\alpha)^2} \leqq (2|F|)^{k^2}.$$

It follows then from the multiplicative property of the differents in relative extensions that $D_{\mathbb{K}}$ is a divisor of the product

$$\prod_{\alpha} D_{\mathbb{Q}(\alpha)}^{k},$$

where $D_{\mathbb{Q}(\alpha)}$ is the discriminant of $\mathbb{Q}(\alpha)$ and the product is taken over all the coefficients of $F(x, y)$. Hence,

$$|D_{\mathbb{K}}| \leqq \prod_{\alpha} |D_{\mathbb{Q}(\alpha)}|^{k} \leqq (2|F|)^{(n+1)(n_1+1)k^3}.$$

Since

$$\Delta_{\mathbb{K}} = \prod_{q \mid D_{\mathbb{K}}} q^{\frac{1}{q-1}} \leqq |D_{\mathbb{K}}|\,,$$

we find

$$\log \Delta_{\mathbb{K}} \leqq (n+1)\,(n_1+1)\,k^3 \log\left(2\overline{|F|}\right),$$

which substituted into (1. 11) gives (1. 12).

The proof of the Theorem is completed.

7. Discussion

Here we consider simple corollaries from the Theorem. They show the directions in which more deep results may be obtained by the same method. We give them in the form of remarks illustrating the contents of the Theorem and do not develop the exact and most general form in which they follow in fact.

In the sequel the expression "x_0 is large enough" means that $H_{\mathbb{K}}(x_0)$ exceeds a value which may be effectively expressed through n, n_1, k, $\overline{|F|}$ and $\Delta_{\mathbb{K}}$ in accordance with our intention to make the remainder term in (1. 11) smaller than $\frac{1}{n}$ or $\frac{1}{nk}$ (we may believe then that the series (1. 8) converges in $x = x_0$).

Suppose x_0 is a unit in $I_{\mathbb{K}}$ and

$$|x_1^{(1)}| < 1\,, \qquad |x_0^{(i)}| \geqq 1 \qquad (i = 2, 3, \ldots, k) \tag{7. 1}$$

(notations as in section 1). In this case $|x_0^{(1)}|^{-1}$ is the only v-component of x_0 which is more than 1, and the $\mathbb{K}$-height of x_0 coincides with it. Hence, the sum in the left-hand side of (1. 11) which contains this v-component, reduces to 1, and if x_0 is large enough the corresponding d_j must be n. We conclude that the polynomial $F(x_0, y)$ is irreducible in $\mathbb{K}[y]$.

Since any positive integer power of x_0 satisfies (7. 1) too, we may replace x_0 by such powers with the same result. In particular, all the solutions of the diophantine equation

$$F(x_0^m, y) = 0 \tag{7. 2}$$

in positive integers m and $y \in I_{\mathbb{K}}$ may be determined effectively.

It is obvious that in fact we may consider the numbers $x_0 \in \mathbb{K}$ which satisfy instead of (7. 1) the more general condition

$$|x_0|_v < 1\,, \qquad |x_0|_w \geqq 1 \qquad (w \neq v)$$

and we shall come to the same conclusion.

We can find by a similar argument that if we take $x_0 \in \mathbb{K}$ and assume that all of the numbers $|x_0|_v$ with $|x_0|_v < 1$ are multiplicatively independent, then the polynomials $F(x_0^m, y)$ will be irreducible in $\mathbb{K}[y]$, provided that m exceeds the bound depending on x_0, $\mathbb{K}$, the degree and the size of F. Indeed, for any positive integer m we have

$$\frac{\log (x_0^m)_v}{\log H_{\mathbb{K}}(x_0^m)} = \frac{\log (x_0)_v}{\log H_{\mathbb{K}}(x_0)}\,.$$

If we fix $\mathbb{K}$ and F, the formulae (1. 11) will give

$$\sum_{v \in V_j} \frac{\log (x_0)_v}{\log H_{\mathbb{K}}(x_0)} - \frac{d_j}{n} = O\left(\frac{1}{\sqrt{m}}\right) \tag{7. 3}$$

with V_j being some subsets of the set V of all v with $|x_0|_v < 1$. The equality (1. 6) shows that $H_{\mathbb{K}}(x_0)$ is a product of $(x_0)_v$ and we shall have a lower bound in terms of x_0, k and n for the absolute value of the left-hand side of (7. 3), if it is not zero. But m can be taken so large that the right-hand side of (7. 3) will be smaller than this bound, and we must have zero in the left-hand side of (7. 3) then. This is possible only in case of $V_j = V$ and $d_j = n$, since we assume the numbers $(x_0)_v$, $v \in V$, multiplicatively independent.

We find again that the equation (7. 2) has only a finite number of solutions in positive integers m and $y \in \mathbb{K}$ and a bound for all solutions may be provided.

Suppose now that x_0 is a power of a rational prime p. We find from (1. 11) that

$$d_j = \frac{n}{k} \sum_{\substack{\mathfrak{p} \mid p \\ f(x_0)_{\mathfrak{p}} \in F_j^{\mathfrak{p}}}} k_{\mathfrak{p}} \qquad (j = 1, 2, \ldots, r). \tag{7. 4}$$

Hence, for any prime divisor $\mathfrak{p}$ over p

$$[f(x_0)_{\mathfrak{p}} : \mathbb{K}] \geqq \frac{n}{k} k_{\mathfrak{p}}, \tag{7. 5}$$

provided x_0 is large enough. In fact a little more follows from (7. 4), since denoting the sum by k', we shall find that $k'(n, k)$ is a multiple of k, and then

$$[f(x_0)_{\mathfrak{p}} : \mathbb{K}] \geqq \frac{n}{(n, k)}$$

where (n, k) is the greatest common divisor of n and k. This in turn shows that if $(n, k) = 1$ then $F(x_0, y)$ is irreducible in $\mathbb{K}[y]$ and if $(n, k) < n$, $F(x_0, y)$ has no linear factor in $\mathbb{K}[y]$.

Now we intend to show how the inequality (7. 5) may be applied to Galois theory. Suppose $F(x, y) \in \mathbb{Q}[x, y]$ and all the roots $\varkappa_v$ of the polynomial $F(0, y)$ are distinct.

Let $f_1(x), \ldots, f_n(x)$ be all the series satisfying $F(x, y) = 0$ with

$$f_v(0) = \varkappa_v \qquad (v = 1, 2, \ldots, n).$$

One can find such rational integers $u_1, \ldots, u_n$ that

$$f^*(x) = u_1 f_1(x) + \cdots + u_n f_n(x)$$

will be a generating element of the field

$$\mathbb{Q}(x; f_1(x), \ldots, f_n(x)).$$

Suppose $n \geqq 3$ and assume that

$$N = [f^*(x) : \mathbb{Q}(x)] = [f^*(x) : \bar{\mathbb{Q}}(x)],$$

i.e. the minimal polynomial $F^*(x, y)$ for $f^*(x)$ over $\mathbb{Q}(x)$ is supposed to be absolutely irreducible. Setting

$$N_1 = [f_1(x) : \mathbb{Q}(x)],$$

$$N_i = [f_i(x) : \mathbb{Q}(x; f_1(x), \ldots, f_{i-1}(x))] \qquad (i = 2, 3, \ldots, n-1),$$

we have

$$N = N_1 N_2 \cdots N_{n-1}. \tag{7. 6}$$

We may apply the Theorem to the polynomial $F^*(x, y)$, the series $f^*(x)$ and the field $\mathbb{K}^* = \mathbb{Q}(\varkappa_1, \ldots, \varkappa_n)$.

Let x_0 be a large enough power of a rational prime p, the series $f_i(x_0)_\mathfrak{p}$ converge for all prime divisors $\mathfrak{p}$ in $I_{\mathbb{K}^*}$ over p. The numbers $f_i(x)_\mathfrak{p}$ are algebraic,

$$N_1' = [f_1(x_0)_\mathfrak{p} : \mathbb{Q}] \leqq N_1,$$

$$N_i' = [f_i(x_0)_\mathfrak{p} : \mathbb{Q}(f_1(x_0)_\mathfrak{p}, \ldots, f_{i-1}(x_0)_\mathfrak{p})] \leqq N_i \qquad (i = 2, 3, \ldots, n-1)$$

and we have

$$N' = [f^*(x_0)_\mathfrak{p} : \mathbb{Q}] = N_1' N_2' \cdots N_{n-1}'.$$

It follows from (7. 5) that we have

$$N' \geqq \frac{N k_\mathfrak{p}}{\deg \mathbb{K}^*} \tag{7. 7}$$

for any $\mathfrak{p}$ over p. Hence, if p is such that $k_\mathfrak{p} = \deg \mathbb{K}^*$ for some $\mathfrak{p}$, we shall have

$$N' = N, \qquad N_i' = N_i \qquad (i = 1, 2, \ldots, n-1)$$

and we find the isomorphism of Galois groups

$$\mathrm{Gal}\,[F(x_0, y)/\mathbb{Q}] \cong \mathrm{Gal}\,[F(x, y)/\mathbb{Q}(x)].$$

In particular, if $\mathbb{K}^* = \mathbb{Q}$, this is true for all p with large enough x_0.

Suppose $\mathbb{K}^* = \mathbb{Q}(\varkappa_1)$. Then $\deg \mathbb{K}^* = n = N_1$ and we find from (7. 7)

$$N' \geqq \frac{N}{N_1} = N_2 \cdots N_{n-1}.$$

If the polynomial $F(x_0, y)$ has a linear factor $y - y_0$ in $\mathbb{Q}[y]$ there has to exist such a prime divisor $\mathfrak{p}$ over p that $y_0 = f_1(x_0)_\mathfrak{p} \in \mathbb{Q}$ and we shall find

$$[f^*(x_0)_\mathfrak{p} : \mathbb{Q}] \leqq [\mathbb{Q}(f_2(x_0)_\mathfrak{p}, \ldots, f_{n-1}(x_0)_\mathfrak{p}) : \mathbb{Q}]$$
$$= N_2' \cdots N_{n-1}' \leqq N_2 \cdots N_{n-1}.$$

Hence,

$$N' = N_2 \cdots N_{n-1}, \qquad N_i' = N_i \qquad (i = 2, 3, \ldots, n-1),$$

$$\mathrm{Gal}\left[\frac{F(x_0, y)}{y - y_0} \Big/ \mathbb{Q}\right] \cong \mathrm{Gal}\left[\frac{F(x, y)}{y - f_1(x)} \Big/ \mathbb{Q}(x; f_1(x))\right].$$

If the polynomial $\dfrac{F(x, y)}{y-f_1(x)}$ is irreducible over $\mathbb{Q}(x; f_1(x))$, both last Galois groups are transitive and the polynomial $\dfrac{F(x_0, y)}{y-y_0}$ must be irreducible over $\mathbb{Q}$. Hence, $F(x_0, y)$ splits in $\mathbb{Q}[y]$ into a linear factor and a factor of degree $n-1$ whose Galois group may be determined.

For the related topics see the works of Fried [3] and Cohen [1].

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Sur une somme liée à la fonction de Möbius

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1. Introduction

On considère les fonctions arithmétiques

$$M(n, z) := \sum_{d \mid n, d \leqq z} \mu(d) \qquad \text{et} \qquad M(n) := \max_{z} |M(n, z)| .$$

L'objet principal de cet article est l'étude de la somme

$$\sum_{n \leqq x} M(n, z)^2 = x S(z) + O(z^2)$$

avec

$$S(z) := \sum_{m, n \leqq z} \frac{\mu(m)\, \mu(n)}{[m, n]} .$$

Notre résultat essentiel est la convergence de $S(z)$ vers une limite strictement positive.

Les premiers résultats connus portent sur la fonction $M(n)$. La majoration fournie par le théorème de Sperner (cf. [1] et [3]):

$$\forall n,\ M(n) \leqq \binom{\omega(n)}{[\frac{1}{2}\omega(n)]} \leqq \sqrt{\frac{2}{\pi}}\, \frac{2^{\omega(n)}}{\sqrt{\omega(n)}},$$

est optimale. Mais des majorations de la forme

$$\forall \varepsilon > 0, \qquad M(n) \leqq (\alpha + \varepsilon)^{\omega(n)},$$

valides pour presque tout n, ont été établies avec les valeurs successives suivantes de α : $\sqrt{2}$ [4], $\frac{3}{e} = 1{,}1036\ldots$ [6], $\left(\frac{3}{e}\right)^{\frac{\log 2}{\log 3}} = 1{,}0641\ldots$ (conséquence du résultat de Hall et Tenenbaum [7] sur la majoration de l'ordre normal de $\Delta(n)$, le nombre maximal de diviseurs de n dans un intervalle $[t, 2t[$). Erdös conjecture qu'il est possible de prendre $\alpha = 1$.

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La fonction $M(n, z)$ est „presque toujours nulle“, comme le précisent les deux théorèmes suivants:

(i) pour presque tout n et tout $\varepsilon>0$,

$$\sum_{m\leqq n}\frac{1}{m}M(n,m)^2\leqq(2+\varepsilon)^{\omega(n)}\ [5]$$

(ii) dens $\{n: M(n,z)\neq 0\}\ll(\log z)^{-\gamma_0}$ avec $\gamma_0=1-\frac{e}{2}\log 2=0{,}0579\ldots$ [5], puis

$$\gamma_0=1-\frac{\log(e\log 2)}{\log 2}=0{,}0860\ldots,\ [9].$$

La majoration de la somme $\sum_{n\leqq x}M(n,z)^2$ intervient dans la méthode de Vaughan pour l'estimation de formes bilinéaires arithmétiques, utilisée notamment dans [10] pour donner une nouvelle démonstration du théorème de Bombieri-Vinogradov:

$$\sum_{q\leqq Q}\max_{\substack{a,X\leqq Y\\(a,q)=1}}\left|\psi(X;q,a)-\frac{X}{\varphi(q)}\right|\ll_A Y(\log Y)^{-A}+Y^{\frac{1}{2}}Q(\log YQ)^4.$$

Notre théorème permet de remplacer l'exposant 4 du logarithme par toute constante $>\frac{5}{2}$.

Les auteurs tiennent à remercier E. Fouvry pour son aide lors de l'élaboration de ce travail.

2. Enoncé et démonstration

Théorème. *Pour tout réel $x\geqq 1$, on définit la quantité*

$$S(x):=\sum_{m,n\leqq x}\frac{\mu(m)\,\mu(n)}{[m,n]}.$$

Alors, lorsque x tend vers l'infini, $S(x)$ tend vers une limite L et l'on a les expressions suivantes:

(a) $$L=\prod_p\left(1-\frac{2}{p^2}+\frac{1}{p^3}\right)\cdot\sum_{j=1}^{\infty}\log\frac{j+1}{j}\sum_{m,n\leqq j}\frac{\mu(m)\,\mu(n)}{mn}\prod_{p\mid mn}\left(1+\frac{1}{p}-\frac{1}{p^2}\right)^{-1};$$

(b) $$L=\frac{6}{\pi^2}\sum_{j=1}^{\infty}\log\frac{j+1}{j}\sum_{m,n\leqq j}\frac{\mu(mn)}{mn}\prod_{p\mid mn}\left(1+\frac{1}{p}\right)^{-1};$$

(c) $$L=\frac{1}{\pi}\int_0^{\infty}\prod_p\left(1-\frac{1}{p}\right)\left(1+\frac{|1-p^{it}|^2-1}{p}\right)\cdot\frac{dt}{t^2}.$$

Le point de départ de la démonstration consiste à transformer l'expression de $S(x)$: on a

$$(1) \qquad S(x)=\sum_{m,n\leqq x}\frac{\mu(m)\,\mu(n)}{mn}\sum_{d\mid(m,n)}\varphi(d)=\sum_{d\leqq x}\frac{\varphi(d)}{d^2}\,m\left(d,\frac{x}{d}\right)^2,$$

où l'on a posé

$$m(d,u):=\sum_{n\leqq u}\frac{\mu(dn)}{n}=\mu(d)\sum_{\substack{n\leqq u\\(n,d)=1}}\frac{\mu(n)}{n}.$$

De l'identité

$$\sum_{n=1}^{\infty}\mu(dn)\,n^{-s}=\mu(d)\prod_{p\mid d}(1-p^{-s})^{-1}\,\zeta(s)^{-1},$$

on déduit, par la méthode classique d'intégration complexe, l'existence d'une constante positive c telle que l'on ait pour $d\geqq 1$, $u\geqq 1$, et, $0<\varepsilon<1$,

$$|m(d,u)|\ll_\varepsilon \lambda_\varepsilon(d)\exp\{-c\sqrt{\log u}\}$$

avec

$$\lambda_\varepsilon(d):=\prod_{p\mid d}(1+p^{-1+\varepsilon}).$$

En reportant dans (1), il vient

$$S(x)=\sum_{\sqrt{x}<d\leqq x}\frac{\varphi(d)}{d^2}\,m\left(d,\frac{x}{d}\right)^2+O_\varepsilon\left(e^{-c\sqrt{\log x}}\sum_{d\leqq\sqrt{x}}\frac{\lambda_\varepsilon(d)^2}{d}\right)=\sum_{j=1}^{\infty}a(j,x)+o(1)$$

avec

$$(2) \qquad a(j,x):=\sum_{\max(\sqrt{x},x/j+1)<d\leqq x/j}\varphi(d)\,d^{-2}m(d,j)^2\ll_\varepsilon j^{-1}e^{-2c\sqrt{\log j}}.$$

Avec la notation $\chi(a,b)=\begin{cases}1 & \text{si}\quad (a,b)=1\\ 0 & \text{si}\quad (a,b)>1\end{cases}$, on a

$$a(j,x)=\sum_{m,n\leqq j}\frac{\mu(m)\,\mu(n)}{mn}\sum_{\max(\sqrt{x},x/j+1)<d\leqq\sqrt{x}}\mu(d)^2\,\varphi(d)\,d^{-2}\chi(d,mn).$$

La somme intérieure peut être évaluée à l'aide d'un théorème taubérien (comme celui de Ikehara) ou par une méthode élémentaire (comme celle du lemme 4 de [2]). On en déduit que la quantité $a(j,x)$ tend vers une limite $a(j)$ dont la valeur est donnée par

$$a(j)=\prod_p\left(1-\frac{2}{p^2}+\frac{1}{p^3}\right)\cdot\log\frac{j+1}{j}\cdot\sum_{m,n\leqq j}\frac{\mu(m)\,\mu(n)}{mn}\prod_{p\mid mn}\left(1+\frac{1}{p}-\frac{1}{p^2}\right)^{-1}.$$

La majoration (2) permet d'appliquer le théorème de Lebesgue sur la convergence dominée. Cela montre l'existence de la limite de $S(x)$ et la première formule du théorème.

En utilisant, au lieu de (1), l'expression

$$S(x)=\sum_{d\leqq x}\frac{\mu(d)^2}{d}\sum_{\substack{m,n\leqq x/d\\(d,mn)=1}}\frac{\mu(mn)}{mn},$$

on montre, avec des calculs similaires, la seconde formule du théorème, i.e. $L=\sum_{j=1}^{\infty} b(j)$, où $b(j)=\lim_{x\to\infty} b(j,x)$ et

$$b(j,x):=\sum_{x/j+1<d\leqq x/j}\frac{\mu(d)}{d^2}\sum_{\substack{m,n\leqq x/d\\(d,mn)=1}}\frac{\mu(mn)}{mn}.$$

Maintenant, introduisons pour chaque couple (σ, t) de nombres réels la fonction fortement multiplicative $n\mapsto f(n;\sigma,t)$ définie par

$$f(p;\sigma,t)=|1-p^{-\sigma-it}|^2$$

pour chaque p premier. On a l'égalité

$$\int_0^{\infty} M(n,e^u)\,e^{-\sigma u}e^{-iut}du=\frac{1}{\sigma+it}\sum_{d\mid n}\mu(d)\,d^{-\sigma-it};$$

en l'interprétant comme une relation de Fourier pour chaque valeur des paramètres $n\geqq 1$ et $\sigma>0$, on peut écrire la formule de Plancherel

$$\int_0^{\infty} M(n,e^u)^2\,e^{-2\sigma u}du=\frac{1}{2\pi}\int_{-\infty}^{+\infty} f(n;\sigma,t)\frac{dt}{\sigma^2+t^2}. \tag{3}$$

D'autre part, en utilisant la convergence absolue de la série $\sum_{n=1}^{\infty} g(n)\,n^{-1}$, où g est définie par l'identité $f(n;\sigma,t)=\sum_{d\mid n} g(d)$, on établit que, lorsque x tend vers l'infini, la relation asymptotique

$$\sum_{n\leqq x} f(n;\sigma,t)\sim x\prod_p\left(1+\frac{|1-p^{-\sigma-it}|^2-1}{p}\right)$$

a lieu uniformément pour $\sigma\geqq\sigma_0>0$ et $t\in\mathbb{R}$.

En sommant (3) pour $n\leqq x$ et en utilisant l'évaluation

$$\sum_{n\leqq x} M(n,e^u)^2=\sum_{m,n\leqq e^u}\mu(m)\,\mu(n)\left[\frac{x}{[m,n]}\right]=xS(e^u)+O(e^{2u}),$$

on déduit de ce qui précède que l'on a pour $\sigma>1$

$$x\int_0^{\infty} S(e^u)\,e^{-2\sigma u}du+O\left(\int_0^{\infty} e^{-2(\sigma-1)u}du\right)\sim\frac{x}{\pi}\int_0^{\infty}\prod_p\left(1+\frac{|1-p^{-\sigma-it}|^2-1}{p}\right)\frac{dt}{\sigma^2+t^2}.$$

En faisant tendre x vers l'infini, il vient

$$\int_0^{\infty} S(e^u)\,e^{-2\sigma u}du=\frac{1}{\pi}\int_0^{\infty}\prod_p\left(1+\frac{|1-p^{-\sigma-it}|^2-1}{p}\right)\frac{dt}{\sigma^2+t^2}.$$

Les deux membres de cette égalité sont des fonctions analytiques de σ, celle-ci est donc encore valable pour $\sigma>0$. De plus, on a alors

$$\lim_{x\to\infty} S(x) = \lim_{\sigma\to 0+} 2\sigma \int_0^\infty S(e^u)\, e^{-2\sigma u} du$$

$$= \lim_{\sigma\to 0+} \zeta(1+2\sigma)^{-1} \int_0^\infty S(e^u)\, e^{-2\sigma u} du$$

$$= \lim_{\sigma\to 0+} \frac{1}{\pi} \int_0^\infty \prod_p (1-p^{-1-2\sigma}) \left(1+\frac{|1-p^{-\sigma-it}|^2-1}{p}\right) \frac{dt}{\sigma^2+t^2}.$$

En remarquant que l'expression sous l'intégrale est, pour $\sigma>0$,

$$\ll |\zeta(\sigma+1+it)|^{-2} \frac{1}{\sigma^2+t^2} \ll \min\left(1, \frac{(\log t)^2}{t^2}\right),$$

on déduit finalement du théorème de Lebesgue la troisième formule de notre théorème.

3. Etude numérique

Considérons la première formule du théorème: $L=\sum_{j=1}^{\infty} a(j)$. Comme les $a(j)$ sont des limites de termes positifs, on peut utiliser cette formule pour minorer numériquement L. Les calculs effectués sur ordinateur ont fourni notamment les valeurs suivantes:

$$\sum_{j=1}^{1030} a(j) = 0{,}440693525\ldots$$

$$\sum_{j=1}^{2060} a(j) = 0{,}440714260\ldots$$

$$\sum_{j=1}^{4120} a(j) = 0{,}440723596\ldots.$$

L'ensemble des résultats numériques suggère $L \# 0{,}440729$. Il n'existe actuellement aucune majoration explicite de $m(x)=\sum_{n\leqq x} \frac{\mu(n)}{n}$, ni a fortiori de $m(d,j)$, qui laisse le moindre espoir d'obtenir une majoration de L en utilisant les expressions (a) ou (b) du théorème. Une majoration directe de l'intégrale (c) nécessiterait une minoration explicite de $|\zeta(1+it)|$. Il n'en existe actuellement aucune qui permette d'obtenir une majoration acceptable. Nous signalons cependant que R. R. Hall [8] a obtenu un majorant inférieur à 1 en utilisant notamment la formule de Plancherel:

$$L < \frac{1}{2\pi} \int_{-\infty}^{+\infty} |\zeta(1+it)|^{-2} \frac{dt}{1+t^2} = \int_1^\infty \frac{M^2(x)}{x^3} dx < 0{,}947.$$

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The Prym variety for a cyclic unramified cover of a hyperelliptic Riemann surface

By *John Ries* at Binghamton

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1. Given a Galois cover $\pi: D \to C$ of closed Riemann surfaces, the complex structure of D is determined by that of C. Thus $J(D)$, the Jacobi variety of D, is determined. The induced map $\tilde{\pi}: J(C) \to J(D)$ maps $J(C)$ onto a subtorus of $J(D)$ and the orthogonal complement of this subtorus with respect to the polarization on $J(D)$ is the Prym variety associated to this cover. If π is an unramified abelian cover, then the Prym variety is composed of the periods of the Prym differentials associated to the non-trivial irreducible representations of the covering group. The problem is to study how special properties of C are reflected in the Prym variety. Here we review the apparatus of [9], [10] and prove a result when C is a hyperelliptic Riemann surface. At a later time we will use the results contained here to investigate theta-constant relations and give an example where the Prym variety can be characterized by them.

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2. Preliminaries and notations. Suppose $A = \mathbb{C}^n/\mathfrak{L}$ is an abelian variety, where $\mathfrak{L}$ is a lattice of rank $2n$ in $\mathbb{C}^n$. We will be mostly concerned with representing maps, etc. on the real form of A [7], [8]. The natural equivalence $\mathfrak{L} \cong H_1(A, \mathbb{Z})$ gives an $\mathbb{R}$-linear isomorphism $\mathbb{C}^n \cong H_1(A, \mathbb{R})$. We call $H_1(A, \mathbb{R})/H_1(A, \mathbb{Z})$ the real form of A;

$$H^1(A, \mathbb{R})/H^1(A, \mathbb{Z})$$

gives a natural real form for $\hat{A}$, the dual of A. Let n_A denote multiplications by n in A and $A_n = \ker n_A$; let m denote addition, that is, $m(x, y) = x + y$.

Now suppose Θ is a divisor giving A a polarization. Then Θ determines the map $\lambda: A \to \hat{A}$ by $\lambda(x) =$ (the divisor class of) $\Theta_x - \Theta$. Conversely, λ determines Θ up to linear equivalence and translation. Also associated to Θ is its cohomology class in $H^2(A, \mathbb{Z})$, considered as a skew-symmetric bilinear form Q on $H_1(A, \mathbb{Z})$. This gives the real form for λ as $\lambda: H_1(A, \mathbb{R}) \to H^1(A, \mathbb{R})$ defined by $\lambda(\gamma) = Q(\cdot, \gamma)$. If, for a subset $\mathfrak{M} \subset H_1(A, \mathbb{R})$ we define

$$\mathfrak{M}^{\perp} = \{\gamma \in H_1(A, \mathbb{R}) \mid Q(\gamma, \delta) \in \mathbb{Z} \quad \text{for all} \quad \delta \in \mathfrak{M}\},$$

then

$$K(\lambda) = \ker \lambda = H_1(A, \mathbb{Z})^{\perp}/H_1(A, \mathbb{Z}).$$

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Associated to the form Q are its elementary divisors $e=(e_1,\ldots,e_n)$, where e_i are non-negative integers, $e_i \mid e_{i+1}$ for $i=1,\ldots,k-1$ and $e_i=0$ for $i=k+1,\ldots,n$. Call e the type of Θ. If Θ is effective, then

$$\dim H^0(A, \mathcal{O}(\Theta)) = e_1 \cdots e_k .$$

Let $|\Theta| = \mathbb{P}H^0(A, \mathcal{O}(\Theta))$ be the linear series of effective divisors linearly equivalent to Θ. If also $k=n$, then Θ is also ample, λ is an isogeny, and

$$K(\lambda) \cong (\mathbb{Z}_{e_1})^2 \times \cdots \times (\mathbb{Z}_{e_n})^2 .$$

An ample polarization $\lambda_A : A \to \hat{A}$ induces a polarization $\lambda_{\hat{A}} : \hat{A} \to \hat{\hat{A}} = A$ on $\hat{A}$ such that $\lambda_{\hat{A}} \cdot \lambda_A = e_{n_A}$. This is easily seen by [10] and the fact that $K(\lambda_A)$ is isotropic in $K(e_n \lambda_A)$.

A map $f: A \to B$ between two abelian varieties induces the maps f_* and f^* on homology and cohomology, respectively. The map f^* is the real form of the dual map $\hat{f}: \hat{B} \to \hat{A}$. If B has a polarization given by λ_B or a divisor Σ_B, then A inherits the induced polarization λ_A defined by $\lambda_A = \hat{f} \cdot \lambda_B \cdot f$ or $\Sigma_A = f^{-1}(\Sigma_B)$ (if this is defined). If $f: A \to B$ is a map between principally polarized abelian varieties, we define the transpose of f by $\tilde{f} = \lambda_A^{-1} \cdot \hat{f} \cdot \lambda_B$ from B to A.

3. Abelian varieties. The treatment here is based on [9] and the proofs are similar. Suppose (A, Θ) is a principally polarized abelian variety. Let $i: X \to A$ be the inclusion of a subtorus. Then $\ker \hat{i}$ is connected and we define another subtorus of A,

$$P = P(X, A, \lambda_A) = \lambda_A^{-1}(\ker \hat{i}),$$

the "orthogonal complement" of X with respect to λ_A. The dimension of P is the codimension of X in A.

Let j be the inclusion of P in A and $\sigma: X \times P \to A$ be defined by $\sigma = m \cdot (i, j)$. By definition $\hat{i} \cdot \lambda_A \cdot j = 0$ and hence $\hat{j} \cdot \lambda_A \cdot i = 0$. We pull back the polarization λ_A to $X \times P$ via σ, to obtain

$$\lambda_{X \times P} = \hat{\sigma} \cdot \lambda_A \cdot \sigma = \begin{pmatrix} \hat{i} \cdot \lambda_A \cdot i & \hat{i} \cdot \lambda_A \cdot j \\ \hat{j} \cdot \lambda_A \cdot i & \hat{j} \cdot \lambda_A \cdot j \end{pmatrix} = \begin{pmatrix} \lambda_X & 0 \\ 0 & \lambda_P \end{pmatrix}$$

which is the product of the induced polarizations on X and P. Hence

$$K(\lambda_{X \times P}) = K(\lambda_X) \times K(\lambda_P) \qquad \text{and} \qquad \operatorname{Ker} \sigma = \{(x, -x) \mid x \in X \cap P\}$$

must be a maximal isotropic subgroup. Since the induced polarizations are ample, all these groups are finite and σ is an isogeny. It follows easily that $K(\lambda_X) = K(\lambda_P) = X \cap P$.

Let $d = \text{order}\,(X \cap P)$ and let $\Sigma_X = i^{-1}(\Theta)$ and $\Sigma_P = j^{-1}(\Theta)$ be divisors belonging to the polarizations on X and P, resp. We can translate Θ so that both of these are defined. Then $|\Sigma_X|$ and $|\Sigma_P|$ are both $\mathbb{P}^{d-1}$ on which $X \cap P$ acts as a group of linear transformations. By definition of P, $i^{-1}(\Theta_y)$ is linearly equivalent to Σ_X for any $y \in P$ for which this divisor is defined; denote it by $\delta(y)$. Let $B = \bigcap \{\Sigma \mid \Sigma \in |\Sigma_P|\}$ be the base locus of the linear series $|\Sigma_P|$ and let $\phi_P : P \sim B \to |\Sigma_P|^*$ be the natural map sending y to the hyperplane of divisors containing y. Then in the same way as [9], we can show

Proposition. *δ is defined exactly on $P \sim B$ and there is an isomorphism $f: |\Sigma_P|^* \to |\Sigma_X|$ such that $\delta = f \cdot \phi_P$. All maps are equivariant with respect to the actions of $X \cap P$.*

Remark. The first statement says that $y \in B$ iff $X \subset \Theta_y$.

Suppose now that $\alpha : A \to A$ is a polarization preserving automorphism of A, that is, $\hat{\alpha} \cdot \lambda_A \cdot \alpha = \lambda_A$. Let n be the order of α. Then

$$0 = 1 - \alpha^n = (1-\alpha)(1+\alpha+\cdots+\alpha^{n-1}).$$

We take $X = \text{image}\,(1+\alpha+\cdots+\alpha^{n-1}) = \ker(1-\alpha)^\circ$ (connected component of the identity in $\ker(1-\alpha)$). Points of X are fixed by α, hence $\alpha \cdot i = i$. Dualizing, we have $\hat{i} \cdot \hat{\alpha} = \hat{i}$ and then

$$\hat{i} \cdot \lambda_A \cdot (1-\alpha) = \hat{i} \cdot \lambda_A - \hat{i} \cdot \lambda_A \cdot \alpha = \hat{i} \cdot \lambda_A - \hat{i} \cdot \hat{\alpha} \cdot \lambda_A \cdot \alpha = \hat{i} \cdot \lambda_A - \hat{i} \cdot \lambda_A = 0.$$

Thus, $\text{image}\,(1-\alpha) \subset P = P(X, A, \lambda_A)$. Since they have the same dimension we have

$$P = \text{image}\,(1-\alpha) = \ker(1+\alpha+\cdots+\alpha^{n-1})^\circ.$$

Let $P_0 = \ker(1-\alpha) \cap P$. If $y \in P_0$, then $0 = (1+\alpha+\cdots+\alpha^{n-1})\, y = ny$ and hence $P_0 \subset P_n$. Thus,

$$\ker(1-\alpha) = X + P_0,$$

$$\ker(1+\alpha+\cdots+\alpha^{n-1}) = X_n + P.$$

Since α preserves the principal polarization of A, $\alpha(\Theta)$ must be linearly equivalent, and hence equal, to a translate of Θ. Suppose $\alpha(\Theta) = \Theta_z$ for some $z \in A$ and write $z = x + y$, for $x \in X$, $y \in P$. Then,

$$\begin{aligned} i^{-1}(\Theta) = i^{-1}(\alpha(\Theta)) &= i^{-1}(\Theta_{x+y}) \\ &\equiv i^{-1}(\Theta_x) \qquad \text{since } y \in P \\ &= i^{-1}(\Theta)_x \qquad \text{since } x \in X. \end{aligned}$$

Hence $x \in K(\lambda_X) = X \cap P$ and $z \in P$. Choose $v \in P$ so that $v - \alpha v = z$; recall that $(1-\alpha)$ is an isogeny of P. Then,

$$\alpha(\Theta_v) = \alpha(\Theta)_{\alpha v} = \Theta_{z+\alpha v} = \Theta_v.$$

That is, α restricts to an automorphism of the divisor Θ_v. Since α commutes with $(1-\alpha)$, α restricts to a polarization preserving automorphism of P. We will now show that we can translate Σ_P so that α restricts to an automorphism of each divisor in $|\Sigma_P|$.

Proposition. *Suppose α is a polarization preserving automorphism of the polarized abelian variety (P, Σ). Suppose $K(\lambda_P) \subset \ker(1-\alpha)$ and $1-\alpha$ is an isogeny. Then there is a $y \in P$ so that the induced linear map $\alpha : |\Sigma_y| \to |\Sigma_y|$ is the identity.*

Proof. α preserves the polarization implies that there is a $z \in P$ so that $\alpha(\Sigma) \equiv \Sigma_z$. Again, choose $v \in P$ so that $v - \alpha v = z$ and we have $\alpha(\Sigma_v) \equiv \Sigma_v$. Thus assume $\alpha(\Sigma) \equiv \Sigma$. Then α induces a linear isomorphism of $|\Sigma|$. Since α fixes $K(\lambda_P)$, this linear map commutes with the action of $K(\lambda_P)$ as linear isomorphisms of $|\Sigma|$.

Choose two maximal isotropic subgroups H_1, H_2 of $K(\lambda_P)$ such that $H_1 \cap H_2 = \{0\}$. Choose $\Sigma' \in |\Sigma|$ so that $\Sigma'_z = \Sigma'$ for all $z \in H_1$. Then, $|\Sigma|^{H_1}$, the set of divisors invariant under translation by H_1, consists of the translates of Σ' by H_2. These points are in general position in $|\Sigma|$.

Since $\alpha(\Sigma')=\alpha(\Sigma'_z)=\alpha(\Sigma')_z$ for any $z\in H_1$, $\alpha(\Sigma')\in|\Sigma|^{H_1}$. Thus, there is a $z_2\in H_2$ such that $\alpha(\Sigma')=\Sigma'_{z_2}$. Similarly, if $\Sigma''\in|\Sigma|^{H_2}$, then there is a $z_1\in H_1$ such that

$$\alpha(\Sigma'')=\Sigma''_{z_1}.$$

Choose $v\in P$ so that $v-\alpha v=z_1+z_2$. Then

$$\alpha(\Sigma'_v)=\Sigma'_{z_2+\alpha v}=\Sigma'_{z_1+z_2+\alpha v}=\Sigma'_v,$$

and similarly, $\alpha(\Sigma''_v)=\Sigma''_v$. Thus α fixes the points in $|\Sigma_v|^{H_1}\cup|\Sigma_v|^{H_2}$. The key here is that Σ''_v cannot lie in the span of any proper subset of $|\Sigma_v|^{H_1}$, since otherwise it could not be invariant under H_2. But the divisors fixed by α must comprise a union of linear subspaces spanned by subsets of $|\Sigma_v|^{H_1}$, since these points yield a basis of eigenvectors for any matrix representing α. Thus α must induce the identity on $|\Sigma_v|$.

This proposition implies, after a translation we may arrange, that $\phi_P\cdot\alpha=\phi_P$.

4. We want to apply the above to $J(D)$ when D has an automorphism, which thus induces a polarization preserving automorphism of $J(D)$. We discuss more general covers first, and then return to cyclic covers.

Suppose $\pi: D\to C$ is an unramified Galois cover of the Riemann surface C of genus g with deck group G. Let n be the order of G. Then the maps

$$\pi_*:\pi_1(D,p)\to\pi_1\big(C,\pi(p)\big)\qquad\text{and}\qquad\pi_*:H_1(D,\mathbb{Z})\to H_1(C,\mathbb{Z})$$

fit into a commutative diagram, with all rows and collumns exact.

(1)

$$\begin{array}{ccccccccccc}
 & & & & & & 0 & & & & \\
 & & & & & & \downarrow & & & & \\
 & & & & 0 & & 0 & & K & & \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & \longrightarrow & \pi_1(D,p)' & \longrightarrow & \pi_1\big(C,\pi(p)\big)' & \longrightarrow & H & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & \longrightarrow & \pi_1(D,p) & \longrightarrow & \pi_1\big(C,\pi(p)\big) & \longrightarrow & G & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & L & \longrightarrow & H_1(D,\mathbb{Z}) & \longrightarrow & H_1(C,\mathbb{Z}) & \longrightarrow & G_1 & \longrightarrow & 0 \\
 & & & & \downarrow & & \downarrow & & \downarrow & & \\
 & & & & 0 & & 0 & & 0 & &
\end{array}$$

Here $'$ denotes commutator subgroup and K, L are defined by the diagram. A routine diagram chase yields

$$L\cong K\cong\big[\pi_1\big(C,\pi(p)\big)'\cap\pi_*\big(\pi_1(D,p)\big)\big]/\pi_*\big(\pi_1(D,p)'\big)$$

and $G' \cong H/K$. Hence $G_1 \cong G/G'$ is the maximal abelian quotient of G (see [2]). We now assume G is abelian. Then $G = G_1$, $H = K$ and $\pi_1(C, \pi(p))' \subset \pi_*(\pi_1(D, P))$.

We identify $H_1(C, \mathbb{R})/H_1(C, \mathbb{Z})$ (resp. $H^1(C, \mathbb{R})/H^1(C, \mathbb{Z})$) as the real form of $J(C)$ (resp. $\hat{J}(C)$). The map π_* on homology (resp. π^* on cohomology) gives the map $\pi_* : J(D) \to J(C)$ (resp. $\pi^* : \hat{J}(C) \to \hat{J}(D)$). We recall that the principal polarization λ_C of $J(C)$ as the Jacobi variety of C is induced from the intersection pairing $\langle , \rangle_C$ as a skew symmetric bilinear form on $H_1(C, \mathbb{Z})$.

Let $\mathfrak{L} = \pi_* H_1(D, \mathbb{Z})$, $\mathfrak{L}$ is a sublattice of $H_1(C, \mathbb{Z})$ with quotient G. Then $\mathfrak{L}^\perp$ is a lattice in $H_1(C, \mathbb{R})$ containing $H_1(C, \mathbb{Z})$ and the quotient is naturally isomorphic to $\hat{G}$, the character group of G, since the intersection pairing is non-degenerate. This also displays $\hat{G}$ as a subgroup of $J(C)$. Since $\mathfrak{L}^\perp$ determines $\mathfrak{L}$ which determines $\pi_1(D, p)$ as a subgroup of $\pi_1(C, \pi(p))$, we have

Proposition 1. *There is a one-to-one correspondence between unramified abelian covers of C with deck group G and injections $\hat{G} \to J(C)$.*

Remark. This also gives a way to construct the covering: if $\lambda_1, \ldots, \lambda_k$ (note $k \leqq 2g$) is a basis for $\hat{G}$, then λ_i can be represented as $\frac{1}{n_i}\gamma_i$ for some $n_i \in \mathbb{Z}^+$ and $\gamma_i \in H_1(C, \mathbb{Z})$. Then a loop σ in $\pi_1(C, \pi(p))$ lifts to a closed curve on D at p (and hence belongs to $\pi_1(D, p)$) iff $\langle \sigma, \gamma_i \rangle_C \equiv 0 (n_i)$ for each i.

For a general cover π, the map $\tilde{\pi} = \lambda_D^{-1} \cdot \pi^* \cdot \lambda_C$ is the adjoint of π_* with respect to the intersection pairing [7], that is, for any $\gamma \in H_1(C, \mathbb{R})$ and $\delta \in H_1(D, \mathbb{R})$,

$$\langle \gamma, \pi_* \delta \rangle_C = \langle \tilde{\pi}\gamma, \delta \rangle_D .$$

Note also that $\pi_* \cdot \tilde{\pi} = n_{J(C)}$, where n is the degree of π.

Lemma [2]. $\hat{G} = \ker \tilde{\pi}$.

Proof. $\lambda \in \mathfrak{L}^\perp$ iff $\langle \lambda, \delta \rangle_C \in \mathbb{Z}$ for all $\delta \in \mathfrak{L}$ iff $\langle \lambda, \pi_* \sigma \rangle_C = \langle \tilde{\pi}\lambda, \sigma \rangle_D \in \mathbb{Z}$ for all $\sigma \in H_1(D, \mathbb{Z})$. But $H_1(D, \mathbb{Z})^\perp = H_1(D, \mathbb{Z})$, since the polarization is principal, and hence this is equivalent to $\tilde{\pi}\lambda \in H_1(D, \mathbb{Z})$.

Thus $\tilde{\pi}$ is an isogeny of $J(C)$ onto a subtorus X of $J(D)$, with real form $H_1(C, \mathbb{R})/\mathfrak{L}^\perp$. We define the *Prym variety* associated to the cover $\pi : D \to C$ by

$$P = \mathrm{Prym}\,(C, \pi) = P\big(X, J(D), \lambda_D\big) .$$

By Riemann-Hurwitz formula, the genus of D is $n(g-1)+1$ and hence

$$\dim P = (n-1)(g-1) .$$

In the diagram (1), $L \otimes \mathbb{R}/L$ gives the real form of P, and since G preserves L and the polarization of $J(D)$, G restricts to a group of polarization preserving automorphisms of P.

The map $\tilde{\pi} : H_1(C, \mathbb{R}) \to H_1(D, \mathbb{R})$ is also the real form of the inclusion of X into $J(D)$. For a moment, denote $\mathfrak{L}^\perp$ by $\mathfrak{M}$. Then,

$$K(\lambda_X) = \{\gamma \in H_1(C, \mathbb{R}) \mid \langle \tilde{\pi}\gamma, \tilde{\pi}\sigma \rangle_D \in \mathbb{Z} \text{ for all } \sigma \in \mathfrak{M}\}/\mathfrak{M} .$$

But,

$$\langle \tilde{\pi}\gamma, \tilde{\pi}\sigma \rangle_D = \langle \gamma, \pi_* \tilde{\pi}\sigma \rangle_C = n \langle \gamma, \sigma \rangle_C .$$

Thus, with $\perp$ being taken now with respect to the bilinear form on $H_1(C, \mathbb{R})$ (or $J(C)$) belonging to $n\lambda_C$, we have

$$K(\lambda_X) = \mathfrak{M}^{\perp}/\mathfrak{M} = (\mathfrak{M}^{\perp}/\mathfrak{L})/(\mathfrak{M}/\mathfrak{L}) = \hat{G}^{\perp}/\hat{G}, \qquad \text{in} \quad J(C).$$

This allows us to calculate $K(\lambda_P)$ and hence the type e of the polarization of P.

5. Lifting automorphisms. We now want to give a criterion for lifting an automorphism of C to D. (See [8] for similar condition for dropping automorphisms.) Let $f: C \to C$ be an automorphism of C. Then f induces $f_*: \pi_1(C, p) \to \pi_1(C, f(p))$ and $f_*: H_1(C, \mathbb{Z}) \to H_1(C, \mathbb{Z})$. From [1], we know that f will lift to a map $f': D \to D$ (which will then be holomorphic) such that $\pi \cdot f' = f \cdot \pi$ iff f_* maps $\pi_*(\pi_1(D, p_0))$ to $\pi_*(\pi_1(D, p_1))$ for some choice of $p_0, p_1 \in D$ such that $\pi(p_0) = p$, $\pi(p_1) = f(p)$. Assuming that π is an unramified abelian cover, an easy diagram chase in (1) shows that f lifts iff f_* restricts to an automorphism of $\mathfrak{L}$ and hence also of $\mathfrak{L}^{\perp}$. Hence

Proposition 2. *If, in proposition* 1 π *corresponds to the subgroup* $\hat{G}$ *in* $J(C)$, *then* f *lifts to an automorphism of* D *iff* $f_*(\hat{G}) = \hat{G}$ *in* $J(C)$.

Define another abelian variety Y with real form $H_1(C, \mathbb{R})/\mathfrak{L}$. Then the identity map on $H_1(C, \mathbb{R})$ induces an isogeny $Y \xrightarrow{\pi'} J(C)$ with kernel naturally isomorphic to G. The inverse image of C in Y is D and G acts on D by translation in Y. The automorphism f_* on $J(C)$ is a group homomorphism and may not map C to itself. To be precise, suppose $p_0 \in C$ is fixed and define a canonical embedding $\psi: C \to J(C)$ by $\psi(p) = [p - p_0]$. Now define the affine map on $J(C)$, $F(x) = f_*(x) + [fp_0 - p_0]$ for $x \in J(C)$. If $f^m = \mathrm{id}$, then $F^m = \mathrm{id}$. Also,

$$F(\psi(p)) = f_*\psi(p) + [fp_0 - p_0] = [fp - fp_0] + [fp_0 - p] = [fp - p_0] = \psi(fp).$$

Thus F maps C to itself in $J(C)$. If now $f_*(\hat{G}) = \hat{G}$, then f_* lifts to an automorphism f' of Y. Let k be any point of Y so that $\pi'(k) = [fp_0 - p_0]$, and define $F'(y) = f'(y) + k$, for $y \in Y$. Since $\pi' \cdot F' = F \cdot \pi'$, F' maps D to itself and F' and G generate a subgroup of $\mathrm{Aut}(D)$, $f^m = \text{identity}$ implies that

$$F'^m(y) = y + (k + f'k + \cdots + f'^{m-1}k) = y + g,$$

where $g \in G$. If p_0 is a fixed point of f on C, then $[fp_0 - p_0] = 0$ and k may be chosen to be 0 as well. If f has no fixed points, then this is impossible, but g may still be 0.

The above construction can be done with any subgroup H of $\mathrm{Aut}(C)$ preserving $\hat{G}$ yielding a group G' of automorphisms of D given by an extension

$$1 \to G \to G' \to H \to 1. \tag{2}$$

Example 1. Suppose C is hyperelliptic and f is the hyperelliptic involution. Then the map $f_* = F$ is $-\mathrm{id}$ on $J(C)$. This clearly preserves any subgroup of $J(C)$ and hence the hyperelliptic involution lifts to any unramified abelian cover; $f' = -\mathrm{id}$ on Y implies $k + f'k = 0$. Thus all lifts of f are involutions on D. For $g \in G$, as automorphism of D, $f' \cdot g \cdot f' = g^{-1}$.

Example 2. If $\hat{G} = J(C)_r$, then all of Aut (C) lifts to the cover determined by $\hat{G}$.

Example 3. Using the representation of the group of automorphisms of the Klein surface of genus 3 in [11], we find exactly 2 choices of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ in $J(C)$ preserved by Aut (C). These give 2 Riemann surfaces of genus 17 covering the Klein surface and having maximal automorphism groups. It is stated in [14], and seems difficult to show here, that the corresponding extensions (2) are not split. These two groups and $J(C)_2$ give the only unramified abelian covers of exponent 2 of the Klein surface to which all its automorphisms lift.

Remark. If an automorphism f of C does not preserve $\hat{G}$, then we have that the covers of C determined by $\hat{G}$ and $f_*(\hat{G})$ are equivalent.

6. $G = \mathbb{Z}_q$. Now assume $G = \mathbb{Z}_q$ with q prime.

Lemma. $X \cap P = P_0$.

This can be seen by examining the representation of G on $H_1(D, \mathbb{Z})$ [6]. Let α denote a generator of G. Then the lemma implies that $X = \ker(1-\alpha)$ in $J(D)$.

By the above, we know that $K(\lambda_X) \cong \hat{G}^{\perp}/\hat{G} \cong \mathbb{Z}_q^{2g-2}$. Hence $K(\lambda_P) \cong \mathbb{Z}_q^{2g-2}$. This immediately implies that the type of λ_P is $e = (1, \ldots, 1, q, \ldots, q)$ with 1 repeated $(q-2)(g-1)$ times and q repeated $g-1$ times. For $q=2$, λ_P is thus twice a principal polarization; this case has been extensively studied. The polarizations are not principal for any other q.

Proposition 3. *For $q = 3$, (P, λ_P) is self-dual, that is, (P, λ_P) and $(\hat{P}, \lambda_{\hat{P}})$ are isomorphic as polarized abelian varieties, where $\lambda_{\hat{P}}$ is the induced polarization on $\hat{P}$.*

Proof. Since $K(\lambda_P) = P_0$, the isogeny $1-\alpha$ induces an isomorphism of P and $\hat{P}$ for any q. [10]. To finish the proof, we must show that the pull-back of λ_P by $1-\alpha$ is $3\lambda_P$.

$$\begin{aligned}(1-\hat{\alpha}) \cdot \lambda_P \cdot (1-\alpha) &= \lambda_P - \hat{\alpha} \cdot \lambda_P - \lambda_P \cdot \alpha + \hat{\alpha} \cdot \lambda_P \cdot \alpha \\ &= \lambda_P - \lambda_P \cdot \alpha^{q-1} - \lambda_P \cdot \alpha + \lambda_P = \lambda_P \cdot (2 - \alpha - \alpha^{q-1}).\end{aligned}$$

For $q=3$, $\alpha^2 + \alpha = -1$ and the proof is complete.

In general $\lambda_{\hat{P}}$ is not equivalent to λ_P and P has many polarizations. The question remains: is the isomorphism mentioned in the above proof, denote it by $\phi : P \to \hat{P}$, a principal polarization? That it is not can be seen as follows. Since $\lambda_P = \phi \cdot (1-\alpha)$, we have $(1-\hat{\alpha}) \cdot \hat{\phi} = \hat{\lambda}_P = \lambda_P = \phi \cdot (1-\alpha)$. Hence $\hat{\phi} \cdot (1-\alpha^{q-1}) = \phi \cdot (1-\alpha)$. Since $1-\alpha$ is an isogeny we have $\hat{\phi} = -\phi \cdot \alpha \neq \phi$. Thus ϕ cannot be a polarization. In fact, P cannot have a principal polarization unless End (P) is strictly larger than $\mathbb{Z}[\alpha]$.

Now suppose C is hyperelliptic and q is an odd prime. From (2), we see that G' is D_{2q}, the dihedral group of order $2q$. Since the involutions in G' are all conjugate, the quotients of D by these are all biholomorphic. Let C_0 be the quotient of D by f' and $\pi_0 : D \to C_0$ the covering map. Since D_{2q} is the union of $\mathbb{Z}_q$ and $q\mathbb{Z}_2$'s, we can use Accola's Theorem [5] to obtain

$$g_0 = g(C_0) = \frac{1}{2}(q-1)(g-1) = \frac{1}{2}\dim P.$$

Recall the results from [6], [9] on two-sheeted covers. Since π_0 is ramified, $\tilde{\pi}_0$ is an inclusion of $J(C_0)$ into $J(D)$, and $J(C_0) = \text{image } (1+f')$ implies $1+f' = \pi_{0*}$.

Pulling back the polarization λ_D to $J(C_0)$ yields

$$\hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \tilde{\pi}_0 = 2\lambda_{C_0}.$$

Now let $2r = q+1$. Consider the two maps

$$J(D) \xrightarrow{A} J(C_0) \times J(C_0) \xrightarrow{B} P,$$

defined by $A(x) = \big((1+f')\,x,\ -(1+f')\,\alpha^r x\big)$ and $B(u, v) = u + \alpha^r v$. A is well-defined by the above remarks and $B \cdot A(x) = (1-\alpha)\,x$. Hence A is onto and B is well-defined. If $(1-\alpha)\,x = 0$, then $x \in X$. But $f' = -\mathrm{id}$ on X. Hence $(1+f')\,x = 0$ and $A(x) = (0, 0)$. Thus, B is an isomorphism.

We will pull λ_P back to $J(C_0) \times J(C_0)$ via B to obtain a polarization λ equivalent to λ_P. We factor B as $B = m \cdot (1, \alpha^r) \cdot (\tilde{\pi}_0, \tilde{\pi}_0)$. Then,

$$\lambda = \hat{B} \cdot \lambda_D \cdot B = \begin{pmatrix} \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \hat{\tilde{\pi}}_0 & \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \alpha^r \cdot \tilde{\pi}_0 \\ \hat{\tilde{\pi}}_0 \cdot \hat{\alpha}^r \cdot \lambda_D \cdot \tilde{\pi}_0 & \hat{\tilde{\pi}}_0 \cdot \hat{\alpha}^r \cdot \lambda_D \cdot \alpha^r \cdot \tilde{\pi}_0 \end{pmatrix}.$$

The terms on the main diagonal are $2\lambda_{C_0}$ by the above.

Lemma. *$\beta = \alpha^r + \alpha^{r-1}$ restricts to an automorphism of $J(C_0)$.*

Proof. β commutes with f', hence with $1+f'$. Thus it induces a map from $J(C_0)$ to itself. Suppose $\beta x = 0$ for some $x \in J(C_0)$. Then $0 = \alpha^r x + \alpha^{r-1} x = \alpha^{r-1}(\alpha x + x)$. Thus, $x = (-\alpha)\,x = (-\alpha)^q x = -x$. Hence x is a point of order 2 in P fixed by α. But α fixes only points of order q. Thus $x = 0$ and β is an automorphism. Now we compute the other terms in λ. Recall the notation $\tilde{\pi}_0 = \lambda_D^{-1} \cdot \pi_0^* \cdot \lambda_{C_0}$; hence $\pi_{0*} = \lambda_{C_0}^{-1} \cdot \hat{\tilde{\pi}}_0 \cdot \lambda_D$.

First we show that the off-diagonal terms are equal. Since $f' \cdot \tilde{\pi}_0 = \tilde{\pi}_0$ and hence

$$\begin{aligned} \hat{\tilde{\pi}}_0 \cdot \hat{\alpha}^r \cdot \lambda_D \cdot \tilde{\pi}_0 &= \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \alpha^{r-1} \cdot f' \cdot \tilde{\pi}_0 = \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot f' \cdot \alpha^r \cdot \tilde{\pi}_0 \\ &= \hat{\tilde{\pi}}_0 \cdot \hat{f'} \cdot \lambda_D \cdot \alpha^r \cdot \tilde{\pi}_0 = \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \alpha_r \cdot \tilde{\pi}_0. \end{aligned}$$

On the other hand,

$$\begin{aligned} \hat{\tilde{\pi}}_0 \cdot \hat{\alpha}^r \cdot \lambda_D \cdot \tilde{\pi}_0 + \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \alpha^r \cdot \tilde{\pi}_0 &= \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot (\alpha^{r-1} + \alpha^r) \cdot \tilde{\pi}_0 \\ &= \hat{\tilde{\pi}}_0 \cdot \lambda_D \cdot \tilde{\pi}_0 \cdot \beta = 2\lambda_{C_0} \cdot \beta \end{aligned}$$

Thus, the off-diagonal terms are each $\lambda_{C_0} \cdot \beta$. Note that β satisfies $\lambda_{C_0} \cdot \beta = \hat{\beta} \cdot \lambda_{C_0}$ and does not preserve the polarization λ_{C_0} on $J(C_0)$ (unless $q = 3$). $\alpha^r \cdot B = B \cdot \begin{pmatrix} 0 & -1 \\ 1 & \beta \end{pmatrix}$ gives the automorphism α^r on $J(C_0) \times J(C_0)$.

Summarizing, we have

Theorem 1. *Let C be a hyperelliptic Riemann surface, with hyperelliptic involution f. Suppose $\pi: D \to C$ is an unramified $\mathbb{Z}_q$-cover and α generates the covering group. Then*

(i) *f lifts to an involution f' of D and α, f' generate a dihedral group D_{2q} of automorphisms of D;*

(ii) *if C_0 is the quotient of D by f', then*

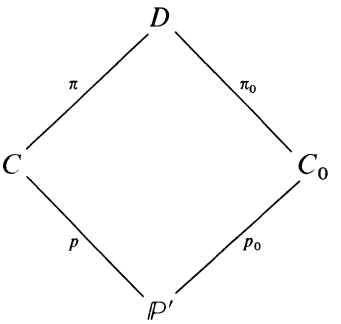
$$Prym\,(C, \pi) \cong \big(J(C_0) \times J(C_0), \lambda\big),$$

where

$$\lambda = \begin{pmatrix} 2\lambda_{C_0} & \lambda_{C_0} \cdot \beta \\ \lambda_{C_0} \cdot \beta & 2\lambda_{C_0} \end{pmatrix}$$

and $\beta = \alpha^r + \alpha^{r-1}$ is an automorphism of $J(C_0)$.

7. We can give a description of the surfaces that arise in theorem 1. Consider the tower of covers where $p \cdot \pi = p_0 \cdot \pi_0$ projects to the quotient of D by all of D_{2q}. C is hyperelliptic of genus g, so p is branched over $2g+2$ points of $\mathbb{P}'$, denoted $\lambda_1, \ldots, \lambda_{2g+2}$. Since π is unramified, $p \cdot \pi$ has q ramification points of index 2 over each λ_i.



The map p_0 is q-sheeted and hence, by the Riemann-Hurwitz formula, must have $\frac{1}{2}(q-1)$ ramification points of index 2 and one unramified point over each λ_i. Thus we have the cycle structure of the local monodromies for the cover p_0. To say more we borrow the apparatus of Hurwitz spaces from [3], [4], [12], [13].

Let Y be $\mathbb{P}'$ punctured at $\lambda_1, \ldots, \lambda_{2g+2}$ and let $\lambda_0 \in Y$. Any q-sheeted cover of $\mathbb{P}'$ branched only over the λ_i is determined by a homomorphism ψ from $\pi_1(Y, \lambda_0)$ to S_q, the symmetric group of $\{1, 2, \ldots, q\}$. The Hurwitz space $H(q, 2g+2)$ is the variety composed of these homomorphisms ψ modulo inner automorphisms of S_q for arbitrary choice of $2g+2$ distinct points in $\mathbb{P}'$. If we choose a standard set of generators $\sigma_1, \ldots, \sigma_{2g+2}$ for $\pi_1(Y, \lambda_0)$ (see [3]) and a numbering of the sheets of C_0 over λ_0, then ψ is determined by the q-tuple $(t_1, \ldots, t_{2g+2})$ of involutions in S_q such that $t_i = \psi(\sigma_i)$, subject to the relation $t_1 \cdots t_{2g+2} = 1$. Let D'_{2q} be a dihedral subgroup of S_q of symmetries of a regular q-gon.

Theorem 2. *The locus of surfaces arising in theorem* 1 *is the component* H *of* $H(q, 2g+2)$ *determined by* $t_i \in D'_{2q}$ *for each* i *and the* t_i *generate* D'_{2q}.

Proof. Given a cover with the monodromy description in theorem 2, the homomorphism ψ is onto D'_{2q} and $\ker \psi$ determines a D_{2q}-cover of Y, and hence of $\mathbb{P}'$. Since the local monodromies are involutions, the automorphisms of order q have no fixed points and we have the situation in theorem 1.

Conversely, suppose the homomorphism ψ is associated to the cover p_0 arising from the surface C_0 obtained in theorem 1. Let $C'_0 = p_0^{-1}(Y)$, $D' = \pi_0^{-1}(C'_0)$, c_k be the point on C'_0 over λ_0 on the sheet labeled k, and d a point on D' over c_1. Set $S^{(k)}$ to be the subgroup of S_q isomorphic to S_{q-1} consisting of the elements fixing k. Then we have $\pi_1(C'_0, c_k) = \psi^{-1}(S^{(k)})$ for each k. Their intersection is $\ker \psi$, a normal subgroup of $\pi_1(Y, \lambda_0)$. Since the $\pi_1(C'_0, c_k)$ are conjugate to each other, any subgroup of $\pi_1(C'_0, c_1)$ normal in $\pi_1(Y, \lambda_0)$ must be contained in $\ker \psi$. Thus, $\pi_1(D', d) \subset \ker \psi$ since $D' \to Y$ is a Galois cover. But $\pi_1(D', d)$ is a subgroup of $\pi_1(C'_0, c_1)$ of index 2, hence $\pi_1(D', d) = \ker \psi$ and ψ has the properties of theorem 2.

Thus, we have a bijection between unramified $\mathbb{Z}_q$-covers of hyperelliptic surfaces of genus g and points of H. We will complete the proof by showing that H is irreducible. We do that in the usual way by showing that any such $2g+2$-tuple $(t_1, \ldots, t_{2g+2})$ of involutions from D'_{2q} can be changed to a normal form under the action of the fundamental group of the variety parametrizing un-ordered $2g+2$-tuples of distinct points from $\mathbb{P}'$. We refer to [3] for details and only state here that this action is generated by the following operations

$$A_i(t_1, \ldots, t_{2g+2}) = (t_1, \ldots, t_{i-1}, t_i t_{i+1} t_i^{-1}, t_i, t_{i+2}, \ldots, t_{2g+2}),$$
$$A_i^{-1}(t_1, \ldots, t_{2g+2}) = (t_1, \ldots, t_{i-1}, t_{i+1}, t_{i+1}^{-1} t_i t_{i+1}, t_{i+2}, \ldots, t_{2g+2}).$$

Let γ_0 be any involution in D'_{2q} and δ any element of order q. Set $\gamma_k = \gamma_0 \delta^k$ for $k = 0, \ldots, q-1$. All indices of γ's and powers of δ are now understood mod q. Then,

$$\gamma_i \gamma_j \gamma_i = \gamma_0 \delta^i \gamma_0 \delta^j \gamma_0 \delta^i = \gamma_0 \delta^{2i-j} = \gamma_{2i-j}.$$

Thus, $A_1(\gamma_i, \gamma_j) = (\gamma_{2i-j}, \gamma_i)$. Note that A_1 preserves the product of γ_i and γ_j, hence $i-j$ is fixed. If $\gamma_i = \gamma_j$, then A_1 has no effect. Otherwise, A_1 permutes the q pairs with fixed $i-j \neq 0$. Similarly, it is easy to see that A_1, A_2, acting on the triple $(\gamma_i, \gamma_j, \gamma_k)$, fix $i+k-j$. If $i=j=k=a$, then A_1 and A_2 have no effect. Otherwise they permute the q^2-1 remaining triples with $i+k-j=a$ fixed. With these simple observations we can obtain a normal form.

Suppose $t_1 = \gamma_0$ and that we have put $t_1 = \cdots = t_n = \gamma_0$ using A_i's. Then $n \leqq 2g$ since the product must be the identity. If $n = 2g$, then $t_{2g+1} = t_{2g+2} \neq \gamma_0$, so suppose $n < 2g$. If $t_{n+1} \neq t_{n+2}$, then we can use A_{n+1} repeatedly to transform t_{n+1} into γ_0. Suppose $t_{n+1} = t_{n+2} \neq \gamma_0$. Then we can use A_n, A_{n+1} on $(\gamma_0, t_{n+1}, t_{n+1})$ to insure that $t_{n+1} = t_{n+2} \neq t_{n+3}$. Then we can use A_{n+1} and A_{n+2} on $(t_{n+1}, t_{n+1}, t_{n+3})$ to transform t_{n+1} into γ_0. Clearly this procedure will end with

$$(t_1, \ldots, t_{2g+2}) = (\gamma_0, \ldots, \gamma_0, \gamma_1, \gamma_1).$$

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Some results on the coefficients of the biquadratic theta series

By *Toshiaki Suzuki* at Okinawa

1. Introduction

In the investigation of the reciprocity law of an algebraic number field, T. Kubota introduced a new analytic method and gave a generalization of the classical theta series ([2], [3]). This generalized theta series is defined for a totally imaginary number field containing the n-th roots of unity, and is deeply connected with the reciprocity law of the n-th power residue symbol and Gauss sums of the number field. If $n=2$, the generalized theta series essentially comes to be the classical theta series. In case $n=3$ for $\mathbb{Q}(\sqrt{-3})$, T. Kubota conjectured that the coefficients of the generalized theta series are essentially cubic Gauss sums ([4]), which is fully proved by S. J. Patterson in [5].

In this paper, we investigate the case $n=4$ for the Gaussian field $\mathbb{Q}(i)$.

If $z \in \mathbb{C}$, let $e(z)=\exp(2\pi i(z+\bar{z}))$. Then $e(z)$ is an additive character of the complex number field $\mathbb{C}$. Let $\mathfrak{o}$ be the ring of Gaussian integers. Considering $\mathfrak{o}$ a lattice of $\mathbb{C}$, let $\mathfrak{o}^*$ be the dual lattice of $\mathfrak{o}$ with respect to $e(z)$.

Put $\lambda=1+i$. For $c \in \mathfrak{o}$ such that $(c, \lambda)=1$, and $\mu \in \mathfrak{o}^*$, we define

$$g(c, \mu)=\sum_{\delta \bmod c}\left(\frac{\delta}{c}\right) e\left(\frac{\delta}{c}\mu\right)$$

where $\left(\frac{\delta}{c}\right)$ is the biquadratic residue symbol in $\mathbb{Q}(i)$ and δ runs through a complete set of residues (mod c) prime to c.

We consider Dirichlet series

$$\begin{aligned}\psi_1(s, \mu)&=\sum_{c\equiv 1 \,(\mathrm{mod}\, 4)} g(c, \mu)\, N(c)^{-s},\\ \psi_2(s, \mu)&=-\sum_{c\equiv 1+\lambda^3 \,(\mathrm{mod}\, 4)} g(c, \mu)\, N(c)^{-s}\end{aligned} \tag{1.1}$$

where $s \in \mathbb{C}$, $N(c)=c\bar{c}$ and c runs through all the elements of $\mathfrak{o}$ satisfying $c \equiv 1 \,(\mathrm{mod}\, 4)$ (resp. $c \equiv 1+\lambda^3 \,(\mathrm{mod}\, 4)$); those are absolutely convergent for $\mathrm{Re}(s)>\frac{3}{2}$.

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It is known that $\psi_1(s, \mu)$ and $\psi_2(s, \mu)$ are holomorphically continued to the region $\mathrm{Re}(s) > 1$ except possibly a simple pole at $s = \frac{5}{4}$.

Let $\psi_1(\mu)$ and $\psi_2(\mu)$ be the residues of $\psi_1(s, \mu)$ and $\psi_2(s, \mu)$ at $s = \frac{5}{4}$, respectively. We shall see that $\psi_1(\mu)$ and $\psi_2(\mu)$ are the coefficients of the generalized theta series for the Gaussian field, i.e. the coefficients of the biquadratic theta series.

The purpose of this paper is to give some information about $\psi_1(\mu)$ and $\psi_2(\mu)$. The main results are stated in Theorem 1, Theorem 2, Theorem 3, Theorem 4, Theorem 5 and Theorem 6. It is easily observed that the method, which is hinted by T. Kubota ([4]) and is used by S. J. Patterson ([5]), is not applicable to the cases $n \geqq 4$. So we make use of a new method. But we have not yet obtained $\psi_1(\mu)$ and $\psi_2(\mu)$ completely.

2. $\Gamma(N)$

We identify the upper half space $H = \mathbb{C} \times \mathbb{R}_+^\times$ with the space of all matrices of the form $u = \begin{pmatrix} z & -v \\ v & \bar{z} \end{pmatrix}$ $(z = x + iy \in \mathbb{C},\ v \in \mathbb{R}_+^\times)$. Let $\tilde{w} = \begin{pmatrix} w & 0 \\ 0 & \bar{w} \end{pmatrix}$ for $w \in \mathbb{C}$. Then $\mathrm{SL}_2(\mathbb{C})$ acts on H by

$$\sigma(u) = (\tilde{a}u + \tilde{b})(\tilde{c}u + \tilde{d})^{-1}$$

where $\sigma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{C})$. The action is transitive and the stabilizer of a point is $\mathrm{SU}_2(\mathbb{C})$. We write $v(u) = v\left(u = \begin{pmatrix} z & -v \\ v & \bar{z} \end{pmatrix}\right)$, then

$$v(\sigma u) = j(\sigma, u)^{-1} v(u)$$

where $j(\sigma, u) = \det(\tilde{c}u + \tilde{d})$.

Let $N \in \mathfrak{o}$ and define

$$\Gamma(N) = \left\{ \sigma \in \mathrm{SL}_2(\mathfrak{o}) : \sigma \equiv \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \bmod N \right\}.$$

Then $\Gamma(N)$ acts discontinuously on H, and has a fundamental domain whose volume with respect to the invariant measure $v^{-3}\,dx\,dy\,dv$ of H is finite.

Now let $(-)$ be the biquadratic residue symbol in $\mathbb{Q}(i)$ and define

$$\chi(\sigma) = \left(\frac{c}{a}\right) \qquad (c \neq 0),$$

$$= 1 \qquad (c = 0)$$

for $\sigma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(N)$.

Here we recall the reciprocity law of the biquadratic residue symbol in $\mathbb{Q}(i)$, which we will make use of repeatedly.

If $\alpha, \beta \equiv 1\ (2)$, then

$$\left(\frac{\beta}{\alpha}\right)\left(\frac{\alpha}{\beta}\right)^{-1}=\left(\frac{\alpha, \beta}{\lambda}\right) \tag{2.1}$$

where the Hilbert symbol $\left(\frac{\alpha, \beta}{\lambda}\right)$ satisfies the following:

$$\begin{aligned}\left(\frac{\alpha, \beta}{\lambda}\right)&=1 \qquad (\alpha \equiv 1\ (4)),\\ &=1 \qquad (\alpha \equiv -1\ (4),\ \beta \equiv -1\ (4)),\\ &=-1 \qquad (\alpha \equiv -1\ (4),\ \beta \equiv 1+\lambda^3\ (4)),\\ &=-1 \qquad (\alpha \equiv 1+\lambda^3\ (4),\ \beta \equiv 1+\lambda^3\ (4));\end{aligned} \tag{2.2}$$

$$\left(\frac{\alpha, \beta}{\lambda}\right)=\left(\frac{\beta, \alpha}{\lambda}\right);\quad \left(\frac{\alpha\alpha', \beta}{\lambda}\right)=\left(\frac{\alpha, \beta}{\lambda}\right)\left(\frac{\alpha', \beta}{\lambda}\right).$$

If $\delta = 1+\alpha_4\lambda^4+\alpha_5\lambda^5+\alpha_6\lambda^6+\cdots$ $(\alpha_4, \alpha_5, \alpha_6, \ldots = 0, 1)$, then

$$\left(\frac{\lambda}{\delta}\right)=i^{-\alpha_4+2\alpha_6},\quad \left(\frac{i}{\delta}\right)=(-1)^{\alpha_4+\alpha_5}. \tag{2.3}$$

If $\delta = 1+\lambda^3+\alpha_4\lambda^4+\alpha_5\lambda^5+\alpha_6\lambda^6+\cdots$ $(\alpha_4, \alpha_5, \alpha_6, \ldots = 0, 1)$, then

$$\left(\frac{\lambda}{\delta}\right)=-i^{-\alpha_4+2\alpha_6},\quad \left(\frac{i}{\delta}\right)=i(-1)^{\alpha_4+\alpha_6}. \tag{2.4}$$

Furthermore, we note that if $\alpha \equiv 0\ (2)$, $\gamma \equiv 0\ (4)$, then

$$\left(\frac{\alpha}{\beta}\right)=\left(\frac{\alpha}{\beta+\alpha\gamma}\right). \tag{2.5}$$

Lemma 1. *Let* $\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}, \begin{pmatrix}a & b\\ c & d\end{pmatrix} \in \mathrm{SL}_2(\mathfrak{o})$. *If* $\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix} \in \Gamma(\lambda^4)$, *then*

$$\begin{aligned}\chi\left(\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}\right)&=\left(\frac{\alpha}{\alpha a+\beta c}\right)^{-1}\left(\frac{c}{\alpha a+\beta c}\right),\\ &=\left(\frac{\beta}{\alpha a+\beta c}\right)^{-1}\left(\frac{a}{\alpha a+\beta c}\right).\end{aligned}$$

Furthermore,

$$\begin{aligned}\chi\left(\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}\right)&=\left(\frac{\beta c}{\alpha}\right)^{-1}\left(\frac{c}{\alpha a+\beta c}\right) \qquad (\textit{if } \alpha \equiv 1\ (2)),\\ &=\left(\frac{\alpha}{\beta c}\right)^{-1}\left(\frac{c}{\alpha a+\beta c}\right) \qquad (\textit{if } \alpha \equiv 0\ (2),\ a \equiv 0\ (4)),\\ &=\left(\frac{\alpha}{\alpha a+\beta c}\right)^{-1}\left(\frac{\alpha a}{c}\right) \qquad (\textit{if } \gamma \equiv 1\ (2)),\end{aligned}$$

$$=\left(\frac{\alpha}{\alpha a+\beta c}\right)^{-1}\left(\frac{c}{\alpha a}\right) \qquad (\textit{if } c\equiv 0(2),\ \beta\equiv 0(4)).$$

This lemma is easily verified by (2. 1) and (2. 5). It follows from this lemma that, if $\lambda^4|N$, χ is a character on $\Gamma(N)$. In the following we assume that $\lambda^4|N$, $\frac{N}{\lambda^4}\equiv 1(\lambda^3)$.

Let κ be a cusp of $\Gamma(N)$, i.e. $\kappa\in\mathbb{Q}(i)$ or $\kappa=\infty\left(=\frac{1}{0}\right)$. We write $\kappa=\frac{\alpha}{\gamma}$ $((\alpha,\gamma)=1)$. If $(\alpha,\lambda)=1$, then we may assume $\alpha\equiv 1(\lambda^3)$. Also, if $(\alpha,\lambda)\neq 1$, then we may assume $\gamma\equiv 1(\lambda^3)$. From the assumptions we have

Lemma 2. *Two cusps* $\kappa=\frac{\alpha}{\gamma}$, $\kappa'=\frac{\alpha'}{\gamma'}$ *are equivalent each other under* $\Gamma(N)$ *if and only if* $\alpha\equiv\alpha'(N)$, $\gamma\equiv\gamma'(N)$.

Using this lemma, we can pick out representatives of inequivalent cusps of $\Gamma(N)$. In the case $N=\lambda^4$, we can do more explicitly.

For every cusp $\kappa=\frac{\alpha}{\gamma}$ of $\Gamma(N)$, we put

$$\Gamma_\kappa=\{\sigma\in\Gamma(N)\colon \sigma\cdot\kappa=\kappa\}$$

and choose

$$\sigma_\kappa=\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}\in \mathrm{SL}_2(\mathfrak{o}).$$

Noting that $\kappa=\sigma_\kappa(\infty)$ and $\Gamma_\infty=\left\{\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}\colon b\in N\mathfrak{o}\right\}$, we get

$$\Gamma_\kappa=\sigma_\kappa\Gamma_\infty\sigma_\kappa^{-1}=\left\{\sigma_\kappa\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}\sigma_\kappa^{-1}\colon b\in N\mathfrak{o}\right\}.$$

A cusp κ is said to be essential if the restriction of χ to Γ_κ is trivial. In other words, a cusp κ of $\Gamma(N)$ is essential if and only if

$$\chi\left(\sigma_\kappa\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}\sigma_\kappa^{-1}\right)=1 \quad \text{for every} \quad b\in N\mathfrak{o}.$$

Using lemma 1, we see that, for $\kappa=\frac{\alpha}{\gamma}$,

$$\chi\left(\sigma_\kappa\begin{pmatrix}1 & b\\ 0 & 1\end{pmatrix}\sigma_\kappa^{-1}\right)=\left(\frac{\gamma}{1-\gamma\alpha b}\right)\left(\frac{\alpha}{1-\gamma\alpha b}\right)^{-1}.$$

Therefore, by the assumption that $\lambda^4|N$, $\lambda^5\nmid N$, the notion "essential" in $\Gamma(N)$ depends only on the equivalence class under $\Gamma(\lambda^4)$. Hence the set of all essential cusps of $\Gamma(N)$ does not depend on N.

Now, by a straigtforward calculation, we have

Proposition 1. *A set of representatives of inequivalent essential cusps of* $\Gamma(\lambda^4)$ *consists of*

$$0, \frac{\lambda^4}{1+\lambda^3},$$

$$\lambda^2, i\lambda^2, \frac{\lambda^2}{1+\lambda^3}, \frac{i\lambda^2}{1+\lambda^3}, \lambda^3, \frac{\lambda^3}{1+\lambda^3},$$

$$1, \frac{1}{-1}, 1+\lambda^3, \frac{1+\lambda^3}{-1}, \frac{1}{1+\lambda^3}, \frac{1}{-1-\lambda^3}, \frac{1+\lambda^3}{1-\lambda^3}, \frac{1+\lambda^3}{-1+\lambda^3},$$

$$\infty, \frac{1+\lambda^3}{\lambda^4},$$

$$\frac{1}{\lambda^2}, \frac{1+\lambda^3}{\lambda^2}, \frac{1}{i\lambda^2}, \frac{1+\lambda^3}{i\lambda^2}, \frac{1}{\lambda^3}, \frac{1+\lambda^3}{\lambda^3}.$$

We classify all the essential cusps of $\Gamma(\lambda^4)$ into several types and choose σ_κ according to the following table:

type		$\kappa=\frac{\alpha}{\gamma}$	$\sigma_\kappa\equiv\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}$
A-type	A_0-type	$\alpha\equiv 0(4),\ \gamma\equiv 1(\lambda^3)$	$\beta\equiv -1(\lambda^3),\ \delta\equiv 0(4)$
	A_1-type	$\alpha\equiv 0(2),\ \alpha\not\equiv 0(4),\ \gamma\equiv 1(\lambda^3)$	$\beta\equiv -1(\lambda^3),\ \delta\equiv 0(4)$
B-type		$\alpha\equiv 1(\lambda^3),\ \gamma\equiv 1(2)$	$\beta\equiv 1(2),\quad \delta\equiv 0(4)$
C-type	C_0-type	$\alpha\equiv 1(\lambda^3),\ \gamma\equiv 0(4)$	$\beta\equiv 0(4),\quad \delta\equiv 1(\lambda^3)$
	C_1-type	$\alpha\equiv 1(\lambda^3),\ \gamma\equiv 0(2),\ \gamma\not\equiv 0(4)$	$\beta\equiv 0(4),\quad \delta\equiv 1(\lambda^3)$

(Table 1)

Next we classify all the essential cusps of $\Gamma(N)$. A essential cusp κ of $\Gamma(N)$ is said to be of $(A, *)$-type if κ is of A-type as an essential cusp of $\Gamma(\lambda^4)$. $(A_0, *)$-type, $(A_1, *)$-type, $(B, *)$-type, etc. are defined similarly. Furthermore, $(*, A)$-type, $(*, B)$-type, etc. are defined by:

type		$\kappa=\frac{\alpha}{\gamma}$ $\left(N'=\frac{N}{\lambda^4}\right)$
$(*, A)$-type	$(*, A_0)$-type	$\alpha\equiv 0(N')$
	$(*, A_1)$-type	$\alpha\not\equiv 0(N')$, $(\alpha, N')\neq 1$
$(*, B)$-type		$(\alpha, N')=1$, $(\gamma, N')=1$
$(*, C)$-type		$(\alpha, N')=1$, $(\gamma, N')\neq 1$

(Table 2)

3. Eisenstein series and theta series

Let $P(N)=\{\kappa_1, \kappa_2, \kappa_3, \ldots\}$ be a set of representatives of inequivalent essential cusps of $\Gamma(N)$. Put $\sigma_{\kappa_i}=\sigma_i$ and $\Gamma_{\kappa_i}=\Gamma_i$. Then, for each cusp κ_i, the Eisenstein series

$$E_i(u, s, \Gamma(N))=\sum_{\sigma\in\Gamma_i\backslash\Gamma(N)}\bar{\chi}(\sigma)\, v(\sigma_i^{-1}\sigma(u))^s \qquad (s\in\mathbb{C}) \tag{3.1}$$

is defined; it is convergent for $\mathrm{Re}(s)>2$ and is an eigenfunction of the Laplacian of H. Furthermore,

$$E_i(\sigma u, s, \Gamma(N))=\chi(\sigma)\, E_i(u, s, \Gamma(N)) \quad \text{for every} \quad \sigma\in\Gamma(N). \tag{3.2}$$

Let

$$K(w, s)=\int_C (x^2+y^2+1)^{-s}\, e(-wz)\, dV(z)=(2\pi)^s\, |w|^{s-1}\, \Gamma(s)^{-1}\, K_{s-1}(4\pi|w|) \tag{3.3}$$

where $dV(z)=dx\,dy$ $(z=x+iy\in\mathbb{C})$ and K_{s-1} is the modified Bessel function. Let $V(N)$ be the volume of $\mathbb{C}/N\mathfrak{o}$ with respect to $dV(z)$.

For two cusps κ_i and κ_j, we put

$$M_{ij}(N)=\left\{(c, d)\in\mathfrak{o}'\times\mathfrak{o}:\ \sigma_i\begin{pmatrix}* & *\\ c & d\end{pmatrix}\sigma_j^{-1}\in\Gamma(N)\right\} \tag{3.4}$$

where $\mathfrak{o}'=\mathfrak{o}-\{0\}$. If $(c, d)\in M_{ij}(N)$, we define

$$\bar{\chi}_{ij}(c, d)=\bar{\chi}\left(\sigma_i\begin{pmatrix}* & *\\ c & d\end{pmatrix}\sigma_j^{-1}\right). \tag{3.5}$$

Then, by a standard argument, $E_i(\sigma_j u, s, \Gamma(N))$ has a Fourier expansion of the form

$$\begin{aligned} E_i(\sigma_j u, s, \Gamma(N)) &= \delta_{ij}v^s+\psi_{ij}(s, \Gamma(N))v^{2-s} \\ &\quad +\sum_{\substack{\mu\in\mathfrak{o}^*\\ \mu\neq 0}}\psi_{ij}(s, \mu, \Gamma(N))v^{2-s}K\left(\frac{\mu v}{N}, s\right)e\left(\frac{\mu z}{N}\right) \end{aligned} \tag{3.6}$$

with

$$\delta_{ij}=\begin{cases}1 & (i=j),\\ 0 & (i\neq j),\end{cases} \tag{3.7}$$

$$\psi_{ij}(s, \Gamma(N))=\pi(s-1)^{-1}V(N)^{-1}\sum_{(c,d)\in M_{ij}(N)}\bar{\chi}_{ij}(c, d)\,N(c)^{-s} \tag{3.8}$$

and

$$\psi_{ij}(s, \mu, \Gamma(N))=V(N)^{-1}\sum_{(c,d)\in M_{ij}(N)}\bar{\chi}_{ij}(c, d)\,e\left(\frac{d\mu}{cN}\right)N(c)^{-s}. \tag{3.9}$$

Now we introduce two important concepts associated with $E_i(u, s, \Gamma(N))$, i.e. "coefficient" and "singular value". We put

$$\psi_i(s, \mu, \Gamma(N))=\psi_{ij}(s, \mu, \Gamma(N)) \qquad (\kappa_j=\infty); \tag{3.10}$$

we call it *the coefficient* of $E_i(u, s, \Gamma(N))$. And also, we say $\psi_{ij}(s, \Gamma(N))$ *the singular value* of $E_i(u, s, \Gamma(N))$ at the cusp κ_j.

It is a remarkable fact that several important results about $E_i(u, s, \Gamma(N))$ (and also those about $\psi_i(s, \mu, \Gamma(N))$) are derived from the properties of $\psi_{ij}(s, \Gamma(N))$. Meanwhile, as we see later, it is not difficult to investigate $\psi_{ij}(s, \Gamma(N))$.

For instance, it is easily observed that $\psi_{ij}(s, \Gamma(N))$ are holomorphically continued to the region $\mathrm{Re}(s)>1$ except a possibly simple pole at $s=\frac{5}{4}$. Then, we can deduce the similar fact about $E_i(u, s, \Gamma(N))$ (and also that about $\psi_i(s, \mu, \Gamma(N))$).

For a linear combination $E(u, s, \Gamma(N))=\sum_i c_iE_i(u, s, \Gamma(N))$ $(c_i\in\mathbb{C})$ of Eisenstein series, we define its coefficient by

$$\psi(s, \mu)=\sum_i c_i\psi_i(s, \mu, \Gamma(N)) \tag{3.11}$$

and its singular value at $\kappa_j\in P(N)$ by

$$\psi(s, \kappa_j)=\sum_i c_i\psi_{ij}(s, \Gamma(N)). \tag{3.12}$$

Furthermore, we can define the singular value $\psi(s, \kappa)$ at every essential cusp κ of $\Gamma(N)$.

Let $\Theta(u, \Gamma(N))$ be the residue of $E(u, s, \Gamma(N))$ at $s=\frac{5}{4}$; it is said to be *a theta series* for $\Gamma(N)$. We define the coefficient of $\Theta(u, \Gamma(N))$ by

$$\psi(\mu)=\operatorname*{Res}_{s=\frac{5}{4}}\psi(s, \mu) \tag{3.13}$$

and the singular value of $\Theta(u, \Gamma(N))$ at $\kappa_j\in P(N)$ by

$$\psi(\kappa_j)=\operatorname*{Res}_{s=\frac{5}{4}}\psi(s, \kappa_j). \tag{3.14}$$

Then, by (3. 3) and (3. 6), the theta series has the following Fourier expansion:

$$\Theta(u, \Gamma(N)) = \psi(\infty) v^{\frac{3}{4}} + (2\pi)^{\frac{5}{4}} \Gamma\left(\frac{5}{4}\right)^{-1} |N|^{-\frac{1}{4}} \times \sum_{\substack{\mu \in \mathfrak{o}^* \\ \mu \neq 0}} \psi(\mu) |\mu|^{\frac{1}{4}} v K_{\frac{1}{4}}\left(4\pi \left|\frac{\mu}{N}\right| v\right) e\left(\frac{\mu z}{N}\right). \tag{3.15}$$

Especially, if $N = \lambda^4$,

$$\Theta(u, \Gamma(\lambda^4)) = \psi(\infty) v^{\frac{3}{4}} + 2^{\frac{3}{4}} \pi^{\frac{5}{4}} \Gamma\left(\frac{5}{4}\right)^{-1} \times \sum_{\substack{\mu \in \mathfrak{o}^* \\ \mu \neq 0}} \psi(\mu) |\mu|^{\frac{1}{4}} v K_{\frac{1}{4}}(\pi |\mu| v)\, e\left(\frac{\mu z}{\lambda^4}\right). \tag{3.16}$$

The argument, which will be used in this paper, depends chiefly on the following fact:

$$\Theta(u, \Gamma(N)) = 0 \quad \text{if and only if} \quad \psi(\kappa_j) = 0 \quad \text{for all} \quad \kappa_j \in P(N).$$

More precisely, $\psi(\mu) = 0$ for all $\mu \in \mathfrak{o}^*$ $(\mu \neq 0)$ if and only if $\psi(\kappa_j) = 0$ for all $\kappa_j \in P(N)$.

We remark that, if $N | N_1$, $\Theta(u, \Gamma(N))$ can be considered a theta series for $\Gamma(N_1)$ and its singular values $\psi(\kappa_j)$ at $\kappa_j \in P(N_1)$ are canonically defined.

4. Eisenstein series for $\Gamma(\lambda^4)$ and their coefficients

In this section we consider Eisenstein series for $\Gamma(\lambda^4)$ and express their coefficients in terms of $\psi_1(s, \mu)$ and $\psi_2(s, \mu)$ defined by (1. 1).

Owing to lemma 1 the following lemmas are easily verified.

Lemma 3. *Let* $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathrm{SL}_2(\mathfrak{o})$ *so that* $\frac{\alpha}{\gamma}$ *is a essential cusp and* $(\gamma, \lambda) = 1$. *If* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathfrak{o})$, $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(\lambda^4)$, *then*

$$\bar{\chi}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \left(\frac{\alpha}{\gamma}\right)^{-1} \left(\frac{d}{c}\right).$$

Lemma 4. *Let* $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \in \mathrm{SL}_2(\mathfrak{o})$ *so that* $\frac{\alpha}{\gamma}$ *is a essential cusp and* $(\gamma, \lambda) \neq 1$. *If* $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathfrak{o})$, $\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(\lambda^4)$, *then*

$$\bar{\chi}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \left(\frac{-\gamma}{\alpha}\right)^{-1} \left(\frac{c}{d}\right).$$

For $\kappa_i = \dfrac{\alpha}{\gamma} \in P(\lambda^4)$, we put

$$
\begin{aligned}
E(u, s, \kappa_i) &= V(\lambda^4)\left(\frac{\alpha}{\gamma}\right) E_i(u, s, \Gamma(\lambda^4)) \qquad (\text{if } (\gamma, \lambda) = 1), \\
&= V(\lambda^4)\left(\frac{-\gamma}{\alpha}\right) E_i(u, s, \Gamma(\lambda^4)) \qquad (\text{if } (\gamma, \lambda) \neq 1)
\end{aligned} \tag{4.1}
$$

where $E_i(u, s, \Gamma(\lambda^4))$ is the Eisenstein series defined by (3. 1). Let $\psi(s, \mu, \kappa)$ $(\kappa \in P(\lambda^4))$ be the coefficient of $E(u, s, \kappa)$ $(\kappa \in P(\lambda^4))$ according to (3. 10) and (3. 11).

First we suppose that $\kappa = \dfrac{\alpha}{\gamma}$ is of A-type or of B-type. From (3. 4), (3. 5), (3. 9), (3. 10) and lemma 3,

$$
\psi(s, \mu, \kappa) = \sum_{c \equiv -\gamma(4)} \sum_{\substack{d \bmod 4c \\ d \equiv \alpha(4)}} \left(\frac{d}{c}\right) e\left(\frac{d\mu}{c\lambda^4}\right) N(c)^{-s}.
$$

We write $d = -\hat{\gamma}ac + \delta\lambda^4$ where $\hat{\gamma} \in \mathfrak{o}$ satisfying $\hat{\gamma} \equiv \gamma^{-1}(4)$ and δ runs through a complete set of residues (mod c) prime to c. Then

$$
\begin{aligned}
\psi(s, \mu, \kappa) &= e(-\hat{\gamma}\alpha\mu\lambda^{-4}) \sum_{c \equiv -\gamma(4)} \sum_{\delta \bmod c} \left(\frac{\delta}{c}\right) e\left(\frac{\delta\mu}{c}\right) N(c)^{-s} \\
&= e(-\hat{\gamma}\alpha\mu\lambda^{-4}) \sum_{c \equiv -\gamma(4)} g(c, \mu)\, N(c)^{-s}.
\end{aligned}
$$

Hence we have

Proposition 2.

$$
\begin{aligned}
&\psi(s, \mu, \lambda^2) = e\left(-\frac{\mu}{\lambda^2}\right)\psi_1(s, \mu), \\
&\psi(s, \mu, i\lambda^2) = e\left(-\frac{i\mu}{\lambda^2}\right)\psi_1(s, \mu), \\
&\psi(s, \mu, \lambda^3) = e\left(-\frac{\mu}{\lambda}\right)\psi_1(s, \mu), \\
&\psi(s, \mu, 0) = \psi_1(s, \mu); \\
&\psi\left(s, \mu, \frac{\lambda^2}{1+\lambda^3}\right) = e\left(-\frac{\mu}{\lambda^2}\right)\psi_2(s, \mu), \\
&\psi\left(s, \mu, \frac{i\lambda^2}{1+\lambda^3}\right) = e\left(-\frac{i\mu}{\lambda^2}\right)\psi_2(s, \mu), \\
&\psi\left(s, \mu, \frac{\lambda^3}{1+\lambda^3}\right) = e\left(-\frac{\mu}{\lambda}\right)\psi_2(s, \mu), \\
&\psi\left(s, \mu, \frac{\lambda^4}{1+\lambda^3}\right) = \psi_2(s, \mu);
\end{aligned}
$$

$$\psi(s,\mu,1)=e\left(-\frac{\mu}{\lambda^4}\right)\psi_1(s,\mu),$$

$$\psi\left(s,\mu,\frac{1}{-1}\right)=e\left(\frac{\mu}{\lambda^4}\right)\psi_1(s,\mu),$$

$$\psi(s,\mu,1+\lambda^3)=e\left(-(1+\lambda^3)\frac{\mu}{\lambda^4}\right)\psi_1(s,\mu),$$

$$\psi\left(s,\mu,\frac{1+\lambda^3}{-1}\right)=e\left((1+\lambda^3\ \frac{\mu}{\lambda^4}\right)\psi_1(s,\mu);$$

$$\psi\left(s,\mu,\frac{1}{1+\lambda^3}\right)=e\left(-(1+\lambda^3)\frac{\mu}{\lambda^4}\right)\psi_2(s,\mu),$$

$$\psi\left(s,\mu,\frac{1}{-1-\lambda^3}\right)=-e\left((1+\lambda^3)\frac{\mu}{\lambda^4}\right)\psi_2(s,\mu),$$

$$\psi\left(s,\mu,\frac{1+\lambda^3}{1-\lambda^3}\right)=e\left(-\frac{\mu}{\lambda^4}\right)\psi_2(s,\mu),$$

$$\psi\left(s,\mu,\frac{1+\lambda^3}{-1+\lambda^3}\right)=-e\left(\frac{\mu}{\lambda^4}\right)\psi_2(s,\mu).$$

Next suppose that $\kappa=\frac{\alpha}{\gamma}$ is of C-type. Then, from (3.4), (3.5), (3.9), (3.10) and lemma 4,

$$\psi\left(s,\mu,\frac{\alpha}{\gamma}\right)=\sum_{c\equiv-\gamma(4)}\ \sum_{\substack{d \bmod 4c\\ d\equiv\alpha(4)}}\left(\frac{c}{d}\right)e\left(\frac{d\mu}{c\lambda^4}\right)N(c)^{-s}$$

where $\gamma\equiv\lambda^2,\ i\lambda^2,\ \lambda^3,\ 0(4),\ \alpha\equiv 1,\ 1+\lambda^3(4)$.

We write $c=i^a\lambda^b c'$ where $0\leqq a\leqq 3,\ b\geqq 2,\ c'\equiv 1(\lambda^3)$. Since $(c',\lambda)=1$, there exist $A,\ B\in\mathfrak{o}$ such that $\lambda^{b+4}A+c'B=1$. Let $d=\lambda^{b+4}Ad_1+c'B\delta$ where d_1 runs through a complete set of residues (mod c') prime to c' and δ runs through a set of residues (mod λ^{b+4}) subject to $\delta\equiv\alpha(4)$.

Using (2.1), we get

$$\left(\frac{c}{d}\right)=\left(\frac{i^a\lambda^b}{d}\right)\left(\frac{c'}{d}\right)=\left(\frac{i^a\lambda^b}{\delta}\right)\left(\frac{d_1}{c'}\right)\left(\frac{c',\alpha}{\lambda}\right)$$

and so

$$\sum_{\substack{d \bmod 4c\\ d\equiv\alpha(4)}}\left(\frac{c}{d}\right)e\left(\frac{d\mu}{c\lambda^4}\right)=\left(\frac{c',\alpha}{\lambda}\right)\sum_{\substack{\delta \bmod \lambda^{b+4}\\ \delta\equiv\alpha(4)}}\left(\frac{i^a\lambda^b}{\delta}\right)e\left(\frac{B\delta\mu}{i^a\lambda^{b+4}}\right)$$

$$\times\sum_{d_1 \bmod c'}\left(\frac{d_1}{c'}\right)e\left(\frac{Ad_1\mu}{i^ac'}\right)$$

$$= \left(\frac{c', \alpha}{\lambda}\right) \sum_{\substack{\delta \bmod \lambda^{b+4} \\ \delta \equiv \alpha c' (4)}} \left(\frac{i^a \lambda^b}{\delta}\right) e\left(\frac{\delta \mu}{i^a \lambda^{b+4}}\right)$$

$$\times \sum_{d_1 \bmod c'} \left(\frac{d_1}{c'}\right) e\left((-1)^a \lambda^{2b} \frac{\mu d_1}{c'}\right)$$

$$= \left(\frac{c', \alpha}{\lambda}\right) \sum_{\substack{\delta \bmod \lambda^{b+4} \\ \delta \equiv \alpha c' (4)}} \left(\frac{i^a \lambda^b}{\delta}\right) e\left(\frac{\delta \mu}{i^a \lambda^{b+4}}\right) g\left(c', (-1)^a \lambda^{2b} \mu\right).$$

Here we put

$$\Gamma_1(i^a \lambda^b, \mu) = \sum_{\substack{\delta \bmod \lambda^{b+4} \\ \delta \equiv 1 (4)}} \left(\frac{i^a \lambda^b}{\delta}\right) e\left(\frac{\delta \mu}{i^a \lambda^{b+4}}\right),$$

$$\Gamma_2(i^a \lambda^b, \mu) = \sum_{\substack{\delta \bmod \lambda^{b+4} \\ \delta \equiv 1 + \lambda^3 (4)}} \left(\frac{i^a \lambda^b}{\delta}\right) e\left(\frac{\delta \mu}{i^a \lambda^{b+4}}\right).$$

Then we obtain:

$$\psi\left(s, \mu, \frac{1}{\lambda^2}\right) = \sum_{a=0,1} \Gamma_1\left((-1)^a \lambda^2, \mu\right) 2^{-2s} \psi_1(s, \mu)$$

$$- \sum_{a=0,1} \Gamma_2\left((-1)^a \lambda^2, \mu\right) 2^{-2s} \psi_2(s, \mu),$$

$$\psi\left(s, \mu, \frac{1}{i\lambda^2}\right) = \sum_{a=0,1} \Gamma_1\left((-1)^a i\lambda^2, \mu\right) 2^{-2s} \psi_1(s, -\mu)$$

$$- \sum_{a=0,1} \Gamma_2\left((-1)^a i\lambda^2, \mu\right) 2^{-2s} \psi_2(s, -\mu),$$

$$\psi\left(s, \mu, \frac{1}{\lambda^3}\right) = \sum_{0 \leqq a \leqq 3} \Gamma_1\left(i^a \lambda^3, \mu\right) 2^{-3s} \psi_1\left(s, (-1)^a \lambda^2 \mu\right)$$

$$- \sum_{0 \leqq a \leqq 3} \Gamma_2\left(i^a \lambda^3, \mu\right) 2^{-3s} \psi_2\left(s, (-1)^a \lambda^2 \mu\right),$$

$$\psi(s, \mu, \infty) = \sum_{\substack{0 \leqq a \leqq 3 \\ b \geqq 4}} \Gamma_1\left(i^a \lambda^b, \mu\right) 2^{-bs} \psi_1\left(s, (-1)^a \lambda^{2b} \mu\right)$$

$$- \sum_{\substack{0 \leqq a \leqq 3 \\ b \geqq 4}} \Gamma_2\left(i^a \lambda^b, \mu\right) 2^{-bs} \psi_2\left(s, (-1)^a \lambda^{2b} \mu\right);$$

$$\psi\left(s, \mu, \frac{1+\lambda^3}{\lambda^2}\right) = \sum_{a=0,1} \Gamma_2\left((-1)^a \lambda^2, \mu\right) 2^{-2s} \psi_1(s, \mu)$$

$$+ \sum_{a=0,1} \Gamma_1\left((-1)^a \lambda^2, \mu\right) 2^{-2s} \psi_2(s, \mu),$$

$$\psi\left(s, \mu, \frac{1+\lambda^3}{i\lambda^2}\right) = \sum_{a=0,1} \Gamma_2\left((-1)^a i\lambda^2, \mu\right) 2^{-2s} \psi_1(s, -\mu)$$

$$+ \sum_{a=0,1} \Gamma_1\left((-1)^a i\lambda^2, \mu\right) 2^{-2s} \psi_2(s, -\mu),$$

$$\psi\left(s, \mu, \frac{1+\lambda^3}{\lambda^3}\right)=\sum_{0\leqq a\leqq 3} \Gamma_2(i^a\lambda^3, \mu)2^{-3s}\psi_1(s, (-1)^a\lambda^2\mu)$$
$$+\sum_{0\leqq a\leqq 3} \Gamma_1(i^a\lambda^3, \mu)2^{-3s}\psi_2(s, (-1)^a\lambda^2\mu),$$

$$\psi\left(s, \mu, \frac{1+\lambda^3}{\lambda^4}\right)=\sum_{\substack{0\leqq a\leqq 3\\ b\geqq 4}} \Gamma_2(i^a\lambda^b, \mu)2^{-bs}\psi_1(s, (-1)^a\lambda^{2b}\mu)$$
$$+\sum_{\substack{0\leqq a\leqq 3\\ b\geqq 4}} \Gamma_1(i^a\lambda^b, \mu)2^{-bs}\psi_2(s, (-1)^a\lambda^{2b}\mu).$$

We write $\mu=i^a\lambda^\beta\mu_1$ or $i^\alpha\lambda^\beta\mu_2$ where $0\leqq\alpha\leqq 3$, $\beta\geqq -2$, $\mu_1\equiv 1(4)$, $\mu_2\equiv 1+\lambda^3(4)$. Then,

$$\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta\mu_1)=\left(\frac{i^a\lambda^b}{\mu_1}\right)^{-1}\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta),$$

$$\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta\mu_2)=\left(\frac{i^a\lambda^b}{\mu_2}\right)^{-1}\Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta),$$

$$\Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta\mu_1)=\left(\frac{i^a\lambda^b}{\mu_1}\right)^{-1}\Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta),$$

$$\Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta\mu_2)=\left(\frac{i^a\lambda^b}{\mu_2}\right)^{-1}\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta).$$

If we put $\delta=1+\alpha_4\lambda^4+\alpha_5\lambda^5+\cdots+\alpha_{b+3}\lambda^{b+3}$ $(\alpha_4, \ldots, \alpha_{b+3}=0, 1)$, then by (2.3),

$$\left(\frac{i^a\lambda^b}{\delta}\right)e\left(\frac{\delta i^\alpha\lambda^\beta}{i^a\lambda^{b+4}}\right)=i^{(2a-b)\alpha_4}(-1)^{a\alpha_5}(-1)^{b\alpha_6}$$
$$\times e\left(\frac{i^{\alpha-a}}{\lambda^{b+4-\beta}}\right)e\left(\frac{\alpha_4 i^{\alpha-a}}{\lambda^{b-\beta}}\right)\cdots e\left(\frac{\alpha_{b+3}i^{\alpha-a}}{\lambda^{1-\beta}}\right).$$

And also, if we put $\delta=1+\lambda^3+\alpha_4\lambda^4+\cdots+\alpha_{b+3}\lambda^{b+3}$ $(\alpha_4, \ldots, \alpha_{b+3}=0, 1)$, then by (2.4),

$$\left(\frac{i^a\lambda^b}{\delta}\right)e\left(\frac{\delta i^\alpha\lambda^\beta}{i^a\lambda^{b+4}}\right)=i^a(-1)^b\, i^{(2a-b)\alpha_4}(-1)^{a\alpha_5}(-1)^{b\alpha_6}$$
$$\times e\left(\frac{(1+\lambda^3)i^{\alpha-a}}{\lambda^{b+4-\beta}}\right)e\left(\frac{\alpha_4 i^{\alpha-a}}{\lambda^{b-\beta}}\right)\cdots e\left(\frac{\alpha_{b+3}i^{\alpha-a}}{\lambda^{1-\beta}}\right).$$

Hence we get

$$\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta)=e\left(\frac{i^{\alpha-a}}{\lambda^{b-\beta+4}}\right)S_4S_5\cdots S_{b+3},$$

$$\Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta)=i^a(-1)^b\, e\left(\frac{(1+\lambda^3)i^{\alpha-a}}{\lambda^{b-\beta+4}}\right)S_4S_5\cdots S_{b+3}$$

where

$$S_4=1+i^{(2a-b)}\, e\left(\frac{i^{\alpha-a}}{\lambda^{b-\beta}}\right),$$

$$S_5 = 1 + (-1)^a e\left(\frac{i^{\alpha-a}}{\lambda^{b-\beta-1}}\right),$$

$$S_6 = 1 + (-1)^b e\left(\frac{i^{\alpha-a}}{\lambda^{b-\beta-2}}\right),$$

$$S_n = 1 + e\left(\frac{i^{\alpha-a}}{\lambda^{b-\beta-n+4}}\right) \qquad (7 \leqq n \leqq b+3).$$

By a straightforward calculation, we see that $S_4 S_5 \cdots S_{b+3} = 2^b$ in the following cases:

(1) $b-\beta=5$, $\alpha \equiv 1(4)$, $a \equiv 0, 3(4)$, $b \equiv 3(4)$;
$b-\beta=5$, $\alpha \equiv 1(4)$, $a \equiv 1, 2(4)$, $b \equiv 1(4)$;
$b-\beta=5$, $\alpha \equiv 3(4)$, $a \equiv 0, 3(4)$, $b \equiv 1(4)$;
$b-\beta=5$, $\alpha \equiv 3(4)$, $a \equiv 1, 2(4)$, $b \equiv 3(4)$;

(2) $b-\beta=4$, $b \equiv 0(2)$, $a \equiv 1(2)$, $\frac{b}{2} \equiv \alpha+1(2)$;

(3) $b-\beta=3$, $b \equiv 2(4)$, $a \equiv 0(2)$;

(4) $b-\beta \leqq 2$, $b \equiv 0(4)$, $a \equiv 0(2)$.

And, if the case is otherwise, then $S_4 S_5 \cdots S_{b+3} = 0$.

Therefore, if $2 \leqq b \leqq 3$ or $b=4$, $-2 \leqq \beta \leqq 1$, $\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta)$ and $\Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta)$ are given by:

$$\begin{aligned}
\Gamma_1(\lambda^2, \lambda^{-1}) &= e^{\frac{\pi i}{4}} 2^2, & \Gamma_2(\lambda^2, \lambda^{-1}) &= -e^{\frac{\pi i}{4}} 2^2,\\
\Gamma_1(\lambda^2, i\lambda^{-1}) &= e^{-\frac{\pi i}{4}} 2^2, & \Gamma_2(\lambda^2, i\lambda^{-1}) &= e^{-\frac{\pi i}{4}} 2^2,\\
\Gamma_1(\lambda^2, -\lambda^{-1}) &= e^{-\frac{\pi i}{4}} 2^2, & \Gamma_2(\lambda^2, -\lambda^{-1}) &= -e^{-\frac{\pi i}{4}} 2^2,\\
\Gamma_1(\lambda^2, -i\lambda^{-1}) &= e^{\frac{\pi i}{4}} 2^2, & \Gamma_2(\lambda^2, -i\lambda^{-1}) &= e^{\frac{\pi i}{4}} 2^2,\\
\Gamma_1(-\lambda^2, \lambda^{-1}) &= e^{-\frac{\pi i}{4}} 2^2, & \Gamma_2(-\lambda^2, \lambda^{-1}) &= e^{-\frac{\pi i}{4}} 2^2,\\
\Gamma_1(-\lambda^2, i\lambda^{-1}) &= e^{\frac{\pi i}{4}} 2^2, & \Gamma_2(-\lambda^2, i\lambda^{-1}) &= -e^{\frac{\pi i}{4}} 2^2,\\
\Gamma_1(-\lambda^2, -\lambda^{-1}) &= e^{\frac{\pi i}{4}} 2^2, & \Gamma_2(-\lambda^2, -\lambda^{-1}) &= e^{\frac{\pi i}{4}} 2^2,\\
\Gamma_1(-\lambda^2, -i\lambda^{-1}) &= e^{-\frac{\pi i}{4}} 2^2, & \Gamma_2(-\lambda^2, -i\lambda^{-1}) &= -e^{-\frac{\pi i}{4}} 2^2;
\end{aligned}$$

$$\begin{aligned}
\Gamma_1(i\lambda^2, \lambda^{-2}) &= 2^2, & \Gamma_2(i\lambda^2, \lambda^{-2}) &= -2^2,\\
\Gamma_1(i\lambda^2, -\lambda^{-2}) &= 2^2, & \Gamma_2(i\lambda^2, -\lambda^{-2}) &= 2^2,\\
\Gamma_1(-i\lambda^2, \lambda^{-2}) &= 2^2, & \Gamma_2(-i\lambda^2, \lambda^{-2}) &= -2^2,\\
\Gamma_1(-i\lambda^2, -\lambda^{-2}) &= 2^2, & \Gamma_2(-i\lambda^2, -\lambda^{-2}) &= 2^2;
\end{aligned}$$

$$\Gamma_1(\lambda^3, i\lambda^{-2}) = e^{\frac{\pi i}{8}} 2^3, \qquad \Gamma_2(\lambda^3, i\lambda^{-2}) = ie^{\frac{\pi i}{8}} 2^3,$$
$$\Gamma_1(-i\lambda^3, i\lambda^{-2}) = e^{-\frac{\pi i}{8}} 2^3, \qquad \Gamma_2(-i\lambda^3, i\lambda^{-2}) = ie^{-\frac{\pi i}{8}} 2^3,$$
$$\Gamma_1(i\lambda^3, -i\lambda^{-2}) = e^{-\frac{\pi i}{8}} 2^3, \qquad \Gamma_2(i\lambda^3, -i\lambda^{-2}) = -ie^{-\frac{\pi i}{8}} 2^3,$$
$$\Gamma_1(-\lambda^3, -i\lambda^{-2}) = e^{\frac{\pi i}{8}} 2^3, \qquad \Gamma_2(-\lambda^3, -i\lambda^{-2}) = -ie^{\frac{\pi i}{8}} 2^3;$$

$$\Gamma_1(i\lambda^4, i) = e^{\frac{\pi i}{4}} 2^4, \qquad \Gamma_2(i\lambda^4, i) = e^{\frac{\pi i}{4}} 2^4,$$
$$\Gamma_1(i\lambda^4, -i) = e^{-\frac{\pi i}{4}} 2^4, \qquad \Gamma_2(i\lambda^4, -i) = -e^{-\frac{\pi i}{4}} 2^4,$$
$$\Gamma_1(-i\lambda^4, i) = e^{-\frac{\pi i}{4}} 2^4, \qquad \Gamma_2(-i\lambda^4, i) = e^{-\frac{\pi i}{4}} 2^4,$$
$$\Gamma_1(-i\lambda^4, -i) = e^{\frac{\pi i}{8}} 2^4, \qquad \Gamma_2(-i\lambda^4, -i) = -e^{\frac{\pi i}{4}} 2^4;$$

$$\Gamma_1(i\lambda^5, i) = e^{\frac{\pi i}{8}} 2^5, \qquad \Gamma_2(i\lambda^5, i) = -ie^{\frac{\pi i}{8}} 2^5,$$
$$\Gamma_1(-\lambda^5, i) = e^{-\frac{\pi i}{8}} 2^5, \qquad \Gamma_2(-\lambda^5, i) = ie^{-\frac{\pi i}{8}} 2^5,$$
$$\Gamma_1(\lambda^5, -i) = e^{-\frac{\pi i}{8}} 2^5, \qquad \Gamma_2(\lambda^5, -i) = -ie^{-\frac{\pi i}{8}} 2^5,$$
$$\Gamma_1(-i\lambda^5, -i) = e^{\frac{\pi i}{8}} 2^5, \qquad \Gamma_2(-i\lambda^5, -i) = ie^{\frac{\pi i}{8}} 2^5;$$

$$\Gamma_1(i^a\lambda^b, i^\alpha\lambda^\beta) = \Gamma_2(i^a\lambda^b, i^\alpha\lambda^\beta) = 0 \quad \text{(otherwise)}.$$

Now we get

Proposition 3. (1) $\psi\left(s, i^\alpha\lambda^\beta\mu_1, \frac{1}{\lambda^2}\right) = \psi\left(s, i^\alpha\lambda^\beta\mu_2, \frac{1}{\lambda^2}\right) = 0$ *if* $\beta \neq -1$;

$$\psi\left(s, \lambda^{-1}\mu_1, \frac{1}{\lambda^2}\right) = \psi\left(s, -\lambda^{-1}\mu_1, \frac{1}{\lambda^2}\right)$$
$$= \left(\frac{\lambda^2}{\mu_1}\right) 2^{-2s+\frac{5}{2}} \psi_1(s, \lambda^{-1}\mu_1) + \left(\frac{\lambda^2}{\mu_1}\right) i 2^{-2s+\frac{5}{2}} \psi_2(s, \lambda^{-1}\mu_1);$$

$$\psi\left(s, i\lambda^{-1}\mu_1, \frac{1}{\lambda^2}\right) = \psi\left(s, -i\lambda^{-1}\mu_1, \frac{1}{\lambda^2}\right)$$
$$= \left(\frac{\lambda^2}{\mu_1}\right) 2^{-2s+\frac{5}{2}} \psi_1(s, i\lambda^{-1}\mu_1) + \left(\frac{\lambda^2}{\mu_1}\right) 2^{-2s+\frac{5}{2}} \psi_2(s, i\lambda^{-1}\mu_1);$$

$$\psi\left(s, \lambda^{-1}\mu_2, \frac{1}{\lambda^2}\right) = \psi\left(s, -\lambda^{-1}\mu_2, \frac{1}{\lambda^2}\right)$$
$$= -\left(\frac{\lambda^2}{\mu_2}\right) 2^{-2s+\frac{5}{2}} \psi_1(s, \lambda^{-1}\mu_2) - i\left(\frac{\lambda^2}{\mu_2}\right) 2^{-2s+\frac{5}{2}} \psi_2(s, \lambda^{-1}\mu_2);$$

$$\psi\left(s, i\lambda^{-1}\mu_2, \frac{1}{\lambda^2}\right) = \psi\left(s, -i\lambda^{-1}\mu_2, \frac{1}{\lambda^2}\right)$$
$$= \left(\frac{\lambda^2}{\mu_2}\right) 2^{-2s+\frac{5}{2}} \psi_1(s, i\lambda^{-1}\mu_2) + i\left(\frac{\lambda^2}{\mu_2}\right) 2^{-2s+\frac{5}{2}} \psi_2(s, i\lambda^{-1}\mu_2);$$

(2) $\psi\left(s, i^{\alpha}\lambda^{\beta}\mu_0, \frac{1+\lambda^3}{\lambda^2}\right)=0 \quad if \quad \beta \not\equiv -1 \quad (\mu_0 \equiv 1(\lambda^3));$

$$\psi\left(s, \lambda^{-1}\mu_1, \frac{1+\lambda^3}{\lambda^2}\right)=-\psi\left(s, -\lambda^{-1}\mu_1, \frac{1+\lambda^3}{\lambda^2}\right)$$
$$=-i\left(\frac{\lambda^2}{\mu_1}\right)2^{-2s+\frac{5}{2}}\psi_1(s, \lambda^{-1}\mu_1)+\left(\frac{\lambda^2}{\mu_1}\right)2^{-2s+\frac{5}{2}}\psi_2(s, \lambda^{-1}\mu_1);$$

$$\psi\left(s, i\lambda^{-1}\mu_1, \frac{1+\lambda^3}{\lambda^2}\right)=-\psi\left(s, -i\lambda^{-1}\mu_1, \frac{1+\lambda^3}{\lambda^2}\right)$$
$$=-i\left(\frac{\lambda^2}{\mu_1}\right)2^{-2s+\frac{5}{2}}\psi_1(s, i\lambda^{-1}\mu_1)+\left(\frac{\lambda^2}{\mu_1}\right)2^{-2s+\frac{5}{2}}\psi_2(s, i\lambda^{-1}\mu_1);$$

$$\psi\left(s, \lambda^{-1}\mu_2, \frac{1+\lambda^3}{\lambda^2}\right)=-\psi\left(s, -\lambda^{-1}\mu_2, \frac{1+\lambda^3}{\lambda^2}\right)$$
$$=i\left(\frac{\lambda^2}{\mu_2}\right)2^{-2s+\frac{5}{2}}\psi_1(s, \lambda^{-1}\mu_2)-\left(\frac{\lambda^2}{\mu_2}\right)2^{-2s+\frac{5}{2}}\psi_2(s, \lambda^{-1}\mu_2);$$

$$\psi\left(s, i\lambda^{-1}\mu_2, \frac{1+\lambda^3}{\lambda^2}\right)=-\psi\left(s, -i\lambda^{-1}\mu_2, \frac{1+\lambda^3}{\lambda^2}\right)$$
$$=-i\left(\frac{\lambda^2}{\mu_2}\right)2^{-2s+\frac{5}{2}}\psi_1(s, i\lambda^{-1}\mu_2)+\left(\frac{\lambda^2}{\mu_2}\right)2^{-2s+\frac{5}{2}}\psi_2(s, i\lambda^{-1}\mu_2).$$

(3) $\psi\left(s, i^{\alpha}\lambda^{\beta}\mu_0, \frac{1}{i\lambda^2}\right)=0 \quad if \quad \beta \not\equiv -2 \quad or \quad \alpha \equiv 1(2) \quad (\mu_0 \equiv 1(\lambda^3));$

$$\psi\left(s, \lambda^{-2}\mu_1, \frac{1}{i\lambda^2}\right)=\psi\left(s, -\lambda^{-2}, \frac{1}{i\lambda^2}\right)$$
$$=\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}2^{-2s+3}\psi_1(s, \lambda^{-2}\mu_1)-\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}2^{-2s+3}\psi_2(s, \lambda^{-2}\mu_1);$$

$$\psi\left(s, \pm\lambda^{-2}\mu_2, \frac{1}{i\lambda^2}\right)=0.$$

(4) $\psi\left(s, i^{\alpha}\lambda^{\beta}\mu_0, \frac{1+\lambda^3}{i\lambda^2}\right)=0 \quad if \quad \beta \not\equiv -2 \quad or \quad \alpha \equiv 1(2) \quad (\mu_0 \equiv 1(\lambda^3));$

$$\psi\left(s, \lambda^{-2}\mu_1, \frac{1+\lambda^3}{i\lambda^2}\right)=-\psi\left(s, -\lambda^{-2}\mu_1, \frac{1+\lambda^3}{i\lambda^2}\right)$$
$$=-\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}2^{-2s+3}\psi_1(s, \lambda^{-2}\mu_1)-\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}2^{-2s+3}\psi_2(s, \lambda^{-2}\mu_1);$$

$$\psi\left(s, \pm\lambda^{-2}\mu_2, \frac{1+\lambda^3}{i\lambda^2}\right)=0.$$

(5) $\psi\left(s, i^{\alpha}\lambda^{\beta}\mu_0, \frac{1}{\lambda^3}\right)=0 \quad \textit{if} \quad \beta\neq -2 \quad \textit{or} \quad \alpha\equiv 0(2) \quad (\mu_0\equiv 1(\lambda^3))$;

$$\psi\left(s, i\lambda^{-2}\mu_1, \frac{1}{\lambda^3}\right)=\psi\left(s, -i\lambda^{-2}\mu_1, \frac{1}{\lambda^3}\right)$$
$$=\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_1(s, i\mu_1)$$
$$-i\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_2(s, i\mu_1);$$

$$\psi\left(s, i\lambda^{-2}\mu_2, \frac{1}{\lambda^3}\right)=\psi\left(s, -i\lambda^{-2}\mu_2, \frac{1}{\lambda^3}\right)$$
$$=i\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_1(s, i\mu_2)$$
$$-\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_2(s, i\mu_2).$$

(6) $\psi\left(s, i^{\alpha}\lambda^{\beta}\mu_0, \frac{1+\lambda^3}{\lambda^3}\right)=0 \quad \textit{if} \quad \beta\neq -2 \quad \textit{or} \quad \alpha\equiv 0(2) \quad (\mu_0\equiv 1(\lambda^3))$;

$$\psi\left(s, i\lambda^{-2}\mu_1, \frac{1+\lambda^3}{\lambda^3}\right)=-\psi\left(s, -i\lambda^{-2}\mu_1, \frac{1+\lambda^3}{\lambda^3}\right)$$
$$=i\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_1(s, i\mu_1)$$
$$+\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_2(s, i\mu_1);$$

$$\psi\left(s, i\lambda^{-2}\mu_2, \frac{1+\lambda^3}{\lambda^3}\right)=-\psi\left(s, -i\lambda^{-2}\mu_2, \frac{1+\lambda^3}{\lambda^3}\right)$$
$$=\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_1(s, i\mu_2)$$
$$+i\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-3s+3}\psi_2(s, i\mu_2).$$

(7) $\psi(s, i^{\alpha}\lambda^{\beta}\mu_0, \infty)=0 \quad \textit{if} \quad -2\leqq\beta\leqq -1 \quad (\mu_0\equiv 1(\lambda^3))$;

$\psi(s, \pm\mu_0, \infty)=0 \quad (\mu_0=1(\lambda^3))$;

$$\psi(s, i\mu_1, \infty)=\psi(s, -i\mu_1, \infty)$$
$$=\left(\frac{i}{\mu_1}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_1(s, i\mu_1)+\left(\frac{i}{\mu_1}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_2(s, i\mu_1)$$

$$+\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_1(s, i\lambda^{-2}\mu_1)$$

$$-i\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_2(s, i\lambda^{-2}\mu_1);$$

$$\psi(s, i\mu_2, \infty)=\psi(s, -i\mu_2, \infty)$$

$$=i\left(\frac{i}{\mu_2}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_1(s, i\mu_2)+i\left(\frac{i}{\mu_2}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_2(s, i\mu_2)$$

$$-i\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_1(s, i\lambda^{-2}\mu_2)$$

$$+\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_2(s, i\lambda^{-2}\mu_2);$$

$$\psi(s, i^\alpha\lambda\mu_0, \infty)=0 \quad (0\leqq\alpha\leqq 3,\ \mu_0\equiv 1(\lambda^3)).$$

(8) $$\psi\left(s, i^\alpha\lambda^\beta\mu_0, \frac{1+\lambda^3}{\lambda^4}\right)=0 \quad \textit{if} \quad -2\leqq\beta\leqq -1 \quad (\mu_0\equiv 1(\lambda^3));$$

$$\psi\left(s, \pm\mu_0, \frac{1+\lambda^3}{\lambda^4}\right)=0 \quad (\mu_0\equiv 1(\lambda^3));$$

$$\psi\left(s, i\mu_1, \frac{1+\lambda^3}{\lambda^4}\right)=-\psi\left(s, -i\mu_1, \frac{1+\lambda^3}{\lambda^4}\right)$$

$$=\left(\frac{i}{\mu_1}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_1(s, i\mu_1)-\left(\frac{i}{\mu_1}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_2(s, i\mu_1)$$

$$-i\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{\lambda}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_1(s, i\lambda^{-2}\mu_1)$$

$$-\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{\lambda}{\mu_1}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_2(s, i\lambda^{-2}\mu_1);$$

$$\psi\left(s, i\mu_2, \frac{1+\lambda^3}{\lambda^4}\right)=-\psi\left(s, -i\mu_2, \frac{1+\lambda^3}{\lambda^4}\right)$$

$$=i\left(\frac{i}{\mu_2}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_1(s, i\mu_2)-i\left(\frac{i}{\mu_2}\right)^{-1} 2^{-4s+\frac{9}{2}}\psi_2(s, i\mu_2)$$

$$+\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{\lambda}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_1(s, i\lambda^{-2}\mu_2)$$

$$+i\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}}-\left(\frac{\lambda}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-5s+5}\psi_2(s, i\lambda^{-2}\mu_2);$$

$$\psi\left(s, i^\alpha\lambda\mu_0, \frac{1+\lambda^3}{\lambda^4}\right)=0 \quad (0\leqq\alpha\leqq 3,\ \mu_0\equiv 1(\lambda^3)).$$

5. Singular values of Eisenstein series for $\Gamma(\lambda^4)$

Recall the Eisenstein series $E(u, s, \kappa_i)$ defined by (4.1). In view of (3.8) and (3.12), the singular value $\psi(s, \kappa_i, \kappa_j)$ of $E(u, s, \kappa_i)$ at $\kappa_j \in P(\lambda^4)$ is given by

$$(5.1)\quad \psi(s, \kappa_i, \kappa_j) = \pi(s-1)^{-1}\left(\frac{\alpha}{\gamma}\right) \sum_{(c,d)\in M_{ij}(\lambda^4)} \bar{\chi}_{ij}(c, d)\, N(c)^{-s} \qquad \left(\text{if } \kappa_i = \frac{\alpha}{\gamma},\ (\gamma, \lambda) = 1\right),$$

$$= \pi(s-1)^{-1}\left(\frac{-\gamma}{\alpha}\right) \sum_{(c,d)\in M_{ij}(\lambda^4)} \bar{\chi}_{ij}(c, d)\, N(c)^{-s} \qquad \left(\text{if } \kappa_i = \frac{\alpha}{\gamma},\ (\gamma, \lambda) \neq 1\right).$$

We put $\xi(s) = \sum_{c\equiv 1\,(\lambda^3)} N(c)^{-s}$.

Lemma 5. *Suppose* $\gamma \equiv 0(2)$ *and* $\alpha \equiv 1(\lambda^3)$, *then*

$$\sum_{c\equiv\gamma(4)} \sum_{\substack{d \bmod 4c \\ d\equiv\alpha(4)}} \left(\frac{c}{d}\right) N(c)^{-s} = 2(2^{4s-4}-1)^{-1}\, \xi(4s-4)\, \xi(4s-3)^{-1} \qquad (\textit{if } \alpha\equiv 1(4),\ \gamma\equiv 0(4)),$$

$$= 0 \qquad (\textit{otherwise}).$$

Proof. We write $c = i^a\lambda^b c'$ where $0 \leqq a \leqq 3$, $b \geqq 2$, $c' \equiv 1(\lambda^3)$. Then, by the argument used in Section 4,

$$\sum_{\substack{d \bmod 4c \\ d\equiv\alpha(4)}} \left(\frac{c}{d}\right) = \left(\frac{c', \alpha}{\lambda}\right) \sum_{\substack{\delta \bmod \lambda^{b+4} \\ \delta\equiv\alpha(4)}} \left(\frac{i^a\lambda^b}{\delta}\right) \sum_{d_1 \bmod c'} \left(\frac{d_1}{c'}\right).$$

Using (2.3) and (2.4),

$$\sum_{\substack{\delta \bmod \lambda^{b+4} \\ \delta\equiv\alpha(4)}} \left(\frac{i^a\lambda^b}{\delta}\right) = 2^b \qquad (\text{if } a = 0,\ 4|b),$$

$$= \left(\frac{\alpha, -1}{\lambda}\right) 2^b \qquad (\text{if } a = 2,\ 4|b),$$

$$= 0 \qquad (\text{if otherwise}).$$

Hence, if $\gamma \not\equiv 0(4)$ or $\alpha \equiv 1 + \lambda^3(4)$,

$$\sum_{c\equiv\gamma(4)} \sum_{\substack{d \bmod 4c \\ d\equiv\alpha(4)}} \left(\frac{c}{d}\right) N(c)^{-s} = 0.$$

Furthermore, we have

$$\sum_{d_1 \bmod c'} \left(\frac{d_1}{c'}\right) = \Phi(c') \qquad (\text{if } c' \text{ is a biquadrate}),$$

$$= 0 \qquad (\text{if otherwise})$$

where $\Phi(\ )$ is the Euler function on $\mathfrak{o}$.

Then, $\sum_{\substack{d \bmod 4c \\ d\equiv\alpha(4)}} \left(\frac{c}{d}\right) \neq 0$ implies $c' \equiv 1(4)$, that means $\left(\frac{c', \alpha}{\lambda}\right) = 1$.

Therefore, if $\alpha \equiv 1(4)$ and $\gamma \equiv 0(4)$,

$$\sum_{c\equiv 0(4)} \sum_{\substack{d \bmod 4c \\ d\equiv 1(4)}} \left(\frac{c}{d}\right) N(c)^{-s} = 2 \sum_{b\geqq 1} 2^{4b(1-s)} \sum_{c\equiv 1(\lambda^3)} \Phi(c)\, N(c)^{3-4s}$$
$$= 2(2^{4s-4}-1)^{-1}\, \xi(4s-4)\, \xi(4s-3)^{-1}.$$

Lemma 6. *Suppose* $\gamma \equiv 1(2)$, *then*

$$\sum_{c\equiv \gamma(4)} \sum_{\substack{d \bmod 4c \\ d\equiv \alpha(4)}} \left(\frac{d}{c}\right) N(c)^{-s} = \xi(4s-4)\, \xi(4s-3)^{-1} \qquad (\textit{if } \gamma \equiv \pm 1(4)),$$
$$= 0 \qquad (\textit{if } \gamma \equiv \pm(1+\lambda^3)\,(4)).$$

Proof. We have

$$\sum_{\substack{d \bmod 4c \\ d\equiv \alpha(4)}} \left(\frac{d}{c}\right) = \Phi(c) \qquad (\text{if } c \text{ or } -c \text{ is a biquadrate}),$$
$$= 0 \qquad (\text{if otherwise}).$$

Then, $\sum_{\substack{d \bmod 4c \\ d\equiv \alpha(4)}} \left(\frac{d}{c}\right) \neq 0$ implies $c \equiv \pm 1(4)$.

Therefore, our assertion follows.

We put $\sigma_i = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, $\sigma_j = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}$ according to Table 1. Then we see that

$$M_{ij}(\lambda^4) = \{(c, d);\quad c \equiv -\gamma\alpha' + \alpha\gamma'(4),\quad d \equiv -\gamma\beta' + \alpha\delta'(4)\}$$

and

$$\bar{\chi}_{ij}(c, d) = \bar{\chi}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}^{-1}\right)$$
$$= \bar{\chi}\left(\begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} a\delta' - b\gamma' & -a\beta' + b\alpha' \\ c\delta' - d\gamma' & -c\beta' + d\alpha' \end{pmatrix}\right).$$

Let's consider several cases separately, depending on the types of κ_i and κ_j.

[1] Both $\kappa_i = \dfrac{\alpha}{\gamma}$ and $\kappa_j = \dfrac{\alpha'}{\gamma'}$ are of A-type.

In this case, if $(c, d) \in M_{ij}(\lambda^4)$, $c \equiv 0(2)$ and $d \equiv \gamma\gamma'^{-1}(4)$. By (2. 1), (2. 5), lemma 3 and table 1,

$$\bar{\chi}_{ij}(c, d) = \left(\frac{\alpha}{\gamma}\right)^{-1} \left(\frac{-c\beta' + d\alpha'}{d\gamma' - c\delta'}\right)$$
$$= \left(\frac{\alpha}{\gamma}\right)^{-1} \left(\frac{\gamma'}{d\gamma' - c\delta'}\right)^{-1} \left(\frac{c}{d\gamma' - c\delta'}\right)$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{c\delta'}{\gamma'}\right)^{-1}\left(\frac{c}{d\gamma'}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right) \qquad \text{(because } d\gamma'-c\delta'\equiv\gamma(4),\ \delta'\equiv 0(4)\text{)}$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{c}{d}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right).$$

Hence, by lemma 5, we have

$$\text{(5. 2)} \qquad \psi(s, \kappa_i, \kappa_j)=\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)2(2^{4s-4}-1)^{-1}\,\xi(4s-4)\,\xi(4s-3)^{-1} \quad (\text{if } \kappa_i=\kappa_j),$$

$$=0 \quad (\text{if } \kappa_i\neq\kappa_j).$$

[2] $\kappa_i=\dfrac{\alpha}{\gamma}$ is of A-type and $\kappa_j=\dfrac{\alpha'}{\gamma'}$ is of B-type.

In this case, if $(c, d)\in M_{ij}(\lambda^4)$, $c\equiv 1(2)$. By (2. 1) and lemma 3,

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{-c\beta'+d\alpha'}{d\gamma'-c\delta'}\right)$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{d\gamma'-c\delta'}\right)^{-1}\left(\frac{c}{d\gamma'-c\delta'}\right)$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{c\delta'}{\gamma'}\right)^{-1}\left(\frac{d\gamma'}{c}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right)\left(\frac{c, \gamma}{\lambda}\right) \qquad \text{(because } d\gamma'-c\delta'\equiv\gamma(4)\text{)}$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{d}{c}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right)\left(\frac{c, \gamma}{\lambda}\right)\left(\frac{c, \gamma'}{\lambda}\right).$$

By lemma 6 and a straightforward calculation, we have

$$\text{(5. 3)} \qquad \psi(s, \kappa_i, \kappa_j)=\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\rho_{ij}\,\xi(4s-4)\,\xi(4s-3)^{-1}$$

where ρ_{ij} is given by:

κ_i \ κ_j	1	$\frac{1}{-1}$	$1+\lambda^3$	$-1-\lambda^3$	$\frac{1}{1+\lambda^3}$	$\frac{1}{-1-\lambda^3}$	$\frac{1+\lambda^3}{1-\lambda^3}$	$\frac{1+\lambda^3}{-1+\lambda^3}$
λ^2			1	1			-1	-1
$\frac{\lambda^2}{1+\lambda^3}$	1	-1			1	-1		
$i\lambda^2$	1	1			-1	-1		
$\frac{i\lambda^2}{1+\lambda^3}$			1	-1			1	-1

κ_i \ κ_j	1	$\frac{1}{-1}$	$1+\lambda^3$	$-1-\lambda^3$	$\frac{1}{1+\lambda^3}$	$\frac{1}{-1-\lambda^3}$	$\frac{1+\lambda^3}{1-\lambda^3}$	$\frac{1+\lambda^3}{-1+\lambda^3}$
λ^3			1	1			1	1
$\frac{\lambda^3}{1+\lambda^3}$	-1	1			1	-1		
0	1	1			1	1		
$\frac{\lambda^4}{1+\lambda^3}$			-1	1			1	-1

(Table 3) (We omit zeros)

[3] $\kappa_i=\frac{\alpha}{\gamma}$ is of A-type and $\kappa_j=\frac{\alpha'}{\gamma'}$ is of C-type.

Let's consider $\sigma_j'=\sigma_j\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}$ $(k\in\mathfrak{o})$. Even if we replace σ_j by σ_j' in $E(\sigma_j u, s, \kappa_j)$, the constant term of the Fourier expansion of $E(\sigma_j u, s, \kappa_i)$ is invariant. So we have

$$\psi(s,\kappa_i,\kappa_j)=\pi(s-1)^{-1}\left(\frac{\alpha}{\gamma}\right)\sum_{(c,d)\in M_{ij}'(\lambda^4)}\bar{\chi}_{ij}'(c,d)\,N(c)^{-s}$$

with

$$M_{ij}'(\lambda^4)=\left\{(c,d)\in\mathfrak{o}'\times\mathfrak{o}:\ \sigma_i\begin{pmatrix}* & *\\ c & d\end{pmatrix}\sigma_j'^{-1}\in\Gamma(\lambda^4)\right\}$$

and

$$\bar{\chi}_{ij}'(c,d)=\bar{\chi}\left(\sigma_i\begin{pmatrix}* & *\\ c & d\end{pmatrix}\sigma_j'^{-1}\right).$$

We take $k\in\mathfrak{o}$ so that $k\equiv\frac{\alpha\delta'}{\gamma\alpha'}(4)$. Put $\sigma_j'=\begin{pmatrix}\alpha' & \beta''\\ \gamma' & \delta''\end{pmatrix}$. Then if $(c,d)\in M_{ij}'(\lambda^4)$, $c\equiv-\gamma\alpha'(4)$ and $d\equiv 0(4)$. By (2.1), (2.5) and lemma 3,

$$\begin{aligned}\bar{\chi}_{ij}'(c,d)&=\bar{\chi}\left(\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}\alpha' & \beta''\\ \gamma' & \delta''\end{pmatrix}^{-1}\right)\\
&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{-c\beta''+d\alpha'}{d\gamma'-c\delta''}\right)\\
&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{d\gamma'-c\delta''}\right)^{-1}\left(\frac{c}{d\gamma'-c\delta''}\right)\\
&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{-c\delta''}\right)^{-1}\left(\frac{d\gamma'}{c}\right)\left(\frac{c,\gamma}{\lambda}\right)\quad\text{(because } d\equiv 0(4),\ \gamma'\equiv 0(2),\ d\gamma'-c\delta''\equiv\gamma(4)\text{)}\end{aligned}$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\delta''}\right)^{-1}\left(\frac{d}{c}\right)\left(\frac{\alpha',\gamma}{\lambda}\right) \qquad \text{(because } c\equiv-\gamma\alpha'(4)\text{)}$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{\gamma'}{1+\frac{k\gamma'}{\delta'}}\right)^{-1}\left(\frac{d}{c}\right)\left(\frac{\alpha',\gamma}{\lambda}\right) \qquad \text{(because } \delta''=\delta'+k\gamma'\text{)}$$

$$=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{\gamma'}{1+\frac{\alpha\gamma'}{\gamma\alpha'}}\right)^{-1}\left(\frac{d}{c}\right)\left(\frac{\alpha',\gamma}{\lambda}\right).$$

By lemma 6 and a straightforward calculation, we have

$$\psi(s,\kappa_i,\kappa_j)=\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\rho_{ij}\,\xi(4s-4)\,\xi(4s-3)^{-1} \tag{5.4}$$

where ρ_{ij} is given by:

κ_i \ κ_j	∞	$\frac{1}{\lambda^2}$	$\frac{1}{i\lambda^2}$	$\frac{1}{\lambda^3}$	$\frac{1+\lambda^3}{\lambda^2}$	$\frac{1+\lambda^3}{i\lambda^2}$	$\frac{1+\lambda^3}{\lambda^3}$	$\frac{1+\lambda^3}{\lambda^4}$
0	1	1	1	1				
λ^2	1	−1	−1	1				
$i\lambda^2$	1	−1	1	−1				
λ^3	1	1	−1	−1				
$\frac{\lambda^2}{1+\lambda^3}$					1	1	−1	−1
$\frac{i\lambda^2}{1+\lambda^3}$					1	−1	1	−1
$\frac{\lambda^3}{1+\lambda^3}$					−1	1	1	−1
$\frac{\lambda^4}{1+\lambda^3}$					−1	−1	−1	−1

(Table 4) (We omit zeros)

[4] $\kappa_i=\dfrac{\alpha}{\gamma}$ is of B-type and $\kappa_j=\dfrac{\alpha'}{\gamma'}$ is of A-type.

If $(c, d)\in M_{ij}(\lambda^4)$, then $c\equiv 1(2)$ and

$$\begin{aligned}\bar{\chi}_{ij}(c, d)&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{-c\beta'+d\alpha'}{d\gamma'-c\delta'}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{d\gamma'-c\delta'}\right)^{-1}\left(\frac{c}{d\gamma'-c\delta'}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{c\delta'}{\gamma'}\right)^{-1}\left(\frac{d\gamma'}{c}\right)\left(\frac{-\gamma, \gamma'}{\lambda}\right)\left(\frac{c, \gamma}{\lambda}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{d}{c}\right)\left(\frac{-\gamma, \gamma'}{\lambda}\right)\left(\frac{c, \gamma}{\lambda}\right)\left(\frac{c, \gamma'}{\lambda}\right).\end{aligned}$$

Hence,

$$\psi(s, \kappa_i, \kappa_j)=\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\rho_{ij}\,\xi(4s-4)\,\xi(4s-3)^{-1} \tag{5. 5}$$

where ρ_{ij} is given by:

κ_i \ κ_j	0	λ^2	$i\lambda^2$	λ^3	$\frac{\lambda^4}{1+\lambda^3}$	$\frac{\lambda^2}{1+\lambda^3}$	$\frac{i\lambda^2}{1+\lambda^3}$	$\frac{\lambda^3}{1+\lambda^3}$
1	1		1			1		-1
$\frac{1}{-1}$	1		1			-1		1
$1+\lambda^3$		1		1	-1		1	
$-1-\lambda^3$		1		1	1		-1	
$\frac{1}{1+\lambda^3}$	1		-1			1		1
$\frac{1}{-1-\lambda^3}$	1		-1			-1		-1
$\frac{1+\lambda^3}{1-\lambda^3}$		-1		1	1		1	
$\frac{1+\lambda^3}{-1+\lambda^3}$		-1		1	-1		-1	

(Table 5) (We omit zeros)

[5] $\kappa_i=\dfrac{\alpha}{\gamma}$ is of B-type and $\kappa_j=\dfrac{\alpha'}{\gamma'}$ is of B-type.

If $(c, d)\in M_{ij}(\lambda^4)$, then $c\equiv 0(2)$, $d\equiv\gamma\gamma'^{-1}(4)$ and

$$\begin{aligned}\bar{\chi}_{ij}(c, d)&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{-c\beta'+d\alpha'}{d\gamma'-c\delta'}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{d\gamma'-c\delta'}\right)^{-1}\left(\frac{c}{d\gamma'-c\delta'}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{-c\delta'}{\gamma'}\right)^{-1}\left(\frac{c}{d\gamma'}\right)\left(\frac{\gamma, \gamma'}{\lambda}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{c}{d}\right)\left(\frac{-\gamma, \gamma'}{\lambda}\right).\end{aligned}$$

Hence, we have

$$\begin{aligned}(5.6)\qquad \psi(s, \kappa_i, \kappa_j)&=\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)2(2^{4s-4}-1)^{-1}\,\xi(4s-4)\,\xi(4s-3)^{-1}\qquad(\kappa_i=\kappa_j),\\&=0\qquad(\kappa_i\neq\kappa_j).\end{aligned}$$

[6] $\kappa_i=\dfrac{\alpha}{\gamma}$ is B-type and $\kappa_j=\dfrac{\alpha'}{\gamma'}$ is of C-type.

If $(c, d)\in M_{ij}(\lambda^4)$, then $c\equiv 1(2)$, $d\equiv 1(2)$ and

$$\begin{aligned}\bar{\chi}_{ij}(c, d)&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{-c\beta'+d\alpha'}{d\gamma'-c\delta'}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\delta'}{d\gamma'-c\delta'}\right)^{-1}\left(\frac{d}{d\gamma'-c\delta'}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{d\gamma'}{\delta'}\right)^{-1}\left(\frac{-c\delta'}{d}\right)\left(\frac{\delta', \gamma}{\lambda}\right)\left(\frac{d, \gamma}{\lambda}\right)\\&=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{d}{c}\right)\left(\frac{\delta', \gamma}{\lambda}\right)\left(\frac{d, \gamma}{\lambda}\right)\left(\frac{d, \delta'}{\lambda}\right)\left(\frac{d, -c}{\lambda}\right).\end{aligned}$$

Hence, we have

$$(5.7)\qquad \psi(s, \kappa_i, \kappa_j)=\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\rho_{ij}\,\xi(4s-4)\,\xi(4s-3)^{-1}$$

where ρ_{ij} is given by:

κ_i \ κ_j	∞	$\frac{1}{\lambda^2}$	$\frac{1}{i\lambda^2}$	$\frac{1}{\lambda^3}$	$\frac{1+\lambda^3}{\lambda^4}$	$\frac{1+\lambda^3}{\lambda^2}$	$\frac{1+\lambda^3}{i\lambda^2}$	$\frac{1+\lambda^3}{\lambda^3}$
1	1		1			1		-1
$\frac{1}{-1}$	1		1			-1		1
$1+\lambda^3$	1		-1			1		1
$\frac{1+\lambda^3}{-1}$	1		-1			-1		-1
$\frac{1}{1+\lambda^3}$		1		1	-1		1	
$\frac{1}{-1-\lambda^3}$		1		1	1		-1	
$\frac{1+\lambda^3}{1-\lambda^3}$		1		-1	-1		-1	
$\frac{1+\lambda^3}{-1+\lambda^3}$		1		-1	1		1	

(Table 6) (We omit zeros)

[7] $\kappa_i=\frac{\alpha}{\gamma}$ is of C-type and $\kappa_j=\frac{\alpha'}{\gamma'}$ is of A-type.

Let $\sigma_j'=\begin{pmatrix}\alpha' & \beta''\\ \gamma' & \delta''\end{pmatrix}=\sigma_j\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}$ $(k\in\mathfrak{o})$. By a similar argument to [3], we have

$$\psi(s,\kappa_i,\kappa_j)=\pi(s-1)^{-1}\left(\frac{-\gamma}{\alpha}\right)\sum_{(c,d)\in M_{ij}'(\lambda^4)}\bar{\chi}_{ij}'(c,d)\,N(c)^{-s}$$

with

$$M_{ij}'(\lambda^4)=\left\{(c,d)\in\mathfrak{o}'\times\mathfrak{o}:\ \sigma_i\begin{pmatrix}* & *\\ c & d\end{pmatrix}\sigma_j'^{-1}\in\Gamma(\lambda^4)\right\}$$

and $\bar{\chi}_{ij}'(c,d)=\bar{\chi}\left(\sigma_i\begin{pmatrix}* & *\\ c & d\end{pmatrix}\sigma_j'^{-1}\right)$.

We take $k \in \mathfrak{o}$ so that $k \equiv \dfrac{\gamma\beta'}{\alpha\gamma'}(4)$. Then, if $(c, d) \in M'_{ij}(\lambda^4)$, $c \equiv \alpha\gamma'(4)$ and $d \equiv 0(4)$. By lemma 5,

$$\begin{aligned}
\bar{\chi}'_{ij}(c, d) &= \bar{\chi}\left(\begin{pmatrix}\alpha & \beta\\ \gamma & \delta\end{pmatrix}\begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}\alpha' & \beta''\\ \gamma' & \delta''\end{pmatrix}^{-1}\right)\\
&= \left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{c\delta'' - d\gamma'}{-c\beta'' + d\alpha'}\right)\\
&= \left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\alpha'}{-c\beta'' + d\alpha'}\right)^{-1}\left(\frac{c}{-c\beta'' + d\alpha'}\right)\\
&= \left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\alpha'}{c\beta''}\right)^{-1}\left(\frac{d\alpha'}{c}\right)\left(\frac{c, \alpha}{\lambda}\right)\\
&= \left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{\alpha'}{1 + \dfrac{\alpha'\gamma}{\gamma'\alpha}}\right)^{-1}\left(\frac{c, \alpha}{\lambda}\right)\left(\frac{d}{c}\right).
\end{aligned}$$

Hence, by lemma 6, we have

$$\psi(s, \kappa_i, \kappa_j) = \pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\rho_{ij}\,\xi(4s-4)\,\xi(4s-3)^{-1} \tag{5. 8}$$

where ρ_{ij} is given by:

κ_i \ κ_j	0	λ^2	$i\lambda^2$	λ^3	$\dfrac{\lambda^4}{1+\lambda^3}$	$\dfrac{\lambda^2}{1+\lambda^3}$	$\dfrac{i\lambda^2}{1+\lambda^3}$	$\dfrac{\lambda^3}{1+\lambda^3}$
∞	1	1	1	1				
$\dfrac{1}{\lambda^2}$	1	−1	−1	1				
$\dfrac{1}{i\lambda^2}$	1	−1	1	−1				
$\dfrac{1}{\lambda^3}$	1	1	−1	−1				
$\dfrac{1+\lambda^3}{\lambda^4}$					1	1	1	1
$\dfrac{1+\lambda^3}{\lambda^2}$					1	−1	−1	1

κ_i \ κ_j	0	λ^2	$i\lambda^2$	λ^3	$\frac{\lambda^4}{1+\lambda^3}$	$\frac{\lambda^2}{1+\lambda^3}$	$\frac{i\lambda^2}{1+\lambda^3}$	$\frac{\lambda^3}{1+\lambda^3}$
$\frac{1+\lambda^3}{i\lambda^2}$					1	-1	1	-1
$\frac{1+\lambda^3}{\lambda^3}$					1	1	-1	-1

(Table 7) (We omit zeros)

[8] $\kappa_i=\frac{\alpha}{\gamma}$ is of C-type and $\kappa_j=\frac{\alpha'}{\gamma'}$ is of B-type.

If $(c, d)\in M_{ij}(\lambda^4)$, then $c\equiv 1(2)$ and

$$\begin{aligned}\bar{\chi}_{ij}(c,d)&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{c\delta'-d\gamma'}{-c\beta'+d\alpha'}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\alpha'}{-c\beta'+d\alpha'}\right)^{-1}\left(\frac{c}{-c\beta'+d\alpha'}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{-c\beta'}{\alpha'}\right)^{-1}\left(\frac{d\alpha'}{c}\right)\left(\frac{\alpha,\alpha'}{\lambda}\right)\left(\frac{c,\alpha}{\lambda}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{d}{c}\right)\left(\frac{c,\alpha'}{\lambda}\right)\left(\frac{\alpha,\alpha'}{\lambda}\right)\left(\frac{c,\alpha}{\lambda}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{d}{c}\right)\left(\frac{\alpha',\gamma'}{\lambda}\right)\left(\frac{c,\alpha'}{\lambda}\right)\left(\frac{\alpha,\alpha'}{\lambda}\right)\left(\frac{c,\alpha}{\lambda}\right).\end{aligned}$$

Hence, by lemma 6, we have

(5. 9) $$\psi(s,\kappa_i,\kappa_j)=\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\rho_{ij}\,\xi(4s-4)\,\xi(4s-3)^{-1}$$

where ρ_{ij} is given by:

κ_i \ κ_j	1	$\frac{1}{-1}$	$1+\lambda^3$	$-1-\lambda^3$	$\frac{1}{1+\lambda^3}$	$\frac{1}{-1-\lambda^3}$	$\frac{1+\lambda^3}{1-\lambda^3}$	$\frac{1+\lambda^3}{-1+\lambda^3}$
∞	1	1	1	1				
$\frac{1}{\lambda^2}$					1	1	1	1
$\frac{1}{i\lambda^2}$	1	1	-1	-1				

κ_i \ κ_j	1	$\frac{1}{-1}$	$1+\lambda^3$	$-1-\lambda^3$	$\frac{1}{1+\lambda^3}$	$\frac{1}{-1-\lambda^3}$	$\frac{1+\lambda^3}{1-\lambda^3}$	$\frac{1+\lambda^3}{-1+\lambda^3}$
$\frac{1}{\lambda^3}$					1	1	-1	-1
$\frac{1+\lambda^3}{\lambda^4}$					1	-1	1	-1
$\frac{1+\lambda^3}{\lambda^2}$	-1	1	-1	1				
$\frac{1+\lambda^3}{i\lambda^2}$					-1	1	1	-1
$\frac{1+\lambda^3}{\lambda^3}$	1	-1	-1	1				

(Table 8) (We omit zeros)

[9] $\kappa_i=\dfrac{\alpha}{\gamma}$ is of C-type and $\kappa_j=\dfrac{\alpha'}{\gamma'}$ is of C-type.

If $(c, d)\in M_{ij}(\lambda^4)$, then $c\equiv 0(2)$, $d\equiv\alpha\alpha'^{-1}(4)$ and

$$\begin{aligned}\bar{\chi}_{ij}(c,d)&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{c\delta'-d\gamma'}{-c\beta'+d\alpha'}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\alpha'}{-c\beta'+d\alpha'}\right)^{-1}\left(\frac{c}{-c\beta'+d\alpha'}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{-c\beta'}{\alpha'}\right)^{-1}\left(\frac{c}{d\alpha'}\right)\left(\frac{\alpha',\alpha}{\lambda}\right)\\&=\left(\frac{-\gamma}{\alpha}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{c}{d}\right)\left(\frac{\alpha',\alpha}{\lambda}\right).\end{aligned}$$

Hence, by lemma 5, we have

$$\begin{aligned}(5.10)\qquad \psi(s,\kappa_i,\kappa_j)&=\pi(s-1)^{-1}\left(\frac{\alpha',-1}{\lambda}\right)\left(\frac{\gamma'}{\alpha'}\right)2(2^{4s-1}-1)^{-1}\,\xi(4s-4)\,\xi(4s-3)^{-1}\\&\qquad(\text{if }\ \kappa_i=\kappa_j),\\&=0\qquad(\text{if }\ \kappa_i\neq\kappa_j).\end{aligned}$$

For future use, we require the following proposition.

Proposition 4. (1) *If* $\frac{\alpha'}{\gamma'}$ *is any essential cusp of* $\Gamma(\lambda^4)$, $\psi\left(s, 0, \frac{\alpha'}{\gamma'}\right)$ *is given by:*

$\frac{\alpha'}{\gamma'}$	$\psi\left(s, 0, \frac{\alpha'}{\gamma'}\right)$
$\alpha' \equiv 0(4),\ \gamma' \equiv 1(4)$	$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right) 2(2^{4s-4}-1)^{-1}\, \xi(4s-4)\, \xi(4s-3)^{-1}$
$\alpha' \equiv 1(4)$	$\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right) \xi(4s-4)\, \xi(4s-3)^{-1}$
otherwise	0

(2) *If* $\frac{\alpha'}{\gamma'}$ *is any essential cusp of* $\Gamma(\lambda^4)$, $\psi\left(s, \frac{\lambda^4}{1+\lambda^3}, \frac{\alpha'}{\gamma'}\right)$ *is given by:*

$\frac{\alpha'}{\gamma'}$	$\psi\left(s, \frac{\lambda^4}{1+\lambda^3}, \frac{\alpha'}{\gamma'}\right)$
$\alpha' \equiv 0(4),\ \gamma' \equiv 1+\lambda^3(4)$	$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right) 2(2^{4s-4}-1)^{-1}\, \xi(4s-4)\, \xi(4s-3)^{-1}$
$\alpha' \equiv 1+\lambda^3(4)$	$-\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right) \xi(4s-4)\, \xi(4s-3)^{-1}$
otherwise	0

6. Residues at $s=\frac{5}{4}$ (I)

Let's consider the residues $\Theta(u, \kappa_i)$ of $E(u, s, \kappa_i)$ $(\kappa_i \in P(\lambda^4))$ at $s=\frac{5}{4}$.

First, we will find several relations among $\Theta(u, \kappa_i)$ $(\kappa_i \in P(\lambda^4))$. For the purpose we make use of our principle: let $\Theta(u)$ be a theta series for $\Gamma(\lambda^4)$; then $\Theta(u)=0$ if and only if its singular values $\psi(\kappa_j)$ vanish for all $\kappa_j \in P(\lambda^4)$.

In view of (3. 14), the singular values $\psi(\kappa_i, \kappa_j)$ of $\Theta(u, \kappa_i)$ $(\kappa_j \in P(\lambda^4))$ are derived from (5. 2), (5. 3),..., (5. 10):

$$\psi(\kappa_i, \kappa_j) = 4\pi\left(\frac{\alpha'}{\gamma'}\right) \operatorname*{Res}_{s=1} \xi(s)\, \xi(2)^{-1} \rho_{ij} \qquad \left(\text{if } \kappa_j = \frac{\alpha'}{\gamma'},\ (\gamma', \lambda)=1\right),$$

$$=4\pi\left(\frac{\gamma'}{\alpha'}\right)\operatorname*{Res}_{s=1}\xi(s)\,\xi(2)^{-1}\rho_{ij}\qquad\left(\text{if }\ \kappa_j=\frac{\alpha'}{\gamma'},\ (\gamma',\lambda)\neq 1\right)$$

where ρ_{ij} is given by Table 3, Table 4,..., Table 8 in the case that κ_i and κ_j are of different type, and

$$\begin{aligned}\rho_{ij}&=2 && (\kappa_i=\kappa_j\text{: } A\text{-type or } B\text{-type}),\\ &=\left(\frac{\alpha',-1}{\lambda}\right)2 && (\kappa_i=\kappa_j\text{: } C\text{-type}),\\ &=0 && (\kappa_i\neq\kappa_j)\end{aligned}$$

in the case that κ_i and κ_j are of the same type.

By a straightforward calculation, we can find 16 linear relations, which are independent of each other, among the 24 column vectors $(\psi(\kappa_i,\kappa_j))_{1\leqq j\leqq 24}$ $(1\leqq i\leqq 24)$. In other words, we get the following relations, which are independent of each other:

$$\Theta(u,0)+\Theta(u,\lambda^2)+\Theta(u,i\lambda^2)+\Theta(u,\lambda^3)=2\Theta(u,\infty),$$

$$\Theta(u,0)+\Theta(u,\lambda^2)-\Theta(u,i\lambda^2)-\Theta(u,\lambda^3)=2\Theta\left(u,\frac{1}{\lambda^3}\right),$$

$$\Theta(u,0)-\Theta(u,\lambda^2)+\Theta(u,i\lambda^2)-\Theta(u,\lambda^3)=2\Theta\left(u,\frac{1}{i\lambda^2}\right),$$

$$\Theta(u,0)-\Theta(u,\lambda^2)-\Theta(u,i\lambda^2)+\Theta(u,\lambda^3)=2\Theta\left(u,\frac{1}{\lambda^2}\right);$$

$$\Theta\left(u,\frac{\lambda^4}{1+\lambda^3}\right)+\Theta\left(u,\frac{\lambda^2}{1+\lambda^3}\right)+\Theta\left(u,\frac{i\lambda^2}{1+\lambda^3}\right)+\Theta\left(u,\frac{\lambda^3}{1+\lambda^3}\right)=2\Theta\left(u,\frac{1+\lambda^3}{\lambda^4}\right),$$

$$\Theta\left(u,\frac{\lambda^4}{1+\lambda^3}\right)+\Theta\left(u,\frac{\lambda^2}{1+\lambda^3}\right)-\Theta\left(u,\frac{i\lambda^2}{1+\lambda^3}\right)-\Theta\left(u,\frac{\lambda^3}{1+\lambda^3}\right)=2\Theta\left(u,\frac{1+\lambda^3}{\lambda^3}\right),$$

$$\Theta\left(u,\frac{\lambda^4}{1+\lambda^3}\right)-\Theta\left(u,\frac{\lambda^2}{1+\lambda^3}\right)+\Theta\left(u,\frac{i\lambda^2}{1+\lambda^3}\right)-\Theta\left(u,\frac{\lambda^3}{1+\lambda^3}\right)=2\Theta\left(u,\frac{1+\lambda^3}{i\lambda^2}\right),$$

$$\Theta\left(u,\frac{\lambda^4}{1+\lambda^3}\right)-\Theta\left(u,\frac{\lambda^2}{1+\lambda^3}\right)-\Theta\left(u,\frac{i\lambda^2}{1+\lambda^3}\right)+\Theta\left(u,\frac{\lambda^3}{1+\lambda^3}\right)=2\Theta\left(u,\frac{1+\lambda^3}{\lambda^2}\right);$$

$$\Theta(u,1)+\Theta\left(u,\frac{1}{-1}\right)+\Theta(u,1+\lambda^3)+\Theta\left(u,\frac{1+\lambda^3}{-1}\right)=2\Theta(u,\infty),$$

$$\Theta(u,1)-\Theta\left(u,\frac{1}{-1}\right)+\Theta(u,1+\lambda^3)-\Theta\left(u,\frac{1+\lambda^3}{-1}\right)=-2\Theta\left(u,\frac{1+\lambda^3}{\lambda^2}\right),$$

$$\Theta(u,1)+\Theta\left(u,\frac{1}{-1}\right)-\Theta(u,1+\lambda^3)-\Theta\left(u,\frac{1+\lambda^3}{-1}\right)=2\Theta\left(u,\frac{1}{i\lambda^2}\right),$$

$$\Theta(u,1)-\Theta\left(u,\frac{1}{-1}\right)-\Theta(u,1+\lambda^3)+\Theta\left(u,\frac{1+\lambda^3}{-1}\right)=2\Theta\left(u,\frac{1+\lambda^3}{\lambda^3}\right);$$

$$\Theta\left(u, \frac{1}{1+\lambda^3}\right)+\Theta\left(u, \frac{1}{-1-\lambda^3}\right)+\Theta\left(u, \frac{1+\lambda^3}{1-\lambda^3}\right)+\Theta\left(u, \frac{1+\lambda^3}{-1+\lambda^3}\right)=2\Theta\left(u, \frac{1}{\lambda^2}\right),$$

$$\Theta\left(u, \frac{1}{1+\lambda^3}\right)-\Theta\left(u, \frac{1}{-1-\lambda^3}\right)+\Theta\left(u, \frac{1+\lambda^3}{1-\lambda^3}\right)-\Theta\left(u, \frac{1+\lambda^3}{-1+\lambda^3}\right)=2\Theta\left(u, \frac{1+\lambda^3}{\lambda^4}\right),$$

$$\Theta\left(u, \frac{1}{1+\lambda^3}\right)+\Theta\left(u, \frac{1}{-1-\lambda^3}\right)-\Theta\left(u, \frac{1+\lambda^3}{1-\lambda^3}\right)-\Theta\left(u, \frac{1+\lambda^3}{-1+\lambda^3}\right)=2\Theta\left(u, \frac{1}{\lambda^3}\right),$$

$$\Theta\left(u, \frac{1}{1+\lambda^3}\right)-\Theta\left(u, \frac{1}{-1-\lambda^3}\right)-\Theta\left(u, \frac{1+\lambda^3}{1-\lambda^3}\right)+\Theta\left(u, \frac{1+\lambda^3}{-1+\lambda^3}\right)=-2\Theta\left(u, \frac{1+\lambda^3}{i\lambda^2}\right)$$

On the other hand, by Proposition 2 and Proposition 3, the coefficients $\psi(\mu, \kappa_i)$ of $\Theta(u, \kappa_i)$ are expressed in terms of $\psi_1(\mu)$ and $\psi_2(\mu)$.

Hence, if we write $\mu = i^\alpha \lambda^\beta \mu_0$ ($\mu_0 = \mu_1$ or μ_2), we get

$$2\psi_1(\lambda^{-1}\mu_1) = \left(\frac{\lambda^2}{\mu_1}\right)\psi_1(\lambda^{-1}\mu_1)+i\left(\frac{\lambda^2}{\mu_1}\right)\psi_2(\lambda^{-1}\mu_1),$$

$$2\psi_1(i\lambda^{-1}\mu_1)=\left(\frac{\lambda^2}{\mu_1}\right)\psi_1(i\lambda^{-1}\mu_1)+i\left(\frac{\lambda^2}{\mu_1}\right)\psi_2(i\lambda^{-1}\mu_1),$$

$$2\psi_1(\lambda^{-1}\mu_2) = -\left(\frac{\lambda^2}{\mu_2}\right)\psi_1(\lambda^{-1}\mu_2)-i\left(\frac{\lambda^2}{\mu_2}\right)\psi_2(\lambda^{-1}\mu_2),$$

$$2\psi_1(i\lambda^{-1}\mu_2)=\left(\frac{\lambda^2}{\mu_2}\right)\psi_1(i\lambda^{-1}\mu_2)+i\left(\frac{\lambda^2}{\mu_2}\right)\psi_2(i\lambda^{-1}\mu_2);$$

$$2\psi_2(\lambda^{-1}\mu_1) = -i\left(\frac{\lambda^2}{\mu_1}\right)\psi_1(\lambda^{-1}\mu_1)+\left(\frac{\lambda^2}{\mu_1}\right)\psi_2(\lambda^{-1}\mu_1),$$

$$2\psi_2(i\lambda^{-1}\mu_1)=-i\left(\frac{\lambda^2}{\mu_1}\right)\psi_1(i\lambda^{-1}\mu_1)+\left(\frac{\lambda^2}{\mu_1}\right)\psi_2(i\lambda^{-1}\mu_1),$$

$$2\psi_2(\lambda^{-1}\mu_2) = i\left(\frac{\lambda^2}{\mu_2}\right)\psi_1(\lambda^{-1}\mu_2)-\left(\frac{\lambda^2}{\mu_2}\right)\psi_2(\lambda^{-1}\mu_2),$$

$$2\psi_2(i\lambda^{-1}\mu_2)=-i\left(\frac{\lambda^2}{\mu_2}\right)\psi_1(i\lambda^{-1}\mu_2)+\left(\frac{\lambda^2}{\mu_2}\right)\psi_2(i\lambda^{-1}\mu_2);$$

$$2\psi_1(\lambda^{-2}\mu_1) = \left(\frac{i\lambda^2}{\mu_1}\right)^{-1}\sqrt{2}\psi_1(\lambda^{-2}\mu_1)-\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}\sqrt{2}\psi_2(\lambda^{-2}\mu_1),$$

$$2\psi_1(\lambda^{-2}\mu_2) = 0;$$

$$2\psi_2(\lambda^{-2}\mu_1) = -\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}\sqrt{2}\psi_1(\lambda^{-2}\mu_1)-\left(\frac{i\lambda^2}{\mu_1}\right)^{-1}\sqrt{2}\psi_2(\lambda^{-2}\mu_1),$$

$$2\psi_2(\lambda^{-2}\mu_2) = 0;$$

$$2\psi_1(i\lambda^{-2}\mu_1)=\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_1(i\mu_1)$$
$$-i\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_2(i\mu_1),$$

$$2\psi_1(i\lambda^{-2}\mu_2)=i\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1}e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_2}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_1(i\mu_2)$$
$$-\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_2}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_2(i\mu_2);$$

$$2\psi_2(i\lambda^{-2}\mu_1)=i\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_1(i\mu_1)$$
$$+\left\{\left(\frac{\lambda^3}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_2(i\mu_1),$$

$$2\psi_2(i\lambda^{-2}\mu_2)=\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1}e^{\frac{\pi i}{8}}-\left(\frac{i\lambda^3}{\mu_2}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_1(i\mu_2)$$
$$+i\left\{\left(\frac{\lambda^3}{\mu_2}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{i\lambda^3}{\mu_2}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{3}{4}}\psi_2(i\mu_2);$$

$$2\psi_1(\mu_0)=0;\quad 2\psi_2(\mu_0)=0;$$

$$2\psi_1(i\mu_1)=\left(\frac{i}{\mu_1}\right)^{-1}2^{-\frac{1}{2}}\psi_1(i\mu_1)+\left(\frac{i}{\mu_1}\right)^{-1}2^{-\frac{1}{2}}\psi_2(i\mu_1)$$
$$+\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{5}{4}}\psi_1(i\lambda^{-2}\mu_1)$$
$$-i\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{5}{4}}\psi_2(i\lambda^{-2}\mu_1),$$

$$2\psi_1(i\mu_2)=i\left(\frac{i}{\mu_2}\right)^{-1}2^{-\frac{1}{2}}\psi_1(i\mu_2)+i\left(\frac{i}{\mu_2}\right)^{-1}2^{-\frac{1}{2}}\psi_2(i\mu_2)$$
$$-i\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_2}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{5}{4}}\psi_1(i\lambda^{-2}\mu_2)$$
$$+\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1}e^{\frac{\pi i}{8}}+\left(\frac{\lambda}{\mu_2}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{5}{4}}\psi_2(i\lambda^{-2}\mu_2);$$

$$2\psi_2(i\mu_1)=\left(\frac{i}{\mu_1}\right)^{-1}2^{-\frac{1}{2}}\psi_1(i\mu_1)-\left(\frac{i}{\mu_1}\right)^{-1}2^{-\frac{1}{2}}\psi_2(i\mu_1)$$
$$-i\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}-\left(\frac{\lambda}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{5}{4}}\psi_1(i\lambda^{-2}\mu_1)$$
$$-\left\{\left(\frac{i\lambda}{\mu_1}\right)^{-1}e^{\frac{\pi i}{8}}-\left(\frac{\lambda}{\mu_1}\right)^{-1}e^{-\frac{\pi i}{8}}\right\}2^{-\frac{5}{4}}\psi_2(i\lambda^{-2}\mu_1),$$

$$2\psi_2(i\lambda_2) = i\left(\frac{i}{\mu_2}\right)^{-1} 2^{-\frac{1}{2}}\psi_1(i\mu_2) - i\left(\frac{i}{\mu_2}\right)^{-1} 2^{-\frac{1}{2}}\psi_2(i\mu_2)$$
$$+\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}} - \left(\frac{\lambda}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-\frac{5}{4}}\psi_1(i\lambda^{-2}\mu_2)$$
$$+i\left\{\left(\frac{i\lambda}{\mu_2}\right)^{-1} e^{\frac{\pi i}{8}} - \left(\frac{\lambda}{\mu_2}\right)^{-1} e^{-\frac{\pi i}{8}}\right\} 2^{-\frac{5}{4}}\psi_2(i\lambda^{-2}\mu_2);$$

$$2\psi_1(i^\alpha\lambda\mu_0) = 0 \ (0 \leqq \alpha \leqq 3); \quad 2\psi_2(i^\alpha\lambda\mu_0) = 0 \ (0 \leqq \alpha \leqq 3).$$

Since $\psi_1(\lambda^4\mu) = \psi_1(\mu)$, $\psi_2(\lambda^4\mu) = \psi_2(\mu)$, $\psi_1(-\mu) = \psi_1(\mu)$ and $\psi_2(-\mu) = -\psi_2(\mu)$, it is enough to consider the cases $-2 \leqq \beta \leqq 1$, $\alpha = 0, 1$.

Now we have

Theorem 1. *Let* $\mu_0 \equiv 1(\lambda^3)$, $\mu_1 \equiv 1(4)$ *and* $\mu_2 \equiv 1 + \lambda^3(4)$. *Then the following hold*:

(1)
$$\psi_1(\lambda^{-1}\mu_1) = i\psi_2(\lambda^{-1}\mu_1) \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_1}\right) = 1\right),$$
$$\psi_1(\lambda^{-1}\mu_1) = \psi_2(\lambda^{-1}\mu_1) = 0 \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_1}\right) = -1\right),$$
$$\psi_1(\lambda^{-1}\mu_2) = \psi_2(\lambda^{-1}\mu_2) = 0 \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_2}\right) = 1\right),$$
$$\psi_1(\lambda^{-1}\mu_2) = i\psi_2(\lambda^{-1}\mu_2) \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_2}\right) = -1\right);$$

(2)
$$\psi_1(i\lambda^{-1}\mu_1) = i\psi_2(i\lambda^{-1}\mu_1) \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_1}\right) = 1\right),$$
$$\psi_1(i\lambda^{-1}\mu_1) = \psi_2(i\lambda^{-1}\mu_1) = 0 \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_1}\right) = -1\right),$$
$$\psi_1(i\lambda^{-1}\mu_2) = i\psi_2(i\lambda^{-1}\mu_2) \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_2}\right) = 1\right),$$
$$\psi_1(i\lambda^{-1}\mu_2) = \psi_2(i\lambda^{-1}\mu_2) = 0 \qquad \left(\text{if } \left(\frac{\lambda^2}{\mu_2}\right) = -1\right);$$

(3)
$$\psi_1(\lambda^{-2}\mu_1) = -(\sqrt{2}+1)\,\psi_2(\lambda^{-2}\mu_1) \qquad \left(\text{if } \left(\frac{i\lambda^2}{\mu_1}\right) = 1\right),$$
$$\psi_1(\lambda^{-2}\mu_1) = (\sqrt{2}-1)\,\psi_2(\lambda^{-2}\mu_1) \qquad \left(\text{if } \left(\frac{i\lambda^2}{\mu_1}\right) = -1\right),$$
$$\psi_1(\lambda^{-2}\mu_2) = \psi_2(\lambda^{-2}\mu_2) = 0;$$

(4) $\psi_1(i\lambda^{-2}\mu_1) = -i\psi_2(i\lambda^{-2}\mu_1)$,
$\psi_1(i\lambda^{-2}\mu_2) = i\psi_2(i\lambda^{-2}\mu_2)$;

(5) $\psi_1(\mu_0) = \psi_2(\mu_0) = 0$;

(6) $\psi_1(i\mu_1) = (1+\sqrt{2})\,\psi_2(i\mu_1)$ $\left(\text{if } \left(\frac{i}{\mu_1}\right)=1\right)$,

$\psi_1(i\mu_1) = (1-\sqrt{2})\,\psi_2(i\mu_1)$ $\left(\text{if } \left(\frac{i}{\mu_1}\right)=-1\right)$,

$\psi_1(i\mu_2) = (1+\sqrt{2})\,\psi_2(i\mu_2)$ $\left(\text{if } \left(\frac{i}{\mu_2}\right)=i\right)$,

$\psi_1(i\mu_2) = (1-\sqrt{2})\,\psi_2(i\mu_2)$ $\left(\text{if } \left(\frac{i}{\mu_2}\right)=-i\right)$;

(7) $\psi_1(i^\alpha\lambda\mu_0) = \psi_2(i^\alpha\lambda\mu_0) = 0 \quad (0 \leqq \alpha \leqq 3)$;

(8) $\psi_1(i\mu_1) = \left(\frac{\lambda}{\mu_1}\right)^{-1}\sqrt{1+\sqrt{2}}\,\psi_1(i\lambda^{-2}\mu_1)$ $\left(\text{if } \left(\frac{i}{\mu_1}\right)=1\right)$,

$\psi_1(i\mu_1) = -i\left(\frac{\lambda}{\mu_1}\right)^{-1}\sqrt{-1+\sqrt{2}}\,\psi_1(i\lambda^{-2}\mu_1)$ $\left(\text{if } \left(\frac{i}{\mu_1}\right)=-1\right)$,

$\psi_1(i\mu_2) = -\left(\frac{\lambda}{\mu_2}\right)^{-1}\frac{1+i}{\sqrt{2}}\sqrt{-1+\sqrt{2}}\,\psi_1(i\lambda^{-2}\mu_2)$ $\left(\text{if } \left(\frac{i}{\mu_2}\right)=i\right)$,

$\psi_1(i\mu_2) = \left(\frac{\lambda}{\mu_2}\right)^{-1}\frac{1-i}{\sqrt{2}}\sqrt{1+\sqrt{2}}\,\psi_1(i\lambda^{-2}\mu_2)$ $\left(\text{if } \left(\frac{i}{\mu_2}\right)=-i\right)$.

Remark. It is interesting that the units in the field of 8-th roots of unity appear in (3), (6), (8) of Theorem 1.

7. Eisenstein series for $\Gamma(\lambda^4 m)$ and their coefficients

Let m be a square-free integer of $\mathfrak{o}$ so that $m \equiv 1 (\lambda^3)$. We consider some special kinds of Eisenstein series for $\Gamma(\lambda^4 m)$.

Let $\kappa_i = \frac{\alpha}{\gamma} \in P(\lambda^4 m)$ be of (A_0, A_0)-type, i.e. $\alpha \equiv 0 (4m)$, and define

(7. 1) $$E(u, s, \gamma \pmod{4m}) = V(\lambda^4 m)\left(\frac{\alpha}{\gamma}\right) E_i(u, s, \Gamma(\lambda^4 m))$$

where $E_i(u, s, \Gamma(\lambda^4 m))$ is the Eisenstein series (with respect to κ_i) defined by (3. 1). Here $E_i(u, s, \gamma \pmod{4m})$ depends on $\gamma \pmod{4m}$, but not on α.

Let $\psi(s, \mu, \gamma \pmod{4m})$ be the coefficient of $E(u, s, \gamma \pmod{4m})$ according to (3. 11). By (3. 4), (3. 5), (3. 9), (3. 10) and lemma 3, we have

$$\psi(s, \mu, \gamma \pmod{4m}) = \sum_{c \equiv -\gamma(4m)} \left(\frac{m}{c}\right) g(c, \mu)\, N(c)^{-s}. \tag{7. 2}$$

Let $\mathfrak{a}$ be an ideal character of $\mathfrak{o}$ defined modulo m, and put:

$$E_1(u, s, \mathfrak{a}) = \sum_{\substack{\gamma \bmod 4m \\ \gamma \equiv 1\,(4)}} \mathfrak{a}(\gamma)\, E(u, s, \gamma \pmod{4m}), \tag{7. 3}$$

$$E_2(u, s, \mathfrak{a}) = \sum_{\substack{\gamma \bmod 4m \\ \gamma \equiv 1+\lambda^3\,(4)}} \mathfrak{a}(\gamma)\, E(u, s, \gamma \pmod{4m}). \tag{7. 4}$$

By (7. 2), we easily get

Proposition 5. *The coefficients* $\psi_1(s, \mu, \mathfrak{a})$ *and* $\psi_2(s, \mu, \mathfrak{a})$ *of* $E_1(u, s, \mathfrak{a})$ *and* $E_2(u, s, \mathfrak{a})$ *are respectively given by*:

$$\psi_1(s, \mu, \mathfrak{a}) = \sum_{\substack{c \equiv 1\,(4) \\ (c, m) = 1}} \mathfrak{a}(c) \left(\frac{m}{c}\right) g(c, \mu)\, N(c)^{-s},$$

$$\psi_2(s, \mu, \mathfrak{a}) = -\sum_{\substack{c \equiv 1+\lambda^3\,(4) \\ (c, m) = 1}} \mathfrak{a}(c) \left(\frac{m}{c}\right) g(c, \mu)\, N(c)^{-s}.$$

Especially,

$$\psi_1(s, \mu, 1_m) = \sum_{\substack{c \equiv 1\,(4) \\ (c, m) = 1}} \left(\frac{m}{c}\right) g(c, \mu)\, N(c)^{-s},$$

$$\psi_2(s, \mu, 1_m) = -\sum_{\substack{c \equiv 1+\lambda^3\,(4) \\ (c, m) = 1}} \left(\frac{m}{c}\right) g(c, \mu)\, N(c)^{-s}$$

where 1_m *is the trivial character modulo* m.

8. Singular values of Eisenstein series for $\Gamma(\lambda^4 m)$

To begin with, we state two lemmas. We may write $m = m_1 m_2$ where $m_1 \equiv m_2 \equiv 1\,(\lambda^3)$, $m_1 \neq 1$, $m_2 \neq 1$.

Lemma 7. *Suppose* $\gamma \equiv 0\,(2)$ *and* $\alpha \equiv 1\,(\lambda^3)$. *Then*,

$$\sum_{c \equiv \gamma(4m)} \sum_{\substack{d \bmod 4mc \\ d \equiv \alpha(4m)}} \left(\frac{c}{d}\right) N(c)^{-s}$$

is given by:

$\gamma \equiv 0(4m),\ \alpha \equiv 1(4)$	$2(2^{4s-4}-1)^{-1} \sum_{m \mid M \mid m^{\infty}} \left(\frac{\alpha}{M}\right) N(M)^{1-s} \sum_{\substack{c \equiv 1\,(\lambda^3) \\ (c,m)=1}} \Phi(c)\, N(c)^{3-4s}$
$\gamma \equiv 0(4m_1),\ \alpha \equiv 1(4)$ $(\gamma, m_2) = 1$	$\sum_{\substack{\pm \lambda^{4b} M c^4 \equiv \gamma(m_2) \\ b \geqq 1,\ m_1 \mid M \mid m_1^{\infty} \\ M \equiv 1\,(\lambda^3),\ c \equiv 1\,(\lambda^3) \\ (c,m)=1}} 2^{4b(1-s)} \left(\frac{\alpha}{M}\right) N(M)^{1-s}\, \Phi(c)\, N(c)^{3-4s}$
$\gamma \equiv 0(4)$ $(\gamma, m) = 1,\ \alpha \equiv 1(4)$	$\sum_{\substack{\pm \lambda^{4b} c^4 \equiv \gamma(m) \\ b \geqq 1,\ c \equiv 1\,(\lambda^3) \\ (c,m)=1}} 2^{4b(1-s)}\, \Phi(c)\, N(c)^{3-4s}$
otherwise	0

Proof. We write $c = i^a \lambda^b M c'$ where $0 \leqq a \leqq 3$, $b \geqq 2$, $M \mid m^{\infty}$, $M \equiv 1(\lambda^3)$, $c' \equiv 1(\lambda^3)$, $(c', m) = 1$. There exist $A_1, A_2, A_3 \in \mathfrak{o}$ such that

$$mMc'A_1 + \lambda^{b+4} c'A_2 + \lambda^{b+4} mMA_3 = 1.$$

Let

$$d = mMc'A_1 d_1 + \lambda^{b+4} c'A_2 d_2 + \lambda^{b+4} mMA_3 d_3$$

where d_1 runs through a set of residues (mod λ^{b+4}) subject to $d_1 \equiv \alpha(4)$, d_2 runs through a set of residues (mod mM) subject to $d_2 \equiv \alpha(m)$ and d_3 runs through a complete set of residues (mod c') prime to c'.

Then,

$$\left(\frac{c}{d}\right) = \left(\frac{i^a \lambda^b}{d}\right)\left(\frac{M}{d}\right)\left(\frac{c'}{d}\right) = \left(\frac{i^a \lambda^b}{d_1}\right)\left(\frac{d_2}{M}\right)\left(\frac{d_3}{c'}\right)\left(\frac{\alpha, Mc'}{\lambda}\right)$$

and so

$$\sum_{\substack{d \bmod 4mc \\ d \equiv \alpha(4m)}} \left(\frac{c}{d}\right) = \left(\frac{\alpha, Mc'}{\lambda}\right) \sum_{\substack{d_1 \bmod \lambda^{b+4} \\ d_1 \equiv \alpha(4)}} \left(\frac{i^a \lambda^b}{d_1}\right) \sum_{\substack{d_2 \bmod mM \\ d_2 \equiv \alpha(m)}} \left(\frac{d_2}{M}\right) \sum_{d_3 \bmod c'} \left(\frac{d_3}{c'}\right)$$

$$= \begin{cases} \left(\frac{\alpha, Mc'}{\lambda}\right) 2^b \left(\frac{\alpha}{M}\right) N(M)\, \Phi(c') & \text{(if } a = 0, 4 \mid b \text{ and } c' \text{ is a biquadrate),} \\ \left(\frac{\alpha, -Mc'}{\lambda}\right) 2^b \left(\frac{\alpha}{M}\right) N(M)\, \Phi(c') & \text{(if } a = 2, 4 \mid b \text{ and } c' \text{ is a biquadrate),} \\ 0 & \text{(if otherwise).} \end{cases}$$

Hence, we write $c = \pm \lambda^{4b} M c_0^4$ where $b \geqq 1$, $c_0 \equiv 1(\lambda^3)$. The condition $c \equiv \gamma(4m)$, respectively, changes to (1) $m \mid M$, (2) $m_1 \mid M$, $M \mid m_1^{\infty}$, $\pm \lambda^{4b} M c_0^4 \equiv \gamma(m_2)$, and (3) $M = 1$, $\pm \lambda^{4b} c_0^4 \equiv \gamma(m)$, according to the cases (1) $\gamma \equiv 0(m)$, (2) $\gamma \equiv 0(m_1)$, $(\gamma, m_2) = 1$, and (3) $(\gamma, m) = 1$. This proves our assertion.

Lemma 8. *Suppose* $\gamma \equiv 1(\lambda^3)$. *Then*

$$\sum_{c \equiv \gamma(4m)} \sum_{\substack{d \bmod 4mc \\ d \equiv \alpha(4m)}} \left(\frac{d}{c}\right) N(c)^{-s} \quad \left(= \sum_{c \equiv -\gamma(4m)} \sum_{\substack{d \bmod 4mc \\ d \equiv \alpha(4m)}} \left(\frac{d}{c}\right) N(c)^{-s}\right)$$

is given by:

$\gamma \equiv 0(m)$	$\sum_{\substack{m \mid M \mid m^\infty \\ M \equiv \gamma(4)}} \left(\frac{\alpha}{M}\right) N(M)^{1-s} \sum_{\substack{c \equiv 1(\lambda^3) \\ (c,m)=1}} \Phi(c)\, N(c)^{3-4s}$
$\gamma \equiv 0(m_1)$ $(\gamma, m_2) = 1$	$\sum_{\substack{Mc^4 \equiv \gamma(m_2) \\ m_1 \mid M \mid m_1^\infty, M \equiv \gamma(4) \\ (c,m)=1}} \left(\frac{\alpha}{M}\right) N(M)^{1-s}\, \Phi(c)\, N(c)^{3-4s}$
$(\gamma, m) = 1$ $\gamma \equiv 1(4)$	$\sum_{\substack{c^4 \equiv \gamma(m) \\ c \equiv 1(\lambda^3), (c,m)=1}} \Phi(c)\, N(c)^{3-4s}$
$(\gamma, m) = 1$ $\gamma \equiv 1 + \lambda^3(4)$	0

Proof. It is sufficient to prove the case $c \equiv \gamma(4m)$. So, we write $c = Mc'$ where $M \mid m^\infty$, $M \equiv 1(\lambda^3)$, $c' \equiv 1(\lambda^3)$, $(c', m) = 1$. Then we have

$$\sum_{\substack{d \bmod 4mc \\ d \equiv \alpha(4m)}} \left(\frac{d}{c}\right) = \left(\frac{\alpha}{M}\right) N(M)\, \Phi(c') \qquad \text{(if } c' \text{ is a biquadrate)},$$

$$= 0 \qquad \text{(if otherwise)}.$$

Hence, we write $c = Mc_0^4$. The condition $c \equiv \gamma(4m)$, respectively, changes to (1) $m \mid M$, $M \equiv \gamma(4)$, (2) $m_1 \mid M$, $M \mid m_1^\infty$, $Mc_0^4 \equiv \gamma(m_2)$, $M \equiv \gamma(4)$, and (3) $M = 1$, $c_0^4 \equiv \gamma(m)$, $\gamma \equiv 1(4)$, according to the cases (1) $\gamma \equiv 0(m)$, (2) $\gamma \equiv 0(m_1)$, $(\gamma, m_2) = 1$, and (3) $(\gamma, m) = 1$.

This proves our assertion.

Let $\psi(s, \gamma \pmod{4m}, \kappa_j)$ be the singular value of $E(u, s, \gamma \pmod{4m})$ at $\kappa_j \in P(\lambda^4 m)$. By (3. 8) and (3. 12),

$$\psi(s, \gamma \pmod{4m}, \kappa_j) = \pi(s-1)^{-1} \left(\frac{\alpha}{\gamma}\right) \sum_{(c,d) \in M_{ij}(\lambda^4 m)} \bar{\chi}_{ij}(c, d)\, N(c)^{-s}$$

where $\kappa_i = \dfrac{\alpha}{\gamma} \in P(\lambda^4 m)$.

If we put $\sigma_i = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, $\sigma_j = \begin{pmatrix} \alpha' & \beta' \\ \gamma' & \delta' \end{pmatrix}$, then

$$M_{ij}(\lambda^4 m) = \{(c, d) : c \equiv -\gamma\alpha'(4m),\ d \equiv -\gamma\beta'(4m)\}.$$

And also $\bar{\chi}_{ij}(c, d)$ is given by [1], [2], [3] in Section 3.

We have to examine seven cases separately, depending on the types of $\kappa_j=\dfrac{\alpha'}{\gamma'}$; the seven cases are

[1] (A_0, A_0)-type,

[2] (A_0, A_1)-type,

[3] (A_0, B)-type, (A_0, C)-type,

[4] $(A_1, *)$-type,

[5] (B, A_0)-type, (C, A_0)-type,

[6] (B, A_1)-type, (C, A_1)-type,

[7] (B, B)-type, (B, C)-type, (C, B)-type, (C, C)-type.

[1] $\kappa_j=\dfrac{\alpha'}{\gamma'}: \alpha' \equiv 0(4m)$.

If $(c, d) \in M_{ij}(\lambda^4 m)$, then $c \equiv 0(4m)$, $d \equiv \gamma\gamma'^{-1}(4m)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{c}{d}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right).$$

Therefore, by lemma 7, we have

$$(8.1)\quad \psi\left(s, \gamma \pmod{4m}, \frac{\alpha'}{\gamma'}\right)$$

$$=\begin{cases} \pi(s-1)^{-1}\left(\dfrac{\alpha'}{\gamma'}\right) 2(2^{4s-4}-1)^{-1} \displaystyle\sum_{m|M|m^{\infty}} \left(\frac{\gamma}{M}\right)\left(\frac{\gamma'}{M}\right)^{-1} N(M)^{1-s} \sum_{\substack{c \equiv 1 (\lambda^3) \\ (c, m)=1}} \Phi(c)\, N(c)^{3-4s} & \\ & \text{(if } \gamma \equiv \gamma'(4)\text{)}, \\ 0 \quad \text{(if } \gamma \not\equiv \gamma'(4)\text{)}. & \end{cases}$$

[2] $\kappa_j=\dfrac{\alpha'}{\gamma'}: \alpha' \equiv 0(4m_1),\ (\alpha', m_2)=1\ (m=m_1 m_2)$.

If $(c, d) \in M_{ij}(\lambda^4 m)$, then $c \equiv 0(4m_1)$, $c \equiv -\gamma\alpha'(m_2)$, $d \equiv \gamma\gamma'^{-1}(4m_1)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{c}{d}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right).$$

Therefore, by lemma 7, we have

$$(8.2)\quad \psi\left(s, \gamma \pmod{4m}, \frac{\alpha'}{\gamma'}\right)$$

$$=\begin{cases} \pi(s-1)^{-1}\left(\dfrac{\alpha'}{\gamma'}\right) \displaystyle\sum_{\substack{\pm \lambda^{4b} M c^4 \equiv -\gamma\alpha'(m_2) \\ b \geqq 1, m_1 | M | m_1^{\infty}, c \equiv 1 (\lambda^3) \\ (c, m)=1}} 2^{4b(1-s)} \left(\frac{\gamma}{M}\right)\left(\frac{\gamma'}{M}\right)^{-1} N(M)^{1-s}\, \Phi(c)\, N(c)^{3-4s} & \\ & \text{(if } \gamma \equiv \gamma'(4)\text{)}, \\ 0 \quad \text{(if } \gamma \not\equiv \gamma'(4)\text{)}. & \end{cases}$$

[3] $\kappa_j=\dfrac{\alpha'}{\gamma'}: \alpha'\equiv 0(4),\ (\alpha', m)=1.$

If $(c, d)\in M_{ij}(\lambda^4 m)$, then $c\equiv 0(4)$, $c\equiv -\gamma\alpha'(m)$, $d\equiv\gamma\gamma'^{-1}(4)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{c}{d}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right).$$

Therefore, by lemma 7, we have

$$(8.3)\quad \psi\left(s, \gamma\ (\mathrm{mod}\,4m), \frac{\alpha'}{\gamma'}\right)$$

$$=\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\sum_{\substack{\pm\lambda^{4b}c^4\equiv -\gamma\alpha'(m)\\ b\geqq 1,\ c\equiv 1\,(\lambda^3)\\ (c,m)=1}} 2^{4b(1-s)}\,\Phi(c)\,N(c)^{3-4s}\qquad (\text{if } \gamma\equiv\gamma'(4)),$$

$$=0\qquad (\text{if } \gamma\equiv\gamma'(4)).$$

[4] $\kappa_j=\dfrac{\alpha'}{\gamma'}: \alpha'\equiv 0(2),\ \alpha'\not\equiv 0(4).$

If $(c, d)\in M_{ij}(\lambda^4 m)$, then $c\equiv 0(2)$, $c\not\equiv 0(4)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\left(\frac{c}{d}\right)\left(\frac{\gamma', -\gamma}{\lambda}\right).$$

Therefore, by lemma 7, we have

$$(8.4)\quad \psi\left(s, \gamma\ (\mathrm{mod}\,4m), \frac{\alpha'}{\gamma'}\right)=0.$$

[5] $\kappa_j=\dfrac{\alpha'}{\gamma'}: \alpha'\equiv 1(\lambda^3),\ \alpha'\equiv 0(m).$

If $(c, d)\in M_{ij}(\lambda^4 m)$, then $c\equiv -\gamma\alpha'(4)$, $c\equiv 0(m)$, $d\equiv\gamma\gamma'^{-1}(m)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{d}{c}\right)\left(\frac{\alpha', \gamma}{\lambda}\right).$$

Therefore, by lemma 8, we have

$$(8.5)\quad \psi\left(s, \gamma\ (\mathrm{mod}\,4m), \frac{\alpha'}{\gamma'}\right)$$

$$=\pi(s-1)^{-1}\left(\frac{\alpha', \gamma}{\lambda}\right)\left(\frac{\gamma'}{\alpha'}\right)\sum_{\substack{m|M|m^\infty\\ M\equiv\gamma\alpha'(4)}}\left(\frac{\gamma}{M}\right)\left(\frac{\gamma'}{M}\right)^{-1}N(M)^{1-s}\sum_{\substack{c\equiv 1\,(\lambda^3)\\ (c,m)=1}}\Phi(c)\,N(c)^{3-4s}.$$

[6] $\kappa_j=\dfrac{\alpha'}{\gamma'}:\alpha'\equiv 1(\lambda^3),\ \alpha'\equiv 0(m_1),\ (\alpha', m_2)=1.$

If $(c, d)\in M_{ij}(\lambda^4 m)$, then $c\equiv -\gamma\alpha'(4m_2)$, $c\equiv 0(m_1)$, $d\equiv\gamma\gamma'^{-1}(m_1)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{d}{c}\right)\left(\frac{\alpha', \gamma}{\lambda}\right).$$

Therefore, by lemma 8, we have

$$(8.6)\quad \psi\left(s, \gamma\ (\mathrm{mod}\ 4m), \frac{\alpha'}{\gamma'}\right)$$
$$=\pi(s-1)^{-1}\left(\frac{\alpha', \gamma}{\lambda}\right)\left(\frac{\gamma'}{\alpha'}\right)\sum_{\substack{Mc^4\equiv\gamma\alpha'(m_2)\\ M\equiv\gamma\alpha'(4), m_1|M|m_1^\infty\\ c\equiv 1(\lambda^3),(c,m)=1}}\left(\frac{\gamma}{M}\right)\left(\frac{\gamma'}{M}\right)^{-1}N(M)^{1-s}\Phi(c)\,N(c)^{3-4s}.$$

[7] $\kappa_j=\dfrac{\alpha'}{\gamma'}:\alpha'\equiv 1(\lambda^3),\ (\alpha', m)=1.$

If $(c, d)\in M_{ij}(\lambda^4 m)$, then $c\equiv -\gamma\alpha'(4m)$ and

$$\bar{\chi}_{ij}(c, d)=\left(\frac{\alpha}{\gamma}\right)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\left(\frac{d}{c}\right)\left(\frac{\alpha', \gamma}{\lambda}\right).$$

Therefore, by lemma 8, we have

$$(8.7)\quad \psi\left(s, \gamma\ (\mathrm{mod}\ 4m), \frac{\alpha'}{\gamma'}\right)$$
$$=\pi(s-1)^{-1}\left(\frac{\alpha', \gamma}{\lambda}\right)\left(\frac{\gamma'}{\alpha'}\right)\sum_{\substack{c^4\equiv\gamma\alpha'(m)\\ c\equiv 1(\lambda^3),(c,m)=1}}\Phi(c)\,N(c)^{3-4s}\qquad (\text{if } \gamma\alpha'\equiv 1(4)),$$
$$=0\qquad (\text{if } \gamma\alpha'\equiv 1+\lambda^3(4)).$$

Now recall $E_1(u, s, \mathfrak{a})$ and $E_2(u, s, \mathfrak{a})$ defined by (7.3) and (7.4). Let $\psi_1(s, \mathfrak{a}, \kappa_j)$ and $\psi_2(s, \mathfrak{a}, \kappa_j)$ be the singular values of $E_1(u, s, \mathfrak{a})$ and $E_2(u, s, \mathfrak{a})$ at $\kappa_j\in P(\lambda^4 m)$, respectively; those are given by

$$\psi_1(s, \mathfrak{a}, \kappa_j)=\sum_{\substack{\gamma\ \mathrm{mod}\ 4m\\ \gamma\equiv 1(4),(\gamma,m)=1}}\mathfrak{a}(\gamma)\,\psi\left(s, \gamma\ (\mathrm{mod}\ 4m), \kappa_j\right),$$

$$\psi_2(s, \mathfrak{a}, \kappa_j)=\sum_{\substack{\gamma\ \mathrm{mod}\ 4m\\ \gamma\equiv 1+\lambda^3(4),(\gamma,m)=1}}\mathfrak{a}(\gamma)\,\psi\left(s, \gamma\ (\mathrm{mod}\ 4m), \kappa_j\right).$$

Define

$$\xi(s, \mathfrak{a})=\sum_{\substack{c\equiv 1(\lambda^3)\\ (c,m)=1}}\mathfrak{a}(c)\,N(c)^{-s}.$$

We note that

$$\sum_{\substack{\gamma \bmod m \\ (\gamma, m)=1}} \sum_{m|M|m^{\infty}} \left(\frac{\gamma}{M}\right)\left(\frac{\gamma'}{M}\right)^{-1} N(M)^{1-s} = \Phi(m) \prod_{p|m} \left(N(p)^{4s-4}-1\right)^{-1}$$

where p denote a prime number of $\mathfrak{o}$.

Hence, by (8. 1), (8. 2),..., (8. 7), we have

Proposition 6. (1) $\psi_1\left(s, 1_m, \frac{\alpha'}{\gamma'}\right)$ *is given by the following*:

if $\frac{\alpha'}{\gamma'}$ *is of* (A_0, A_0)*-type and* $\gamma' \equiv 1(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right) 2(2^{4s-4}-1)^{-1}\, \Phi(m) \prod_{p|m} \left(N(p)^{4s-4}-1\right)^{-1} \xi(4s-4, 1_m)\, \xi(4s-3, 1_m)^{-1};$$

if $\frac{\alpha'}{\gamma'}$ *is of* (A_0, A_1)*-type* $\left(\alpha' \equiv 0(m_1),\ (\alpha', m_2)=1\right)$ *and* $\gamma' \equiv 1(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right) 2(2^{4s-4}-1)^{-1}\, \Phi(m_1) \prod_{p|m_1} \left(N(p)^{4s-4}-1\right)^{-1} \xi(4s-4, 1_m)\, \xi(4s-3, 1_m)^{-1};$$

if $\frac{\alpha'}{\gamma'}$ *is of* (A_0, B)*-type or of* (A_0, C)*-type, and* $\gamma' \equiv 1(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right) 2(2^{4s-4}-1)^{-1}\, \xi(4s-4, 1_m)\, \xi(4s-3, 1_m)^{-1};$$

if $\frac{\alpha'}{\gamma'}$ *is of* $(*, A_0)$*-type and* $\alpha' \equiv 1(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right) \Phi(m) \prod_{p|m} \left(N(p)^{4s-4}-1\right)^{-1} \xi(4s-4, 1_m)\, \xi(4s-3, 1_m)^{-1};$$

if $\frac{\alpha'}{\gamma'}$ *is of* $(*, A_1)$*-type* $\left(\alpha' \equiv 0(m_1),\ (\alpha', m_2)=1\right)$ *and* $\alpha' \equiv 1(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right) \Phi(m_1) \prod_{p|m_1} \left(N(p)^{4s-4}-1\right)^{-1} \xi(4s-4, 1_m)\, \xi(4s-3, 1_m)^{-1};$$

if $\frac{\alpha'}{\gamma'}$ *is of* $(*, B)$*-type or* $(*, C)$*-type, and* $\alpha' \equiv 1(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right) \xi(4s-4, 1_m)\, \xi(4s-3, 1_m)^{-1};$$

if otherwise, then (it is given by) zero.

(2) $\psi_2\left(s, 1_m, \dfrac{\alpha'}{\gamma'}\right)$ *is given by the following*:

if $\dfrac{\alpha'}{\gamma'}$ *is of* (A_0, A_0)*-type and* $\gamma' \equiv 1+\lambda^3(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)2(2^{4s-4}-1)^{-1}\,\Phi(m)\prod_{p|m}\left(N(p)^{4s-4}-1\right)^{-1}\,\xi(4s-4, 1_m)\,\xi(4s-3, 1_m)^{-1};$$

if $\dfrac{\alpha'}{\gamma'}$ *is of* (A_0, A_1)*-type* $\left(\alpha' \equiv 0(m_1),\ (\alpha', m_2)=1\right)$ *and* $\gamma' \equiv 1+\lambda^3(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)2(2^{4s-4}-1)^{-1}\,\Phi(m_1)\prod_{p|m_1}\left(N(p)^{4s-4}-1\right)^{-1}\,\xi(4s-4, 1_m)\,\xi(4s-3, 1_m)^{-1};$$

if $\dfrac{\alpha'}{\gamma'}$ *is of* (A_0, B)*-type or of* (A_0, C)*-type, and* $\gamma' \equiv 1+\lambda^3(4)$, *then*

$$\pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)2(2^{4s-4}-1)^{-1}\,\xi(4s-4, 1_m)\,\xi(4s-3, 1_m)^{-1};$$

if $\dfrac{\alpha'}{\gamma'}$ *is of* $(*, A_0)$*-type and* $\alpha' \equiv 1+\lambda^3(4)$, *then*

$$-\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\Phi(m)\prod_{p|m}\left(N(p)^{4s-4}-1\right)^{-1}\,\xi(4s-4, 1_m)\,\xi(4s-3, 1_m)^{-1};$$

if $\dfrac{\alpha'}{\gamma'}$ *is of* $(*, A_1)$*-type* $\left(\alpha' \equiv 0(m_1),\ (\alpha', m_2)=1\right)$ *and* $\alpha' \equiv 1+\lambda^3(4)$, *then*

$$-\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\Phi(m_1)\prod_{p|m_1}\left(N(p)^{4s-4}-1\right)^{-1}\,\xi(4s-4, 1_m)\,\xi(4s-3, 1_m)^{-1};$$

if $\dfrac{\alpha'}{\gamma'}$ *is of* $(*, B)$*-type or* $(*, C)$*-type, and* $\alpha' \equiv 1+\lambda^3(4)$, *then*

$$-\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\xi(4s-4, 1_m)\,\xi(4s-3, 1_m)^{-1};$$

if otherwise, then (it is given by) zero.

If $\mathfrak{a}^4 \neq 1_m$, then

$$\sum_{\gamma \bmod m}\mathfrak{a}(\gamma)\left(\frac{\gamma}{M}\right)=0$$

for any $M \in \mathfrak{o}$ such that $m|M$, $M|m^\infty$.

Therefore, by (8. 1), (8. 2),..., (8. 7), we have

Proposition 7. *Suppose* $\mathfrak{a}^4 \neq 1_m$.

(1) $\psi_1\left(s, \mathfrak{a}, \frac{\alpha'}{\gamma'}\right)$

$$= \begin{cases} \pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\bar{\mathfrak{a}}(\alpha')\left(1+\bar{\mathfrak{a}}(-1)\right)\left(\bar{\mathfrak{a}}^4(\lambda)2^{4s-4}-1\right)^{-1}\xi(4s-4, \mathfrak{a}^4)\,\xi(4s-3, \mathfrak{a}^4)^{-1} \\ \qquad \left(\text{if } \frac{\alpha'}{\gamma'} \text{ is of } (A_0, B)\text{-type or } (A_0, C)\text{-type, and } \gamma' \equiv 1\,(4)\right), \\ \pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\bar{\mathfrak{a}}(\alpha')\,\xi(4s-4, \mathfrak{a}^4)\,\xi(4s-3, \mathfrak{a}^4)^{-1} \\ \qquad \left(\text{if } \frac{\alpha'}{\gamma'} \text{ is of } (*, B)\text{-type or } (*, C)\text{-type, and } \alpha' \equiv 1\,(4)\right), \\ 0 \quad (\text{if otherwise}). \end{cases}$$

(2) $\psi_2\left(s, \mathfrak{a}, \frac{\alpha'}{\gamma'}\right)$

$$= \begin{cases} \pi(s-1)^{-1}\left(\frac{\alpha'}{\gamma'}\right)\bar{\mathfrak{a}}(\alpha')\left(1+\bar{\mathfrak{a}}(-1)\right)\left(\bar{\mathfrak{a}}^4(\lambda)2^{4s-4}-1\right)^{-1}\xi(4s-4, \mathfrak{a}^4)\,\xi(4s-3, \mathfrak{a}^4)^{-1} \\ \qquad \left(\text{if } \frac{\alpha'}{\gamma'} \text{ is } (A_0, B)\text{-type or } (A_0, C)\text{-type, and } \gamma' \equiv 1+\lambda^3\,(4)\right), \\ -\pi(s-1)^{-1}\left(\frac{\gamma'}{\alpha'}\right)\bar{\mathfrak{a}}(\alpha')\,\xi(4s-4, \mathfrak{a}^4)\,\xi(4s-3, \mathfrak{a}^4)^{-1} \\ \qquad \left(\text{if } \frac{\alpha'}{\gamma'} \text{ is } (*, B)\text{-type or } (*, C)\text{-type, and } \alpha' \equiv 1+\lambda^3\,(4)\right), \\ 0 \quad (\text{if otherwise}). \end{cases}$$

9. Residues at $s = \frac{5}{4}$ (II)

For an ideal character $\mathfrak{a}$ defined modulo m, we defined $E_1(u, s, \mathfrak{a})$ and $E_2(u, s, \mathfrak{a})$ by (7. 3) and (7. 4), respectively.

Now we put

$$\Theta_1(u, \mathfrak{a}) = \operatorname*{Res}_{s=\frac{5}{4}} E_1(u, s, \mathfrak{a}),$$

$$\Theta_2(u, \mathfrak{a}) = \operatorname*{Res}_{s=\frac{5}{4}} E_2(u, s, \mathfrak{a}).$$

Proposition 8. *Suppose* $\mathfrak{a}^4 \neq 1_m$. *Then,*

$$\Theta_1(u, \mathfrak{a}) = \Theta_2(u, \mathfrak{a}) = 0.$$

Proof. The singular values $\psi_1(\mathfrak{a}, \kappa_j)$ and $\psi_2(\mathfrak{a}, \kappa_j)$ $(\kappa_j \in P(\lambda^4 m))$ of $\Theta_1(u, \mathfrak{a})$ and $\Theta_2(u, \mathfrak{a})$ are derived from Proposition 7. Since $\mathfrak{a}^4 \neq 1_m$, the residues of

$$\xi(4s-4),\ \xi(4s-4, \mathfrak{a}^4)\ \xi(4s-3, \mathfrak{a}^4)^{-1}$$

and $(\bar{\mathfrak{a}}^4(\lambda) 2^{4s-4}-1)^{-1}\ \xi(4s-4, \mathfrak{a}^4)\ \xi(4s-3, \mathfrak{a}^4)^{-1}$ at $s=\frac{5}{4}$ are equal to zero. Hence, the singular values $\psi_1(\mathfrak{a}, \kappa_j)$ and $\psi_2(\mathfrak{a}, \kappa_j)$ vanish at all $\kappa_j \in P(\lambda^4 m)$. This proves our proposition.

By Proposition 5 and Proposition 8, we have

Theorem 2. *Suppose* $\mathfrak{a}^4 \neq 1_m$, *then the Dirichlet series*

$$\sum_{\substack{c \equiv 1\,(4) \\ (c, m)=1}} \mathfrak{a}(c)\, g(c, \mu)\, N(c)^{-s} \qquad (\mu \in \mathfrak{o}^*),$$

$$\sum_{\substack{c \equiv 1+\lambda^3\,(4) \\ (c, m)=1}} \mathfrak{a}(c)\, g(c, \mu)\, N(c)^{-s} \qquad (\mu \in \mathfrak{o}^*)$$

are holomorphic in the region $\mathrm{Re}(s)>1$.

Next, recall the theta series $\Theta(u, 0)$ and $\Theta\left(u, \frac{\lambda^4}{1+\lambda^3}\right)$ for $\Gamma(\lambda^4)$ studied in Section 6, and consider them as theta series for $\Gamma(\lambda^4 m)$.

Proposition 9. *We have*

$$\Theta(u, 0) = \prod_{p|m} \left(1+N(p)^{-1}\right) \Theta_1(u, 1_m),$$

$$\Theta\left(u, \frac{\lambda^4}{1+\lambda^3}\right) = \prod_{p|m} \left(1+N(p)^{-1}\right) \Theta_2(u, 1_m).$$

Proof. It is sufficient to see the singular values of both sides. The singular values $\psi(0, \kappa_j)$, $\psi\left(\frac{\lambda^4}{1+\lambda^3}, \kappa_j\right)$ $(\kappa_j \in P(\lambda^4 m))$ of $\Theta(u, 0)$, $\Theta\left(u, \frac{\lambda^4}{1+\lambda^3}\right)$ are derived from Proposition 4. The singular values $\psi_1(1_m, \kappa_j)$, $\psi_2(1_m, \kappa_j)$ $(\kappa_j \in P(\lambda^4 m))$ of $\Theta_1(u, 1_m)$, $\Theta_2(u, 1_m)$ are derived from Proposition 6.

We note that the value of $\Phi(m) \prod_{p|m} (N(p)^{4s-4}-1)^{-1}$ at $s=\frac{5}{4}$ is 1, and that

$$\lim_{s \to \frac{5}{4}} \frac{\xi(4s-4)}{\xi(4s-3)} \frac{\xi(4s-3, 1_m)}{\xi(4s-4, 1_m)} = \prod_{p|m} \left(1+N(p)^{-1}\right).$$

This proves our assertion.

By (3. 15), (3. 16), Proposition 2, Proposition 5 and Proposition 9, we get

Proposition 10. *If* $(\mu, m)=1$, *then*

$$\operatorname*{Res}_{s=\frac{5}{4}} \psi_1(s, \mu, 1_m) = \operatorname*{Res}_{s=\frac{5}{4}} \psi_2(s, \mu, 1_m) = 0.$$

In other words, if $(\mu, m)=1$, *then the Dirichlet series*

$$\sum_{\substack{c\equiv 1\,(4)\\(c,m)=1}} \left(\frac{m}{c}\right) g(c,\mu)\, N(c)^{-s}, \qquad \sum_{\substack{c\equiv 1+\lambda^3\,(4)\\(c,m)=1}} \left(\frac{m}{c}\right) g(c,\mu)\, N(c)^{-s}$$

are holomorphic in the region $\mathrm{Re}(s)>1$.

Proposition 11. *The following relations hold*:

$$\psi_1(\mu)=\prod_{p|m}\left(1+N(p)^{-1}\right)\operatorname*{Res}_{s=\frac{5}{4}}\psi_1(s, m\mu, 1_m),$$

$$\psi_2(\mu)=\prod_{p|m}\left(1+N(p)^{-1}\right)\operatorname*{Res}_{s=\frac{5}{4}}\psi_2(s, m\mu, 1_m).$$

It is easily seen that

$$\psi_1(s, m^5\mu, 1_m)=\psi_1(s, m\mu, 1_m),$$
$$\psi_2(s, m^5\mu, 1_m)=\psi_2(s, m\mu, 1_m).$$

Hence, by Proposition 11, we have

Theorem 3. $\psi_1(m^4\mu)=\psi_1(\mu)$; $\psi_2(m^4\mu)=\psi_2(\mu)$.

By Proposition 10 and Proposition 11, we get

Theorem 4. *If* $(\mu, m)=1$, *then*

$$\psi_1(m^3\mu)=\psi_2(m^3\mu)=0.$$

Making use of Proposition 11, we shall get further information about $\psi_1(\mu)$ and $\psi_2(\mu)$.

Rearranging the right hand sides of (1. 1), we have

$$\psi_1(s,\mu)=\sum_{\substack{M\equiv 1\,(\lambda^3)\\M|m^\infty}} N(M)^{-s} \sum_{\substack{(c,m)=1\\Mc\equiv 1\,(4)}} g(Mc,\mu)\, N(c)^{-s}, \tag{9. 1}$$

$$\psi_2(s,\mu)=-\sum_{\substack{M\equiv 1\,(\lambda^3)\\M|m^\infty}} N(M)^{-s} \sum_{\substack{(c,m)=1\\Mc\equiv 1+\lambda^3\,(4)}} g(Mc,\mu)\, N(c)^{-s}. \tag{9. 2}$$

First, we assume $(\mu, m)=1$. Since

$$g(Mc,\mu)=\left(\frac{M,c}{\lambda}\right)\left(\frac{M}{c}\right)^2 g(M,\mu)\, g(c,\mu) \qquad (\text{if } M|m,\ M|m^\infty),$$
$$=0 \qquad (\text{if } M|m,\ M|m^\infty),$$

it follows from (9. 1) and (9. 2) that

$$\psi_1(s,\mu)=\sum_{\substack{m_1|m\\m_1\equiv 1\,(4)}} g(m_1,\mu)\, N(m_1)^{-s}\,\psi_1(s, mm_1^2\mu, 1_m)$$
$$+\sum_{\substack{m_1|m\\m_1\equiv 1+\lambda^3\,(4)}} g(m_1,\mu)\, N(m_1)^{-s}\,\psi_2(s, mm_1^2\mu, 1_m),$$

$$\psi_2(s,\mu)=\sum_{\substack{m_1\mid m\\ m_1\equiv 1\,(4)}} g(m_1,\mu)\,N(m_1)^{-s}\,\psi_2(s,mm_1^2\mu,1_m)$$
$$-\sum_{\substack{m_1\mid m\\ m_1\equiv 1+\lambda^3\,(4)}} g(m_1,\mu)\,N(m_1)^{-s}\,\psi_1(s,mm_1^2\mu,1_m).$$

On taking residues and by Proposition 11,

$$\prod_{p\mid m}\left(1+N(p)^{-1}\right)\psi_1(\mu)=\sum_{\substack{m_1\mid m\\ m_1\equiv 1\,(4)}} g(m_1,\mu)\,N(m_1)^{-\frac{5}{4}}\,\psi_1(m_1^2\mu)$$
$$+\sum_{\substack{m_1\mid m\\ m_1\equiv 1+\lambda^3\,(4)}} g(m_1,\mu)\,N(m_1)^{-\frac{5}{4}}\,\psi_2(m_1^2\mu),$$

$$\prod_{p\mid m}\left(1+N(p)^{-1}\right)\psi_2(\mu)=\sum_{\substack{m_1\mid m\\ m_1\equiv 1\,(4)}} g(m_1,\mu)\,N(m_1)^{-\frac{5}{4}}\,\psi_2(m_1^2\mu)$$
$$-\sum_{\substack{m_1\mid m\\ m_1\equiv 1+\lambda^3\,(4)}} g(m_1,\mu)\,N(m_1)^{-\frac{5}{4}}\,\psi_1(m_1^2\mu).$$

By induction on the number of primes in m, we have

Theorem 5. (1) *If* $(\mu,m)=1$, $m\equiv 1\,(4)$, *then*

$$\psi_1(m^2\mu)=\bar{g}(m,\mu)\,N(m)^{-\frac{3}{4}}\,\psi_1(\mu),$$
$$\psi_2(m^2\mu)=\bar{g}(m,\mu)\,N(m)^{-\frac{3}{4}}\,\psi_2(\mu).$$

(2) *If* $(\mu,m)=1$, $m\equiv 1+\lambda^3\,(4)$, *then*

$$\psi_1(m^2\mu)=-\bar{g}(m,\mu)\,N(m)^{-\frac{3}{4}}\,\psi_2(\mu),$$
$$\psi_2(m^2\mu)=\bar{g}(m,\mu)\,N(m)^{-\frac{3}{4}}\,\psi_1(\mu).$$

Next, we put

$$g_2(m,\mu)=\sum_{\delta \bmod m}\left(\frac{\delta}{m}\right)^2 e\left(\frac{\delta\mu}{m}\right).$$

Suppose $(\mu,m)=1$. Then, since

$$g(Mc,m\mu)=g_2\left(m_1,\frac{m\mu}{m_1}\right)N(m_1)\,g(c,m\mu)\qquad(\text{if } M=m_1^2,\ m_1\mid m),$$
$$=0\qquad(\text{if otherwise}),$$

it follows from (9. 1) and (9. 2) that

$$\psi_1(s,m\mu)=\sum_{\substack{m_1\mid m\\ m_1\equiv 1\,(\lambda^3)}} g_2\left(m_1,\frac{m\mu}{m_1}\right)N(m_1)^{1-2s}\,\psi_1(s,m\mu,1_m),$$

$$\psi_2(s,m\mu)=\sum_{\substack{m_1\mid m\\ m_1\equiv 1\,(\lambda^3)}} g_2\left(m_1,\frac{m\mu}{m_1}\right)N(m_1)^{1-2s}\,\psi_2(s,m\mu,1_m).$$

On taking residues and by Proposition 11,

$$\prod_{p|m}(1+N(p)^{-1})\,\psi_1(m\mu)=\left(\sum_{m_1|m} g_2\left(m_1,\frac{m\mu}{m_1}\right)N(m_1)^{-\frac{3}{2}}\right)\psi_1(m\mu),$$

$$\prod_{p|m}(1+N(p)^{-1})\,\psi_2(m\mu)=\left(\sum_{m_1|m} g_2\left(m_1,\frac{m\mu}{m_1}\right)N(m_1)^{-\frac{3}{2}}\right)\psi_2(m\mu).$$

Therefore, we have

Theorem 6. *Suppose* $(\mu, m)=1$. *Then,*

$$\psi_1(m\mu)=\psi_2(m\mu)=0$$

unless $g_2\left(m_1,\frac{m\mu}{m_1}\right)=N(m_1)^{\frac{1}{2}}$ *for all* m_1 *such that* $m_1|m$, $m_1\equiv 1(\lambda^3)$.

10. Some remarks

Our final aim is to evaluate $\psi_1(\mu)$ and $\psi_2(\mu)$ for all $\mu\in\mathfrak{o}^*$ $(\mu\neq 0)$, but we have not yet attained it completely.

By Theorem 1 and Theorem 4, $\psi_1(\mu)$ and $\psi_2(\mu)$ vanish unless one of the following holds:

(1) $\mu=\pm i\lambda^{-2+4\beta}mc^2d^4$,

(2) $\mu=\pm\lambda^{-2+4\beta}mc^2d^4$, $m\equiv 1(4)$,

(3) $\mu=\pm i\lambda^{-1+4\beta}mc^2d^4$, $\left(\frac{\lambda^2}{m}\right)=1$,

(4) $\mu=\pm\lambda^{-1+4\beta}mc^2d^4$, $m\equiv 1(4)$, $\left(\frac{\lambda^2}{m}\right)=1$,

(4′) $\mu=\pm\lambda^{-1+4\beta}mc^2d^4$, $m\equiv 1+\lambda^3(4)$, $\left(\frac{\lambda^2}{m}\right)=-1$,

(5) $\mu=\pm i\lambda^{4\beta}mc^2d^4$,

where m, c are prime to each other, $\equiv 1(\lambda^3)$ and square-free, $d\equiv 1(\lambda^3)$ and $\beta\geqq 0$.

Furthermore, by Theorem 1, Theorem 3 and Theorem 5, our problem in each case is reduced to evaluating

(1) $\psi_1(i\lambda^{-2}m)$,

(2) $\psi_1(\lambda^{-2}m)$, $m\equiv 1(4)$,

(3) $\psi_1(i\lambda^{-1}m)$, $\left(\frac{\lambda^2}{m}\right)=1$,

(4) $\psi_1(\lambda^{-1}m)$, $m\equiv 1(4)$, $\left(\frac{\lambda^2}{m}\right)=1$,

(4′) $\psi_1(\lambda^{-1}m)$, $m \equiv 1+\lambda^3 (4)$, $\left(\frac{\lambda^2}{m}\right) = -1$,

(5) $\psi_1(i\lambda^{-2}m)$,

respectively.

Meanwhile, by Theorem 6,

$$\psi_1(i^\alpha \lambda^\beta m) = 0 \qquad (\alpha = 0, 1; \quad \beta = -2, -1)$$

unless m satisfies the condition

$$(*) \qquad g_2\left(m_1, \frac{i^\alpha \lambda^\beta m}{m_1}\right) = N(m_1)^{\frac{1}{2}} \quad \text{for all } m_1 \text{ such that } \quad m_1 | m,\ m_1 \equiv 1 (\lambda^3).$$

For instance, in case of (1), we question whether $\psi_1(i\lambda^{-2}m) \neq 0$ for all m satisfying the condition (*). We can't say anything about this question, but it is not difficult to prove the fact that: for any pair (m_1, m') of square-free integers prime to each other, there exist infinitely many integers m such that $m_1 | m$, $(m, m') = 1$, $\psi_1(i\lambda^{-2}m) \neq 0$.

Similar facts hold for other cases.

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Ein globaler Starrheitssatz für Mumfordkurven

Von *Werner Lütkebohmert* in Münster

J. Tate hat gezeigt, daß eine elliptische Kurve E über einem p-adischen Körper k genau dann eine analytische Uniformisierung der Form $k^\times/\langle q\rangle$ mit $0<|q|<1$ besitzt, wenn die j-Invariante $j(E)$ von E nicht ganzzahlig in k ist. Betrachtet man analytische Familien $X\to S$ von solchen elliptischen Kurven über zusammenhängenden, affinalgebraischen Kurven S, so definieren die j-Invarianten der Fasern von $X\to S$ eine beschränkte holomorphe Funktion $j^{-1}\colon S\to k$. Eine elementare Überlegung zeigt nun, daß j^{-1} konstant sein muß. Folglich ist $X\to S$ eine konstante Familie, d.h. die Fasern von $X\to S$ sind zueinander isomorph.

Die Verallgemeinerung der Tateschen Kurven auf höheres Geschlecht beschreibt D. Mumford in [16]. Dort gibt er eine Charakterisierung derjenigen nichtsingulären, kompletten, algebraischen Kurven über k, die eine Schottky-Uniformisierung Ω_Γ/Γ besitzen; dabei ist $\Gamma\subset \mathrm{PGL}(2,k)$ eine endlich erzeugte, freie, diskontinuierliche Untergruppe und $\Omega_\Gamma\subset \mathbb{P}_1(k)$ die Menge der gewöhnlichen Punkte für Γ. Solche Kurven werden in [8] und [15] Mumfordkurven genannt.

Das Hauptergebnis dieser Arbeit ist der Satz, daß algebraische Familien $X\to S$ von Mumfordkurven über zusammenhängenden, affinalgebraischen Kurven konstant sind. Unsere Beweismethode liefert das Ergebnis jedoch nur für den Fall $g+1<p$, wo g das Geschlecht der Familie und p die Charakteristik des Bewertungsrestklassenkörpers ist.

Herrn G. Faltings möchte ich für zahlreiche informative Diskussionen und sein Interesse am Entstehen dieser Arbeit recht herzlich danken.

1. Formelle Standardmodelle für algebraische Kurven

1.0 In dieser Arbeit sei k stets ein nichtarchimedisch, nichttrivial, vollständig bewerteter Körper. Der Einfachheit halber sei k auch algebraisch abgeschlossen. Ein analytischer oder holomorpher Raum ist stets ein rigid-analytischer Raum im Sinne von Kiehl ([10], [11]), insbesondere sind so vertraute Räume wie der projektive Raum $\mathbb{P}_n$ immer mit der gewöhnlichen holomorphen Struktur versehen.

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1.1 Die Gruppe $\mathrm{PGL}(2, k)$ operiere in üblicher Weise auf $\mathbb{P} = \mathbb{P}_1(k)$. Eine Untergruppe $\Gamma \subset \mathrm{PGL}(2, k)$ heißt *Schottkygruppe*, wenn Γ endlich erzeugt und diskontinuierlich ist und bis auf id keine Elemente endlicher Ordnung besitzt. Man kann zeigen, daß Schottkygruppen frei sind. Bezeichnet man mit $\Omega_\Gamma \subset \mathbb{P}$ die Menge der gewöhnlichen Punkte für Γ, so kann man den Quotientenraum $S_\Gamma = \Omega_\Gamma/\Gamma$ in natürlicher Weise zu einer projektiv-algebraischen, nichtsingulären Kurve vom Geschlecht $g = \mathrm{rank}(\Gamma)$ machen. Algebraische Kurven dieses Typs bezeichnet man als *Mumfordkurven* (vgl. [8]).

1.2 Wir betrachten nun g Paare von abgeschlossenen Kreisen U_i; $i = \pm 1, \ldots, \pm g$; die paarweise disjunkt sind:

$$U_i = \{z \in \mathbb{P} : |z - a_i| \leqq |\varepsilon_i|\},$$
$$\mathring{U}_i = \{z \in \mathbb{P} : |z - a_i| < |\varepsilon_i|\}.$$

Für $i = 1, \ldots, g$ seien $\gamma_i \in \mathrm{PGL}(2, k)$ mit $\gamma_i(\mathbb{P} - U_i) = \mathring{U}_{-i}$. Dann ist

$$\Gamma := \langle \gamma_1, \ldots, \gamma_g \rangle \subset \mathrm{PGL}(2, k)$$

diskontinuierlich und frei vom Rang g. Das System $(\gamma_1, \ldots, \gamma_g)$ heißt *geometrische Basis* für Γ. Es gilt mit $F := \mathbb{P} - \bigcup_i \mathring{U}_i$

$$\Omega_\Gamma = \bigcup_{\gamma \in \Gamma} \gamma(F).$$

$\mathbb{P} - \Omega_\Gamma$ ist nirgends dicht und kompakt. $\Omega_\Gamma \to S_\Gamma = \Omega_\Gamma/\Gamma$ ist die „universelle Überlagerung". Man kann zeigen, daß jede Mumfordkurve in dieser Weise gewonnen werden kann, also daß jede Schottkygruppe eine geometrische Basis besitzt (vgl. [8], Kap. I).

1.3 Sei X eine nichtsinguläre, projektiv-algebraische Kurve, X^{an} die zugehörige analytische Kurve. Eine *formelle Struktur* auf X wird gegeben durch eine Überdeckung $\mathfrak{U} = \{U_1, \ldots, U_n\}$ von X^{an} derart, daß $U_i \cap U_j$ stets endliche Vereinigung von strikten Laurentbereichen von U_i bzw. U_j ist. Ist $\pi_i : U_i \to \tilde{U}_i$ die Projektion auf das zugehörige affine Modell, so lassen sich diese zu einer Projektion $\pi : X \to \tilde{X}_\mathfrak{U}$ auf ein reduziertes Schema $\tilde{X}_\mathfrak{U}$ von lokal-endlichem Typ über dem Bewertungsrestklassenkörper $\tilde{k}$ verkleben. $\tilde{X}_\mathfrak{U}$ bezeichnet man als (die durch $\mathfrak{U}$ gegebene) *Reduktion* von X. Für $x \in X$ bezeichnet man die Teilmenge $X_\mathfrak{U}(x) := \pi^{-1}(\pi(x))$ als die *formelle Faser*, die x enthält. Die formellen Fasern sind zusammenhängende Steinsche Unterräume von X^{an}. Ist die formelle Faser isomorph zu einem offenen Kreis, so nennen wir sie *kreisförmig*. Das ist genau dann der Fall, wenn $\tilde{x} = \pi(x) \in \tilde{X}_\mathfrak{U}$ nicht-singulär ist. Ist die formelle Faser isomorph zu einem Kreisring $\{z : \varepsilon < |z| < 1\}$, so nennen wir sie *ringförmig*. Das ist genau dann der Fall, wenn $\tilde{x} = \pi(x) \in \tilde{X}_\mathfrak{U}$ ein gewöhnlicher Doppelpunkt ist. Der *Formfaktor* ε eines Kreisringes ist offenbar eine biholomorphe Invariante (vgl. [1]).

1.4 Man sagt, eine nichtsinguläre, projektiv-algebraische Kurve X besitzt ein *formelles Standardmodell*, wenn sie eine formelle Struktur besitzt, deren sämtliche Fasern kreisförmig oder ringförmig sind. Eine formelle Struktur ist also ein formelles Standardmodell, wenn die zugehörige Reduktion als Singularitäten höchstens gewöhnliche Doppelpunkte hat. Die Menge der nicht-rationalen, irreduziblen Komponenten der Reduktion $\tilde{X}$ eines formellen Standardmodells von X sind bis auf birationale Äquivalenz eindeutig bestimmt (vgl. [1]).

1.5 Eine nichtsinguläre, projektiv-algebraische Kurve besitzt ein formelles Standardmodell, wenn sie schon über einem diskret bewerteten Unterkörper $k_0 \subset k$ schon definiert ist [3] oder der Körper algebraisch abgeschlossen ist [17].

1.6 Mit Hilfe des formellen Standardmodells kann man nach ([8], [16]) Mumford-Kurven folgendermaßen charakterisieren: Eine nicht-singuläre, projektiv-algebraische Kurve X ist genau dann Mumford-Kurve, wenn X ein formelles Standardmodell derart besitzt, daß sämtliche irreduziblen Komponenten der Reduktion rationale Kurven sind ([8], p. 153).

1.7 Jeder Mumford-Kurve kann man z.B. via Standardmodell einen Graphen zuordnen. — Die Ecken seien die irreduziblen Komponenten der Reduktion und die Verbindungskanten seien die gemeinsamen singulären Punkte. — Reduziert man diesen Graphen, so erhält man einen Graphen, der nur noch von der Kurve, aber nicht mehr von der Wahl des formellen Standardmodells abhängt. Den so erhaltenen, reduzierten Graphen bezeichnet man als *kanonischen Graphen der Mumfordkurve*. Er hat als zyklomatische Zahl gerade das Geschlecht der Kurve.

1.8 Sei X eine nicht-singuläre, projektiv-algebraische Kurve, die ein formelles Standardmodell besitzt. Indem man die formelle Struktur eventuell noch verfeinert, kann man ohne Einschränkung annehmen, daß alle irreduziblen Komponenten der Reduktion $\tilde{X}$ nicht-singulär sind. Sei $\pi: X \to \tilde{X}$ die Reduktionsprojektion, seien $\tilde{X}_1, \ldots, \tilde{X}_n$ die irreduziblen Komponenten von $\tilde{X}$. Sei G der Koinzidenzgraph der irreduziblen Komponenten von $\tilde{X}$, $\lambda: T \to G$ die graphentheoretische universelle Überlagerung und $K \subset T \times T$ die Kantenmenge von T. Wir definieren dann den k-analytischen Raum

$$\Omega = \bigcup_{(st) \in K} X_{st}/\sim,$$

$$X_{st} = \pi^{-1}\left(\tilde{X}_{\lambda(s)} \cup \tilde{X}_{\lambda(t)} - \{\text{singuläre Punkte} \neq \lambda(st)\}\right),$$

$$X_s = \pi^{-1}(\tilde{X}_{\lambda(s)} - \{\text{singuläre Punkte}\}).$$

Dabei sind X_{st} und X_{ts} zu identifizieren, und X_{st} ist mit X_{tr} über X_t für $s \neq r$ zu verheften. Die kanonische Abbildung $\Omega \to X$ ist die universelle Überlagerung im Sinne von k-analytischen Räumen (vgl. [8], IV § 3). Weil Teilbereiche von $\mathbb{P}$ einfach-zusammenhängend bezüglich Überdeckungsketten sind, ist auch Ω einfach-zusammenhängend bezüglich Überdeckungsketten.[1])

Satz 1. *Jede beschränkte, holomorphe Funktion auf Ω ist konstant.*

Beweis. Ist G ein Baum, so ist $X = \Omega$ geschlossen und wir sind fertig. Sonst betrachte die formelle Überdeckung $\mathfrak{V} := \{X_{st} : (st) \in K\}$ von Ω. Sei $\tilde{\Omega}$ die Reduktion von Ω bezüglich $\mathfrak{V}$. Dann gibt es nur endlich viele Isomorphietypen von ringförmigen, formellen Fasern. Folglich gibt es $c < 1$ derart, daß für den Formfaktor ε einer jeden ringförmigen formellen Faser $\varepsilon \leqq c$ gilt. Seien X_t, $t \in T$, wie oben und $\tilde{X}_t$, $t \in T$, die irreduziblen Komponenten von $\tilde{\Omega}$.

---

[1]) Für zwei zusammenhängende Teilbereiche $U, V \subset \Omega$ ist $U \cap V = \emptyset$ oder $U \cap V$ zusammenhängend. Sind weiterhin $U_1, \ldots, U_n \subset \Omega$ zusammenhängende affinoide Teilbereiche mit $U_i \cap U_{i+1} \neq \emptyset$ und

$$f_i \in \mathcal{O}(U_i) \quad \text{mit} \quad f_i = f_{i+1} | U_i \cap U_{i+1}$$

gegeben, so gibt es ein

$$f \in \mathcal{O}(U_1 \cup \cdots \cup U_n) \quad \text{mit} \quad f | U_i = f_i.$$

Sei nun $f \in \mathcal{O}(\Omega)$ beschränkt. Dann wählen wir $c_t \in k$ mit $|c_t| = ||f|X_t||$. Im Fall $c_t \neq 0$ setzen wir $f_t := c_t^{-1} f|X_t$. f_t induziert eine meromorphe Funktion $\tilde{f}_t$ auf $\tilde{X}_t$. Ist f nicht konstant, so kann man ohne Einschränkung annehmen, daß f auf T_{t_0} eine Nullstelle hat. Dann ist $\tilde{f}_{t_0}$ aber auch nicht konstant. Also hat $\tilde{f}_{t_0}$ eine Polstelle $\tilde{p}$. Nun muß $\tilde{p}$ aber ein singulärer Punkt von $\tilde{\Omega}$ sein, etwa $\tilde{p} \in \tilde{X}_{t_0} \cap \tilde{X}_{t_1}$ mit $t_0 \neq t_1$. Weil die formelle Faser von $\tilde{p}$ ringförmig mit Formfaktor $\varepsilon \leqq c$ ist, folgt

$$|c_{t_1}| \geqq \varepsilon^{-1} |c_{t_0}| \geqq c^{-1} |c_{t_0}|.$$

Weil $\tilde{f}_{t_1}$ jetzt in $\tilde{p} \in \tilde{X}_{t_1}$ eine Nullstelle hat, ist $\tilde{f}_{t_1}$ auch nicht konstant. So erhält man t_2 mit $|c_{t_2}| \geqq c^{-2} |c_{t_0}|$ und $\tilde{f}_{t_2} \not\equiv \textit{konstant}$. Iteriert man dieses Vorgehen, so findet man wegen $c < 1$ Bereiche, auf denen f beliebig große Werte annimmt.

2. Unverzweigte Überlagerungen p-adischer Kreise

Im folgenden sei k algebraisch abgeschlossen, p-adischer Körper, $\tilde{k}$ sein Bewertungsrestklassenkörper mit $p = \operatorname{char} \tilde{k} > 0$.

2.1 Sei L ein algebraisch abgeschlossener Körper, $\varphi: X \to \mathbb{P}_1(L)$ eine endliche Überlagerung nichtsingulärer, irreduzibler, algebraischer Varietäten. Wenn

$$\operatorname{grad} \varphi < \operatorname{char}(L)$$

ist, so gilt für φ die Hurwitzsche Geschlechterformel

$$2g - 2 = -2 \cdot \operatorname{grad}(\varphi) + b,$$

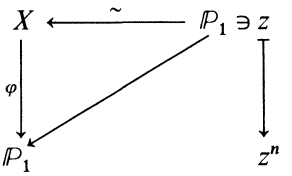
$$b = \sum_{x \in X} (e_x - 1),\ e_x = \text{Multiplizität von } \varphi \text{ in } x,$$

$$g = \text{Geschlecht von } X.$$

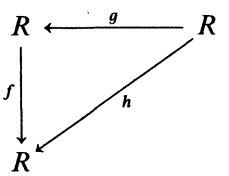
Daraus folgt sofort:

a) Ist φ höchstens über ∞ verzweigt, so ist φ Isomorphismus.

b) Ist φ höchstens über $\{0, \infty\}$ verzweigt, so besteht $\varphi^{-1}(0)$ und $\varphi^{-1}(\infty)$ aus genau einem Punkt, und es gibt ein kommutatives Diagramm



2.2 Sei $R = \{z \in k: |z| = 1\}$. Ist $f: R \to R$ endlicher Morphismus vom Grad n mit $(n, p) = 1$, so gibt es ein kommutatives Diagramm



wobei g Isomorphismus und $h(z) = z^n$ ist.

Beweis. Weil $f(Z)=\sum a_\nu Z^\nu \in \mathcal{O}(R)$ nullstellenfrei ist, gibt es $m \in \mathbb{Z}$ mit $|a_m|=1>|a_\nu|$ für $\nu \neq m$. Dann ist $|m|=n$. Ohne Einschränkung sei $m=n$ und $a_n=1$. Dann folgt die Behauptung durch Iteration mit dem folgenden Lemma.

Lemma. *Sei* $0<c<1$. *Dann gibt es zu*

$$f(Z)=Z^n+\sum a_\nu Z^\nu \in \mathcal{O}(R) \quad \textit{mit} \quad |a_\nu| \leqq c$$

ein

$$g(Z)=Z+\sum b_\nu Z^\nu \in \mathcal{O}(R) \quad \textit{mit} \quad |b_\nu| \leqq c$$

derart, daß für $h(Z):=f\circ g(Z)$ *gilt*

$$h(Z)=Z^n+\sum c_\nu Z^\nu \in \mathcal{O}(R) \quad \textit{mit} \quad |c_\nu| \leqq c^2.$$

Beweis. Setze $b_\nu=(-n)^{-1}a_{\nu+n-1}$. Wegen $(n,p)=1$ gilt $|b_\nu| \leqq c$, und somit gilt

$$\begin{aligned} f\circ g(Z) &\equiv (Z+\sum b_\nu Z^\nu)^n+\sum a_\nu Z^\nu && \bmod c^2 \\ &\equiv (Z^n+\sum n b_\nu Z^{\nu+n-1})+\sum a_\nu Z^\nu && \bmod c^2 \\ &\equiv Z^n+\sum (n b_\nu + a_{\nu+n-1}) Z^{\nu+n-1} && \bmod c^2 \\ &\equiv Z^n && \bmod c^2. \end{aligned}$$

Beispiel. $\varphi: R \longrightarrow R$ definiert durch $\varphi(z)=z^p+cz$ mit $|p|<|c|<1$ ist endlich und unverzweigt, aber φ ist nicht isomorph zu einer Potenzüberlagerung.

2.3 Sei $0 \leqq c<1$ und ${}_cR_1:=\{z \in E,\ c \leqq |z| \leqq 1\}$. Seien X reduzierter, affinoider Raum und $\varphi: X \longrightarrow {}_cR_1$ endliche Überlagerung. Weiter sei

$${}_aX_b:=\varphi^{-1}({}_aR_b) \quad \text{für} \quad c \leqq a \leqq b \leqq 1.$$

Dann gibt es $c<a<1$, so daß ${}_aX_a$ isomorph zu einer disjunkten Summe von Kreisringen ist.

Wir beweisen die Behauptung durch Induktion über $\operatorname{grad}\varphi$. Für $\operatorname{grad}\varphi=1$ ist die Aussage offensichtlich richtig. Im folgenden sei also $\operatorname{grad}\varphi>1$.

A) Nach Induktionsvoraussetzung kann man ohne Einschränkung annehmen, daß ${}_b\tilde{X}_b$ für jedes $c<b \leqq 1$ irreduzibel ist. Wenn nämlich ${}_b\tilde{X}_b$ nicht irreduzibel ist, so ist ${}_aX_a$ für hinreichend großes $a<b$ nicht zusammenhängend! Ist nämlich ${}_b\tilde{X}_b$ nicht irreduzibel, so gibt es wegen $\dim {}_b\tilde{X}_b=1$ endlich viele Punkte $\tilde{Q}=\{\tilde{q}_1,\ldots,\tilde{q}_r\}$ in ${}_b\tilde{X}_b$, so daß ${}_b\tilde{X}_b-\tilde{Q}$ nicht zusammenhängend ist. Bezeichnen wir mit $Q \subset {}_bX_b$ die formelle Faser von $\tilde{Q}$, so ist ${}_bX_b-Q$ ein nicht zusammenhängender affinoider Teilbereich. Nach [2], Korollar 1.3 gibt es $a<b$, so daß ${}_aX_b-Q$ nicht zusammenhängend ist. Dann ist auch ${}_aX_d$ für $a \leqq d<b$ nicht zusammenhängend. Für $a \leqq d<b$ zerfällt dann $\varphi: {}_aX_d \longrightarrow {}_aR_d$ in eine Summe von zwei Überlagerungen. Folglich ist man nach Induktionsvoraussetzung fertig.

B) Wegen der Gradgleichung $[{}_1X_1 : {}_1R_1] = [{}_1\tilde{X}_1 : {}_1\tilde{R}_1]$ gibt es eine Funktion $\xi \in \mathcal{O}(X)$ mit $|\xi| = 1$, so daß das Minimalpolynom von $\widetilde{\xi|_1 X_1}$ Grad größer als 1 hat. Betrachte dann die Abbildung

$$\pi := (\varphi, \xi) : X \longrightarrow {}_cR_1 \times E,$$
$$\pi(X) = V(P) \text{ für das Minimalpolynom von } \xi,$$
$$P = T^n + a_1 T^{n-1} + \cdots + a_n \in \mathcal{O}({}_cR_1)[T],\ |a_\nu| \leqq 1.$$

Indem man ξ eventuell noch durch einen Faktor der Form Z^e, $e \in \mathbb{Z}$, abändert, darf man annehmen, daß gilt

$$\tilde{P} := \widetilde{P|_1 R_1} \times E \in \tilde{k}[Z][T].$$

Nach A) dürfen wir ${}_1\tilde{X}_1$ als irreduzibel voraussetzen. Wegen $\tilde{\pi}({}_1\tilde{X}_1) = V(\tilde{P})$ ist $\tilde{P}$ ein Primärpolynom, nämlich eine Potenz des Minimalpolynoms von $\tilde{\xi}$. Insbesondere ist $\tilde{P}$ nicht Potenz eines linearen Polynoms in T. Nun setzen wir

$$a_\nu = \sum_{\mu \in Z} c_{\nu\mu} Z^\mu,$$
$$r(\nu) := \operatorname{Min}\{\mu \text{ mit } |c_{\nu\mu}| = 1;\ \deg {}_Z\tilde{P}\},$$
$$Q(Z, T) := T^n + c_{1r(1)} Z^{r(1)} T^{n-1} + \cdots + c_{nr(n)} Z^{r(n)}.$$

Falls für ein ν alle $|c_{\nu\mu}| < 1$ sind, so sei in $Q(Z, T)$ der Koeffizient von $T^{n-\nu}$ als 0 definiert.

a) Es gibt $\varepsilon < 1$ derart, daß

$$|a_\nu(z) - c_{\nu r(\nu)} z^{r(\nu)}| < |z^{r(\nu)}| \quad \text{für} \quad \varepsilon \leqq |z| < 1.$$

b) Weil man den Betrag der Nullstellen eines Polynoms durch die Beträge der Koeffizienten abschätzen kann, gilt mit

$$\delta := \operatorname{Max}\{\sqrt[\nu]{\varepsilon^{r(\nu)}},\ 1 \leqq \nu \leqq n\},$$
$$N(P) \cap \{|z| = \varepsilon\} \subset \{(z, t),\ |z| = \varepsilon,\ |t| \leqq \delta\} = R_\varepsilon \times E_\delta.$$

c) Seien $e, d \in k^\times$ mit $|e| = \varepsilon$, $|d| = \delta$, so führen wir die neuen Koordinaten auf $R_\varepsilon \times E_\delta$ ein

$$Z_\varepsilon := e^{-1} Z,\ T_\delta := d^{-1} T,$$
$$Q_\varepsilon(Z_\varepsilon, T_\delta) = d^{-n} Q(eZ_\varepsilon, T) = T_\delta^n + \sum_{\nu=1}^{n} c_{\nu r(\nu)} d^{-\nu} e^{r(\nu)} Z_\varepsilon^{r(\nu)} T_\delta^{n-\nu},$$
$$P_\varepsilon(Z_\varepsilon, T_\delta) = d^{-n} P(eZ_\varepsilon, T) = T_\delta^n + \sum_{\nu=1}^{n} a_\nu d^{-\nu} T_\delta^{n-\nu}.$$

Wegen a) und b) gilt $|Q_\varepsilon - P_\varepsilon| < 1$ auf $R_\varepsilon \times E_\delta$.

d) Weil ${}_\varepsilon\tilde{X}_\varepsilon$ irreduzibel ist, ist $\tilde{P}_\varepsilon = \tilde{Q}_\varepsilon$ ein Primärpolynom. Ist $n_1 = \operatorname{Max}\{\nu, \delta^\nu = \varepsilon^{r(\nu)}\}$, so gilt $n_1 = n$, weil sonst $\tilde{Q}_\varepsilon$ zwei verschiedene Primfaktoren hätte.

e) Wegen $0<\varepsilon<1$ gilt somit

$$|d^{-\nu}e^{r(\nu)}|=1 \Leftrightarrow \nu r(n)=nr(\nu).$$

Ist $g=\mathrm{ggT}(n, r(n))$, also $n=gm$ und $r(n)=gr$, so gilt

$$|d^{-\nu}e^{r(\nu)}|=1 \Leftrightarrow \nu=\gamma m,\ r(\nu)=\gamma r \quad \text{für ein} \quad 1\leqq\gamma\leqq g.$$

Somit hat $\tilde{Q}_\varepsilon$ eine Darstellung in der Form

$$\tilde{Q}_\varepsilon(Z_\varepsilon, T_\varepsilon)=\sum_{\gamma=0}^{g}\tilde{q}_\gamma(Z_\varepsilon^r)^\gamma(T_\delta^m)^{g-\gamma},$$

wobei $\tilde{q}_\gamma\in\tilde{k}$ und $\tilde{q}_0=1$ ist. Weil $\tilde{Q}_\varepsilon$ primär ist, gibt es $\tilde{a}\in\tilde{k}$, $\tilde{a}\neq 0$, so daß gilt

$$\tilde{Q}_\varepsilon(Z_\varepsilon, T_\delta)=(T_\delta^m-\tilde{a}Z_\varepsilon^r)^g.$$

Wegen $(m, r)=1$ ist dies aber genau die Primfaktorzerlegung von $\tilde{Q}_\varepsilon$.

f) Ist $g=n$, so ist $m=1$, dann nehmen wir die Variablentransformation

$$T\mapsto T_1+\sqrt[n]{-c_{n,r(n)}}\,Z^r$$

vor. Diese Transformation läßt $R_\varepsilon\times E_\delta$ invariant. Nun wird dabei $\tilde{P}_\varepsilon=\tilde{Q}_\varepsilon$ in T_1^n transformiert. Startet man nun wieder von vorne, so gelangt man zu einem neuen $\delta_1=\delta_1(\varepsilon)<\delta(\varepsilon)$ und $r_1(n)>r(n)$, wobei

$$P_1(Z, T_1)=P(Z, T_1+\sqrt[n]{-c_{nr(n)}}\,Z^r)=T_1^n+a_1^1T_1^{n-1}+\cdots+a_n^1,$$
$$a_n^1=\sum c_{n\mu}^1Z^n,\ r_1(n)=\mathrm{Min}\{\mu:|c_{n\mu}^1|=1\}.$$

Würde sich bei jeder Wiederholung des oben beschriebenen Verfahrens $g=n$ ergeben, so würde das bedeuten, daß es zu jedem $N\in\mathbb{N}$ eine Transformation $T\mapsto T+f(Z)$ für ein Polynom $f(Z)$ gibt, so daß alle Koeffizienten des so aus P gewonnenen Polynoms P_1 Koeffizienten a_ν^1 für $\nu=1,\ldots,n$ derart hätten, daß alle Reduktionen $\tilde{a}_\nu^1$ in Z einen Untergrad größer als N hätten. Das ist aber nur möglich, wenn $\tilde{P}$ Potenz eines linearen Polynoms in T ist. Das war aber durch die Wahl von $\tilde{\xi}$ ausgeschlossen.

g) Also gelangt man nach endlich vielen Schritten zu dem Fall $g<n$, also auch $m>1$. Nachdem man Z um einen skalaren Faktor vom Betrag 1 abgeändert hat, gilt auf ${}_aX_b$ für $a<b<1$ mit hinreichend großem a

$$(*)\qquad \varphi^*(Z^{-r})\,\xi^m=1+\zeta \quad \text{mit} \quad \zeta\in\mathcal{O}({}_aX_b) \quad \text{und} \quad |\zeta|<1.$$

Seien nun $\mu, \rho\in\mathbb{Z}$ mit $1=\mu m+\rho r$. Dann betrachten wir die Abbildung

$$\psi: {}_aX_b\to{}_\alpha R_\beta, \quad \text{wobei} \quad \alpha^m=a, \quad \beta^m=b,$$
$$\psi^*(Z)=\varphi^*(Z^\mu)\,\xi^\rho.$$

Dann gilt wegen $(*)$

$$\psi^*(Z^m)=\varphi^*(Z^{\mu m})\,\xi^{\rho m}=\varphi^*(Z^{\mu m+\rho r})\,(1+\zeta)^\rho$$
$$=\varphi^*(Z)\,(1+\eta) \quad \text{mit} \quad |\eta|<1.$$

Folglich gilt

$$\operatorname{grad}\psi = \operatorname{grad}\tilde{\psi} = \frac{1}{m}\operatorname{grad}\tilde{\varphi} = \frac{1}{m}\operatorname{grad}\varphi < \operatorname{grad}\varphi .$$

Somit folgt die Behauptung nach Induktionsvoraussetzung.

2.4 Nach diesen Vorbereitungen folgt sofort

Satz 2. *Sei E^n der n-dimensionale Einheitspolyzylinder im k^n. Ist $\varphi: X \to E^n$ eine endliche, unverzweigte Überlagerung mit* $\operatorname{grad}\varphi < p$, *so ist φ trivial.*

Beweis. Wir behandeln zuerst den eindimensionalen Fall $E^1 = E$. Wenn $\tilde{\varphi}: \tilde{X} \to \tilde{E}$ unverzweigt ist, so folgt die Behauptung mit 2. 1. Somit genügt es zu zeigen, daß für jedes $\tilde{x} \in \tilde{X}$ die Einschränkung auf die formelle Faser $\varphi: X_+(\tilde{x}) \to E_+(\tilde{\varphi}(\tilde{x}))$ ein Isomorphismus ist. Ohne Einschränkung sei $\tilde{\varphi}(\tilde{x}) = \tilde{0}$. Für hinreichend großes $\varepsilon < 1$ ist

$$X_\varepsilon := \varphi^{-1}(\{z, |z| \leqq \varepsilon\}) \cap X_+(\tilde{x})$$

zusammenhängend. Nach 2. 3 kann man annehmen, daß ${}_\varepsilon X_\varepsilon$ eine Summe von Kreisringen ist. Nach 2. 2 kann man $\varphi | X_\varepsilon$ über ${}_\varepsilon X_\varepsilon$ mit entsprechenden Potenzüberlagerungen verkleben zu einer Überlagerung $X(\varepsilon) \to \mathbb{P}_1$, die höchstens über ∞ verzweigt ist. Nach 2. 1 ist diese Überlagerung ein Isomorphismus, folglich ist $\varphi | X_\varepsilon$ ein Isomorphismus und somit auch $\varphi | X_+(\tilde{x})$. Den höherdimensionalen Fall E^n reduziert man durch Einschränkung auf eine generische Gerade (wie in [14], p. 139ff).

Beispiel. $\varphi: E \to E$, definiert durch $\varphi(z) = z^p + z$, ist endlich vom Grad p und unverzweigt.

Korollar 1. *Sei $0 < c \leqq 1$ und $R = \{z \in k, c \leqq |z| \leqq 1\}$. Ist $\varphi: X \to R$ endlich und unverzweigte Überlagerung mit* $\operatorname{grad}\varphi < p$, *so ist φ isomorph zu einer Summe von Potenzüberlagerungen.*

Beweis. Zunächst sei $c = 1$. Nach Satz 2 ist $\tilde{\varphi}: \tilde{X} \to \tilde{R}$ unverzweigt, somit folgt die Behauptung aus 2. 1. Nun können wir für $c < 1$ die Überlagerung φ über $\{|z| = c\}$ und $\{|z| = 1\}$ mittels 2. 2 mit Potenzüberlagerungen von $\{|z| \leqq c\}$ und $\{|z| \geqq 1\}$ zu einer Überlagerung $\hat{X} \to \mathbb{P}_1$ verheften, die nur über $\{0, \infty\}$ verzweigt ist. Dann folgt die Behauptung wieder mit 2. 1.

Analog zeigt man

Korollar 2. *Sei $\varphi: X \to E$ endliche Überlagerung mit* $\operatorname{grad}\varphi < p$. *Ist φ höchstens über einem Punkt von E verzweigt und ist X zusammenhängend, so ist X isomorph zu einem Kreis.*

2.5 Als Anwendung dieser Sätze erhält man nützliche Ergebnisse für die Konstruktion formeller Standardmodelle.

Definition. Sei $T \subset \mathbb{P} = \mathbb{P}_1$ eine endliche Menge. Eine formelle Überdeckung $\mathfrak{U}$ von $\mathbb{P}$ heißt *T-fein*, wenn gilt:

1) $\mathfrak{U}$ definiert ein formelles Standardmodell von $\mathbb{P}$.

2) Jede formelle Faser von $\mathbb{P}_\mathfrak{U}$ enthält höchstens einen Punkt von T.

3) Jede ringförmige formelle Faser von $\mathbb{P}_\mathfrak{U}$ enthält keinen Punkt von T.

Bemerkung. Zu jeder endlichen Menge T gibt es eine T-feine Überdeckung von $\mathbb{P}$. Hat T mehr als 2 Elemente, so gibt es eine kanonische T-feine Überdeckung $\mathfrak{U}_T$ (vgl. [8], I, § 2).

Satz 3. *Sei X eine nichtsinguläre, projektiv-algebraische Kurve, $\varphi: X \to \mathbb{P}_1$ eine endliche Überlagerung mit $\operatorname{grad}\varphi < p$. Ist $T \subset \mathbb{P}_1$ die Menge der Verzweigungspunkte von φ, so ist für jede T-feine, formelle Überdeckung $\mathfrak{U}$ von $\mathbb{P}_1$ die Überdeckung*

$$\mathfrak{V} := \{V,\ V \text{ Zusammenhangskomponente von } \varphi^{-1}(U) \quad \text{für ein} \quad U \in \mathfrak{U}\}$$

eine formelle Überdeckung von X, die ein formelles Standardmodell von X liefert.

Beweis. Offenbar ist $\mathfrak{V}$ formelle Überdeckung. Jede formelle Faser von $X_{\mathfrak{V}}$ überlagert eine formelle Faser von $\mathbb{P}_{\mathfrak{U}}$. Mit den Korollaren 1, 2 zu Satz 2 folgt, daß die formellen Fasern von $X_{\mathfrak{V}}$ kreis- oder ringförmig sind. Also definiert $\mathfrak{V}$ ein formelles Standardmodell.

Korollar 1. *Ist* $\operatorname{char}\tilde{k} \neq 2$, *so besitzt jede hyperelliptische, nichtsinguläre, algebraische Kurve ein formelles Standardmodell.*

Korollar 2. *Ist X eine nichtsinguläre, projektiv-algebraische Kurve vom Geschlecht g, so besitzt X ein formelles Standardmodell, falls gilt*

a) $\operatorname{char}\tilde{k} = 0$ *oder*

b) $g+1 < p = \operatorname{char}\tilde{k}$ *oder*

c) $\operatorname{char}k = 0$ *und* $\left[\frac{g+3}{2}\right] < p = \operatorname{char}\tilde{k}$.

Der *Beweis* folgt mit Satz 3 und bekannten Resultaten über die Konkretisierung von Kurven ([4], p. 135).

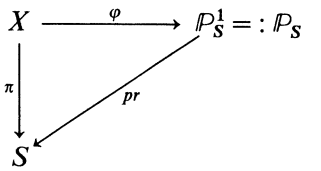
Korollar 1 ist das Hauptergebnis von [1], Korollare 1, 2 a, b sind auch in [8] (Kap. V, § 3, § 4) enthalten.

3. Zerschneidung analytischer Familien

Eine *analytische Familie von Mumfordkurven* $\pi: X \to S$ ist ein eigentlicher, flacher Morphismus zusammenhängender, analytischer Räume, wobei die Fasern $X_s = \pi^{-1}(s)$ Mumfordkurven sind.

Weil π flach ist, ist die Euler-Poincaré-Charakteristik $\mathrm{EP}(\pi, \mathcal{O})(s) = 1 - g(X_s)$ lokal konstant auf S. Für zusammenhängendes S ist somit das Geschlecht $g(X_s) = g$ der Fasern X_s konstant. Wenn $S = \mathrm{Sp}(A)$ ein affinoider Raum ist, so ist π ein algebraischer Morphismus $X \to \mathrm{Spec}(A)$! Man wähle nämlich ein Geradenbündel $\mathscr{L}$ auf X, dessen Einschränkungen $\mathscr{L}(s)$ auf die Fasern X_s sehr ampel sind. Dann ist $\mathscr{E} = \pi_* \mathscr{L}$ ein Vektorraumbündel auf S, und X läßt sich in $\mathbb{P}(\mathscr{E})$ über S einbetten. Nach den GAGA-Sätzen im relativen Fall [13] ist X algebraisch über $\mathrm{Spec}(A)$.

In diesem Abschnitt betrachten wir die folgende Situation:



π analytische Familie von Mumfordkurven,

φ endliche Überlagerung mit $\operatorname{grad}\varphi < p = \operatorname{char}\tilde{k}$,

S reduzierter, affinoider Raum.

Satz 4. *Es gibt eine endliche Überlagerung $S' \to S$ von S, eine zulässige Überdeckung $\{S'_i\}$ von S' durch affinoide Unterräume S'_i von S' und zulässige Überdeckungen $\{X'_{ik}\}$ von $X'_i := X \times_S S'_i$, so daß gilt:*

$$X'_{ik} \cong \{(s,z) \in S'_i \times \mathbb{P} : |z - a^0_{ik}(s)|^{m_0} \leqq |r^0_{ik}(s)|,$$
$$|z - a^\nu_{ik}(s)|^{m_\nu} \geqq |r^\nu_{ik}(s)| \text{ für } 1 \leqq \nu \leqq n_{ik}\},$$

wobei $m_\nu \geqq 1$ natürliche Zahlen und a^ν_{ik}, $r^\nu_{ik} \in \mathcal{O}(S'_i)$ mit folgenden Eigenschaften sind:

$$|a^\nu_{ik}(s) - a^0_{ik}(s)|^{m_0} < |r^0_{ik}(s)| \quad \textit{für} \quad 1 \leqq \nu \leqq n_{ik},$$
$$|a^\nu_{ik}(s) - a^\mu_{ik}(s)|^{m_\nu} > |r^\nu_{ik}(s)| \quad \textit{für} \quad 1 \leqq \nu \neq \mu \leqq n_{ik},$$
$$0 < |r^\nu_{ik}(s)|^{m_0} < |r^0_{ik}(s)|^{m_\nu} \quad \textit{für} \quad 1 \leqq \nu \leqq n_{ik}.$$

Weiterhin gilt für die Zusammenhangskomponenten U_ν von $X_{ik} \cap X_{il}$:

$$U_\nu \cong \{(s,z) \in S'_i \times \mathbb{P} : |r^-_\nu(s)| \leqq |z - a_\nu(s)|^{m_\nu} \leqq |r^+_\nu(s)|\}$$

wobei r^+_ν, r^-_ν, $a_\nu \in \mathcal{O}(S'_i)$ sind mit $0 < |r^-_\nu(s)| < |r^+_\nu(s)|$ für alle $s \in S'_i$.

3.0 Die Basiserweiterung $S' \to S$ mit einer endlichen Überlagerung ist wegen der Existenz gewisser Schnitte zu π notwendig. Man konstruiert solche Überlagerungen folgendermaßen: Ist $\zeta : S \to \mathbb{P}_S$ ein Schnitt zur Projektion, so sei $S' := \pi^{-1}(\zeta(S))$. Dann ist $S' \to S$ eine endliche Überlagerung, und $S' \times_S X \to S'$ gestattet den tautologischen Schnitt, der über ζ liegt.

Im folgenden beweisen wir Satz 4 durch Induktion über $\dim S$. Im Fall $\dim S = 0$ ist die Behauptung wohlbekannt. In den folgenden Absätzen wird der Induktionsschritt gezeigt.

3.1 Ist $N \subset S'$ eine analytische Teilmenge einer endlichen Überlagerung $S' \to S$, über der Überdeckungen der gewünschten Form existieren, so gibt es eine Schlauchumgebung $N \subset U \subset S'$, über der auch solche Überdeckungen existieren.

Man reduziert sofort darauf, daß es global über N eine Zerlegung $\{X_i\}$ von $X \times_S N$ der gewünschten Form gibt und X in einem $\mathbb{P}^r_S$ liegt. Dann existiert ein offener affinoider Bereich $Y_i \subset X$ über S mit $Y_i \times_S N = X_i$.[2]) Die Uniformisierende Z auf X_i läßt sich auf Y_i liften. Ebenso lassen sich alle zur Beschreibung von X_i benötigten Funktionen auf N nach S' liften. Nun wähle man eine Schlauchumgebung U von N in S', so daß über U alle gelifteten Funktionen den geforderten Ungleichungen genügen und Z den Raum $Y_i \times_S U$ in P_i abbildet, wobei $P_i \subset U \times \mathbb{P}^1$ der durch die gelifteten Funktionen beschriebene Bereich ist. Nach [12], Folg. 1. 9 liefert Z nach Verkleinerung von U eine Einbettung. Weil π flach ist und $Z|X_i = Y_i \times_S N$ Isomorphismus ist, liefert Z nach Verkleinerung von U einen Isomorphismus von $Y_i \times_S U$ auf P_i.

Ist $N \subset S$ eine analytische Teilmenge, zu der es eine endliche Überlagerung $N' \to N$ gibt, so daß über N' Überdeckungen von $X \times_S N'$ der gewünschten Form existieren, so gibt es auch eine endliche Überlagerung $S' \to S$, so daß auch über $S' \times_S N$ solche Überdeckungen existieren. Nach den oben gemachten Ausführungen gibt es somit auch eine Schlauchumgebung von $S' \times_S N$ in S', über der Überdeckungen der gewünschten Form existieren.

3.2 Sind $f_0, \ldots, f_l \in \mathcal{O}(S')$ ohne gemeinsame Nullstelle, dann sei für $\delta \geqq 1$

$$V(\delta) := \{s \in S' : |f_i(s)| \leqq \delta |f_0(s)| \quad \text{für} \quad 1 \leqq i \leqq l\}.$$

Gibt es Überdeckungen der gewünschten Form über $V(1)$, so existiert auch ein $\delta > 1$, so daß über $V(\delta)$ solche Überdeckungen existieren.

Diese Aussage folgt mit den Sätzen über abgeschlossene Einbettungen aus [7].

3.3 Wegen 3. 1 können wir nach Induktionsvoraussetzung ohne Einschränkung annehmen, daß die Verzweigungsmenge von φ in $\mathbb{P}_S$ endlichblättrig über S liegt. Dann sei $S' \to S$ eine endliche Überlagerung, so daß die Verzweigungsmenge von $\varphi' := \varphi \times_S S'$ in Divisoren vom Grad 1 zerfällt. Ohne Einschränkung können wir $S = S'$ annehmen. Seien also $\tau_0, \ldots, \tau_n : S \to \mathbb{P}$ holomorph mit

$$\text{Verzweigung}\,(\varphi) = \tau_0(S) \cup \cdots \cup \tau_n(S).$$

Wegen 3. 1 können wir nach Induktionsvoraussetzung annehmen, daß gilt

$$\tau_i(s) \neq \infty \quad \text{für} \quad 0 \leqq i \leqq n \quad , \quad s \in S,$$
$$\tau_i(s) \neq \tau_j(s) \quad \text{für} \quad 0 \leqq i < j \leqq n, \quad s \in S.$$

Dann setzen wir

$$\{\rho_1, \ldots, \rho_k\} := \{\tau_i - \tau_j : 0 \leqq i < j \leqq n\}$$

und für jede Permutation $(j_1, \ldots, j_k)$ von $(1, \ldots, k)$

$$S_{j_1 \ldots j_k} := \{s \in S : |\rho_{j_1}(s)| \leqq \cdots \leqq |\rho_{j_k}(s)|\}.$$

---

[2]) Wie in 3. 5 dargelegt wird, gewinnt man den Bereich X_i als Zusammenhangskomponente des φ-Urbildes eines Teilbereichs P in $N \times \mathbb{P}_1$ ganz spezieller Form. Dann kann man Y_i als Zusammenhangskomponente des φ-Urbildes einer geeigneten Umgebung von P in $S \times \mathbb{P}_1$ wählen.

Dann ist $\{S_{j_1 \dots j_k}\}$ eine Überdeckung von S durch rationale Teilbereiche. Sei nun ohne Einschränkung

$$S = S_{1 \dots k}.$$

Nun setzen wir für $1 \leqq l \leqq k$ und $0 = j_0 < j_1 < \cdots < j_l = k$

$$V^l_{j_0 \dots j_l} := \{s \in S : \begin{cases} |\rho_{j_\lambda + 1}(s)| = \cdots = |\rho_{j_{\lambda+1}}(s)| & \text{für } 0 \leqq \lambda < l \\ |\rho_{j_\lambda}(s)| < |\rho_{j_{\lambda+1}}(s)| & \text{für } 1 \leqq \lambda < l \end{cases}\}.$$

Gilt $V^1 = \cdots = V^{l-1} = \emptyset$ für sämtliche Indizes, so ist V^l für jeden Index ein rationaler Teilbereich von S. Es gilt dann nämlich

$$V^l_{j_0 \dots j_l} = \{s \in S : |\rho_{j_\lambda + 1}(s)| \geqq |\rho_{j_{\lambda+1}}(s)| \quad \text{für} \quad 0 \leqq \lambda < l\}.$$

Wegen 3. 2 genügt es nun, die folgende Behauptung zu beweisen:

3.4 Sei l maximal mit $V^1_\bullet = \cdots = V^{l-1}_\bullet = \emptyset$ für alle Indizes, sei $V = V^l_{j_0 \dots j_l}$ für einen Index $(j_0 \cdots j_l)$. Dann gibt es eine endliche Überlagerung $S' \to S$, so daß über $V' = V \times_S S'$ Überdeckungen der gewünschten Form existieren.

Weil die Behauptung in Satz 4 für Familien vom Geschlecht $g = 0$ klar ist, können wir uns auf den Fall $g \geqq 1$ beschränken. Sei nun $T(s) := \{\tau_0(s), \dots, \tau_n(s)\} \subset \mathbb{P}$ die Verzweigungsmenge von φ_s. Wegen $g \geqq 1$ besteht $T(s)$ aus mindestens 3 Punkten. Dann gibt es eine kanonische $T(s)$-feine Überdeckung (vgl. 2. 5). Sei $A(s)$ der Koinzidenzgraph dieses Standardmodells von $\mathbb{P}$; dieser ist hier ein Baum, welcher nur von den Anordnungsrelationen der Beträge $|\tau_i(s) - \tau_j(s)|$ für $\tau_i(s), \tau_j(s) \in T(s)$ abhängt. Nach Definition von V ist $A(s) = A$ für alle $s \in S$ konstant. Indem man das System eventuell durch geeignete $\tau \in \mathcal{O}(S)$ erweitert, kann man ohne Einschränkung annehmen, daß A wenigstens zwei Ecken hat. Jetzt unterteilt man jede Kante von A zweimal und erhält so einen neuen Baum B. Indem man gewisse Schnitte der $\tau_0, \dots, \tau_n$ mit geeigneten konstanten Faktoren multipliziert und diese neuen Schnitte dem alten System hinzufügt, erhält man ein System $T' = \{\tau'_0, \dots, \tau'_m\}$, das über jedem Punkt $s \in V$ den Baum B als Koinzidenzgraph des zu T' assoziierten Standardmodells liefert.

Wir fassen nun die Ecken von A in natürlicher Weise als Teilmenge der Ecken von B auf. Zwei in A benachbarte Ecken haben in B den Abstand 3. Für $t \in A \subset B$ sei $b_t \subset B$ der Teilbaum, der aus allen Ecken und Kanten von B besteht, die zu t Abstand kleiner gleich 2 haben. Ein solcher Teilbaum definiert dann einen Teilbereich $P_t \subset V \times \mathbb{P}$ von der im Satz beschriebenen Form. Ein Durchschnitt $P_t \cap P_{t'}$, sofern er nicht trivial ist, wird definiert durch eine Kante mit 2 Endpunkten, wobei jeder Endpunkt die Ordnung 2 hat; somit ist er von der im Satz beschriebenen Form für die Durchschnitte. Alle Exponenten m_ν sind in beiden Fällen 1. Bei den zur Beschreibung der P_t erforderlichen Funktionen hat man für die a^i gewisse τ_i und für die Radien gewisse Differenzen $\tau_i - \tau'_j$ zu wählen. Insbesondere sind diese Funktionen global über S definiert.

Nun definieren wir die Überdeckung $\{X_k\}$ von $\pi^{-1}(V)$ als die Familie der Zusammenhangskomponenten von $\varphi^{-1}(P_t)$, wobei $t \in A$ durchläuft. In den folgenden Abschnitten wird gezeigt, daß $\{X_k\}$ nach Basiserweiterung $S' \to S$ und Verfeinerung mit einer Überdeckung von $V' = V \times_S S'$ eine Überdeckung der gewünschten Form liefert.

3.5 Sei X_k eine Zusammenhangskomponente von $\varphi^{-1}(P_t)$. Sei $s \in V$ fest gewählt mit $X_k(s) \neq \emptyset$. Dann ist jede Zusammenhangskomponente U von $X_k(s)$ isomorph zu einem affinoiden Teilbereich von $\mathbb{P}$.

Beweis. Seien $T(s)$ und A wie in 3. 4. Für jede Kante $t_1 t_2$ in A sei $Q_{t_1 t_2}$ der dazu assoziierte Teilbereich von $\mathbb{P}$. (Die Definition von Q_t bzw. $Q_{tt'}$ ist analog zu den Definitionen im Absatz 1. 8.) Nach Satz 3 liefert

$$\mathfrak{W} := \{W\colon W \text{ Zusammenhangskomponente von } \varphi_s^{-1}(Q_{t_1 t_2}) \text{ für Kante } t_1 t_2 \text{ in } A\}$$

ein formelles Standardmodell von X_s. Weil die irreduziblen Komponenten vom Geschlecht größer 0 einer semistabilen Reduktion $(\tilde{X}_s)_{\mathfrak{W}}$ eindeutig durch X_s bestimmt sind und X_s Mumfordkurve ist, sind alle irreduziblen Komponenten von $(\tilde{X}_s)_{\mathfrak{W}}$ rationale Kurven (vgl. 1. 4 und 1. 6).

Für eine Ecke t von A sei Q_t der zu t assoziierte Teilbereich von $\mathbb{P}$. Sei W_t eine Zusammenhangskomponente von $\varphi_s^{-1}(Q_t)$. Weil das affine Modell $\tilde{W}_t$ irreduzibel, nichtsingulär und Zariski-offener, affiner Teil von $(\tilde{X}_s)_{\mathfrak{W}}$ ist, ist $\tilde{W}_t$ eine affine, rationale, nichtsinguläre Kurve. Folglich ist W_t isomorph zu einem affinoiden Teilbereich von $\mathbb{P}$. Nun gibt es ein W_t mit $W_t \subset U$.

Wir bezeichnen mit $\mathring{Q}_{tt'}$ die formelle Faser, die zu der Kante tt' in A gehört. Sei G der Koinzidenzgraph zu $(\tilde{X}_s)_{\mathfrak{W}}$ und a die zu $\tilde{W}_t$ assoziierte Ecke in G. Die formelle Faser U_{ab} zu einer Kante ab in G überlagert ein $\mathring{Q}_{tt'}$. Nach Konstruktion ist $\varphi\colon U_{ab} \to Q_{tt'}$ unverzweigt, also ist U_{ab} isomorph zu einem offenen Kreisring. Weiterhin enthält U jeweils nur eine Seite des Kreisringes U_{ab}, eben genau den, der an W_t grenzt, folglich gilt $U \subset W_t \cup \bigcup_b U_{ab}$, wobei die disjunkte Vereinigung über alle Kanten in G mit Endpunkt a läuft. Weil W_t isomorph zu einem Teilbereich im $\mathbb{P}_1$ ist, gibt es $W_t' \subset U$, das auch isomorph zu einem Teilbereich im $\mathbb{P}_1$ ist, so daß $W_t \Subset W_t'$ im Innern von W_t liegt. Dann ist $W_t' - W_t \subset \bigcup_b U_{ab}$, also ist U isomorph zu einem Raum, der aus W_t' durch Verklebung mit Kreisringen über den Rändern von W_t' entsteht. Dann ist auch U isomorph zu einem affinoiden Teilbereich von $\mathbb{P}_1$ ([8], IV, § 2).

3.6 Für $t \neq t'$ mit $P_t \cap P_{t'} \neq \emptyset$ hat man folgende Beschreibung für

$$P_t \cap P_{t'} = \{(s, z) \in V \times \mathbb{P} : |r^-(s)| \leqq |z - a(s)| \leqq |r^+(s)|\},$$

wobei $a, r^-, r^+ \in \mathcal{O}(S)$ global definiert sind und über V gilt

$$0 < |r^-(s)| < |r^+(s)| \quad \text{für} \quad s \in V.$$

Ohne Einschränkung seien $a \equiv 0$ und $r^+ \equiv 1$. Für jedes $\varphi : X_k \to P_t$ betrachten wir die Einschränkung von φ auf die Ränder $P_t \cap P_{t'}$ von P_t. Sei U eine Zusammenhangskomponente von $\varphi^{-1}(P_t \cap P_{t'})$. Zunächst studieren wir die endliche Überlagerung

$$\varphi\colon U \to P := P_t \cap P_{t'}, \quad n = \operatorname{grad} \varphi.$$

a) Nach Konstruktion ist $\varphi | U$ unverzweigt.

b) Es gibt einen globalen Schnitt $\zeta: S \to \mathbb{P}_s$ mit $\zeta(V) \subset P$, weil die P beschreibenden Funktionen global auf S definiert sind. Dann gibt es eine endliche Überlagerung $S' \to S$, so daß das Urbild $\varphi^{-1}(\zeta(S))$ über S' vollständig in Schnitte $\xi_1, \ldots, \xi_n: S' \to X' = X \times_S S'$ zu $\pi' = \pi \times_S S'$ zerfällt. Im folgenden nehmen wir ohne Einschränkung $S' = S$ und als normal an. Wir zeigen nun, daß $U_s := U \cap \pi^{-1}(s)$ zusammenhängend ist!

Sei $R = \{z \in \mathbb{P}^1: |z| = 1\}$, dann ist $V \times R \subset P$. Ist U zusammenhängend, so ist auch $U' = \varphi^{-1}(V \times R)$ zusammenhängend! Betrachten wir nämlich ein primitives Element ϑ der Erweiterung $\mathcal{O}(U) \supset \mathcal{O}(P)$, und sei $Q(T) \in \mathcal{O}(P)[T]$ das Minimalpolynom von ϑ. Wenn U' nicht zusammenhängend wäre, so würde $Q(T) | V \times R$ in zwei Faktoren $Q_1(T) \cdot Q_2(T)$ sich nichttrivial zerlegen lassen. Die Koeffizienten $q \in \mathcal{O}(V \times R)$ von $Q_i(T)$ sind Laurentreihen

$$q = \sum_{\nu \in \mathbb{Z}} q_\nu Z^\nu \in \mathcal{O}(V) \langle Z, Z^{-1} \rangle.$$

Weil $U_s \to P_s$ unverzweigt ist, also Potenzüberlagerung ist, sind die Zusammenhangskomponenten von U_s' gerade die von U_s auf U_s' geschnittenen Komponenten. Folglich zerfällt die Spezialisierung $Q(s)(T)$ in das Produkt der Spezialisierungen

$$Q(s)(T) = Q_1(s)(T) \cdot Q_2(s)(T).$$

Also konvergieren die Laurentreihen q für jeden Koeffizienten sogar auf P. Das bedeutet aber eine Zerlegung von $Q(T)$ im Widerspruch zur Irreduzibilität von U.

Im folgenden setzen wir zunächst einmal $P = S \times R$. Vorab wollen wir den Spezialfall behandeln, daß $\tilde{S}$ regulär ist. In diesem Fall ist jede formelle Faser von

$$\tilde{S} \times \tilde{R} = \widetilde{S \times R}$$

isomorph zu einem offenen Polyzylinder. Nach Satz 2 gestattet ein Polyzylinder nur die trivialen unverzweigten Überlagerungen, folglich ist $\tilde{\pi}: \tilde{U} \to \tilde{S} \times \tilde{R}$ unverzweigt. Weil $\tilde{\pi}$ Schnitte gestattet, ist generisch $\tilde{U}_\eta \cong \tilde{S}_\eta \times \tilde{R}$ bzw. eine disjunkte Vereinigung von solchen Räumen, und $\tilde{\pi}_\eta$ ist eine Potenzüberlagerung. Weil $\tilde{S}$ regulär ist und ein Schnitt über $\tilde{S}$ zu $\tilde{\pi}$ existiert, gilt $\tilde{U} \cong \tilde{S} \times \tilde{R}$ bzw. eine disjunkte Vereinigung von solchen Exemplaren. Dann ist aber auch $U \cong S \times R$, weil U zusammenhängend ist, und es gibt eine Uniformisierende T auf U mit $T^n = \varphi^* Z$.

Um zu zeigen, daß U_s zusammenhängend ist, genügt es, den Fall zu betrachten, daß S eine nichtsinguläre affinoide Kurve ist. Dann genügt es nach dem Existenzsatz für formelle Standardmodelle aus [17], die beiden Fälle zu behandeln: 1) $\tilde{S}$ ist regulär, 2) $S = \{\varepsilon \leqq |z| \leqq 1\}$ ist ein Kreisring.

Den 1. Fall hatten wir schon erledigt.

Im 2. Fall setze $S_\varepsilon = \{|z| = \varepsilon\}$ und $U_\varepsilon = U | S_\varepsilon$. Nach dem 1. Fall ist $U_\varepsilon \cong S_\varepsilon \times R$ bzw. disjunkte Vereinigung von solchen Exemplaren. Verheftet man U über U_ε mit einer entsprechenden Anzahl von Exemplaren $E_\varepsilon \times R$, wobei $E_\varepsilon = \{|z| \leqq \varepsilon\}$ ist, so erhält man eine unverzweigte Überlagerung $U_1 \to E_1 \times R$, auf die wir wieder unsere Aussage im 1. Fall anwenden können.

c) Mit den Ausführungen in b) läßt sich nun leicht folgern, daß im Fall $P = S \times R$ gilt:

1) Es gibt $f \in \mathcal{O}(S)$ mit $|f| = 1$, derart daß $\tilde{f}$ ein Nichtnullteiler in $\mathcal{O}(S)$ ist, so daß $U_{\tilde{f}} \cong S_{\tilde{f}} \times R$ isomorph ist, wobei $S_{\tilde{f}} = \{s \in S : |f(s)| = 1\}$ ist.

2) Für jede 1-dimensionale, analytische Teilmenge $C \subset S$ ist $U_C \cong C \times R$ mit $U_C := \pi^{-1}(C)$ isomorph.

Wir bezeichnen in 1) die Uniformisierende mit T und normieren sie durch die Bedingungen $T^n = \varphi^* Z$ und $T(\xi_1(s)) = 1$. Durch Betrachtung der affinen Modelle findet man eine endliche Überlagerung

$$\psi : S \to E^m \quad \text{mit} \quad \psi^{-1}(E^{m-1} \times \partial E^1) \subset S_{\tilde{f}}.$$

Sei $e_1, \ldots, e_N \in \mathcal{O}(U)$ eine Basis der Quotientenkörpererweiterung von

$$A := \mathcal{O}(E^m \times R) \to \mathcal{O}(U),$$

die durch r induziert wird; weiter sei $0 \neq \omega \in A$ mit

$$\omega \mathcal{O}(U) \subset A e_1 \oplus \cdots \oplus A e_N.$$

Dann gibt es eine eindeutige Darstellung

$$\omega T = a_1 e_1 + \cdots + a_N e_N,$$

wobei a_i holomorph auf $E^{m-1} \times \partial E^1 \times R$ sind. Bezeichnet man mit

$$(x' = (x_1, \ldots, x_{m-1}), x_m)$$

die Koordinaten auf E^m, so ist $C := \psi^{-1}(\{x' = c\})$ für jedes $c \in E^{m-1}$ eine 1-dimensionale, analytische Menge in S. Nach 2) ist $U_C \cong C \times R$. Das zeigt, daß man die Einschränkung von T auf $U_C \cap U_{\tilde{f}}$ als holomorphe Funktion nach U_C ausdehnen kann. Folglich kann man für eine dichte Menge D in E^{m-1} die Beschränkung der Koeffizienten $a_i | (\{d\} \times \partial E^1)$ für $d \in D$ nach $\{d\} \times E^1$ fortsetzen. Eine elementare Rechnung mit Laurentreihen zeigt dann, daß $a_i \in \mathcal{O}(E^m)$ holomorph auf E^m sind, also auch $\omega T \in \mathcal{O}(U)$ holomorph ist. Man verifiziert sofort $T \in \mathcal{O}(U)$ und damit

$$U \cong S \times R.$$

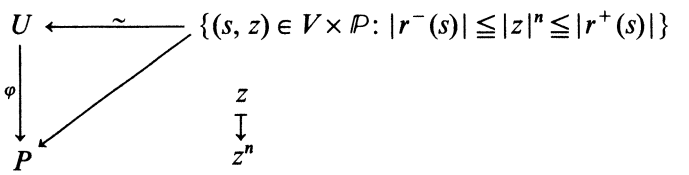
d) Betrachten wir nun ein allgemeines P, wie es in unserem Fall gegeben ist. Jetzt sei $e_1, \ldots, e_n \in \mathcal{O}(U)$ eine Basis der Quotientenkörpererweiterung zu $\varphi^* : \mathcal{O}(P) \to \mathcal{O}(U)$ und $0 \neq \omega \in \mathcal{O}(P)$, so daß

$$\omega \mathcal{O}(U) \subset \mathcal{O}(P)\, e_1 \oplus \cdots \oplus \mathcal{O}(P)\, e_n.$$

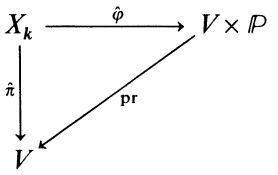
Ist T die Uniformisierende auf $\varphi^{-1}(V \times R)$ mit $T^n = \varphi^* Z$ und $T(\xi_1(s)) = 1$, so gilt

$$\omega T = a_1 e_1 + \cdots + a_n e_n,$$

wobei $a_i \in \mathcal{O}(V \times R)$ ist. Nun konvergiert für jedes $s \in V$ die Laurentreihe zu $a_i(s)$ auf P_s, weil T_s sich auf U_s ausdehnen läßt; somit sind alle Koeffizienten $a_i \in \mathcal{O}(P)$. Damit ist $T \in \mathcal{O}(U)$ und die Überlagerung φ läßt sich darstellen in der Form:



3.7 Betrachten wir nun die Überlagerung $\varphi: X_k \to P_t$. Mit Hilfe der genauen Beschreibung der Überlagerung φ über den Rändern von P_t gemäß 3. 6 d) kann man φ über den Rändern von P_t mit geeigneten Potenzüberlagerungen verheften. So erhält man einen Raum $\hat{X}_k$, eine endliche Überlagerung $\hat{\varphi}$, die φ fortsetzt, und somit einen eigentlichen, flachen Morphismus $\hat{\pi} = \mathrm{pr} \circ \hat{\varphi}$:



Mit 3. 5 folgt, daß die Fasern von $\hat{\pi}$ nichtsinguläre, rationale Kurven sind. Weil $\hat{\pi}$ Schnitte gestattet, ist $\hat{\pi}$ ein lokal-triviales $\mathbb{P}_1$-Bündel. Ohne Einschränkung sei $\hat{X}_k \cong V \times \mathbb{P}_1$. Daraus gewinnt man sofort die Beschreibung von X_k in der gewünschten Form. Die Behauptung in Satz 4 über die Beschreibung der Durchschnitte $X_k \cap X_l$ folgt aus 3. 6 d).

4. Uniformisierung analytischer Familien

Satz 5. *Sei S ein reduzierter, affinoider Raum, $\pi: X \to S$ eine analytische Familie von Mumfordkurven vom Geschlecht g. Im Fall* $\mathrm{char}\,\tilde{k} = p > 0$ *sei $g+1 < p$. Dann gibt es lokal bezüglich der holomorphen Topologie von S endliche Überlagerungen $S' \to S$ und endliche Überdeckungen $\{S'_1, \ldots, S'_r\}$ von S' durch rationale Teilbereiche S'_ρ von S', so daß sich $X'_\rho := X' \times_S S'_\rho$ parametrisiert uniformisieren läßt:*

Für $\rho = 1, \ldots, r$ gibt es

$$\Omega_\rho \subset \mathbb{P}_{S'_\rho} \text{ offene analytische Unterräume,}$$

$$\Omega_\rho \to X'_\rho \quad S'_\rho\text{-Morphismus,}$$

$$\gamma_1^\rho, \ldots, \gamma_g^\rho \colon S'_\rho \to \mathrm{PGL}(2, k) \text{ holomorph,}$$

so daß $\Gamma_\rho := \langle \gamma_1^\rho, \ldots, \gamma_g^\rho \rangle$ in kanonischer Weise holomorph und eigentlich auf Ω_ρ als Gruppe von Decktransformationen zu $\Omega_\rho \to X'_\rho$ operiert und der S'_ρ-Morphismus $\Omega_\rho \to X'_\rho$ in die S'_ρ-Isomorphie

$$\Omega_\rho / \Gamma_\rho \xrightarrow{\sim} X'_\rho$$

übergeht. Für jedes $s \in S'_\rho$ ist $\Omega_\rho(s)$ die Menge der gewöhnlichen Punkte von $\Gamma_\rho(s)$.

Weiter kann man $\gamma_1^\rho, \ldots, \gamma_g^\rho$ noch so wählen, daß $(\gamma_1^\rho(s), \ldots, \gamma_g^\rho(s))$ für jedes $s \in S'_\rho$ eine geometrische Basis (vgl. 1. 2) *von $\Gamma_\rho(s)$ ist.*

Nach Definition operiert Γ_ρ *eigentlich* auf Ω_ρ, wenn es eine endliche Vereinigung von affinoiden Bereichen $U_\rho \subset \Omega_\rho$ gibt, so daß die Translate von U_ρ unter Γ_ρ ganz Ω_ρ überdecken und $U_\rho \cap \gamma U_\rho \neq \emptyset$ nur für endlich viele $\gamma \in \Gamma_\rho$ gilt.

Bezeichnet $\mathfrak{M}_g$ den Modulraum aller nichtsingulären Kurven vom Geschlecht g über k, so ist $\mathfrak{M}_g$ ein quasi-projektives Schema und trägt damit eine wohlbestimmte holomorphe Struktur. Für jede analytische Familie $\pi: X \to S$ ist die kanonische Abbildung $\hat{\pi}: S \to \mathfrak{M}_g$, die $s \in S$ die Isomorphieklasse der Faser X_s zuordnet, nach den GAGA-Sätzen im relativen Fall holomorph. Wir bezeichnen mit $\mathscr{M}_g \subset \mathfrak{M}_g$ die Menge der Mumfordkurven. Nach [5] läßt sich $\mathscr{M}_g$ im Fall $g \geqq 2$ als mengentheoretischer Quotient des Raumes $\mathscr{B}_g \subset k^{3g-3}$ der Isomorphieklassen geometrischer Basen von Schottkygruppen darstellen. Man kann zeigen, daß über $\mathscr{B}_g$ eine universelle Familie von Kurven existiert. Also ist die kanonische Abbildung

$$\varkappa: \mathscr{B}_g \to \mathscr{M}_g \subset \mathfrak{M}_g$$

holomorph. Die Fasern von $\varkappa$ bestehen aus einer positiven, endlichen Anzahl von Punkten. Nach [5] ist $\mathscr{M}_g$ eine disjunkte Vereinigung von Teilmengen $\mathscr{M}_g(G)$, wobei G eine gewisse endliche Menge von Graphen durchläuft. Die Anzahl der Punkte in einer Faser von $\varkappa$ über $\mathscr{M}_g(G)$ ist gerade die Ordnung der Automorphismengruppe des Graphen G. Im allgemeinen ist die Ordnung der $\varkappa$-Fasern nicht konstant.

Korollar. *Die kanonische Abbildung $\hat{\pi}: S \to \mathscr{M}_g$ gestattet lokal bezüglich der holomorphen Topologie von S endliche Überlagerungen $S' \to S$, so daß lokal bezüglich der holomorphen Topologie von S' holomorphe Liftungen zu $\hat{\pi}$ existieren.*

$$\begin{array}{ccc} S'_\rho & \xrightarrow{\hat{\pi}_\rho} & \mathscr{B}_g \\ \downarrow & & \downarrow \varkappa \\ S & \xrightarrow{\hat{\pi}} & \mathscr{M}_g. \end{array}$$

Der *Beweis* zum Korollar folgt aus Satz 5 mit dem folgenden Lemma.

Lemma. *Sei S ein normaler affinoider Raum, $\gamma: S \to \mathrm{PGL}(2, k)$ holomorph und für jedes $s \in S$ sei $\gamma(s)$ hyperbolisch. Dann sind die Funktionen $x, y: S \to \mathbb{P}_1$, die s den anziehenden Fixpunkt $x(s)$ bzw. den abstoßenden Fixpunkt $y(s)$ von $\gamma(s)$ zuordnen, und $t: S \to k^\times$, die s den Multiplikator $t(s)$ von $\gamma(s)$ zuordnet, holomorph.*

Beweis. Sei $S' = \{(s, z) \in S \times \mathbb{P}_1;\ z = \gamma(s)(z)\}$. Dann ist $S' \to S$ eine 2-blättrige unverzweigte Überlagerung von S. Sei nun $T \subset S'$ eine irreduzible Komponente. Die Projektion $Z_1: T \to \mathbb{P}_1$ ist ein Morphismus. Sei $\varphi: T \to S$ die Projektion auf S. Weil T über S wegen $\operatorname{grad} \varphi \leqq 2$ galoissch ist, ist auch $Z_2: T \to \mathbb{P}_1$, der $t \in T$ auf den von $Z_1(t)$ verschiedenen Fixpunkt von $\gamma(\varphi(t))$ abbildet, ein Morphismus. Ohne Einschränkung sei nun $Z_1 \equiv 0$ und $Z_2 \equiv \infty$. Dann definieren wir

$$M = \{(t, z) \in T \times \mathbb{P}_1,\ z = 1\}.$$

Die Abbildung γ definiert einen Morphismus

$$\hat{\gamma}: T \times \mathbb{P}_1 \to T \times \mathbb{P}_1,\ (t, z) \mapsto (t, \gamma(\varphi(t))(z)).$$

Dann ist $\hat{\gamma}(M) \subset T \times \mathbb{P}_1$ eine analytische Teilmenge mit

$$\hat{\gamma}(M) \cap \{(t, z) \in T \times \mathbb{P}_1, |z| = 1\} = \emptyset,$$

weil $\gamma(s)$ für jedes $s \in S$ hyperbolisch ist. Also sind

$$M_1 := \hat{\gamma}(M) \cap \{(t, z);\ |z| \leqq 1\},$$
$$M_2 := \hat{\gamma}(M) \cap \{(t, z);\ |z| \geqq 1\}$$

analytische Teilmengen von $T \times \mathbb{P}_1$. Sei nun $\pi: T \times \mathbb{P}_1 \to T$ die Projektion. Weil T irreduzibel ist, ist $\pi(M_1) = T$ im Fall $M_1 \neq \emptyset$, also ist dann auch $M_2 = \emptyset$, oder aber es liegt der umgekehrte Fall vor. Somit ist $Z_1(t)$ entweder stets der anziehende oder stets der abstoßende Fixpunkt von $\gamma(\varphi(t))$. Folglich ist $\operatorname{grad} \varphi = 1$ und somit $T = S$, weil S normal ist. Dann sind $x, y: S \to \mathbb{P}_1$ holomorph. Normiert man nun $x \equiv 0$ und $y \equiv \infty$, so ist $t(s) = \gamma(s)$ (1) offenbar auch holomorph.

In den folgenden Abschnitten wird der Beweis zu Satz 5 erbracht.

4.0 Sei $\mathscr{L}$ ein holomorphes Geradenbündel auf X, so daß die Einschränkungen $\mathscr{L}(s)$ auf die Fasern X_s sehr ampel sind. Im Fall $g \geqq 2$ kann man $\mathscr{L} = \omega_{X/S}^{\otimes 3}$ wählen, im Fall $g = 1$ findet man lokal bezüglich der holomorphen Topologie Divisoren $D \subset X | S'$, die endlich über S' liegen, hier kann man $\mathscr{L} = \mathscr{O}(3D)$ wählen. Dann ist $\mathscr{E} = \pi_* \mathscr{L}$ ein holomorphes Vektorraumbündel auf S, und man kann $X \to \mathbb{P}(\mathscr{E})$ einbetten. Lokal bezüglich der holomorphen Topologie von S ist $\mathscr{E}$ trivial, ohne Einschränkung sei $\mathscr{E} = \mathscr{O}_S^{N+1}$, also $\mathbb{P}(\mathscr{E}) = \mathbb{P}_S^N$. Zu $x \in X$ und $s = \pi(x)$ gibt es eine Hyperebene $H \subset \mathbb{P}_S^N$ mit $x \in H$, so daß $X_s \cap H$ eine endliche Punktmenge ist. Dann existiert eine Zariski-offene Teilmenge $s \in T \subset S$, so daß $X_T \cap H =: T'$ endlich über T liegt. Nach Basiserweiterung mit T' besitzt $X' = X \times_S T'$ den tautologischen Schnitt. Auf diese Weise konstruiert man nun lokal bezüglich der holomorphen Topologie von S endliche Überlagerungen S', so daß $X' := X \times_S S'$ Schnitte über S' gestattet.

4.1 Sei nun $\sigma: S \to X$ ein Schnitt zu π. Nach den GAGA-Sätzen im relativen Fall [13] ist $X \to S$ ein algebraischer Morphismus. Nach dem Satz von Riemann-Roch gibt es generisch eine nichtkonstante S_η-meromorphe Funktion m auf der nichtsingulären Kurve X_η über S_η, die höchstens in $\sigma(\eta)$ einen Pol der Ordnung $\leqq g+1$ hat. Dann definiert m über einem Zariski-offenen Teil $S^* \subset S$ von S eine endliche Überlagerung $\varphi: X^* \to \mathbb{P}_{S^*}$ mit $\operatorname{grad} \varphi \leqq g+1$, wobei $X^* = X \times_S S^*$ ist.

Daraus folgert man nun mit Satz 4 durch Induktion über $\dim S$ wie in 3. 1, daß es lokal bezüglich der holomorphen Topologie von S endliche Überlagerungen $S' \to S$ gibt, so daß $X' = X \times_S S'$ nach Basiserweiterung mit S' eine Zerschneidung $\{X'_{ih}\}$ gestattet, wie sie in Satz 4 angegeben ist.

4.2 Sei $\pi: X \to S$ eine Familie von Mumfordkurven, die global über S schon eine Zerschneidung $\{X_1, \ldots, X_n\}$ wie in Satz 4 gestattet. Dann definieren wir den folgenden Graphen G, der als Ecken $X_1, \ldots, X_n$ und als verbindende Kanten der Ecken X_i und X_j die Zusammenhangskomponenten von $X_i \cap X_j$ hat. Sei $\lambda: T \to G$ die graphentheoretische universelle Überlagerung von G. Mittels T definieren wir den analytischen Raum

$$\Omega = \bigcup_{t \in T} X_t / \sim .$$

Für eine Ecke $t \in T$ sei dabei X_t isomorph zu dem Glied $X_{\lambda(t)}$ der Zerschneidung $\{X_1, \ldots, X_n\}$ von X, das der Ecke $\lambda(t)$ in G entspricht; wenn t und t' in T benachbart sind, so sind X_t und $X_{t'}$ über der Zusammenhangskomponente von $X_{\lambda(t)} \cap X_{\lambda(t')}$ zu verheften, die der Kante $\lambda(tt')$ in G entspricht. Sei nun $t_0 \in T$ fest gewählt:

$$T_n := \{t \in T: \text{Abstand } (t, t_0) \leqq n\}, \quad \Omega_n := \bigcup_{t \in T_n} X_t.$$

Weiterhin wählen wir drei holomorphe Schnitte mit

$$c_0, c_1, c_\infty : S \to \Omega_0, \quad c_i(s) \neq c_j(s) \quad \text{für} \quad i \neq j \quad \text{und alle} \quad s \in S.$$

Weil die Ränder von Ω_n parametrisierte Kreisringe über S sind, läßt sich Ω_n über allen Rändern mit über S parametrisierten Kreisen zu einem eigentlichen, flachen Morphismus

$$\hat{\pi}_n : \hat{\Omega}_n \to S$$

verheften. Die punktale Rechnung über $s \in S$ zeigt, daß $\hat{\pi}_n$ ein $\mathbb{P}_1$-Bündel ist (vgl. [8], p. 144 ff). Weil $\hat{\pi}_n$ drei sich nicht kreuzende Schnitte gestattet, gibt es genau einen S-Isomorphismus $\hat{F}_n : \hat{\Omega}_n \to \mathbb{P}_S$, der den Schnitt c_i auf i für $i = 0,1, \infty$ wirft. Sei nun

$$F_n := \hat{F}_n | \Omega_n : \Omega_n \to \mathbb{P}_S, \quad F_n(c_i(s)) = (s, i) \quad \text{für} \quad i = 0, 1, \infty.$$

Nach dem Maximumsprinzip gibt es eine Konstante $K < 1$, so daß K größer als die Formfaktoren aller in der Parametrisierung der bei $X_i \cap X_j$ auftretenden Kreisringe ist. Wie im Beweis zu Satz 1 zeigt man:

Ist $f \in \mathcal{O}(\Omega_m)$ mit $f(c_1) = 0$ und $|f|\Omega_m| \leqq 1$, so folgt $|f|\Omega_n| \leqq K^{m-n}$ für $m \geqq n$.

Damit folgt nun:

Die Folge (F_n) konvergiert gegen eine offene Einbettung

$$F := \lim F_n : \Omega \to \mathbb{P}_S.$$

Beweis. Für $i \geqq j \geqq n$ sind F_j, $F_i|\Omega_j$ offene Einbettungen. Auf jedem Rand von Ω_j haben F_j bzw. $F_i|\Omega_j$ den Normgrad $+1$ oder -1. Weil $F_i \circ F_j^{-1}$ Punkte im Innern von $F_j(\Omega_j)$ festläßt, sind F_i und F_j gleichorientiert, haben also auf jedem Rand von Ω_j denselben Normgrad. Also ist der Normgrad von F_i/F_j auf jedem Rand von Ω_j Null. Weil F_i/F_j nullstellenfrei und holomorph auf Ω_j ist, hat somit F_i/F_j wegen $F_i/F_j(c_1) = 1$ die Form $1 + f$ mit $f \in \mathcal{O}(\Omega_j)$ und $|f| < 1$ nach [6], § 1. Nun ist nach Obigem $|f|\Omega_n| \leqq K^{j-n}$. Somit konvergiert die Folge $(F_i|\Omega_n)_{i \geqq n}$ gegen eine offene Einbettung $F|\Omega_n : \Omega_n \to \mathbb{P}_S$ für jedes $n \in \mathbb{N}$.

Im folgenden identifizieren wir Ω mit seinem Bild $F(\Omega) \subset \mathbb{P}_S$.

4. 3 Jede graphentheoretische Decktransformation σ von $T \to G$ induziert einen S-Automorphismus $\hat{\sigma}$ von Ω, der Decktransformation der kanonischen Abbildung $\Omega \to X$ ist. Weil jede Faser Ω_s nach Satz 1 nur die konstanten Funktionen als beschränkte holomorphe Funktionen hat, läßt sich $\hat{\sigma}$ nach [6], § 2 in jeder Faser zu einem Automorphismus γ_s von $\mathbb{P}$ ausdehnen. Somit ist γ_s die Möbiustransformation, die (s, i) auf $F(\hat{\sigma}_s(i))$ für $i = 0, 1, \infty$ wirft. Für jedes $\sigma \in \mathrm{Deck}\,(T/G)$ ist $\gamma : S \to \mathrm{PGL}\,(2, k)$ holomorph.

4. 4 Aus dem in 4. 2 definierten Graphen G erhält man den kanonischen Graphen der Faser X_s folgendermaßen: Für jede Ecke X_i in G ist $X_i(s) \subset \mathbb{P}$ ein zusammenhängender Teilbereich. Jedem $X_i(s)$ ist somit via Standardmodell ein Baum zugeordnet, dessen Endpunkte genau den Kanten in G entsprechen, die in X_i enden. Nun füge man diesen Baum für die Ecke X_i in G ein, indem man seine Endpunkte mit den entsprechenden Kanten verbindet. Die Reduktion dieses Graphen ist der kanonische Graph von X_s. Es ist klar, daß bei diesem Prozeß für verschiedene $s, s' \in S$ auch verschiedene Graphen entstehen können. Nach [5] ist nun bekannt, für welche Systeme $(\sigma_1, \ldots, \sigma_g)$ in $\mathrm{Deck}(T/G)$ die assoziierten Systeme $(\gamma_1, \ldots, \gamma_g)$ in $\mathrm{PGL}(2, \mathcal{O}(S))$ eine geometrische Basis bilden und wie man die Konfiguration der zugehörigen (parametrisierten) Kreise findet.

Wie im punktalen Fall folgt nun die Behauptung des Satzes. Siehe [8], p. 105.

5. Starrheit algebraischer Familien

Eine *algebraische Familie von Mumfordkurven* $\pi: X \to S$ ist ein eigentlicher, flacher Morphismus zusammenhängender Schemata von endlichem Typ über k, wobei alle Fasern X_s Mumfordkurven sind.

Satz 6. *Jede algebraische Familie $\pi: X \to S$ von Mumfordkurven vom Geschlecht g ist konstant, falls $g+1 < p = \mathrm{char}\, \tilde{k}$ gilt.*

Eine Familie von Kurven heißt konstant, wenn die durch π induzierte kanonische Abbildung von S in den Modulraum $\mathfrak{M}_g$ aller Kurven konstant ist.

Beweis. In der Einleitung hatten wir schon den Beweis für den Fall $g = 1$ gegeben. Somit können wir uns jetzt auf den Fall $g \geqq 2$ beschränken. Es genügt, die Behauptung für affin-algebraische, nichtsinguläre Kurven S zu zeigen.

Für eine algebraische Familie zeigt man wie in 4. 0 und 4. 1, daß es über einem Zariski-offenen Teil $S_1 \subset S$ eine endliche Überlagerung durch eine affin-algebraische Kurve S_2 gibt, so daß eine rationale Funktion auf $X_2 = X \times_S S_2$ existiert, die eine algebraische Überlagerung

$$\varphi : X_2 = X \times_S S_2 \to \mathbb{P}_{S_2} \qquad \text{mit} \qquad \mathrm{grad}\, \varphi \leqq g+1 < p$$

liefert. Ohne Einschränkung zerfällt die Verzweigungsmenge von φ über S_2 in Divisoren vom Grad 1. Es gilt also

$$\mathrm{Verzweigung}(\varphi) = \tau_1(S_2) \cup \cdots \cup \tau_n(S_2).$$

Dabei sind $\tau_\nu: S_2 \to \mathbb{P}_{S_2}$ für $\nu = 1, \ldots, n$ reguläre, algebraische Schnitte, die sich auch ohne Einschränkung nicht überkreuzen. Dann existiert eine endliche, algebraische Überlagerung durch eine affin-algebraische Kurve $S' \to S_2$, so daß sich $X' = X \times_S S'$ lokal bezüglich der holomorphen Topologie von S' wie in Satz 4 angegeben zerschneiden läßt.

Der *Beweis* dieser Aussage erfolgt mit dem Beweis zu Satz 4. In dieser Situation muß man die Basis S_2 nur noch wie in 3. 6 b) für die Existenz von gewissen Schnitten zu $\pi_2 = \pi \times_S S_2$ erweitern, die das volle Urbild $\varphi^{-1}(\zeta(S_2))$ beschreiben. Dabei durchläuft ζ eine endliche Familie von algebraischen Schnitten zu $\mathbb{P}_{S_2} \to S_2$, nämlich alle möglichen algebraischen Schnitte τ_j' (vgl. 3. 4), die als Radien in 3. 6 auftreten. Anhand

der Konstruktion in Abschnitt 3 überlegt man sich, daß man nur endlich viele zu betrachten hat, weil man die algebraischen Schnitte $\tau_\nu: S_2 \to \mathbb{P}_{S_2}$ auf die komplette algebraische Kurve $\hat{S}_2$ ausdehnen kann, wobei $\hat{S}_2$ die eindeutig bestimmte singularitätenfreie komplette Kurve ist, die S_2 enthält. Weil φ und alle ζ algebraisch sind, findet man die gewünschten Schnitte schon nach endlicher Basiserweiterung mit einer affin-algebraischen Kurve $S' \to S_2$.

Sei nun $\hat{S}'$ die eindeutig bestimmte, nichtsinguläre, komplette Kurve, die S' enthält. Weiter sei $\hat{\Omega} \to \hat{S}'$ die in 1. 8 definierte universelle Überlagerung, $\Omega \subset \hat{\Omega}$ das volle Urbild von S'. Nach Abschnitt 4 gibt es eine zulässige Überdeckung $\{\Omega_i\}$ von Ω und holomorphe Abbildungen

$$f_i: \Omega_i \to \mathscr{B}_g \subset k^{3g-3}$$

in den Raum der geometrischen Basen von Schottkygruppen, so daß $f_i(s)$ geometrische Basis einer Schottkygruppe $\Gamma(s)$ ist, deren assoziierte Kurve $\Omega_{\Gamma(s)}/\Gamma(s)$ isomorph zu der zu s gehörigen Faser von $X \to S$ ist. Weil $\hat{\Omega} - \Omega$ in $\hat{\Omega}$ diskret liegt, ist mit $\hat{\Omega}$ auch Ω einfach zusammenhängend bezüglich Überdeckungsketten in der holomorphen Topologie. Dann kann man die f_i aber auch so wählen, daß sie sich zu einer globalen Funktion $f: \Omega \to \mathscr{B}_g$ zusammensetzen.

Jetzt betrachten wir diejenigen Komponentenfunktionen von f, die gerade die Multiplikatoren der geometrischen Basis $f(s)$ beschreiben. Weil die Multiplikatoren Betrag kleiner 1 haben, sind diese Komponentenfunktionen beschränkte holomorphe Funktionen auf Ω. Dann lassen sie sich auf $\hat{\Omega}$ fortsetzen. Nach Satz 1 sind sie somit konstant.

Die übrigen Komponenten von f beschreiben das Fixpunktsystem der geometrischen Basis $f(s)$. Aus der Beschreibung von $\mathscr{B}_g$ in [5] geht hervor, daß die Fixpunkte einer geometrischen Basis mit festem Multiplikatorsystem auch nur beschränkt variieren kann. Also sind auch die übrigen Komponentenfunktionen von f beschränkt und damit konstant.

Damit ist Satz 6 im Fall $g+1<p$ bewiesen.

Analog beweist man mit Satz 4 oder folgert mittels GAGA-Sätzen aus Satz 6

Korollar. *Jede analytische Familie $\pi: X \to S$ von Mumfordkurven vom Geschlecht g über einer kompletten, algebraischen Kurve ist konstant, wenn $g+1<p=\operatorname{char} \tilde{k}$ ist.*

Beweis. Ohne Einschränkung sei $g \geqq 2$. Dann ist $\mathscr{E} = \pi_* \omega_{X/S}^{\otimes 3}$ ein holomorphes und somit nach den GAGA-Sätzen ein algebraisches Vektorraumbündel auf S, und X läßt sich in $X \to \mathbb{P}(\mathscr{E})$ einbetten. Weil $S \subset \mathbb{P}^r$ eine projektiv-algebraische Kurve ist, gibt es ein algebraisches Geradenbündel $\mathscr{L}$ auf S und einen Epimorphismus $\mathscr{L}^{n+1} \to \mathscr{E}^* \to 0$ auf das duale Bündel zu $\mathscr{E}$. Nun erhält man eine Einbettung

$$X \to \mathbb{P}(\mathscr{E}) = \mathbb{P}(\mathscr{E}^{**}) \to \mathbb{P}(\mathscr{L}^{n+1}) = \mathbb{P}_S^n \subset \mathbb{P}^r \times \mathbb{P}^n.$$

Dann ist X nach den GAGA-Sätzen projektiv-algebraisch, und π ist als analytischer Morphismus zwischen projektiv-algebraischen Varietäten auch algebraisch. Somit ist $X \to S$ eine algebraische Familie. Daraus folgt dann die Behauptung mit Satz 6.

Es ist klar, daß man ebenso analoge Sätze für Familien von hyperelliptischen Mumfordkurven im Fall $2 \neq \operatorname{char} k$ beweisen kann. Auch läßt sich im Fall $\operatorname{char} k = 0$ die Einschränkung $g+1<p$ mit Hilfe von Sätzen über Konkretisierung von Riemannschen Flächen etwas abschwächen. Prinzipiell sollten aber Satz 6 sowie das Korollar ohne zusätzliche Voraussetzungen über das Geschlecht der Familie gelten.

Ob wie im Fall analytischer Familien von Tate-Kurven auch analytische Familien von Mumford-Kurven über affin-algebraischen Kurven konstant sind, sollte man annehmen, aber es folgt nicht unmittelbar mit unserer Beweismethode.

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On the Jacobian conjecture and the configurations of roots

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Introduction

The well-known Jacobian conjecture states "let k be an algebraically closed field of characteristic zero and $f(x, y)$, $g(x, y)$ two polynomials over k. If

$$f_x g_y - f_y g_x = 1$$

then $k[x, y] = k[f, g]$." In other words a polynomial map from $\mathbb{A}^2$ to $\mathbb{A}^2$ is one-one and onto iff the Jacobian of the map is a nonzero constant. Clearly this conjecture is false if the characteristic p of the field k is positive. For instance $f = x + x^p$, $g = y$ furnish a counter example. Furthermore this conjecture is false in the analytic case. The pair $f = e^x$, $g = ye^{-x}$ furnishes a counter-example. For the history of this problem the reader is referred to [B-C-W.1] and [W.2]. Throughout this article the field k is assumed to be algebraically closed and of characteristic zero.

The treatment in the present article is rather classical. Let K be a valued field and $g(y)$ a polynomial which factors completely over K i.e.

$$g(y) = a_0 \prod (y - \tau_i)$$

where $\tau_i \in K$. Then we may associate the following configuration data of real numbers $\{V_{ij}\}$ with $g(y)$:

$$V_{ij} = |\tau_i - \tau_j|$$

where $|\ |$ denotes the value of the field K. In the case of complex numbers the above data and the leading coefficients a_0 uniquely determine the polynomial $g(y)$ up to the translations, the rotations and the reflection. In the case of a non-archimedean valued field K the above data give some weak but valuable information about the polynomial $g(y)$.

Let the valued field K be $k((t))$ the meromorphic field of one variable and the polynomial $g(y)$ be irreducible over K. Then it can be shown that the configuration data of $g(y)$ over its splitting field is equivalent to the classical data of the characteristic pairs of an algebroid curve.

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In general, given a polynomial $g(x, y) \in k[x, y]$, we may factor $g(x, y)$ completely over the Puiseux field $k\langle\langle t\rangle\rangle$ where $t = x^{-1}$ and $k\langle\langle t\rangle\rangle = \bigcup_{n=1}^{\infty} k((t^{\frac{1}{n}}))$. Note the field $k\langle\langle t\rangle\rangle$ is algebraically closed (cf. [W.1]). Similarly we define $k[\pi]\langle\langle t\rangle\rangle = \bigcup_{n=1}^{\infty} k[\pi]((t^{\frac{1}{n}}))$ with π a symbol. Let us assume that

$$\deg_{x,y} g(x, y) = \deg_y g(x, y).$$

Then it is easy to see that the configuration data determine the number of points of the plane curve $g(x, y) = 0$ at ∞ and the desingularization processes at those points.

Let K be a non-archimedian field and $\{\tau_1, \ldots, \tau_n\}$ n elements in K. It is more convenient to consider the following tree data (cf. § 1) of $\{\tau_1, \ldots, \tau_n\}$ in place of the configuration data. Let D_1 be the minimal disc which contains all τ_i. Let $\{D_{1,i_2}\}$ be the collection of the minimal subdiscs of D_1 with

(1) $$\bigcup D_{1,i_2} \supset \{\tau_i : \tau_i \in D_1\} \qquad (= \{\tau_1, \ldots, \tau_n\}),$$

(2) $$D_{1,i_2} \cap D_{1,j_2} = \emptyset \quad \text{if} \quad i_2 \neq j_2,$$

(3) the distance between D_{1,i_2} and D_{1,j_2} is the radius of D_1 if $i_2 \neq j_2$.

Due to the non-archimedean property of the field K the collection $\{D_{1,i_2}\}$ uniquely exists. We may repeat the above process for D_{1,i_2}. Finally we stop when all discs are with radii zero. In this way we obtain a tree of discs in K. Then the tree data consist of the radii and the numbers of τ_i's in those discs. It is easy to see this tree of numbers is the characteristic pairs for an algebroid curve $g(x, y) = 0$ if we set $K = k\langle\langle x\rangle\rangle$ and $\{\tau_1, \ldots, \tau_n\}$ the complete set of roots of $g(x, y) = 0$ over K.

The major portion of this article consists of the study of the tree data of two polynomials $f(x, y)$ and $g(x, y)$ satisfying the Jacobian condition

$$f_x g_y - f_y g_x = 1.$$

It can be illustrated that the Jacobian condition implies some similarities between the two tree data. Namely consider any disc $D \subset K = k\langle\langle t\rangle\rangle$ where $t = x^{-1}$ with

$$\min_{\tau \in D} \{\operatorname{ord} f(t^{-1}, \tau)\} \leqq 0.$$

Then the number of roots of $f(t^{-1}, y) = 0$ and $g(t^{-1}, y) = 0$ in D are proportional to the degrees of $f(x, y)$ and $g(x, y)$ if both numbers of roots are positive. This qualitative observation is not useful for our purpose. To obtain the useful quantitative properties of those similarities we have to adapt the concept of the approximate roots for the present task (cf. [M.1], [A-M.1]).

In § 2 we show that the Jacobian condition implies the existence of a finite sequence of polynomials monic in y $T_i^{\psi}(f(x, y), g(x, y)) \in k[f(x, y), g(x, y)]$ with y-degree $(-\mu_i)$ (cf. § 2). The polynomials T_i^{ψ} are specialized approximate roots of the defining equation of $f(x, y)$ and $g(x, y)$ over $k[x]$. This sequence $\{-\mu_i\}$ determines and is determined by the characteristic sequence of the rational curve parametrized by $(f(0, y), g(0, y))$ at $y = \infty$. Our analysis of the distribution of roots of $f(t^{-1}, y) = 0$ and $g(t^{-1}, y) = 0$ over $k\langle\langle t\rangle\rangle$ solely depends on the existence of the said polynomials T_i^{ψ}.

In § 3 we relate the sequence of polynomials T_i^{ψ} with the Jacobian condition for $f(x, y)$ and $g(x, y)$. It is easy to see that the task is to study the partial derivatives of $T_i^{\psi}(f, g)$ with respect to f. Proposition 3. 3 shows that if the similarity of the distribution of roots holds for $g(x, y)$, $T_1^{\psi}(f(x, y), g(x, y)), \ldots, T_r^{\psi}(f(x, y), g(x, y))$, then it holds for the extra polynomials $\frac{\partial}{\partial f} T_{r+1}^{\psi}(f(x, y), g(x, y))$. It furnishes an important step in our study. *From the next section on the Jacobian condition is assumed.*

In § 4 we study the basic splitting phenomena and nonsplitting phenomena. Inductively we locate a certain disc D in $k\langle\langle t\rangle\rangle$. The question is to determine if some roots τ_i, τ_j of $g(t^{-1}, y) \prod_{i=1}^{r} T_i^{\psi}(f(t^{-1}, y), g(t^{-1}, y))$ in D have a distance which equals the radius of D. If all distances are less than the radius of D, then the disc D is not one of the discs which determine the tree data of the said polynomial. In this case we have a nonsplitting phenomenon. To locate the tree data we have to further shrink the radius of the disc. Proposition 4. 4 establishes some sufficient numerical criteria for the nonsplitting phenomena. On the other hand if there are some roots τ_i, τ_j having a distance which equals the radius of D, then the disc D is clearly one of the discs which determine the tree data of the said polynomial. In this case we have a splitting phenomenon. Proposition 4. 6 establishes some sufficient numerical criteria for the splitting phenomenon and some information about the size of the disc D and the number of the subdiscs of D in the next step of the decomposition in the construction of the tree of discs. Later on we implicitly establish that the two numerical criteria of Propositions 4. 4 and 4. 6 cover all possibilities.

According to the number of roots in it, the disc D can be classified as either a major disc or a minor disc. A major disc is one which contains more than a predetermined number of roots (cf. Definition 4. 1). Otherwise it is a minor one. In § 5 we study the towers of major discs. Propositions 5. 1, 5. 2 and 5. 3 establish the existence and the critical numerical properties of them. Namely a new sequence $\{v_s, v_{s-1}, \ldots, v_2\}$ of restricted integers which are the relative numbers of roots in a tower of major discs $D_s \supset D_{s-1} \cdots \supset D_2 \supset D_1$ will determine the radii of those discs completely.

In the remaining part of § 5 we show various applications of Proposition 5. 3. Proposition 5. 4 establishes that in the case of $\delta_i > -1$ (i.e. there is only one point at ∞ for $f(x, y)$) there is a reduction of $\deg f(x, y)$ and $\deg g(x, y)$. Proposition 5. 5 establishes the similar result for the case of two characteristic pairs.

In § 6 we study those minor discs. In Proposition 6. 1 we show that in a minor disc the proportional property of the numbers of roots will be carried to its extreme limit. Then we compile the results in Propositions 5. 4 and 6. 1 to be the main theorem of this article. The other propositions in § 6 are dealing with the phenomena of reducing the degrees of the polynomials involved while increasing the Jacobian. This trade-off is sometimes very useful as illustrated in Appendix II.

In this article we make an effort to compare our results and our methods with the published one ([A.1]). The explicit references are included in various places.

In Appendix I we collect all needed propositions of the polynomial solutions of the differential equations involved. There is a simple computer program essentially based on the discussion at the end to determine the possible counterexamples to the Jacobian conjecture for all polynomials of degrees less than or equal to 100. In fact there are only very few possibilities. However the number of coefficients of those possible counterexamples range from 3370 to 7328. In Appendix II we give some theoretic arguments to reduce the number of coefficients to at most 12. Thus it becomes a computable problem to show that the Jacobian conjecture is valid for all polynomials of degrees less than or equal to 100. The direct computation is presented in Appendix II.

The computer program is a routine one and will be furnished to anyone upon request. It uses 147.148 sec of a CDC 6500 and costs only $2.41.

After many years of research we feel little information about the distribution of roots can be added to the present article. As we pointed out before, these distribution of roots correspond to the desingularization at ∞. The geometric approach of desingularization will probably add only little information. Our Appendix II shows that the Jacobian conjecture is quite plausible. Even if it is false a counterexample may require a large number of coefficients and is thus almost uncheckable. If the Jacobian conjecture is valid, the logical sequence to the present article is to show that a part of the data collected thus far is not permissible algebraically. The arguments in Appendix II implicitly imply that the tree data deduced thus far are permissible for two individual polynomials $f(x, y)$ and $g(x, y)$. This is a fact. The only hope, in our opinion, is to put all these tree data in the defining equation of $f(x, y)$ and $g(x, y)$ over $k[x]$ to establish their non-permissibility. However we have had limited success in this complicated task.

The materials of this article are compiled from our research during years 1971—1978. We wish to thank our friends for their encouragement. We wish to express our appreciation to Professor D. Drasin for reading carefully part of our manuscript.

§ 1. The roots configurations and the tree data

Let K be a non-archimedean valued field with value $|\ |$. Let $f(y)$ be a polynomial in $K[y]$ which factors completely with roots $\tau_1, \ldots, \tau_m$. Then the data

$$\{V_{ij} = |\tau_i - \tau_j| \colon\ i, j = 1, \ldots, m\}$$

determine the configuration of the roots of $f(y)$. Equivalently let τ_i be any one of the roots, consider the discs centered at τ_i with various radii. As the radius of the disc shrinks the other roots will gradually be left outside. When the number of roots inside the disc is just about to drop we record the number of roots inside the disc and its radius.

For a fixed τ_i, let $S_i = \{|\tau_i - \tau_j| : j = 1, \ldots, m\}$. We see that the set S_i is the set of these critical radii. Let the elements in S_i be ordered as $W_{i1} > W_{i2} > \cdots > W_{ih} > W_{ih+1} = 0$. Let

$$d_{ij} = \text{card } \{s : |\tau_i - \tau_s| \leqq W_{ij}\},$$

$$D_{ij} = \text{the disc centered at } \tau_i \text{ with radius } W_{ij}.$$

Then the numbers $\{d_{ij}, W_{ij}\}$ will be called the tree data of the polynomial $f(y)$ along its root τ_i.

We may define the tree data of the polynomial $f(y)$ as follows. Let D_1 be the smallest disc which contains all roots of $f(y)$ and let

$$d_1^* = \text{the number of roots in } D_1 = \deg f(y),$$

$$W_1^* = \text{the radius of } D_1.$$

Due to the non-archimedean property of the distance function $|\ |$, the disc D_1 uniquely exists.

Let us consider the partition of all roots of $f(y)$ induced by the following equivalence relation

$$\tau_i \sim \tau_j \Leftrightarrow |\tau_i - \tau_j| < W_1^*.$$

Using the strong triangle inequality it is a simple exercise to verify the above statement. Let the set R_1 of all roots of $f(y)$ be thus decomposed into disjoint union of subsets R_{1j}

$$R_1 = \bigcup R_{1j}.$$

For each R_{1j} let D_{1j} be the smallest disc which contains all elements in it and let

$$d_{1j}^* = \text{the number of roots in } D_{1j},$$

$$W_{1j}^* = \text{the radius of } D_{1j}.$$

Note that the distance between two roots τ_i and τ_j which belong to two distinct D_{1j}'s is W_1^*.

We may repeat the above discussion for each D_{1j} and cover the roots in it by a disjoint union of subdiscs D_{1jr} with the roots from different subdiscs having distances precisely W_{1j}^*. Thus we formulate the following definition.

Definition 1. 1. Let $f(y) \in K[y]$ with roots $\tau_1, \ldots, \tau_m$. The tree data of $f(y)$ consist of $\{d_{1,i_2,\ldots,i_s}^*, W_{1,i_2,\ldots,i_s}^*$ for some $s \leqq m\}$ where $d_{1,i_2,\ldots,i_s}^*$ are positive integers and $W_{1,i_2,\ldots,i_s}^*$ are numbers satisfying the following conditions:

1) $d_1^* = m$.

2) $\sum_j d_{1,i_2,\ldots,i_s,j}^* = d_{1,i_2,\ldots,i_s}^*$.

3) The smallest disc D_1 which contains all roots is of radius W_1^*.

4) For any subindices $1, i_2, \ldots, i_s$ there is a disc $D_{1,\ldots,i_s}$ with radius $W^*_{1,\ldots,i_s}$ which contains $d^*_{1,\ldots,i_s}$ roots of $f(y)$.

5) Any proper subdisc of $D_{1,\ldots,i_s}$ contains less than $d^*_{1,\ldots,i_s}$ roots of $f(Y)$.

6) $D_{1,\ldots,i_s,j} \subsetneqq D_{1,\ldots,i_s}$.

7) Any two roots of $f(y)$ in two $D_{1,\ldots,i_s,j}$'s with distinct last indices have distance $W^*_{1,\ldots,i_s}$.

8) The minimal discs $D_{i_1,\ldots,i_s}$ are with radii zero.

In fact, it is elementary to establish the following proposition. Since it will not be used in this article the proof is omitted.

Proposition 1. 1. *Given a polynomial $f(y)$ which factors completely over a non-archimedean valued field K. Then the roots configuration $V_{ij} = \{|\tau_i - \tau_j| : \tau_i, \tau_j$ roots of $f(y)\}$ and the tree data $\{d^*_{1,\ldots,i_s}, W^*_{1,\ldots,i_s}\}$ determine each other.*

Definition 1. 2. Given a polynomial $f(y) \in K[y]$ with roots $\tau_1, \ldots, \tau_m$. Let the tree data of $f(y)$ be $\{d^*_{1,i_2,\ldots,i_s}, W^*_{1,i_2,\ldots,i_s}\}$ and the associated discs be $\{D_{1,i_2,\ldots,i_s}\}$. Let τ_i be any root of $f(y)$ and $D_1 \supset D_{1,j_2} \supset \cdots \supset D_{1,j_2,\ldots,j_{h+1}} \ni \tau_i$ be the tower of associated discs containing τ_i. Then the tree data along τ_i are

$$\{d^*_1, d^*_{1,j_2}, \ldots, d^*_{1,j_2,\ldots,j_{h+1}};\quad W^*_1, W^*_{1,j_2}, \ldots, W^*_{1,j_2,\ldots,j_{h+1}}\}.$$

Lemma 1. 1. *Given a polynomial $f(y) \in K[y]$ with roots $\tau_1, \ldots, \tau_n$. Let the tree data of $f(y)$ be $\{d^*_{1,i_2,\ldots,i_s}, W^*_{1,i_2,\ldots,i_s}\}$ and the associated discs be $\{D_{1,i_2,\ldots,i_s}\}$. Let D be any disc in K. Then $D \in \{D_{1,i_2,\ldots,i_s}\}$ iff*

(1) $\tau_j \in D$ *for some j,*

(2) *there exist τ_i, τ_j in D with $|\tau_i - \tau_j| =$ radius of D.*

Proof. Suppose that $D \in \{D_{1,i_2,\ldots,i_s}\}$. If D is one of the minimal discs, then D is of radius zero. The above conditions are satisfied with $\tau_i = \tau_j =$ any root in D. On the other hand if D is not among the minimal discs in $\{D_{1,i_1,\ldots,i_s}\}$, say, $D = D_{1,i_1,\ldots,i_s}$, then it follows from the conditions (2) (4) (5) of Definition 1. 1 that

$$D = D_{1,i_1,\ldots,i_s} \supsetneqq D_{1,i_1,\ldots,i_s,j}$$

for two different indices j. The conditions (1) and (2) of this lemma follow from the condition (7) of Definition 1. 1.

Let us consider the converse. Let D be a disc with the conditions (1) & (2) of this lemma fulfilled. Since

$$\begin{aligned}\text{radius of } D = |\tau_i - \tau_j| &\leqq \max\{|\tau_r - \tau_s|;\ r, s = 1, \ldots, m\} \\ &= \text{radius of } D_1\end{aligned}$$

and

$$\tau_j \in D \cap D_1.$$

Then we have

$$D_1 \supset D.$$

Let us use the notations of Definition 1.2 with

$$D_1 \supset D_{1,j_2} \supset \cdots \supset D_{1,j_2,\ldots,j_{h+1}} \ni \tau_i.$$

Let

$$D_{1,j_2,\ldots,j_s} \supsetneqq D \supseteqq D_{1,j_2,\ldots,j_s,j_{s+1}}.$$

We claim

$$D = D_{1,j_2,\ldots,j_{s+1}}.$$

Otherwise suppose for a moment that we have

$$\text{radius of } D > W^*_{1,j_2,\ldots,j_{s+1}} = \text{radius of } D_{1,j_2,\ldots,j_{s+1}}.$$

Then it follows from

$$\tau_i \in D_{1,\ldots,j_s} \setminus D_{1,\ldots,j_{s+1}}$$

and the condition (7) of Definition 1.1 that

$$\text{radius of } D = |\tau_i - \tau_j| = W^*_{1,i_2,\ldots,i_s} = \text{radius of } D_{1,\ldots,i_s}.$$

Then we must have that $D = D_{1,\ldots,i_s}$. A contradiction. Q.E.D.

One famous example of the concept of the tree data can be deduced from the classical characteristic pairs of algebroid curves (cf. [Z.1]).

Remark. Our primary concern in this article is the case that $K = k((t))$ or $K = k\langle\langle t\rangle\rangle$ = the Puiseux field of $k((t))$. We shall use $|\sigma_i - \sigma_j| = 2^{-\operatorname{ord}(\sigma_i - \sigma_j)}$ and denote $-\log_2 W_{ij}$ by M_{ij}. We henceforth reformulate the tree data along τ_i as the set $\{M_{ij}, d_{ij}\}$ with $\{\operatorname{ord}(\tau_i - \tau_j)\colon j = 1, \ldots, m\} = \{M_{i1} < M_{i2} < \cdots < M_{ih} < M_{ih+1} = \infty\}$ and

$$d_{ij} = \operatorname{card}\{s\colon \operatorname{ord}(\tau_i - \tau_s) \geqq M_{ij}\}.$$

The tree data of a polynomial will be reformulated accordingly with order $M_{1,i_2,\ldots,i_s}$ replacing distance $W^*_{1,i_2,\ldots,i_s}$.

The following elementary algebraic tool will be useful.

Definition 1.3. Let K be $k\langle\langle t\rangle\rangle$ and π a symbol. Let $\sigma = \sum_{j<\delta} a_j t^j + \pi t^\delta \in k[\pi]\langle\langle t\rangle\rangle$ with $a_j \in k$. Then σ is said to be a π-approximate root (abbr. π-root) of a polynomial $f(y)$ if in the expression in $k[\pi]\langle\langle t\rangle\rangle$

$$f(\sigma) = f_\sigma(\pi) t^\lambda + \text{higher terms in } t$$

we have $1 \leqq \deg_\pi f_\sigma(\pi) < \infty$. The multiplicity of σ as an π-root of $f(y)$ is defined to be $\deg_\pi f_\sigma(\pi)$ as polynomial in π.

A π-root $\sigma = \sum_{j<\delta} a_j t^j + \pi t^\delta$ is a general point in the disc with radius $2^{-\delta}$ centered at $\sum_{j<\delta} a_j t^j$. Any disc will uniquely determine a π-root as a general point in it. The multiplicity of σ is the number of roots of $f(y)$ in the said disc. More precisely we have

Proposition 1. 2. *The element* $\sigma = \sum_{j<\delta} a_j t^j + \pi t^\delta$ *is an* π*-root of* $f(y) \in K[y]$ *iff there is a root* τ_j *such that* $\operatorname{ord}(\sigma - \tau_j) = \delta$. *Moreover let* $\{\tau_1, \ldots, \tau_s\}$ *be the collection of roots of* $f(y)$ *with* $\operatorname{ord}(\sigma - \tau_i) = \delta$ *and let the coefficient of* t^δ *in* $\tau_1, \ldots, \tau_s$ *be* $\alpha_1, \ldots, \alpha_s$ *respectively. Then, with* $f_\sigma(\pi)$ *as defined in Definition* 1. 3, *we have*

$$f_\sigma(\pi) = c \prod_{j=1}^{s} (\pi - \alpha_j)$$

where c is a nonzero constant.

Proof. Let us extend $\{\tau_1, \ldots, \tau_s\}$ to a complete set of roots $\{\tau_1, \ldots, \tau_m\}$ of $f(y)$. It is clear that $f_\sigma(\pi) = \pi$ (the leading coefficient of $(\sigma - \tau_i)$). If $\operatorname{ord}(\sigma - \tau_i) < \delta$, then the leading coefficient of $(\sigma - \tau_i)$ is a nonzero constant. Moreover $\operatorname{ord}(\sigma - \tau_i) \geqq \delta$ iff $1 \leqq i \leqq s$ and the leading coefficient of $(\sigma - \tau_i)$ is $(\pi - \alpha_i)$ in this case. Our statement follows trivially. Q.E.D.

Corollary. *Let* $\sigma = \sum a_j t^j + \pi t^\delta$ *be a* π*-root of* $f(y)$ *with* $f(\sigma) = f_\sigma(\pi) t^\lambda +$ *higher terms in t. Let* τ_i *be a root of* $f(y)$ *with* $\operatorname{ord}(\sigma - \tau_i) = \delta$. *Then the only permissible degree for* $f_\sigma(\pi)$ *is* d_{ij} *for some j. Moreover* $f_\sigma(\pi)$ *has only one root iff* $\delta \neq M_{ij}$ *for all j.*

Proof. Proposition 1. 2 and Lemma 1. 1.

As in the theory of the classical characteristic pairs for plane algebroid curves we shall introduce the following auxiliary data:

$$q_{i1} = M_{i1}$$
$$q_{ij} = M_{ij} - M_{ij-1} \qquad j = 2, \ldots, h.$$
$$\lambda_{ij} = \sum_{s=1}^{j} q_{is} d_{is}$$

The following proposition and its proof indicate the meaning of the derived numbers q_{ij} and λ_{ij}.

Proposition 1. 3. *Let* $\sigma = \sum a_j t^j + \pi t^\delta$ *be a* π*-root for* $f(y) = \prod (y - \tau_s)$. *Suppose* $\operatorname{ord}(\sigma - \tau_i) = \delta$ *for some i and* $M_{ij} < \delta \leqq M_{ij+1}$. *Then*

$$\operatorname{ord} f(\sigma) = \lambda_{ij} + d_{ij+1}(\delta - M_{ij}) \quad \text{if} \quad j \geqq 1,$$
$$= m\delta \quad \text{if} \quad j = 0.$$

Proof. Let us compute $\operatorname{ord}(\sigma-\tau_s)$ for various s. If $\operatorname{ord}(\tau_i-\tau_s)<\delta$ then $\operatorname{ord}(\sigma-\tau_s)=\operatorname{ord}(\tau_i-\tau_s)$. On the other hand if $\operatorname{ord}(\tau_i-\tau_s)\geqq\delta$ then we have $\operatorname{ord}(\sigma-\tau_s)=\delta$. In the case $j\geqq 1$ we have

$$\begin{aligned}\operatorname{ord} f(\sigma) &= \sum \operatorname{ord}(\sigma-\tau_s)\\ &= (d_{i1}-d_{i2})M_{i1}+\cdots+(d_{ij}-d_{ij+1})M_{ij}+d_{ij+1}\delta\\ &= d_{i1}q_{i1}+d_{i2}q_{i2}+\cdots+d_{ij}q_{ij}+d_{ij+1}(\delta-M_{ij})\\ &= \lambda_{ij}+d_{ij+1}(\delta-M_{ij}).\end{aligned}$$

In the case $j=0$ we have $\operatorname{ord}(\sigma-\tau_s)=\delta$ for all s. We clearly have, with $j=0$, the following

$$\operatorname{ord} f(\sigma)=m\delta. \qquad \text{Q.E.D.}$$

Two natural properties of a system of π-roots are the complete and the coherent conditions (cf. Definition 5 & 8 of [M.1]) which are defined as follows.

Definition 1.4. A system of π-roots $(\sigma_1,\ldots,\sigma_s)$ of $f(y)$ is said to be complete if with

$$\sigma_i=\sum a_{ij}t^j+\pi t^{\delta_i} \quad \text{for} \quad i=1,\ldots,s$$

we have

(1) $$\operatorname{ord}(\sigma_i-\sigma_j)<\min(\delta_i,\delta_j),$$

(2) $$\sum \deg_\pi f_{\sigma_i}(\pi)=\deg f(y)=m.$$

Moreover the complete system is said to be coherent if

(3) $$\operatorname{ord} f(\sigma_i)=\lambda \quad \text{for} \quad i=1,\ldots,s$$

and the common value λ of (3) will be called the accuracy of the coherent complete system.

In the above definition the condition (1) means that the $2^{-\delta_i}$-discs associated with σ_i's are all disjoint. The condition (2) means that the numbers of roots of $f(y)$ in those disjoint $2^{-\delta_i}$-discs add up to the total degree of $f(y)$. The condition (3) means, in case $f(y)$ is monic, the product of the distances between the *general* point σ_i and the roots of $f(y)$ is $2^{-\lambda}$ for all σ_i's.

Recall the following concept of the approximate roots (cf. [M.1]).

Definition. Let $f(y)$ be a monic polynomial of degree $m\cdot d$ over a ring R with d a unit in R. Let $\theta=y^{-1}$ and $R[y]\subset R((\theta))$. Then the d-th root of $f(y)$ exists in $R((\theta))$ and let it be $\theta^{-m}+a_1\theta^{-m+1}+\cdots+a_{-1}\theta+\cdots$. Then the d-th approximate root $f_d(y)$ of $f(y)$ is defined to be $y^m+a_1y^{m-1}+\cdots+a_m=\theta^{-m}+a_1\theta^{-m+1}+\cdots+a_m$.

We shall quote a simpler version of the main result of [M.1] for later references.

Theorem 1. 1. *Let $f(y)$ be monic in $K[y]$ with $K=k\langle\langle t\rangle\rangle$. Let $(\sigma_1,\ldots,\sigma_s)$ be a coherent complete system of π-roots of $f(y)$ with multiplicities $(l_1,\ldots,l_s)$. Let $d|\text{g.c.d.}(l_1,\ldots,l_s)$ and $f_d(y)$ the d-th approximate root of $f(y)$. Then $(\sigma_1,\ldots,\sigma_s)$ is a coherent complete system of π-roots of $f_d(y)$ with multiplicities $\left(\frac{l_1}{d},\ldots,\frac{l_s}{d}\right)$.*

The following further refinement of Theorem 1. 1 and the generalization of the approximate roots are interesting. We shall show some of their applications in Appendix II. A proof of Theorem 1. 2 can be found in [M.4].

Definition 1. 5. Let $\{\sigma_1,\ldots,\sigma_s\}$ be a coherent complete system of π-roots for a polynomial $f(y)$ in $k\langle\langle t\rangle\rangle[y]$. Let the multiplicities be $\{l_1,\ldots,l_s\}$ and the accuracy be λ. Let d be an integer with $d|\text{g.c.d.}(l_1,\ldots,l_s)$. A monic polynomial $h(y)$ is said to be a d-th quasi-approximate root of $f(y)$ with accuracy λ if $\{\sigma_1,\ldots,\sigma_s\}$ is a coherent complete system of π-roots of $h(y)$ with multiplicities $\left\{\frac{l_1}{d},\ldots,\frac{l_s}{d}\right\}$ and accuracy $\frac{\lambda}{d}$.

Theorem 1. 2. *Let $f(y)$ be monic in $k\langle\langle t\rangle\rangle[y]$ and $\{\sigma_1,\ldots,\sigma_s\}$ be a coherent complete system of π-roots with multiplicities $\{l_1,\ldots,l_s\}$ and accuracy λ. Let $d|\text{g.c.d.}(l_1,\ldots,l_s)$ and $h(y)$ a d-th quasi-approximate root of $f(y)$ with accuracy λ. Let*

$$f(y)=(h(y))^d+\sum_{j=1}^{d}h_j(y)\,(h(y)^{d-j})$$

where

$$\deg h_j(y)<\deg h(y)=\frac{\deg f(y)}{d}.$$

Then we have for $i=1,\ldots,s$ and $j=1,\ldots,d$

$$\operatorname{ord}h_j(\sigma_i)\geqq\frac{\lambda}{d}j.$$

§ 2. The characteristic data for a pair of polynomials

The purpose of this section is to introduce the characteristic data for a pair of polynomials $f(x, y)$ and $g(x, y)$. It will be illustrated in the sections 4, 5 & 6 that the Jacobian condition

$$J_{x,y}(f(x, y), g(x, y))=1$$

implies certain relations between the root configurations of $f(x, y)=0$ and $g(x, y)=0$ as equations over $k\langle\langle x^{-1}\rangle\rangle$. Furthermore we shall show that the characteristic data introduced here will control the tree data of the two polynomial equations within certain natural limitations.

Let $f(x, y)$ and $g(x, y)$ be two polynomials monic in y with

$$\deg_y f(x, y)=m,\quad \deg_y g(x, y)=n.$$

Then we may view $(f(x, y), g(x, y))$ as a parametrization over $k(x)$ of a plane rational curve with y as the parameter. Clearly the said curve has only one place at $y=\infty$. Analytically, if we let

$$g(x, y)=\eta^{-n},$$

$$\eta=g(x, y)^{-\frac{1}{n}}=y^{-1}+\alpha_2(x)y^{-2}+\cdots+\in k[x]\,((y^{-1}))=k[x]\,((\eta))$$

then

$$f(x, y)=\eta^{-m}+\sum_{j>-m} f_j(x)\eta^j \in k[x]\,((\eta)).$$

Let Ω_i be the automorphism of $k[x]\,((\eta))$ defined by

$$\Omega_i(x)=x, \quad \Omega_i(\eta)=\omega^i\eta$$

where ω is a fixed n-th primitive root of unity. Suppose

$$y \in k(x, f(x, y), g(x, y)).$$

Since y is integral over $k[x, g(x, y)]$ which is integrally closed, we deduce that $f(x, y)$ satisfies a monic polynomial of degree n over $k[x, g(x, y)]$. The equation is irreducible over $k[x]\,((g^{-1}))=k[x]\,((\eta^n))$. Then the defining equation of $f(x, y)$ over

$$k[x]\,((\eta^n))=k[x]\,((g^{-1}))$$

is the following

$$\prod_{i=1}^{n}(f-\Omega_i f(x, y))=0 \tag{1}$$

where f is a symbol as usual. In fact the above equation is the analytic factorization of the defining equation of the said rational curve at $g=\infty$ or $g^{-1}=0$.

Due to the existence of the automorphisms Ω_i, the tree data associated with the configuration of the roots of the equation (1) over $k[x]\,((\eta))$ are completely symmetric. Hence the tree data along any root will determine the tree data completely. Let the tree data along one root (hence all roots) be (reindexed as) $\{M_j, d_j\}$. We shall call $\{M_j, d_j\}$ the *characteristic data* of the pair of polynomials $(f(x, y), g(x, y))$ with respect to x. Recall (cf. [A-M.1] p. 68)

$$d_1=n, \quad d_{j+1}=\text{g.c.d.}\,(n, M_1, \ldots, M_j),$$

$$M_j=\min\{i, f_i(x)\neq 0, d_j \nmid i\}, \quad M_{h+1}=\infty$$

and the following auxiliary notations

$$n_0=1, \quad n_j=\frac{d_j}{d_{j+1}},$$

$$q_1=M_1, \quad q_j=M_j-M_{j-1},$$

$$\lambda_j=\sum_{1\leqq i\leqq j} q_i d_i, \quad \mu_j=\frac{\lambda_j}{d_j}, \quad \theta_j=\mu_j-M_j.$$

To make some sense out of the long list of notations introduced above, we shall insert the following lemma first observed by S. S. Abhyankar ([A.1], p. 149).

Lemma 2. 1. *Let* $f(x, y)$, $g(x, y) \in k[x, y]$ *be two polynomials monic in* y *with* y*-degree* m, n *respectively. Let*

$$g(x, y) = \eta^{-n}, \quad f(x, y) = \eta^{-m} + \sum_{i > -m} f_i(x)\eta^i.$$

Then $J_{x,y}(f(x, y), g(x, y)) = c$ *a nonzero constant iff*

$$\operatorname{ord}_\eta \left(\sum f_i(x)\eta^i - \sum f(0)\eta^i\right) = n - 1$$

and

$$\deg_x f_{n-1}(x) = 1.$$

Proof. It follows from chain rule that

$$\begin{aligned} J_{x,y}(f(x, y), g(x, y)) &= J_{x,\eta}(f(x, y), g(x, y))\, J_{x,y}(x, \eta) \\ &= -n\left(\sum f_i'(x)\eta^i\right)\eta^{-n-1} \cdot (-\eta^2 + \cdots) \\ &= \quad n\left(\sum f_i'(x)\eta^i\right)(\eta^{-n+1} + \cdots). \end{aligned}$$

The right hand side is in $k[x, y]$ and a unit in $k[x]\,[[\eta]]$, hence a nonzero constant, iff

$$f_i'(x) = 0 \qquad \forall i < n - 1$$

and

$$f_{n-1}'(x) = \text{nonzero constant}. \qquad \text{Q.E.D.}$$

Note that since $f_{n-1}(x) \neq 0$ and $(n, n-1) = 1$, then $M_i \leqq n-1$ for all $i \leqq h$. Thus we know all M_i's terminate at $n-1$ if the Jacobian condition is satisfied. Note that this implies that $f(x, y)$ has n conjugates over $k[x]\,((g^{-1}))$ and $y \in k(x, g(x, y), f(x, y))$. We shall call the reader's attention to our ignorance about the numbers M_i. For instance, after long computation we (as verified by A. Sathaye) established that for the following polynomials over k

$$\begin{aligned} X &= T^6 + \cdots \ = \theta^{-6}, \\ Y &= T^8 + \cdots = \theta^{-8} + \cdots + \theta^{M_2} + \cdots \end{aligned}$$

the value M_2 could never be 13. It shows how little we know about M_i in the polynomial case.

The characteristic data $\{d_i; M_i\}$ are analytic in nature. We shall replace them, and thus algebraicize them, by the approximate roots. This method has been well demonstrated in [A-M.1]. In the present article we are interested in some specializations of the approximate roots. Let

$$F(f, g) = \prod_{i=1}^{n} \left(f - \Omega_i\left(\eta^{-m} + \sum_{j > -m} f_j(x)\eta^j\right)\right) = 0$$

be the defining equation of $f(x, y)$ and $g(x, y)$ over $k[x]$. Let ψ be the specialization $k[x] \to k$ defined by

$$\psi(x) = 0,$$

we shall extend ψ naturally to $k[x][f, g]$ with f, g symbols. Then clearly we have

$$F^{\psi}(f, g) = \prod_{i=1}^{n} [f - \Omega_i(\eta^{-m} + \sum f_j(0)\eta^j)].$$

We shall establish the following proposition.

Proposition 2. 1. *Let $T_r(f, g)$ be the d_r-th approximate root of $F(f, g)$ with f viewed as the variable. Then $T_r^{\psi}(f, g)$ is the d_r-th approximate root of $F^{\psi}(f, g)$ with f viewed as the variable.*

Proof. Recall the definition of the approximate roots in § 1. Let the d_r-th root of $F(f, g)$ in $k[x, g]((f^{-1}))$ be

$$F(f, g)^{\frac{1}{d_r}} = f^{\frac{n}{d_r}} + \sum_{i>0} \alpha_i(x, g) f^{\frac{n}{d_r} - i}.$$

Then we have

$$F^{\psi}(f, g)^{\frac{1}{d_r}} = f^{\frac{n}{d_r}} + \sum_{i>0} \alpha_i(0, g) f^{\frac{n}{d_r} - i}.$$

Naturally we conclude the following

$$T_r^{\psi}(f, g) = f^{\frac{n}{d_r}} + \sum_{\frac{n}{d_r} \geqq i > 0} \alpha_i(0, g) f^{\frac{n}{d_r} - i}$$

$$= \text{the } d_r\text{-th approximate root of } F^{\psi}(f, g). \qquad \text{Q.E.D.}$$

The advantage of the specialization $\psi: x \longrightarrow 0$ is that the polynomials $T_r^{\psi}(f, g)$ are in $k[f, g]$ while the polynomials $T_r(f, g)$ are with coefficients in $k[x]$. Thus the differentiation of the polynomial $T_r(f(x, y), g(x, y))$ with respect to x which will be used later on will be rather complicated. The most useful information provided by the polynomials $T_r(f, g)$ are the y-degrees of the polynomials $T_r(f(x, y), g(x, y))$. This valuable information is largely retained by the simpler polynomials $T_r^{\psi}(f, g)$. To be more precise let e be an integer defined by (cf. Lemma 2. 1).

$$e = \operatorname{ord}_{\eta}(\sum f_j(x)\eta^j - \sum f_j(0)\eta^j).$$

We are primarily interested in $T_r^{\psi}(f, g)$ for $M_r \leqq e$. Let us prove the following proposition (cf. [A.1], p. 155).

Proposition 2. 2. *Let e be an integer defined by*

$$e = \operatorname{ord}_{\eta}(\sum f_j(x)\eta^j - \sum f_j(0)\eta^j).$$

Let $M_r \leqq e$. Then we have

(1) $\deg_y T_r^{\psi}(f(x, y), g(x, y)) = \deg_y T_r(f(x, y), g(x, y)) = -\mu_r$.

If $M_r < e$, then we have

(2) $T_r(f(x, y), g(x, y))$ *and* $T_r^{\psi}(f(x, y), g(x, y))$ *are monic in y.*

If $M_r = e$, then we have for some nonzero constant c the following

(3) $T_r^{\psi}(f(x, y), g(x, y)) = cf_e(x)y^{-\mu_r} +$ *lower terms in y.*

Proof. Let

$$\sigma = \eta^{-m} + \sum_{-m<j<M_r} f_j(x)\eta^j + \pi\eta^{M_r}. \tag{4}$$

Due to the restriction of $M_r \leqq e$, i.e. $f_j(x) \in k$, we have that σ is also a π-root of the equation

$$F^{\psi}(f, g) = 0.$$

Let us set

$$\bar{\sigma} = \eta^{-m} + \sum_{-m<j<M_r} f_j(x)\eta^j.$$

Then the greatest common divisor of all exponents of the terms with nonzero coefficients in $\bar{\sigma}$ and the integer n is d_r according to the definition of d_r. The automorphisms Ω_j will produce $\left(\frac{n}{d_r}\right)$ distinct conjugates of $\bar{\sigma}$. Let the corresponding π-roots for $F(f, g)$ in the M_r-discs be $\{\sigma_1, \ldots, \sigma_{\frac{n}{d_r}}\}$. Note that σ_i is simply the summation of a conjugate of $\bar{\sigma}$ and $\pi\eta^{M_r}$. Then this system of π-roots is coherent and complete for $F(f, g)$ with multiplicities $\{d_r, \ldots, d_r\}$ and accuracy λ. It follows from Proposition 1.3 that

$$\lambda = \lambda_r = \mu_r d_r.$$

As we pointed out before, the inequality

$$M_r \leqq e$$

implies $f_j(x) \in k$ in the expression (4). Then the above reasoning works for $F^{\psi}(f, g)$. Thus the same π-root system $\{\sigma_1, \ldots, \sigma_{\frac{n}{d_r}}\}$ is coherent and complete for $F^{\psi}(f, g)$ with identical multiplicities $\{d_r, \ldots, d_r\}$ and $\lambda = \lambda_r = \mu_r d_r$.

Let us consider the polynomials $T_r(f, g)$ and $T_r^{\psi}(f, g)$ with f viewed as the variable. It follows from Theorem 1.1 that the π-roots system $\{\sigma_1, \ldots, \sigma_{\frac{n}{d}}\}$ is coherent and complete for $T_r^{\psi}(f, g)$ (resp. $T_r(f, g)$) with multiplicities $\{1, \ldots, 1\}$. Its accuracy can be easily determined to be $\frac{\lambda_r}{d_r} = \mu_r$ as follows. Let $\{\tau_1, \ldots, \tau_n\}$ be the complete set of roots of $F(f, g)$ (resp. $F^{\psi}(f, g)$) and $\{\tau_1^*, \ldots, \tau_{\frac{n}{d_r}}^*\}$ be the complete set of roots of $T_r(f, g)$ (resp. $T_r^{\psi}(f, g)$). Since the M_r-discs associated with σ_i's are all disjoint (cf. Definition 1.4), then it follows from the strong triangle inequality that for τ_s, τ_s^* belonging to the M_r-disc associated with σ_s with $s \neq i$ we always have

$$\operatorname{ord}(\sigma_i - \tau_s) = \operatorname{ord}(\sigma_i - \sigma_s), \quad \operatorname{ord}(\sigma_i - \tau_s^*) = \operatorname{ord}(\sigma_i - \sigma_s)$$

while in each M_r-disc there are d_r τ_j's and only one τ_j^*. A simple computation verifies that the accuracy is indeed $\frac{\lambda_r}{d_r} = \mu_r$.

Let us reorder if necessary the complete set of roots $\{\tau_1^*, \ldots, \tau_{\frac{n}{d_r}}^*\}$ so that

$$\operatorname{ord}(\sigma_i - \tau_i^*) = M_r.$$

We claim the following

$$\sigma_i - \tau_i^* = \pi\eta^{M_r} + \text{higher terms in } \eta.$$

Otherwise the leading coefficient is of the form $(\pi - \alpha)$ with some nonzero α which is the coefficient of η^{M_r} in τ_i^*. Note that the automorphisms Ω_j will leave g fixed and change η to $\omega^j\eta$. Thus the automorphisms Ω_j will produce too many roots for $T_r(f, g)$ (resp. $T_r^{\psi}(f, g)$). A contradiction. Let $\sigma_1 = \sigma$ be the one defined by equation (4). Then it follows from the fact $f_{M_r}(x) \neq 0$ that

$$\operatorname{ord}\left(f(x, y) - \tau_1^*\right) = M_r = \operatorname{ord}\left(\sigma_1 - \tau_1^*\right).$$

Moreover it follows from the disjointness of the M_r-discs associated with σ_i's that for $i \neq 1$

$$\operatorname{ord}\left(f(x, y) - \tau_i^*\right) = \operatorname{ord}\left(\sigma_1 - \tau_i^*\right).$$

Thus we conclude

$$\begin{aligned}\deg_y T_r\left(f(x, y), g(x, y)\right) &= -\operatorname{ord}_{\eta} T_r\left(f(x, y), g(x, y)\right) \\ &= -\sum \operatorname{ord}\left(f(x, y) - \tau_i^*\right) \\ &= -\sum \operatorname{ord}\left(\sigma_1 - \tau_i^*\right) = -\mu_r\end{aligned}$$

and

$$\deg_y T_r^{\psi}\left(f(x, y), g(x, y)\right) = -\mu_r.$$

We have established the conclusion (1).

To prove the conclusions (2) and (3) we observe that the leading coefficient $\left(f(x, y) - \tau_1^*\right)$ is $f_{M_r}(x)$ while the leading coefficients of $\left(f(x, y) - \tau_i^*\right)$ for $i \neq 1$ are all nonzero constants. Thus the product of all leading coefficients is either a nonzero constant if $f_{M_r}(x)$ is a constant or of the form $cf_{M_r}(x)$. We shall remind the reader that the term monic means the coefficient of the highest term is a unit in our terminology.

Q.E.D.

In the tradition of Pythagoras we compute numbers. In fact we show that there is a sequence of polynomials $T_r^{\psi}\left(f(x, y), g(x, y)\right)$ with y-degree $(-\mu_r)$. One can convince oneself that the sequences $\{n, \mu_1, \ldots, \mu_r\}$ and $\{n, M_1, \ldots, M_r\}$ determine each other. Especially one has

$$\text{g.c.d.}\{n, \mu_1, \ldots, \mu_r\} = \text{g.c.d.}\{n, M_1, \ldots, M_r\}.$$

Moreover the existence of such a sequence of polynomials $T_r^{\psi}\left(f(x, y), g(x, y)\right)$ implies that the characteristic data $\{d_i, M_i\}$ can be recovered by (cf. [A-M.1], [S.1], [AN.1]) the following formula:

$$d_1 = n, \quad M_1 = \mu_1$$

$$d_{j+1} = \text{g.c.d.}(n, \mu_1, \ldots, \mu_j), \quad M_j = \frac{1}{d_j - d_{j+1}}\left[\mu_j d_j - \sum_{i=1}^{j-1} (d_i - d_{i+1}) M_i\right].$$

The mere y-degrees of the polynomials $T_r^{\psi}\left(f(x, y), g(x, y)\right)$ will form a foundation of our investigation of the Jacobian problem.

§ 3. The partial derivatives of T_i^{ψ}

Let $H(f, g)$ be a polynomial in $k[f, g]$. We may compute the Jacobian of $H(f(x, y), g(x, y))$ and $g(x, y)$ by the chain rule as follows

$$\begin{aligned} J_{x,y}(H(f(x, y), g(x, y)), g(x, y)) &= J_{f,g}(H(f, g), g)\, J_{x,y}(f(x, y), g(x, y)) \\ &= \left(\frac{\partial}{\partial f} H(f, g)\right) J_{x,y}(f(x, y), g(x, y)) \\ &= H(f(x, y), g(x, y))_f \cdot J_{x,y}(f(x, y), g(x, y)). \end{aligned}$$

We are primarily interested in the case that the Jacobian condition is satisfied

$$J_{x,y}(f(x, y), g(x, y)) = 1$$

and for $H(f, g) = T_r^{\psi}(f, g)$ as introduced in the last section. In those cases the Jacobians become the partial derivatives. We shall deduce the necessary information about the forms and the y-degrees of the partial derivatives in this section.

Let us list the three basic arithmetic properties of the polynomials $T_r(f, g)$ and the sequence $\{n, \mu_1, \ldots, \mu_r\}$ as follows (cf. [A-M.1], p. 51 and Proposition 2. 2):

(1) g.c.d. $(n, \mu_1, \ldots, \mu_j) = d_{j+1} < d_j$,

(2) $T_r(f, g) = f^{\frac{n}{d_r}} +$ lower terms in f,

(3) $T_r(f(x, y), g(x, y)) = c_r y^{-\mu_r} +$ lower terms in y for all r with $M_r < e$.

For the definition of the number e the reader is referred to Proposition 2. 2. Recall from § 2

$$n_0 = \frac{n}{d_1} = 1, \quad n_j = \frac{d_j}{d_{j+1}},$$

$$\theta_j = \mu_j - M_j.$$

Then clearly we have

$$\deg_f T_r(f, g) = \frac{n}{d_r} = \prod_{j=1}^{r-1} n_j. \tag{4}$$

We shall use the three basic arithmetic properties listed above and the technique of the decimal expansions to establish a canonical expansion for $T_{r+1}(f, g)$ in terms of $T_j(f, g)$ for $j \leqq r$.

We may use the integers $1, n_1, n_1 \cdot n_2, \ldots, \prod_{j=1}^{r-1} n_j$ to express any nonnegative integer l uniquely as

$$l = \sum_{i=1}^{r} \alpha_i \prod_{j=1}^{i-1} n_j$$

with the following restrictions

$$0 \leqq \alpha_i < n_i \quad \text{for} \quad i < r, \tag{5}$$

$$0 \leqq \alpha_r < \infty. \tag{6}$$

The above expansion of the nonnegative integer l is not different in principle from the usual decimal expansions. We shall use the following common vector notation

$$\alpha=(\alpha_1,\ldots,\alpha_r),\quad T_\alpha=\prod_{i=1}^{r} T_i^{\alpha_i}.$$

Then it follows from equation (4) that

$$\deg T_\alpha=\sum \alpha_i \prod_{j=1}^{i-1} n_j. \tag{7}$$

We shall establish the following lemma (cf. [A-M.1], p. 40).

Lemma 3. 1. *Let $H(f, g)$ be a polynomial in $k[x]\,[f, g]$. Then $H(f, g)$ can be uniquely expressed over $k[x]$ as follows*

$$H(f, g)=\sum_{j,\alpha} c_{j,\alpha} g^j T_\alpha \tag{8}$$

with $\alpha=(\alpha_1,\ldots,\alpha_r)$ satisfying the restriction (5) & (6) *in the preceding paragraph. Moreover we have*

$$\begin{aligned}\deg_f H(f, g)&=\max\{\deg_f T_\alpha\colon c_{j,\alpha}\neq 0 \text{ for some } j\}\\&=\max\left\{\sum \alpha_i \prod_{j=1}^{i-1} n_j\colon c_{j,\alpha}\neq 0\right\}.\end{aligned}$$

Proof. Let us use the expansion technique for the following integer l

$$\deg_f H(f, g)=l=\sum \alpha_i \prod_{j=1}^{i-1} n_j.$$

We have the following expression for $H(f, g)$

$$\begin{aligned}H(f, g)&=\sum_j (c_{j,l} g^j) f^l+\text{lower terms in } f\\&=\sum_j c_{j,\alpha} g^j T_\alpha+\text{lower terms in } f\end{aligned}$$

where $c_{j,l}=c_{j,\alpha}$ for all j. The existence of the expression (8) can be easily established by applying the preceding process to the polynomial

$$H(f, g)-\sum c_{j,\alpha} g^j T_\alpha$$

which is of f-degree less than l.

Note that in the expression (8) the terms with distinct α indices will have distinct f-degrees. This is simply due to the fact that the f-degrees of the terms of the right hand side of the equation (7) are the expansions of distinct nonnegative integers if the α's are distinct.

For the uniqueness of the expression (8) it suffices, as usual, to give a prove for the zero polynomial. Let

$$0=\sum_{j,\alpha} c^*_{j,\alpha} g^j T_\alpha.$$

Using the preceding argument about f-degrees we separate the above equation into many equations for fixed α's as follows

$$0=\sum_{j} c^*_{j,\alpha} g^j T_\alpha$$

or

$$0=\sum_{j} c^*_{j,\alpha} g^j.$$

Certainly we have for all j and all α

$$0=c^*_{j,\alpha}.$$ Q.E.D.

The first half of the following proposition follows from § 7.2 of [A-M.1]. For completeness we give a short argument.

Proposition 3. 1. *The polynomial $T_{r+1}(f, g)$ has a unique expression over $k[x]$ of the following form*

$$T_{r+1}(f, g)=T_r(f, g)^{n_r}+\sum_{j,\alpha} c_{j,\alpha} g^j T_\alpha \tag{9}$$

with $\alpha=(\alpha_1, \ldots, \alpha_r)$ satisfying the restrictions (5) *and* (6). *Moreover let r be an integer with $M_r<e$ (cf. Proposition* 2. 2), *we have the following for $c_{j,\alpha}\neq 0$*

(1) $\alpha_r<n_r$,

(2) $\deg_y g^j(x, y)\, T_\alpha(f(x, y), g(x, y))\leqq(-\mu_r)n_r$.

Furthermore there is a unique j and a unique α for the equality to hold in (2) *and this particular α is with $\alpha_r=0$.*

Proof. Applying Lemma 3. 1 to the polynomial

$$T_{r+1}(f, g)-T_r(f, g)^{n_r}$$

we establish the first part of the proposition.

Since both polynomials $T_{r+1}(f, g)$ and $T_r(f, g)^{n_r}$ have leading coefficient 1 and are of degree $\left(\frac{n}{d_{r+1}}\right)$ in f then we have

$$\deg_f\left(T_{r+1}(f, g)-T_r(f, g)^{n_r}\right)=\frac{n}{d_{r+1}}<\prod_{j=1}^{r} n_j.$$

Thus it follows from Lemma 3. 1 that in the expression (9) we always have

$$\deg_f T_\alpha=\sum \alpha_i \prod_{j=1}^{i-1} n_j<\prod_{j=1}^{r} n_j.$$

Hence we conclude easily for $c_{j,\alpha} \neq 0$ the conclusion (1)

$$\alpha_r < n_r.$$

Let us replace the symbols f and g in all terms of the expression (9) by $f(x, y)$ and $g(x, y)$ and compute the y-degrees of the resulting polynomials. We have (cf. Proposition 2. 2)

$$\deg_y T_{r+1}\big(f(x, y), g(x, y)\big) = -\mu_{r+1} = (-\mu_r)n_r - q_{r+1} < (-\mu_r)n_r = \deg_y T_r\big(f(x, y), g(x, y)\big)^{n_r} \tag{10}$$

and for the other terms we have

$$\deg_y c_{j,\alpha} g^j(x, y)\, T_\alpha\big(f(x, y), g(x, y)\big) = jn + \sum (-\mu_i)\alpha_i \tag{11}$$

where $\alpha = (\alpha_1, \ldots, \alpha_r)$. We shall prove that any two terms with distinct j and α indices will have distinct y degrees. The proof is arithmetic in nature. Let j^* and α^* be the other indices with

$$\deg_y g^j(x, y)\, T_\alpha\big(f(x, y), g(x, y)\big) = \deg_y g^{j^*}(x, y)\, T_{\alpha^*}\big(f(x, y), g(x, y)\big).$$

Then we have

$$j \cdot n + \sum (-\mu_i)\alpha_i = j^* n + \sum (-\mu_i)\alpha_i^*. \tag{12}$$

Recall

$$d_{j+1} = \text{g.c.d.}\,(n, \mu_1, \ldots, \mu_j).$$

It is easy to see that the residue

$$\frac{-\mu_r}{d_{r+1}} \bmod \left(\frac{d_r}{d_{r+1}}\right) = \frac{-\mu_r}{d_{r+1}} \bmod (n_r)$$

is a unit in $\mathbb{Z}/(n_r)$. Let us divide the numbers $n, -\mu_i, \ldots, -\mu_r$ in the equation (12) by d_{r+1} and then mod the equation by $\dfrac{d_r}{d_{r+1}} = n_r$. The residue equation is of the following form

$$\bar{\alpha}_r\left(\overline{\frac{-\mu_r}{d_{r+1}}}\right) \equiv \bar{\alpha}_r^*\left(\overline{\frac{-\mu_r}{d_{r+1}}}\right)$$

or

$$\bar{\alpha}_r \equiv \bar{\alpha}_r^* \bmod n_r.$$

Recall

$$0 \leqq \alpha_r, \quad \alpha_r^* < n_r.$$

We thus conclude

$$\alpha_r = \alpha_r^*.$$

Now we may throw away α_r and α_r^* from the equation (12) and proceed likewise. It is routine to show

$$\begin{aligned} \alpha_{r-1} &= \alpha_{r-1}^*, \\ &\vdots \\ \alpha_1 &= \alpha_1^*, \\ j &= j^*. \end{aligned}$$

Thus we establish that distinct j and α indices in the equation (1) will produce terms with distinct y-degrees. In light of the equation (10) we must have the conclusion (2) and the uniqueness of a particular term of y-degree $(-\mu_n)n_r$. We are left to show that this particular term is with $\alpha_r=0$. Let it be with indices j and $\alpha=(\alpha_1,\ldots,\alpha_r)$. Then we need to solve the following arithmetic equation

$$jn+\sum(-\mu_i)\alpha_i=(-\mu_n)n_r. \tag{13}$$

Let us divide the numbers $n, -\mu_1,\ldots,-\mu_r$ in the above equation by d_{r+1}. The residue equation $\bmod \frac{d_r}{d_{r+1}}$ is of the following form

$$\bar{\alpha}_r\left(\overline{\frac{-\mu_n}{d_{r+1}}}\right)\equiv 0.$$

Since the residue of $\frac{-\mu_n}{d_{r+1}}$ is a unit and

$$0\leqq\alpha_r<n_r$$

we must have

$$\alpha_r=0. \qquad \text{Q.E.D.}$$

Remark. Proposition 3. 1 remains valid after applying the specialization ψ with

$$\psi: x\longrightarrow 0.$$

In fact the specialized proposition is the one interesting to us in the present article.

We shall give an arithmetic proof of the following proposition about the forms and the y-degrees of the partial derivatives of $T_r(f, g)$ (cf. [A.1], p. 155).

Proposition 3. 2. *We have the following*

(1) $T_1\big(f(x, y), g(x, y)\big)_f=1,$

(2) $T_{r+1}\big(f(x, y), g(x, y)\big)_f=n_r T_r\big(f(x, y), g(x, y)\big)^{n_r-1}\cdot T_r\big(f(x, y), g(x, y)\big)_f$
$+\sum C_{i,\beta}\, g^i(x, y)\, T_\beta\big(f(x, y), g(x, y)\big)$

where the terms under the summation sign satisfy

$$\deg_y T_{r+1}(f, g)_f=\deg_y T_r(f, g)^{n_r-1}\, T_r(f, g)_f$$
$$>\deg_y g^i(x, y)\, T_\beta\big(f(x, y), g(x, y)\big)$$

for $r\geqq 1$.

(3) $\deg_y T_{r+1}\big(f(x, y), g(x, y)\big)_f=(-\mu_{r+1})+M_{r+1}=\theta_{r+1}$

for $r\geqq 1$.

Proof. The statement (1) is trivial. Let us prove the statements (2) and (3) for the case $r=1$. The equation (9) of Proposition 3.1 is of the following form

$$T_2(f,g)=T_1(f,g)^{n_1}+\sum C_{j,\alpha}g^jT_\alpha$$

where $\alpha=(\alpha_1)$ and the unique j,α which gives the maximal y-degree is with $\alpha=(0)$. So we conclude

$$T_2(f,g)_f=n_1T_1(f,g)^{n_1-1}+\sum \alpha_1 C_{j,\alpha}g^jT_{(\alpha_1-1)}.$$

Multiply the above equation by $T_1(f,g)$ and the right hand side is restored to the original form except for the change of coefficients and the dropping out of the unique term in $\sum C_{i,\alpha}g^iT_\alpha$ which gives the maximal y-degree and all other terms with $\alpha=0$. Thus we conclude

$$\begin{aligned}\deg_y T_2(f,g)_f&=\deg_y T_1(f,g)^{n_1-1}=(-\mu_1)(n_1-1)\\&=-q_1n_1+M_1=-q_1n_1-M_2+M_1+M_2\\&=-q_1\left(\frac{n}{d_2}\right)+q_2+M_2=-\mu_2+M_2=\theta_2\end{aligned}$$

which proves statements (2) and (3) for $r=1$.

In general let us assume that the statements (2) and (3) have been proved for all smaller indices and we shall proceed for the index $r+1$. From the equation (9) of Proposition 3.1 we have the following

$$T_{r+1}(f,g)_f=n_rT_r(f,g)^{n_r-1}T_r(f,g)_f+\sum C_{j,\alpha}(g^jT_\alpha)_f.$$

Note that the unique term giving the maximal y-degree in $\sum C_{j,\alpha}g^jT_\alpha$ of the equation (9) is with $\alpha=(\alpha_1,\ldots,\alpha_{r-1},0)$. We have

$$\deg_y T_r(f,g)^{n_r-1}T_r(f,g)_f=(-\mu_r)n_r+M_r$$

by mathematical induction. And the term $(g^jT_\alpha)_f$ can be spread out as

$$(g^jT_\alpha)_f=\sum \alpha_i g^jT_{\alpha^{(i)}}(T_i)_f$$

where $\alpha=(\alpha_1,\ldots,\alpha_r)$ and $\alpha^{(i)}=(\alpha_1,\ldots,\alpha_i-1,\ldots,\alpha_r)$. We have by mathematical induction

$$\deg_y g^jT_{\alpha^{(i)}}(T_i)_f=\deg_y g^jT_\alpha+M_i \tag{14}$$

and there are only two possibilities; either

$$\deg_y g^jT_\alpha<(-\mu_r)n_r=\deg_y T_r(f,g)^{n_r}$$

or

$$\deg_y g^jT_\alpha=(-\mu_r)n_r=\deg_y T_r(f,g)^{n_r}.$$

In the second case the corresponding α_r is zero, we may assume the following for the equation (14)

$$M_i<M_r.$$

In either case we always have

$$\deg_y g^j T_{\alpha(i)}(T_i)_f < \deg_y T_r(f,g)^{n_r-1}(T_r)_f .$$

In other words we must have

$$\begin{aligned}\deg_y T_{r+1}(f,g)_f &= \deg_y T_r(f,g)^{n_r-1}(T_r)_f \\ &= (-\mu_r)n_r + M_r = -\mu_{r+1} + M_{r+1} = \theta_{r+1}\end{aligned}$$

which establishes statement (3). To finish statement (2) we note that in the following expression (cf. Lemma 3. 1)

$$\sum C_{j,\alpha}(g^j T_\alpha)_f = \sum C_{i,\beta} g^i T_\beta$$

the y-degree of the left hand side is less than $\deg_y T_{r+1}(f,g)_f$ as proved above and the y-degrees of all terms on the right side are distinct (cf. the proof of Lemma 3. 1). Thus we must have

$$\deg_y T_{r+1}(f,g)_f > \deg_y g^i T_\beta$$

for each individual term $g^i T_\beta$. Q.E.D.

Remark. For r with $M_{r+1} \leqq e$ we may specialize $\psi: x \to 0$ and thus replace $T_i(f,g)$ by $T_i^\psi(f,g)$ for all $i \leqq r+1$ in proposition 3. 2. All conclusions are valid. In fact, we are only interested in $T_i^\psi(f,g)$ for the obvious reason that $T_i^\psi(f,g) \in k[f,g]$.

We are in a position to connect $T_i^\psi(f,g)$ with the concept of π-root introduced in § 1. For this purpose we give the following definition

Definition 3. 1. Let $h_1(y),\ldots,h_s(y)$ be polynomials in $K[y]$ where $K=k\langle\langle t\rangle\rangle$. A common π-root $\sigma=\sum \alpha_j t^j + \pi t^\delta$ of $h_1(y),\ldots,h_s(y)$ is said to be a distribution detector if given

(1) $\deg_y h_i(y) = p_i$,

(2) $h_i(\sigma) = h_{i\sigma}(\pi)t^{\lambda_i^*}$ + higher terms in t

we have for some λ and $c_{ij} \in k$ the following

(3) $\lambda_i^* = \lambda p_i$,

(4) $h_{i\sigma}(\pi)^{p_j} = c_{ij} h_{j\sigma}(\pi)^{p_i}$.

Remark. Note that condition (4) is equivalent to the statement that with $p = \text{g.c.d.}(p_1,\ldots,p_s)$ there exists a polynomial $h(\pi)$ of degree p such that

$$h_{i\sigma}(\pi) = c_i h(\pi)^{r_i}$$

with

$$p_i = p r_i .$$

The existence of a distribution detector $\sigma=\sum \alpha_j t^j + \pi t^\delta$ implies that the roots of $h_1(y),\ldots,h_s(y)$ in the δ-disc centered at $\sum \alpha_j t^j$ are all distributed proportionally according to the degrees of $h_i(y)$.

Now we are in a position to state the first connection between the approximate roots $T_i^{\psi}(f, g)$ and the concept of π-root introduced in § 1. Namely we have

Proposition 3. 3. *Let* $g(y)=g(x, y)$, $T_i^{\psi}(y)=T_i^{\psi}\big(f(x, y), g(x, y)\big)$ *and*

$$T_i^{\psi}(y)_f=T_i^{\psi}\big(f(x, y), g(x, y)\big)_f$$

be considered as polynomials in $k\langle\langle t\rangle\rangle[y]$ *with* $x=t^{-1}$. *Suppose* $\sigma=\sum\alpha_j t^j+\pi t^{\delta}$ *is a distribution detector for* $g(y)$, $T_i^{\psi}(y)$, $i=1,\ldots, r$ *where* r *is an integer with* $M_r\leqq e$ *and* $\operatorname{ord} g(\sigma)<0$. *Then* σ *is a distribution detector after we add* $T_{r+1}^{\psi}(y)_f$ *to the above list of polynomials.*

Proof. Let λ be the rational negative number satisfying

$$\operatorname{ord} g(\sigma)=n\lambda<0.$$

Then we have for $i=1,\ldots, r$

$$\operatorname{ord} T_i^{\psi}(\sigma)=(-\mu_i)\lambda<0$$

and

$$\operatorname{ord} T_{\alpha}^{\psi}(\sigma)=\sum(-\mu_i\alpha_i)\lambda=\big(\deg_y T_{\alpha}^{\psi}(y)\big)\lambda.$$

We shall establish for $i=1,\ldots, r+1$

$$\operatorname{ord} T_i^{\psi}(\sigma)_f=\deg_y^{\psi}\big(T_i(y)\big)_f\lambda.$$

Note that for $i=1$ both sides of the above equation are zeroes hence equal. In general let us consider the following equation mentioned in the conclusion (2) of Proposition 3.2 (cf. the Remark after it).

$$(14)^* \qquad T_{r+1}^{\psi}(y)_f=n_r T_r^{\psi}(y)^{n_r-1}T_r^{\psi}(y)_f+\sum C_{i,\beta}\, g^i(y)\, T_{\beta}^{\psi}(y).$$

By mathematical induction we have

$$\operatorname{ord} T_r^{\psi}(\sigma)_f=\deg_y\big(T_r^{\psi}(y)\big)_f\lambda$$

and, note $\lambda<0$, the following for the terms in equation (14)*

$$\begin{aligned}\operatorname{ord}\big(T_r^{\psi}(\sigma)^{n_r-1}T_r^{\psi}(\sigma)_f\big)&=\deg_y\big(T_r^{\psi}(y)^{n_r-1}T_r^{\psi}(y)_f\big)\lambda\\&<\big(\deg_y g^i(y)\, T_{\beta}(y)\big)\lambda=\operatorname{ord} g^i(\sigma)\, T_{\beta}(\sigma).\end{aligned}$$

Thus

$$\begin{aligned}\operatorname{ord}\big(T_{r+1}^{\psi}(\sigma)\big)_f&=\operatorname{ord}\big(T_r^{\psi}(\sigma)^{n_r-1}T_r^{\psi}(\sigma)_f\big)\\&=\deg_y\big(T_r^{\psi}(y)^{n_r-1}T_r^{\psi}(y)_f\big)\lambda=\big(\deg_y T_{r+1}^{\psi}(y)_f\big)\lambda.\end{aligned}$$

We have verified the condition (3) in Definition 3. 1. To verify the condition (4) we note that the leading coefficient $T(\pi)$ of $T_{r+1}^{\psi}(\sigma)_f$ is the leading coefficient of $T_r^{\psi}(\sigma)^{n_r-1}T_r^{\psi}(\sigma)_f$. We know that there is a polynomial $h(\pi)$ (cf. the remark after Definition 3. 1) such that the leading coefficients of $T_r^{\psi}(\sigma)$ and $T_r^{\psi}(\sigma)_f$ are of the form $c_i h(\pi)^{r_i}$ where r_i equals (the y-degree of the corresponding polynomial in y)·(a common constant q). Thus $T(\pi)$, the leading coefficient of $T_{r+1}^{\psi}(\sigma)_f$, equals $\operatorname{ch}(\pi)^{r^*}$ where r^* equals $\big(\deg_y T_{r+1}^{\psi}(y)_f\big)q$. The condition (4) of Definition 3. 1 is thus verified. Q.E.D.

§ 4. The splitting and nonsplitting phenomena

From this section on we shall assume that the following Jacobian condition is satisfied for a pair of polynomials monic in y, $f(x, y)$ and $g(x, y)$ in $k[x, y]$ if the opposite is not explicitly stated.

Jacobian condition:

$$J_{x,y}(f, g) = \det \begin{vmatrix} \dfrac{\partial f(x, y)}{\partial x}, & \dfrac{\partial f(x, y)}{\partial y} \\ \dfrac{\partial g(x, y)}{\partial x}, & \dfrac{\partial g(x, y)}{\partial y} \end{vmatrix} = 1.$$

As usual we set $t = x^{-1}$ and consider the polynomials $f(x, y)$ and $g(x, y)$ as polynomials in y with coefficients in $k\langle\langle t\rangle\rangle$. We shall write $f(x, y)$ as $f(y)$ and $g(x, y)$ as $g(y)$ to simplify our notations. Our problem is to investigate the tree data of $f(y)$, $g(y)$ and $T_i^{\psi}(y) = T_i^{\psi}(f(x, y), g(x, y))$.

As we pointed out in section 1, the algebraic tool for our investigation is the notion of π-root $\sigma \in k[\pi]\langle\langle t\rangle\rangle$ for a polynomial $h(y)$ with

$$\sigma = \sum_{j<\delta} a_j t^j + \pi t^{\delta}.$$

Then we have established in § 1 that

(1) The number of roots of $h(y) = 0$ inside the disc centered at $\sum a_j t^j$ with the logarithmic radius δ is the π-degree of the leading coefficient of $h(\sigma)$ i.e. the multiplicity of σ as a π-root of $h(y) = 0$ (cf. Proposition 1. 2).

(2) The leading coefficient of $h(\sigma)$ has two distinct roots for π iff the disc centered at $\sum a_j t^j$ with radius δ is a disc associated with the tree data of $h(y) = 0$ (cf. Lemma 1. 1). Hence, if the leading coefficient of $h(\sigma)$ has only one root, then we have to shrink the radius to locate the disc associated with the tree data.

(3) Let $(\pi - b)$ be a factor of the leading coefficient of $h(\sigma)$ with multiplicity l. Then there are precisely l roots $\tau_1, \ldots, \tau_l$ of the equation $h(y) = 0$ (cf. Proposition 1. 2) which satisfy

$$\operatorname{ord}(\tau_i - \sum a_j t^j + b t^{\delta}) > \delta$$

thus

$$\operatorname{ord}(\tau_i - \tau_j) > \delta.$$

In view of the discussion preceding Definition 1. 1 there is a disc associated with the tree data and of radius less than δ which contains those l roots $\tau_1, \ldots, \tau_l$. Note that the size of this disc is not determined.

To summarize the preceding discussions we have *the nonsplitting phenomena* (cf. Proposition 4.4): If the leading coefficient of $h(\sigma)$ has only one root, then the δ-disc has to be shrunk to determine the tree data. *The splitting phenomena* (cf. Proposition 4.6): If the leading coefficient of $h(\sigma)$ has distinct roots $b_1,\ldots, b_s$ with multiplicities $l_1,\ldots, l_s$ then the δ-disc can be covered by s proper subdisc containing $l_1,\ldots, l_s$ roots respectively (cf. Definition 1.1).

In this section we shall determine the conditions for the nonsplitting phenomena and the splitting phenomena for the polynomials $f(y)$, $g(y)$ and $T_i^{\psi}(y)$. For the splitting phenomena we shall further determine the necessary restrictions on the number of roots and the corresponding multiplicities.

Firstly we shall establish the following simple proposition.

Proposition 4.1. *Let* $x=t^{-1}$ *and* $\sigma=\sum a_j t^j+\pi t^{\delta}$ *be a* π*-root of* $g(y)=g(x,y)$. *Given any polynomial* $h(y)=h\big(f(x,y), g(x,y)\big)\in k[f(x,y), g(x,y)]$ *with*

$$g(\sigma)=g_\sigma(\pi)t^{\lambda_g}+\textit{higher terms in } t,$$
$$h(\sigma)=h_\sigma(\pi)t^{\lambda_h}+\textit{higher terms in } t$$

we have

$$(1)\qquad J_{t,\pi}\big(h(\sigma), g(\sigma)\big)=[\lambda_h h_\sigma(\pi)\, g'_\sigma(\pi)-\lambda_g\, g_\sigma(\pi)\, h'_\sigma(\pi)]\, t^{\lambda_g+\lambda_h-1}+\cdots$$
$$=-h_f(\sigma)t^{-2+\delta}.$$

Proof. The first equality follows trivially from the definition of Jacobians. For the second equality we have by the chain rule the following

$$J_{t,\pi}\big(h(\sigma), g(\sigma)\big)=J_{f,g}(h,g)\, J_{x,y}(f,g)\, J_{t,\pi}(x,\sigma)$$
$$=h_f(\sigma)\cdot 1\cdot J_{t,\pi}(x,\sigma)=-h_f(\sigma)t^{-2+\delta}. \qquad \text{Q.E.D.}$$

The ordinary differential operator involving two polynomials $h_\sigma(\pi)$ and $g_\sigma(\pi)$ in the parenthesis of the equation (1) is of some interest to us. In Appendix I we shall give a detailed discussion of the polynomial solutions of some equations involving this differential operator. For the convenience of our discussions we shall introduce the following definition.

Definition 4.1. Let a, b be constants in k and $p(\pi)$, $q(\pi)$ be two polynomials in $k[\pi]$. Then the symbol $D\big(a, b, p(\pi), q(\pi)\big)$ is defined as

$$D\big(a, b, p(\pi), q(\pi)\big)=ap(\pi)\, q'(\pi)-bq(\pi)\, p'(\pi).$$

In Proposition 3.3 we have established that under the condition $\operatorname{ord} g(\sigma)<0$ if σ is a distribution detector for $g(y)$, $T_1^{\psi}(y),\ldots, T_r^{\psi}(y)$ then it is a distribution detector for $T_{r+1}(y)_f$. The following proposition will allow us to proceed one step further if certain conditions are satisfied.

Proposition 4. 2. *Let* $\sigma=\sum a_j t^j+\pi t^\delta$ *be a distribution detector* (*cf. Definition* 3. 1) *for* $g(y)=g(x, y)$, $T_i^\psi(y)=T_i^\psi\big(f(x, y), g(x, y)\big)$ *where* $i=1,\ldots, r$. *Let*

$$\deg_y g(y)=n.$$

Suppose for some $\lambda<0$ *we have*

(1) $\operatorname{ord} g(\sigma)=n\lambda$,

(2) $\operatorname{ord} T_{r+1}^\psi(\sigma)=(-\mu_{r+1})\lambda$,

(3) $\lambda<\dfrac{-1+\delta}{n-M_{r+1}}$.

Then σ *is a distribution detector for* $g(y)$, $T_i^\psi(y)$, $i=1,\ldots, r+1$.

Proof. We observe the following

$$\lambda<\frac{-1+\delta}{n-M_{r+1}} \Leftrightarrow n\lambda+(-\mu_{r+1})\lambda-1<-2+\delta+(-\mu_{r+1}+M_{r+1})\lambda.$$

Note it follows from the proof of Proposition 3. 3 and Definition 3. 1 and Proposition 3. 2 that

$$\operatorname{ord} T_{r+1}^\psi(\sigma)_f=\big(\deg_y T_{r+1}^\psi(y)_f\big)\lambda=(-\mu_{r+1}+M_{r+1})\lambda.$$

Thus the left hand side of equation (1) of Proposition 4. 1 with $h=T_{r+1}^\psi$ has a smaller formal order. We must then have

$$D\big(-\mu_{r+1}\lambda, n\lambda, T_{r+1,\sigma}^\psi(\pi), g_\sigma(\pi)\big)=0. \tag{4}$$

Let

$$\deg g_\sigma(\pi)=n^*$$

and

$$\frac{n\lambda}{n^*}=c\neq 0.$$

Then the above ordinary differential equation (4) can be reduced to the following form with $a=-\mu_{r+1}\dfrac{\lambda}{c}$

$$cD\big(a, n^*, T_{r+1,\sigma}^\psi(\pi), g_\sigma(\pi)\big)=0. \tag{5}$$

It follows from Proposition A. 1 of Appendix I that

$$\deg T_{r+1,\sigma}^\psi(\pi)=\frac{(-\mu_{r+1})\lambda}{c}=(-\mu_{r+1})^*$$

and

$$g_\sigma(\pi)^{(-\mu_{r+1})^*}=C_1\big(T_{r+1,\sigma}^\psi(\pi)\big)^{n^*}$$

where $(-\mu_{r+1})^*$ is thus defined and C_1 is a nonzero constant. Our proposition follows from Definition 3. 1. Q.E.D.

Before handling the general nonsplitting phenomena, we shall state the following proposition which is illustrative (cf. [A.1], pp. 158, 159). Recall that M_h is the last M_i in our notation.

Proposition 4. 3. *Suppose that* $\deg g(x, y) = \deg_y g(x, y) = n > 1$, *and* $M_i \neq n-2$ *for* $i = 1, \ldots, h$. *Then the highest homogeneous forms of* $g(x, y)$, $T_i^{\psi}(f(x, y), g(x, y))$ *for all* i *with* $M_i < n-2$ *are powers of a common linear form* (*i.e. one point at infinite*).

Proof. Let δ be the number defined as

$$\delta = \min \left\{ \operatorname{ord} \tau : \tau \text{ root of } g(y) \prod_{i=1}^{h} T_i^{\psi}(y) \right\}.$$

Let σ be the π-root of $g(y) \prod_{i=1}^{h} T_i^{\psi}(y)$ defined as

$$\sigma = \pi t^{\delta}.$$

Let us consider the possibility of $\delta > -1$. With this possibility we know that all roots of the equation $g(y) = 0$ are in the disc centered at zero with logarithmic radius $\delta > -1$. Then we set

$$\sigma^* = \pi t^{-1}$$

and conclude (cf. Proposition 1. 3) that

$$g(\sigma^*) = \pi^n t^{-n} + \cdots.$$

The above statement is equivalent to the degree n homogeneous form of $g(x, y)$ being y^n. Similarly the highest homogeneous forms of $T_i(f(x, y), g(x, y))$ are powers of y^n. Our proposition is thus trivial. Let us henceforth assume that

$$\delta \leqq -1.$$

There are two cases i.e. (1) $M_h \neq n-1$, thus $M_h < n-2$ (cf. Lemma 2. 1) (2) $M_h = n-1$.

Case 1. We shall apply Proposition 4. 2. It follows straightforward from the definition of δ that we have

$$\begin{aligned}
&\deg g(\pi) = n, \\
&\operatorname{ord} g(\sigma) = n\delta, \\
&\deg T_{i,\sigma}^{\psi}(\pi) = \deg_y T_i^{\psi}(y) = -\mu_i \quad \text{for} \quad i = 1, \ldots, h, \\
&\operatorname{ord} T_i^{\psi}(\sigma) = (-\mu_i)\delta \qquad\quad \text{for} \quad i = 1, \ldots, h.
\end{aligned}$$

We shall set

$$\lambda = \delta.$$

Now it is simply assumed that

$$M_i < n-2 \quad \text{for} \quad i = 1, \ldots, h.$$

Then we must have

$$\lambda = \delta < \frac{-1}{n - M_i - 1}$$

which is equivalent to

$$\lambda = \delta < \frac{-1 + \delta}{n - M_i}.$$

Thus the conditions of Proposition 4. 2 are verified. The conclusion of Proposition 4. 2 states that σ is a distribution detector for $g(y)$, $T_1(y), \ldots, T_h(y)$. Recall

$$\text{g.c.d.}(n, -\mu_1, \ldots, -\mu_h) = 1.$$

Thus the polynomials $g_\sigma(\pi)$, $T^{\psi}_{1,\sigma}(\pi), \ldots, T_{h,\sigma}(\pi)$ are powers of a common linear polynomial (cf. Definition 3. 1).

We claim that

$$\delta = -1.$$

Otherwise suppose for a moment that

$$\delta < -1.$$

Due to the assumption

$$\deg_y g(x, y) = \deg g(x, y)$$

it is elementary to see

$$g(\sigma) = \pi^n t^{n\delta} + \cdots.$$

Thus the common linear polynomial for $g_\sigma(\pi)$, $T^{\psi}_{1,\sigma}(\pi), \ldots, T^{\psi}_{h,\sigma}(\pi)$ must be simply π. It contradicts the very definition of δ. Hence our claim has been proved. Let

$$g(x, y) = g_n(x, y) + \text{lower terms in } x, y.$$

Then it is easy to see

$$g(t^{-1}, \pi t^{-1}) = g_n(1, \pi) t^{-n} + \text{higher terms in } t.$$

Moreover we have proved that the polynomial $g_n(1, \pi) = g_\sigma(\pi)$ is a power of a linear polynomial in π. It follows trivially that the highest homogeneous forms of $g(x, y)$, $T_i(f(x, y), g(x, y))$ are powers of a linear form.

Case 2. $M_h = n - 1$. The following Proposition 4. 4 will establish that the polynomials $g_\sigma(\pi)$, $T^{\psi}_{1,\sigma}(\pi), \ldots, T^{\psi}_{h,\sigma}(\pi)$ are powers of a common linear polynomial $(\pi - a)$. Then the last paragraph of the proof for Case 1 can be used verbatim to finish Case 2. Q.E.D.

The following proposition gives some sufficient conditions for the nonsplitting phenomena.

Proposition 4. 4. *Suppose that for some* r, λ, v, ε *and a* π*-root* $\sigma=\sum a_j t^j+\pi t^\delta$ *of* $g(y)$ *we have the following*:

(1) $\deg_y g(y)=n>1$.

(2) $\operatorname{ord} g(\sigma)=n\lambda<0$,

$\operatorname{ord} T_i^\psi(\sigma)=(-\mu_i)\lambda \qquad i=1,\ldots,r-1$.

(3) $\lambda<\dfrac{-1+\delta}{n-M_i}$.

(4) $\deg g_\sigma(\pi)=v\left(\dfrac{n}{d_r}\right)$.

(5) $\deg T_{r,\sigma}^\psi(\pi)=v\left(\dfrac{-\mu_r+M_r-n}{d_r}\right)+1$.

(6) $\operatorname{ord} T_r^\psi(\sigma)=(-\mu_r+M_r-n)\lambda-1+\delta$

$=(\deg T_{r,\sigma}^\psi(\pi)+\varepsilon)\left(\dfrac{d_r}{v}\right)\lambda$.

(7) *The number* ε *defined above satisfies* $\varepsilon>-1$.

Then $g_\sigma(\pi), T_{1,\sigma}^\psi(\pi),\ldots, T_{r,\sigma}^\psi(\pi)$ *are powers of a common linear polynomial in* π.

Remark. This proposition has been applied to the second case of the preceding proposition with $r=h$, $M_r=n-1$, $\lambda=\delta$, $v=d_h$ and $\varepsilon=-\dfrac{1}{\delta}$.

Proof of the proposition. It follows from the conditions (1), (2), (3), and Proposition 4. 2 that σ is a distribution detector for $g(y), T_1^\psi(y),\ldots, T_{r-1}^\psi(y)$. Thus the π-degrees of the leading coefficients of $g(\sigma), T_1^\psi(\sigma),\ldots, T_{r-1}^\psi(\sigma)$ are proportional to

$$n:-\mu_1:\cdots:-\mu_{r-1}.$$

Note that g.c.d. $(n,-\mu_1,\ldots,-\mu_{r-1})=d_r$. Hence the said degrees must be $v\left(\dfrac{n}{d_r}\right)$, $v\left(\dfrac{-\mu_1}{d_r}\right),\ldots, v\left(\dfrac{-\mu_{r+1}}{d_r}\right)$ for the integer v satisfying condition (4). It follows from the definition of the distribution detectors that there is a polynomial $p(\pi)$ of degree V such that

$$g_\sigma(\pi)=C_0\,p(\pi)^{\frac{n}{d_r}},$$

$$T_{1,\sigma}^\psi(\pi)=C_1\,p(\pi)^{\frac{-\mu_1}{d_r}},$$

$$\vdots$$

$$T_{r-1,\sigma}^\psi(\pi)=C_{r-1}\,p(\pi)^{\frac{-\mu_{r-1}}{d_r}},$$

$$(T_{r,\sigma}^\psi(\pi))_f=C_r\,p(\pi)^{\frac{-\mu_r+M_r}{d_r}}.$$

In view of the condition (6) after comparing the orders on both sides of equation (1) of Proposition 4.1 with $h(f, g) = T_r^{\psi}(f, g)$ we conclude that $T_{r,\sigma}^{\psi}(\pi)$ satisfies the following differential equation

$$\begin{aligned}(\deg T_{r,\sigma}^{\psi}(\pi)+\varepsilon)\, T_{r,\sigma}^{\psi}(\pi)\, g_{\sigma}'(\pi) - \deg g_{\sigma}(\pi)\, g_{\sigma}(\pi)\, (T_{r,\sigma}^{\psi}(\pi))' \\ = cp(\pi)^{\frac{-\mu_n+M_r}{d_r}}.\end{aligned}$$

After simplifying, the above equation becomes

$$\begin{aligned}(\deg T_{r,\sigma}^{\psi}(\pi)+\varepsilon)\, T_{r,\sigma}^{\psi}(\pi)\, p'(\pi) - vp(\pi)\, (T_{r,\sigma}^{\psi}(\pi))' \\ = c^* p(\pi)^{\frac{-\mu_r+M_r-n}{d_r}+1}.\end{aligned}$$

It follows from Proposition A.2 of Appendix I that the polynomials $p(\pi)$ and $T_{r,\sigma}^{\psi}(\pi)$ are powers of a common linear polynomial in π. Our proposition is thus established.

Q.E.D.

Remark. With a verbatim proof, for a slightly general Jacobian condition of the following form (cf. Appendix II)

$$J_{x,y}(f(x, y), g(x, y)) = x^l$$

Proposition 4.4 is still valid with the condition (6) replaced by the following

$$\begin{aligned}(6)^* \qquad \operatorname{ord} T_r^{\psi}(\sigma) &= (-\mu_r + M_r - n)\lambda - 1 + \delta - l \\ &= (\deg T_{r,\sigma}^{\psi}(\pi)+\varepsilon)\left(\frac{d_r}{v}\right)\lambda.\end{aligned}$$

The following proposition will be proved after Proposition 4.6. The purpose of inserting Proposition 4.5 is to illustrate the splitting phenomena. One interpretation of Proposition 4.5 is that there are at most two points at infinite for $f(x, y) = 0$. Note that the conditions of Proposition 4.5 complete the conditions of Proposition 4.3 (cf. [A.1], p. 139).

Proposition 4.5. *Suppose that* $\deg g(x, y) = \deg_y g(x, y) = n > 1$ *and* $M_r = n-2$ *for some* r. *Then the highest homogeneous forms of* $g(x, y)$, $T_i(f(x, y), g(x, y))$ *for* $i < r$ *are powers of a common form which has at most two distinct linear factors. Moreover the multiplicities for the two factors are different.*

The following proposition gives some sufficient conditions for the general splitting phenomena.

Proposition 4. 6. *Suppose that for some* r, λ, v *and a* π*-root* $\sigma=\sum a_j t^j+\pi t^\delta$ *of* $g(y)$ *we have the following*:

(1) $\deg_y g(y)=n>1$,

(2) $\operatorname{ord} g(\sigma)=n\lambda<0$, $\operatorname{ord} T_i^\psi(\sigma)=(-\mu_i)\lambda \qquad i=1,\ldots,r$,

(3) $\lambda=\dfrac{-1+\delta}{n-m_r}<\dfrac{-1+\delta}{n-m_i} \qquad i=1,\ldots,r-1$,

(4) $\deg g_\sigma(\pi)=v\left(\dfrac{n}{d_r}\right)$,

(5) $\deg T_{r,\sigma}^\psi(\pi)=v\left(\dfrac{-\mu_r}{d_r}\right)$

$$=v\left(\frac{-\mu_r+M_r-n}{d_r}\right)+v\left(\frac{n-M_r}{d_r}\right).$$

Then we have, if $r\geqq 2$, *the following*:

(1) σ *is a distribution detector for* $g(y)$, $T_1^\psi(y),\ldots,T_{r-1}^\psi(y)$. *The leading coefficients of* $g(\sigma)$, $T_1^\psi(\sigma),\ldots,T_{r-1}^\psi(\sigma)$ *are powers of a common polynomial* $p(\pi)$ *of degree* v,

(2) $T_{r,\sigma}^\psi(\pi)=p(\pi)^{\frac{-\mu_r+M_r-n}{d_r}}\cdot q(\pi)$ *where* $q(\pi)$ *is a polynomial of degree* $v\left(\dfrac{n-M_r}{d_r}\right)$,

(3) $q(\pi)$ *has all distinct roots*,

(4) *all roots of* $p(\pi)$ *are roots of* $q(\pi)$,

(5) $p(\pi)$ *is not a power of* $q(\pi)$.

Moreover if $r=1$ *then* $g_\sigma(\pi)$ *and* $T_{1,\sigma}^\psi(\pi)$ *satisfy the following differential equation*

$$D\left(n,\ -M_1,\ g_\sigma(\pi),\ T_{1,\sigma}^\psi(\pi)\right)=\text{nonzero constant}.$$

Proof. Let us consider the case $r\geqq 2$. The conclusion (1) follows from Proposition 4. 2 and Definition 3. 1.

Let us apply Proposition 4. 1 for $h(y)=T_r^\psi(y)$. It follows from the conditions (2) and (3) that we have

$$\lambda=\frac{-1+\delta}{n-m_r} \Leftrightarrow (n-m_r)\lambda=-1+\delta$$

$$\Leftrightarrow n\lambda+(-\mu_r)\lambda-1=-2+\delta+(-\mu_r+M_r)\lambda$$

$\Leftrightarrow$ the order of both sides of the equation (1) of proposition 4. 1 are equal.

Thus we conclude that the polynomials $g_\sigma(\pi)$ and $T^\psi_{r,\sigma}(\pi)$ must satisfy the following differential equation

$$(6) \qquad D\big(n\lambda, (-\mu_r)\lambda, g_\sigma(\pi), T^\psi_{r,\sigma}(\pi)\big) = C \text{ the leading coefficient of } T^\psi_r(\pi)_f.$$

Note that the conclusion (1) and Proposition 3.3 imply

$$g_\sigma(\pi) = C_0\, p(\pi)^{\frac{n}{d_r}},$$
$$\big(T^\psi_{r,\sigma}(\pi)\big)_f = \text{the leading coefficient of } T^\psi_r(\sigma)_f$$
$$= C_r\, p(\pi)^{\frac{-\mu_r + M_r}{d_r}}.$$

The differential equation (6) can be simplified as follows

$$D\left(v\left(\frac{-\mu_r}{d_r}\right)v,\ p(\pi),\ T^\psi_{r,\sigma}(\pi)\right) = C^* p(\pi)^{\frac{-\mu_r + M_r - n}{d_r}+1}.$$

Note that

$$\deg T^\psi_{r,\sigma}(\pi) = v\left(\frac{-\mu_r}{d_r}\right) = v\left(\frac{-\mu_r + M_r - n}{d_r}\right) + v\left(\frac{n - M_r}{d_r}\right).$$

Then the conclusions (2), (3), (4) and (5) follow at once from Propositions A.3 and A.4 of Appendix 1.

Let us consider the case $r=1$. As in the preceding argument we shall apply Proposition 4.1 for $h(y) = T^\psi_1(y)$. Note that

$$T^\psi_1 = f + \text{polynomial in } g.$$

It follows

$$(T^\psi_1)_f = 1.$$

It is elementary to show that orders of both sides of the equation (1) of Proposition 4.1 are equal. Thus the leading coefficients $g_\sigma(\pi)$ and $T_{1,\sigma}(\pi)$ must satisfy the said equation.

Q.E.D.

Remark. With a verbatim proof, for a slightly general Jacobian condition of the following form (cf. Appendix II)

$$J_{x,y}\big(f(x,y), g(x,y)\big) = x^l$$

Proposition 4.6 is still valid with the condition (3) replaced by the following:

$$(3)^* \qquad \lambda = \frac{-1-l+\delta}{n-m_r} < \frac{-1-l+\delta}{n-m_i} \qquad i = 1, \ldots, r-1.$$

Proof of Proposition 4. 5. Let δ be the number defined as

$$\delta = \min\left\{\operatorname{ord}\tau\colon\ \tau \text{ root of } g(y)\prod_{i=1}^{r} T_i^{\psi}(y)\right\}.$$

Using the first paragraph of the proof of Proposition 4. 3 verbatim we reduce the proof to the case that

$$\delta \leqq -1.$$

We claim

$$\delta = -1.$$

Otherwise assume for a moment

$$\delta < -1.$$

Then we shall apply Proposition 4. 2 with $(r-1)$ replacing r, $\lambda = \delta$ and $\sigma = \pi t^{\delta}$. The conditions (1) and (2) are trivially verified. To verify the condition (3) we observe

$$\lambda = \delta < \frac{-1+\delta}{n-M_r} = \frac{-1+\delta}{2} \Leftrightarrow \delta < -1.$$

Thus σ is a distribution detector for $g(y)$, $T_1(y),\ldots, T_r(y)$. With the assumption that

$$\deg g(x, y) = \deg_y g(x, y)$$

we see at once

$$g(\sigma) = \pi^n t^{n\delta} + \cdots.$$

It follows from the definition of the distribution detectors (cf. Definition 3. 1) that

$$T_i^{\psi}(\sigma) = \pi^{-\mu_i} t^{-\mu_i \delta} + \cdots.$$

It thus contradicts the definition of δ. Our claim is established.

We shall now apply Proposition 4. 6 with $r = r$, $\lambda = \delta = -1$, $v = d_r$ and $\sigma = \pi t^{\delta}$. It is trivial to verify all conditions. Note that $r \geqq 2$. Otherwise we have $M_1 = n - 2 \geqq 0$. Note that we then have from the definition of M_1

$$n \nmid M_1 \Rightarrow M_1 > 0,$$
$$\deg_y T_1^{\psi}(f(x, y), g(x, y)) = -\mu_1 = -M_1 < 0.$$

A contradiction. The polynomial $q(\pi)$ in the conclusion (2) is of degree

$$\deg q(\pi) = v\left(\frac{n-M_r}{d_r}\right) = 2.$$

Thus the polynomial $q(\pi)$ has two distinct roots. Since the polynomial $p(\pi)$ has at most two roots (cf. the conclusion (4)) with distinct multiplicities (cf. the conclusion (5)). Thus our proposition follows from the conclusion (2). Q.E.D.

§ 5. The major discs

In this section we shall study the root configurations of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$ for $r=1,\ldots,h$. Propositions 4.4 and 4.6 are our main tools. Topologically in both propositions we investigate certain δ-discs. Under the conditions of Proposition 4.4 we prove that all roots of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$ in it have distances less than $2^{-\delta}$ or the order of their differences are bigger than δ. In the terminology of Section 1 this particular δ-disc is not one of the discs associated with the tree data of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$ (cf. Definition 1.1). In other words we have to further shrink the size of the discs to locate the tree data. After finitely many shrinking steps we expect that the conditions of Proposition 4.6 will be satisfied. Proposition 4.6 implies that there are roots of $g(y)\prod_{i=1}^{r} T_i(y)$ in the corresponding δ-disc with distances equal to $2^{-\delta}$. Thus the δ-disc $D_{1,i_2,\ldots,i_r}$ is associated with the tree data of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$. Moreover if $r\geqq 2$ then the roots in it will be covered by subdiscs $D_{1,i_2,\ldots,i_r,j}$ satisfying the criterion of Definition 1.1. Furthermore Proposition 4.6 specifies that the number of subdiscs $D_{1,i_2,\ldots,i_r,j}$ is the degree of $q(\pi)$, and the number of roots in those subdiscs are unequal. If we choose one of the subdiscs with the number of roots above a certain number (cf. Definition 5.1) and drop $T_r^{\psi}(y)$ from the product $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$, i.e. we shall consider $g(y)\prod_{i=1}^{r-1} T_i^{\psi}(y)$, then the phenomena of shrinking the size of the disc until the conditions of Proposition 4.6 being met will reappear again. This way we plan to get certain information of the root configurations of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$, especially $g(y)\,T_1^{\psi}(y)$. The most important result of this section is Proposition 5.3 which establishes the existence of major towers ending in D_1 (cf. Definition 5.1). The complicated formula of δ_r can be used to show the reductions of "one point at infinite" (cf. Proposition 5.4) and "the two characteristic pair case" (cf. Proposition 5.5).

We shall start our process by establishing the following proposition.

Proposition 5.1. *Suppose that $g(x,y)$ is monic in y with y-degree $n>1$. Let δ_s be defined for $s=h-1$, h as follows*

$$\delta_s=\min\left\{\operatorname{ord}(\tau_i-\tau_j):\ \tau_i,\tau_j \text{ roots of } g(y)\prod_{i=1}^{s} T_i^{\psi}(y)=0\right\}.$$

Then we have:

(1) *If $M_h=n-1$ then $\delta_h=\delta_{h-1}=\dfrac{-1}{n-M_{h-1}-1}$. Moreover the conditions of Proposition 4.6 are satisfied with σ the unique π-root in the δ_{h-1}-disc D_{h-1} containing all roots of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$, $\delta=\delta_{h-1}=\lambda$, $r=h-1$ and $v=d_{h-1}$.*

(2) *If* $M_h<n-1$, *then* $\delta_h=\dfrac{-1}{n-M_h-1}$. *Moreover the conditions of Proposition* 4.6 *are satisfied with* σ *the unique* π*-root in the* δ_h*-disc* D_h *containing all roots of* $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$, $\delta=\delta_h=\lambda$, $r=h$ *and* $v=d_h$.

Definition – Remark. The conclusion (1) implies that if $M_h=n-1$, then we should drop it from our own consideration. The "effective" characteristic pairs exclude M_h if $M_h=n-1$. The last effective characteristic pair is denoted by M_s in this article.

Proof. (1) Let us assume for a moment that

$$\delta_h<\frac{-1}{n-M_{h-1}-1}.$$

Then the conditions of Proposition 4.4 can be verified with σ the unique π-root of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$ in the δ_h-disc and let

$$\delta=\delta_h=\lambda<0,\quad r=h,\quad v=d_h,\quad \varepsilon=-\frac{1}{\delta_h}.$$

The conclusion of Proposition 4.4 implies that

$$\operatorname{ord}(\tau_i-\tau_j)>\delta=\delta_h$$

for all τ_i, τ_j roots of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$. It contradicts the definition of δ_h. Thus we must have

$$\delta_h\geqq\frac{-1}{n-M_{h-1}-1}=\delta^*.$$

Now we shall verify the conditions of Proposition 4.6 for σ the unique π-root of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$ in the δ^*-disc and let

$$\delta=\delta^*=\lambda<0,\quad r=h-1,\quad v=d_{h-1}.$$

This verification is elementary. Note the polynomial factor $q(\pi)$ of $T^{\psi}_{h-1,\sigma}(\pi)$ in Proposition 4.6 is with degree as follows

$$\deg q(\pi)=n-M_{h-1}\geqq 2$$

and has distinct roots. Thus we have

$$\delta_h=\delta_{h-1}=\frac{-1}{n-M_{h-1}-1}$$

and the conclusion (1) is finished.

(2) Let us assume for a moment that

$$\delta_h<\frac{-1}{n-M_h-1}.$$

Then the conditions of Proposition 4.2 can be easily verified with σ the unique π-root of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$ in the δ_h-disc and let

$$\delta=\delta_h=\lambda<0, \quad r=h, \quad v=d_h.$$

Note that

$$\text{g.c.d.}(n, -\mu_1, \ldots, -\mu_h)=1.$$

The conclusion of Proposition 4.2 implies that the leading coefficients of $g(\sigma)$, $T_1^{\psi}(\sigma), \ldots, T_h^{\psi}(\sigma)$ are powers of a polynomial of degree 1. Thus

$$\text{ord}(\tau_i-\tau_j)>\delta=\delta_h$$

for all roots τ_i, τ_j of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$. It contradicts the definition of δ_h. Thus we must have

$$\delta_h \geqq \frac{-1}{n-M-1}=\delta^*.$$

Now we shall verify the conditions of Proposition 4.6 for σ the unique π-root of $g(y)\prod_{i=1}^{h} T_i^{\psi}(y)$ in the δ^*-disc and let

$$\delta=\delta^*=\lambda<0, \quad r=h, \quad v=d_h.$$

This verification is elementary. Note the polynomial factor $q(\pi)$ of $T_h^{\psi}(\pi)$ in Proposition 4.6 is with degree as follows

$$\deg q(\pi)=n-M_h\geqq 2$$

and has distinct roots. Thus we have

$$\delta_h=\frac{-1}{n-M_h-1}$$

and our proposition is finished. Q.E.D.

In fact later on we shall prove that the possibility of

$$\delta_h=\frac{-1}{n-M_1-1}$$

i.e. $h=2$ in the first case, $h=1$ in the second case, will imply that

$$h[f, g]=h[T_1, g]=h[x, y].$$

However we postpone the proof of the above statement to avoid the repetition of certain arguments. Let us examine the conclusion of Proposition 4.6 for $h-1\geqq 2$ in the first case and $h\geqq 2$ in the second case. Let $s=h-1$ in the first case and $s=h$ in the second case, then we have the following:

(1) All roots of $p(\pi)$ are roots of $q(\pi)$ which is a polynomial of degree $n-M_s$ and with distinct roots,

(2) the polynomial $p(\pi)$ with degree d_s is not a power of $q(\pi)$.

Thus among the roots of $p(\pi)$ there must be some with multiplicities bigger than $\frac{d_s}{n-M_s}$. Let c_s be any one of them. Then we shall establish

Proposition 5. 2. *Suppose that $g(x, y)$ is monic in y with y-degree $n>1$. Let M_s be the largest M_i which is less than $n-1$. Let δ_s be defined as in Proposition* 5. 1 *and suppose $s\geqq 2$. Let $\sigma\sum a_j t^j+\pi t^{\delta_s}$ be the unique π-root of $g(y)\prod_{i=1}^{s} T_i^{\psi}(y)$ in the δ_s-disc D_s and $\pi-c_s$ be a factor of $p(\pi)$ (cf. Proposition* 4. 6) *with multiplicity $V_s>\frac{d_s}{m-M_s}$. Let τ and δ_{s-1} be defined as follows*

$$\tau=\sum a_j t^j+c_s t^{\delta_s}\in D_s,$$

$$\delta_{s-1}=\min\left\{\operatorname{ord}(\tau_i-\tau_j):\ \tau_i\tau_j \text{ roots of } g(y)\prod_{i=1}^{s} T_i^{\psi}(y) \text{ with } \operatorname{ord}(\tau_i-\tau)>\delta_s\right\}$$

$$=\text{the logarithmic radius of } D_{s-1} \text{ which contains all roots } \tau_i \text{ of } g(y)\prod_{i=1}^{s} T_i^{\psi}(y) \text{ with } \operatorname{ord}(\tau_i-\tau)>\delta_s.$$

Then we have the following

$$\delta_{s-1}=1-\frac{(n-M_{s-1})\,[V_s(n-M_s)-d_s]}{(n-M_s-1)\,[V_s(n-M_{s-1})-d_s]}.$$

Moreover the conditions of Proposition 4. 6 *are satisfied with σ^* the unique π-root of $g(y)\prod_{i=1}^{s-1} T_i(y)$ in the δ_{s-1}-disc D_{s-1}, $\delta=\delta_{s-1}=\lambda$, $r=s-1$ and $v=V_s\left(\frac{d_{s-1}}{d_s}\right)$.*

The above proposition is a special case of Proposition 5. 3 which will be stated and proved in a moment. We shall prove the following numerical lemmas. The readers are advised to skip them for the time being and return to them when are needed.

Lemma 5. 1. *Let L, V_s be numbers with $L\leqq M_s<n-1$ and $V_s(n-M_s)>d_s$. Let $\delta(L)$, δ_s and $\lambda(L)$ be defined as*

$$\delta(L)=\delta^*=1-\frac{(n-L)\,[V_s(n-M_s)-d_s]}{(n-M_s-1)\,[V_s(n-L)-d_s]},$$

$$\delta_s=\delta(M_s)=1-\frac{n-M_s}{n-M_s-1}=\frac{-1}{n-M_s-1},$$

$$\lambda(L)=\lambda^*=d_s\delta_s+V_s(\delta^*-\delta_s).$$

Then $\delta(L)$ is a monotonically decreasing function of L and we have

$$\left(\frac{(n-L)}{d_s}\right)\lambda^*=-1+\delta^*<0.$$

Proof. It is easy to see that $\delta(L)$ is a monotonically decreasing function of L. To establish

$$-1+\delta^*<0$$

we simply observe the following

$$-1+\delta^*=-\frac{(n-L)\,[V_s(n-M_s)-d_s]}{(n-M_s-1)\,[V_s(n-L)-d_s]}$$

and

$$n-L\geqq(n-M_s)>(n-M_s)-1>0,$$

$$V_s(n-L)-d_s>V_s(n-M_s)-d_s>0.$$

We shall establish the equality

$$\begin{aligned}\frac{n-L}{d_s}\lambda^*=-1+\delta^* &\Leftrightarrow (n-L)\,[(d_s-V_s)\delta_s+V_s\delta^*]=(-1+\delta^*)d_s\\ &\Leftrightarrow (n-L)\,[(d_s-V_s)\,(1-\delta_s)+V_s(1-\delta^*)-d_s]=(1-\delta^*)d_s\\ &\Leftrightarrow [(n-L)\,V_s-d_s]\,(1-\delta^*)=(n-L)d_s-(n-L)\,(d_s-V_s)\,(1-\delta_s)\\ &\qquad=\frac{(n-L)\,[d_s(n-M_s-1)-(d_s-V_s)\,(n-M_s)]}{n-M_s-1}\\ &\qquad=\frac{(n-L)\,[V_s(n-M_s)-d_s]}{n-M_s-1}\end{aligned}$$

The last one is equivalent to the definition of $\delta(L)=\delta^*$. Q.E.D.

The following lemma is a generalization of the preceding lemma and will be used in the proof of Proposition 5. 3.

Lemma 5. 2. *Let L and V_i for $i=r,\ r+1,\ldots,s$ be numbers with*

$$L\leqq M_r<\cdots<M_s<n-1$$

and

$$V_i(n-M_i)>d_i.$$

Let $\delta(L)$, δ_i and $\lambda(L)$ be defined as

$$\delta(L)=\delta^*=1-\frac{(n-L)\prod_{j=r}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)\,[V_r(n-L)-d_r]\prod_{j=r+1}^{s}[V_j(n-M_{j-1})-d_j]},$$

$$\delta_i=1-\frac{(n-M_i)\prod_{j=i+1}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)\prod_{j=i+1}^{s}[V_j(n-M_{j-1})-d_j]},$$

$$\lambda(L)=\lambda^*=d_r\delta_s+\sum_{i=r+1}^{s}V_i\left(\frac{d_r}{d_i}\right)(\delta_{i-1}-\delta_i)+V_r(\delta^*-\delta_r).$$

Then $\delta(L)$ is a monotonically decreasing function of L and we have

$$\left(\frac{n-L}{d_r}\right)\lambda^* = -1+\delta^* < 0.$$

Proof. It is easy to see that $\delta(L)$ is a monotonically decreasing function of L. To establish

$$-1+\delta^* < 0$$

we observe that

$$-1+\delta^* = \frac{-(n-L)\prod_{j=r}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)[V_r(n-L)-d_r]\prod_{j=r+1}^{s}[V_j(n-M_{j-1})-d_j]}$$

and

$$(n-L) \geqq (n-M_s) > (n-M_s-1) > 0,$$
$$V_r(n-L)-d_r \geqq V_r(n-M_s)-d_r > 0,$$
$$V_j(n-M_{j-1})-d_j > V_j(n-M_j)-d_j > 0$$

which imply

$$-1+\delta^* < 0.$$

We shall establish the equality

$$\left(\frac{n-L}{d_s}\right)\lambda^* = -1+\delta^*.$$

Note the following identities:

$$\delta_{i-1}-\delta_i = \frac{\prod_{j=i+1}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)\prod_{j=i+1}^{s}[V_j(n-M_{j-1})-d_j]}\left\{(n-M_i)-\frac{(n-M_{i-1})[V_i(n-M_i)-d_i]}{V_i(n-M_{i-1})-d_i}\right\}$$

$$= \frac{d_i(M_i-M_{i-1})\prod_{j=i+1}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)\prod_{j=i}^{s}[V_j(n-M_{j-1})-d_j]}.$$

Substitute them into the following summation

$$\frac{n-L}{d_r}\lambda^* = \frac{n-L}{d_r}\left[d_r\delta_s + V_s\frac{d_r}{d_s}(\delta_{s-1}-\delta_s)+\cdots\right.$$
$$\left.+V_{r+1}\frac{d_r}{d_{r+1}}(\delta_r-\delta_{r+1})+V_r(\delta^*-\delta_r)\right].$$

Note that

$$\delta_s+\frac{V_s}{d_s}(\delta_{s-1}-\delta_s)=\frac{-1}{n-M_s-1}+\frac{V_s(M_s-M_{s-1})}{(n-M_s-1)\,[V_s(n-M_{s-1})-d_s]}$$

$$=-\frac{V_s(n-M_s)-d_s}{(n-M_s-1)\,[V_s(n-M_{s-1})-d_s]}.$$

After repeating the above computation $s-r$ times we see at once that

$$\left(\frac{n-L}{d_r}\right)\lambda^*=-1+\delta^*. \qquad \text{Q.E.D.}$$

Now we shall treat the generalization of Proposition 5.1 and 5.2. Inductively suppose that for some $r\geqq 2$ we identify a tower of *major discs* $D_s\supsetneqq D_{s-1}\supsetneqq\cdots\supsetneqq D_r$ defined as

Definition 5.1. A tower of major discs $D_s\supsetneqq D_{s-1}\supsetneqq\cdots\supsetneqq D_r$ satisfies the following four criteria with a sequence of integers $\{V_i: i=(r+1,\ldots,s+1\}$:

(1) In the disc D_i the polynomials $g(y)$ have precisely $\frac{n}{d_{i+1}}V_{i+1}$ roots. $T_j^{\psi}(y)$ has precisely $\frac{-\mu_j}{d_{i+1}}V_{i+1}$ roots for $j=1,\ldots,i$,

(2) the numbers V_i satisfy

$$\frac{V_{i+1}d_i}{d_{i+1}}\geqq V_i>\frac{d_i}{n-M_i} \quad \text{for} \quad i=r+1,\ldots,s, \quad V_{s+1}=d_{s+1},$$

(3) the disc D_i is with logarithmic radius δ_i where for $i=r,\ldots,s$

$$\delta_i=1-\frac{(n-M_i)\prod_{j=i+1}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)\prod_{j=i+1}^{s}[V_j(n-M_{j-1})-d_j]},$$

(4) for $i=r,\ldots,s$, let $\sigma_i=\sum\alpha_j t^j+\pi t^{\delta_i}$ be the unique general point in the disc D_i then the conditions of Proposition 4.6 are satisfied for $\sigma=\sigma_i$, $\delta=\delta_i$ and $v=V_{i+1}\left(\frac{d_i}{d_{i+1}}\right)$.

Note that Proposition 5.1, which has been proved, implies that the minimal disc D_s which contains all roots of $g(y)\prod_{i=1}^{h}T_i^{\psi}(y)$ satisfies the criteria (1), (2), (3) and (4). Proposition 5.2 simply states that among the disjoint minimal subdiscs of D_s (cf. Definition 1.1) which cover the roots of $g(y)\prod_{i=1}^{s}T_i^{\psi}(y)$ in D_s any subdisc can be used as D_{s-1} if it contains more than the average number of roots. The existence of such a subdisc is ensured by Proposition 4.6.

Proposition 5. 3. *Suppose that $g(x, y)$ is monic in y with y-degree $n>1$ and for $r\geqq 2$ a tower of major discs $D_s \supsetneqq D_{s-1} \supsetneqq \cdots \supsetneqq D_r$ is constructed. Let $\pi - C_r$ be a factor of $p(\pi)$ as in the conclusions of Proposition* 4. 6 *with multiplicity V_r satisfying*

$$\deg p(\pi) = V_{r+1}\left(\frac{d_r}{d_{r+1}}\right) \geqq V_r > \frac{d_r}{n-M_r}.$$

Let τ and δ_{r-1} be defined as follows

$$\tau = \sum \alpha_j t^j + c_r t^{\delta_r} \in D_r,$$

$$\delta_{r-1} = \min\left\{\operatorname{ord}(\tau_i - \tau_j):\ \tau_i, \tau_j \text{ roots of } g(y)\prod_{i=1}^{r} T_i^{\psi}(y) \text{ with } \operatorname{ord}(\tau_i - \tau) > \delta_r \text{ and } \operatorname{ord}(\tau_j - \tau) > \delta_r\right\}$$

$$= \text{the logarithmic radius of } D_{r-1} \text{ which is the minimal disc containing all roots } \tau_i \text{ of } g(y)\prod_{i=1}^{r} T_i^{\psi}(y) \text{ with } \operatorname{ord}(\tau - \tau_i) > \delta_r.$$

Then we have the following:

$$\delta_{r-1} = 1 - \frac{(n-M_{r-1})\prod_{j=r}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)\prod_{j=r}^{s}[V_j(n-M_{j-1})-d_j]}.$$

Moreover the tower $D_s \supsetneqq \cdots \supsetneqq D_r \supsetneqq D_{r-1}$ is a tower of major discs.

Proof. The criterion (2) of Definition 5.1 is automatic. Let us handle the remaining ones. Let $\delta(L)$ denote

$$\delta(L) = 1 - \frac{(n-L)\prod_{j=r}^{s}[V_j(n-M_j)-d_j]}{(n-M_s-1)[V_r(n-L)-d_r]\prod_{j=r+1}^{s}[V_j(n-M_{j-1})-d_j]}.$$

Firstly we claim the following

$$\delta_{r-1} \geqq \delta(M_{r-1}).$$

Let us assume the contrary for a moment. i.e.

$$\delta(M_r) = \delta_r < \delta_{r-1} < \delta(M_{r-1}).$$

Due to the monotony of $\delta(L)$ as a function of L, we have for some L^* with

$$M_r > L^* > M_{r-1}$$

the equality

$$\delta_{r-1} = \delta(L^*) = \delta^*.$$

Let σ^* be the unique π-root of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$ in the δ_{r-1}-disc with

$$\operatorname{ord}(\sigma^*-\tau)>\delta_r.$$

We shall verify the conditions of Proposition 4. 4 with $\sigma=\sigma^*$, $r=r$, $v=V_r$ to produce a contradiction. Note that it follows from Proposition 1. 3 that

$$\begin{aligned}\operatorname{ord} g(\sigma^*) &= n\delta_s+\sum_{i=r+1}^{s} V_i\left(\frac{n}{d_i}\right)(\delta_{i-1}-\delta_i)+V_r\left(\frac{n}{d_r}\right)(\delta^*-\delta_r)\\ &=\left(\frac{n}{d_r}\right)\left[d_r\delta_s+\sum V_i\left(\frac{d_r}{d_i}\right)(\delta_{i-1}-\delta_i)+V_r(\delta^*-\delta_r)\right]\\ &=\frac{n}{d_r}\lambda^*=n\left(\frac{\lambda^*}{d_r}\right)=n\lambda.\end{aligned}$$

Then it follows from Lemma 5. 2 that

$$\lambda^*<0 \quad \text{or} \quad \lambda<0.$$

It is straightforward to verify

$$\operatorname{ord} T_i^{\psi}(\sigma)=(-\mu_i)\lambda \qquad i=1,\ldots,r-1.$$

Thus the condition (2) of Proposition 4. 4 is established. Note that it follows from Lemma 5. 2 that

$$\lambda=\frac{\lambda^*}{d_r}=\frac{-1+\delta^*}{n-L}<\frac{-1+\delta^*}{n-M_{r-1}}<\cdots<\frac{-1+\delta^*}{n-M_1}<0.$$

Thus the condition (3) of Proposition 4. 4 is finished. The conditions (4) and (5) are trivial. Let us consider condition (6) of Proposition 4. 4. It follows from Proposition 1. 3, criterion (4) of Definition 5. 1 and Proposition 4. 6 that

$$\begin{aligned}\operatorname{ord} T_r^{\psi}(\sigma) &= (-\mu_r)\delta_s+\sum_{i=r+1}^{s} V_i\left(\frac{-\mu_r}{d_i}\right)(\delta_{i-1}-\delta_i)\\ &\quad+\left[V_r\left(\frac{-\mu_r+M_r-n}{d_r}\right)+1\right](\delta^*-\delta_r)\\ &=(-\mu_r+M_r-n)\frac{\lambda^*}{d_r}+\frac{n-M_r}{d_r}\left[d_r\delta_s+\sum_{i=r+1}^{s} V_i\left(\frac{d_r}{d_i}\right)(\delta_{i-1}-\delta_i)\right]+\delta^*-\delta_r.\end{aligned}$$

Using lemma 5. 2 with $L=M_r$, we have $\delta(M_r)=\delta_r$ and

$$\frac{n-M_r}{d_r}\left[d_r\delta_s+\sum_{i=r+1}^{s} V_i\left(\frac{d_r}{d_i}\right)(\delta_{i-1}-\delta_i)\right]=-1+\delta_r.$$

We thus deduce

$$\begin{aligned}\operatorname{ord} T_r^{\psi}(\sigma) &= (-\mu_r + M_r - n)\frac{\lambda^*}{d_r} - 1 + \delta^* \\ &= (-\mu_r + M_r - n)\lambda - 1 + \delta^* \\ &= \left(\deg T_{r,\sigma}^{\psi}(\pi) + \varepsilon\right)\left(\frac{d_r}{V_r}\right)\lambda\end{aligned}$$

where

$$\varepsilon = \left(\frac{1+\delta^*}{\lambda d_r}\right) V_r - 1 = \frac{-1+\delta^*}{\lambda^*} V_r - 1 > -1.$$

We have verified all conditions of Proposition 4. 4. The conclusion of it means

$$\operatorname{ord}(\tau_i - \tau_j) > \delta_{r-1}$$

for all τ_i, τ_j roots of $g(y)\prod_{i=1}^{s} T_i^{\psi}(y)$ satisfying

$$\operatorname{ord}(\tau_i - \tau) > \delta_r, \quad \operatorname{ord}(\tau_j - \tau) > \delta_r.$$

It contradicts the very definition of δ_{r-1}. Our claim has thus been established.

Let δ^{**} denote $\delta(M_{r-1})$. Let σ^{**} be the unique π-root of $g(y)\prod_{i=1}^{r} T_i^{\psi}(y)$ in the δ^{**}-disc with

$$\operatorname{ord}(\sigma^{**} - \tau) > \delta_r.$$

We shall verify the conditions of Proposition 4. 6 (the splitting phenomena) with $\sigma = \sigma^{**}$, $\delta = \delta^{**}$, $V = V_r\left(\frac{d_{r-1}}{d_r}\right)$ and $r-1$ replacing r. Note that

$$\begin{aligned}\operatorname{ord} g(\sigma^{**}) &= \frac{n}{d_{r-1}}\left[d_{r-1}\delta_s + \sum_{i=r+1}^{s} V_i\left(\frac{d_{r-1}}{d_i}\right)(\delta_{i-1} - \delta_i) + V_r\left(\frac{d_{r-1}}{d_r}\right)(\delta^{**} - \delta_r)\right] \\ &= \frac{n}{d_{r-1}}\lambda^{**} = n\left(\frac{\lambda^{**}}{d_{r-1}}\right) < 0\end{aligned}$$

and

$$\operatorname{ord} T_i^{\psi}(\sigma^{**}) = (-\mu_i)\left(\frac{\lambda^{**}}{d_{r-1}}\right) \qquad i = 1, \ldots, r-1.$$

Conditions (1) and (2) are verified. Moreover it follows from Lemma 5. 2 that

$$\frac{\lambda^{**}}{d_{s-1}} = \frac{-1+\delta^{**}}{n - M_{s-1}} < \frac{-1+\delta^{**}}{n - M_i} \qquad i = 1, \ldots, s-2.$$

We thus have condition (3). The conditions (4) and (5) are trivial. Note that the polynomials $q(\pi)$ and $p(\pi)$ in Proposition 4. 6 are of degree

$$\deg q(\pi) = V_r\left(\frac{n - M_{r-1}}{d_r}\right) > 1, \quad \deg p(\pi) = V_r\left(\frac{d_{r-1}}{d_r}\right).$$

Moreover the polynomial $q(\pi)$ has distinct roots. It implies that at least two roots of $T^{\psi}_{r-1}(y)$ in the δ^{**}-disc have distance $2^{-\delta^{**}}$. Thus we conclude

$$\delta_{r-1} = \delta(M_{r-1}) = \delta^{**}.$$

The criteria (1), (3) and (4) follow for the extended tower. Q.E.D.

From the second criterion of Definition 5.1 we can find the smallest disc which contains all roots of $g(y)\, T^{\psi}_1(y)$. Namely (cf. [A.1], p. 14) "one point at infinite".

Proposition 5.4. *Suppose that $g(x, y)$ is monic in y with y-degree $n>1$ and a tower of major discs $D_s \supsetneq D_{s-1} \supsetneq \cdots \supsetneq D_1$ is constructed. Then the smallest disc which contains all roots of $g(y)\, T^{\psi}_1(y)$ is the D_i with*

$$i = \max\left\{r: \frac{V_{r+1} d_r}{d_{r+1}} > V_r\right\}.$$

Moreover if $\delta_i > -1$ then either $k[x, y] = k[T^{\psi}_1, g] = k[f, g]$ or there exists an automorphism of $k[x, y]$ which reduces the degrees of $T^{\psi}_1(f(x, y), g(x, y))$, $g(x, y)$ and $f(x, y)$ simultaneously.

Proof. As proved in Proposition 5.3 the numbers $\frac{V_r d_{r-1}}{d_r}$ are the degrees of $p(\pi)$'s involved. Clearly $\frac{V_r d_{r-1}}{d_r} = V_{r-1}$ iff $p(\pi)$ has only one root which means the leading coefficients $g_\sigma(\pi)$ and $T^{\psi}_{1,\sigma}(\pi)$ have only one root. It is thus evident that D_i is the smallest disc which contains all roots of $g(y)\, T^{\psi}_1(y)$.

Suppose that we have

$$\delta_i > -1 \quad \text{and} \quad \frac{V_{j+1} d_j}{d_{j+1}} = V_j \quad \text{for} \quad j = i+1, \ldots, s.$$

Then it follows from the definition of δ_i that

$$\delta_i = 1 - \frac{(n - M_i) \prod_{j=i+1}^{s} [V_j(n - M_j) - d_j]}{(n - M_s - 1) \prod_{j=i+1}^{s} [V_j(n - M_{j-1}) - d_j]} = 1 - \frac{n - M_i}{n - M_i - 1} = \frac{-1}{n - M_i - 1}.$$

There are two possibilities $i=1$ or $i>1$. In the first case we shall establish that $k[x, y] = k[T^{\psi}_1, g] = k[f, g]$. In the second case we shall show the existence of certain automorphisms of $k[x, y]$.

Let the number $n - M_i - 1$ be l. Suppose that $i=1$. Then Proposition 4.6 implies that the leading coefficients $g_\sigma(\pi)$ and $T^{\psi}_{1,\sigma}(\pi)$, which are polynomials in π, satisfy the following differential equation (cf. Definition 4.1)

(1) $$D(n, -\mu_1, g_\sigma(\pi), T^{\psi}_{1,\sigma}(\pi)) = \text{nonzero constant}.$$

Note that $M_1 = \mu_1 = -\deg_y T_1(y)$ a negative integer. We thus have

$$l = n - M_i - 1 \geqq n.$$

It follows from Proposition A.5 of Appendix 1 that the polynomial $g_\sigma(\pi)$ has all distinct roots. Any root of $g(y)$ with nonzero coefficient of $t^{\frac{1}{l}}$ will produce a different root for $g(y)$ under the automorphism over $k((t))$ of the following form

$$t^{\frac{1}{l}} \to \omega t^{\frac{1}{l}}$$

where ω is a l-th primitive root of unity. Thus we must have

$$n = l = n - M_1 - 1 \quad \text{or} \quad M_1 = -1.$$

Hence T_1^ψ is a monic polynomial of degree 1 in y and an automorphism of $k[x, y]$ will send T_1^ψ to y. A simple computation on the Jacobian shows at once that under the same automorphism

$$g(x, y) \to x + \text{polynomial in } y.$$

Thus we have

$$k[x, y] = k[T_1^\psi, g] = k[f, g].$$

Let us consider the possibility of $i > 1$. Let σ_i be the unique π-root of $g(y)\ T_1^\psi(y)$ in the δ_i-disc D_i and

$$g(\sigma) = g_\sigma(\pi) t^{n\delta_i} + \cdots, \quad T_1^\psi(\sigma) = T_{1,\sigma}^\psi(\pi)^{-\mu_1 \delta_i} + \cdots.$$

Then Propositions 5.3 and 4.6 show that $g_\sigma(\pi)$ and $T_{1,\sigma}^\psi(\pi)$ are powers of a common polynomial $p(\pi)$. Our definition of δ_i implies that $p(\pi)$ has more than one root. Let $\pi - a$ be a factor of $p(\pi)$ with $a \neq 0$. Let the unique π-root σ_i be

$$\sigma_i = \sum \alpha_j t^j + \pi t^{\delta_i}.$$

Since a noninteger exponent j in the above expression will produce different roots for $g(y)\ T_1^\psi(y)$ as shown in the previous case, then all exponents must be integers. In other words for some polynomial $h(x)$ we have

$$\sigma = h(x) + \pi t^{\delta_i}.$$

The following automorphism

$$x \to x, \quad y \to y - h(x)$$

will reduce σ to the following simple form

$$\sigma \to \pi t^{\delta_i}$$

and without changes in the y-degrees of $g(y)$ and $T_i^\psi(y)$. We shall simply assume from the beginning, that the π-root σ is of the form

$$\sigma = \pi t^{\delta_i}.$$

It follows from

$$\text{ord } g(t^{-1}, \pi t^{\delta_i}) = n\delta_i = -\frac{n}{l}$$

that we have

$$\text{ord } g(t^{-l}, \pi t^{-1}) = -n \quad \text{or} \quad \deg g(x^l, \pi x) = n.$$

Similarly we have for the nonzero a defined above

$$\text{ord } g(t^{-l}, at^{-1}) > -n \quad \text{or} \quad \deg g(x^l, ax) < n.$$

This way we have established that the highest weighted form of $g(x, y)$ with x of weight l and y of weight 1 has a factor of the form

$$a^l x - y^l.$$

An automorphism of $k[x, y]$ defined by

$$x \to x + \frac{1}{a^l} y^l, \quad y \to y$$

will save the weight of every term while sending $a^l x - y^l$ to $a^l x$. Thus under the automorphism the polynomial $g(x, y)$ will have x as a factor for the highest weighted form of weight n. This proves at once the (unweighted) degree of $g(x, y)$ must drop at least 1. Similar arguments work for T_1^{ψ}. As for $f(x, y)$ we note that (cf. Proposition 2.2)

$$T_1^{\psi} = f + \text{polynomial in } g$$

and

$$\deg T_1^{\psi}(f(x, y), g(x, y)) = \deg_y T_1^{\psi} = -M_1 \notin (n).$$

Thus if both degrees of $g(x, y)$ and T_1^{ψ} drops, then the degree of $f(x, y)$ must drop too. Q.E.D.

The following lemma will specify the implications of Proposition 5.4 in terms of the forms of the polynomial $g(x, y)$ (cf. [A.1], p. 139).

Lemma 5.3. *Suppose that $g(x, y)$ is monic in y with $\deg g(x, y) = \deg_y g(x, y) = n > 1$. Then the smallest disc which contains all roots of $g(y)$ $T_1^{\psi}(y)$ is of logarithmic radius -1 iff $M_s = n-2$ and the highest homogeneous form of $g(x, y)$ has two roots with one root having a multiplicity $\frac{n}{d_s} v_s$ where v_s satisfies*

$$d_s > v_s > \frac{d_s}{2}.$$

Proof. Necessary. Let $D_s \supset \cdots \supset D_1$ be any tower of major discs with logarithmic radii $\delta_s < \cdots < \delta_1$. The proof of Proposition 5.4 shows for some i the following

$$-1 = \delta_i = 1 - \frac{n - M_i}{n - M_i - 1} = \frac{-1}{n - M_i - 1}.$$

We conclude that

$$n-M_i-1=1, \quad n-2\geqq M_s\geqq M_i=n-2.$$

Thus

$$M_s=n-2.$$

It follows from Proposition 5.4 and 5.3

$$\frac{V_{s+1}d_s}{d_{s+1}}>V_s>\frac{d_s}{n-M_s}=\frac{d_s}{2}.$$

Recall

$$V_{s+1}=d_{s+1}.$$

We thus have

$$d_s>V_s>\frac{d_s}{2}.$$

The criteria (4) of Definition 5.1 and Proposition 4.6 specify the degree of the polynomial $q(\pi)$ (cf. Proposition 4.6) is $n-M_s=2$. Then the polynomial $p(\pi)$ has two roots with one of the multiplicities being V_s. It is routine to deduce the stated properties of the highest homogeneous form of $g(x, y)$ (cf. Propositions 4.3 and 4.5).

Sufficient. It follows from Proposition 5.3 that there is a tower of major discs $D_s\supset D_{s-1}\supset\cdots\supset D_1$ with the corresponding V_s given by this lemma. Thus the index i as specified in Proposition 5.4 is

$$i=\max\left\{r\colon \frac{V_{r+1}d_r}{d_{r+1}}>V_r\right\}=s$$

and the corresponding logarithmic radius δ_i is

$$\delta_s=\frac{-1}{n-M_s-1}=-1. \qquad \text{Q.E.D.}$$

The following proposition shows that there is a reduction for the two characteristic pairs case.

Proposition 5.5. *Suppose that $g(x, y)$ is monic in y with y-degree $n>2$. If there are only two effective characteristic pairs i.e. $s=2$, then either $k[x, y]=k[T_1^{\psi}, g]=k[f, g]$ or there exists an automorphism of $k[x, y]$ which reduces the degrees of $T_1^{\psi}(f(x, y), g(x, y))$, $g(x, y)$ and $f(x, y)$ simultaneously.*

Proof. It follows from Proposition 5.4 that we may assume that the smallest disc D_i which contains all roots of $g(y)\,T_1^{\psi}(y)$ is of logarithmic radius -1 i.e. for $i=1$ or 2 the following

$$\delta_i=-1=\frac{-1}{n-M_i-1}.$$

We shall deduce a contradiction. It follows from $M_1 = \mu_1 = -\deg_y T_1^{\psi}(y)$ a negative integer that we must have $i = 2$ and

$$M_2 = n-2, \quad \delta_2 = -1.$$

Hence we have a tower of major discs $D_2 \supset D_1$ (cf. Proposition 5. 3) with the corresponding numbers V_2, V_3 satisfying

$$\frac{V_3 d_2}{d_3} > V_2 > \frac{d_2}{n-M_2} \quad \text{or} \quad d_2 > V_2 > \frac{d_2}{n-M_2} = \frac{d_2}{2}.$$

Then it follows from Proposition 5. 3 that the logarithmic radius δ_1 of D_1 is

$$\delta_1 = 1 - \frac{(n-M_1)\,[2V_2 - d_2]}{[V_2(n-M_1) - d_2]} = \frac{(n-M_1)\,(d_2 - V_2) - d_2}{(n-M_1)\,V_2 - d_2}.$$

Recall

$$d_2 = \text{g.c.d.}\,(n, M_1).$$

Let us write

$$n = d_2 n^*, \quad -M_1 = d_2 m^*, \quad U_2 = d_2 - V_2 < V_2$$

and write δ_1 in its reduced form as

$$\delta_1 = \frac{(n^* + m^*)\,U_2 - 1}{(n^* + m^*)\,V_2 - 1} = \frac{B}{A}.$$

Clearly we have

$$0 < \delta_1 < 1$$

i.e. the rational number δ_1 is a true fraction and $A > 1$. Let σ be the π-root in the δ_1-disc D_1 and

$$g(\sigma) = g_\sigma(\pi)\,t^{n\lambda} + \cdots, \quad T_1^{\psi}(\sigma) = T_{1,\sigma}^{\psi}(\pi)\,t^{-M_1\lambda} + \cdots.$$

Then it follows from Proposition 5. 3 and Proposition 4. 6 that $g_\sigma(\pi)$ and $T_{1,\sigma}^{\psi}(\pi)$ satisfy the following differential equation

$$D\big(n, -M_1, g_\sigma(\pi), T_{1,\sigma}^{\psi}(\pi)\big) = \text{nonzero constant}.$$

Proposition A.5 implies that

$$\big(g_\sigma(\pi), T_{1,\sigma}^{\psi}(\pi)\big) = 1$$

and

$$\left(\frac{d}{d\pi}\, g_\sigma(\pi), \frac{d}{d\pi}\, T_{1,\sigma}^{\psi}(\pi)\right) = 1.$$

The conjugations of $k((t^{\frac{1}{A}}))$ over $k((t))$ show us that if the reduced denominator A of δ_1 is not a factor of $\deg g_\sigma(\pi) = n^* V_2$ then π is a factor of $g_\sigma(\pi)$. Moreover if A is a factor of $n^* V_2$ then $g_\sigma(\pi) \in k[\pi^A]$ and π is a factor of $\frac{d}{d\pi} g_\sigma(\pi)$. In view of the two co-prime conditions above we must have

$$A | n^* V_2 \quad \text{and} \quad A \nmid m^* V_2 \quad \text{or} \quad A \nmid n^* V_2 \quad \text{and} \quad A | m^* V_2 .$$

Moreover we always have from the very definition of A the following

$$A | (n^* + m^*) V_2 - 1 .$$

Let us consider the possibility of

$$(1) \qquad A | n^* V_2 \quad \text{and} \quad A | m^* V_2 - 1 .$$

The other possibility can be treated verbatim. Let

$$n^* V_2 = AD, \quad m^* V_2 - 1 = AE.$$

The second equality implies

$$(A, V_2) = 1 .$$

Thus we must have from the first equality

$$V_2 | D .$$

Let us consider the following equation

$$1 - \delta_1 = \frac{(n^* + m^*)(V_2 - U_2)}{(n^* + m^*) V_2 - 1} = (n^* + m^*) \frac{C}{A}$$

where

$$(n^* + m^*) V_2 - 1 = A(D + E), \quad V_2 - U_2 = C(D + E).$$

Then we deduce the following

$$V_2 > V_2 - U_2 = C(D + E) \geqq D \geqq V_2 .$$

A contradiction. Q.E.D.

The following proposition will take Proposition 5.5 as a special case and gives a different proof of the reduction of the two characteristic pair case.

Proposition 5.6. *Suppose that $s \geqq 2$ and that a tower of major discs*

$$D_s \supsetneqq D_{s-1} \supsetneqq \cdots \supsetneqq D_1$$

has been constructed. If the corresponding π-root σ_1, which is the unique one in D_1, is of the following form

$$\sigma_1 = \pi t^{\delta_1}$$

then either $k[x, y] = k[T_1^\psi, g] = k[f, g]$ or there exists an automorphism $k[x, y] \to k[x, y]$ which will reduce the degrees of $f(x, y)$, $g(x, y)$ and $T_1^\psi(f(x, y), g(x, y))$ simultaneously.

Proof. It follows from Proposition 5.4 and Lemma 5.3 that we shall only consider the case $\delta_s = -1$, $M_s = n-2$ and $d_s > v_s$. It suffices to produce a numerical contradiction for this case. Note the following

$$\delta_{s-1} = 1 - \frac{(n-M_{s-1})\,[V_s(n-M_s)-d_s]}{(n-M_s-1)\,[V_s(n-M_{s-1})-d_s]}$$

$$= 1 - \frac{(n-M_{s-1})\,[2V_s-d_s]}{V_s(n-M_{s-1})-d_s} = \frac{(d_s-V_s)\,(n-M_{s-1})-d_s}{V_s(n-M_{s-1})-d_s}.$$

Recall that

$$V_s(n-M_{s-1}) > V_s(n-M_s) > d_s,$$
$$d_s > V_s, \quad d_s | (n-M_{s-1}).$$

Then we conclude

$$\delta_1 \geqq \delta_{s-1} \geqq 0.$$

We shall establish $\delta_1 < 0$ to produce a contradiction.

First we shall point out a trivial observation about the Jacobian problem. Namely the polynomials $g(x, y)$ and $T_1^{\psi}(f(x, y), g(x, y))$ have no common polynomial factor. Otherwise let it be $h(x, y)$. A simple computation will show at once that

$$J_{x,y}(T_1^{\psi}, g) \in (h(x, y)).$$

A contradiction. Especially y is not a factor of either $T_1^{\psi} + c_1$ or $g + c_2$, say $g + c_2$, for all possible constants c_1 and c_2. We conclude at once that there must be a term of the form x^l in $g(x, y)$ for $l \geqq 1$. Then we have

$$\operatorname{ord} g(\sigma_1) = n\lambda \leqq -l.$$

Note that the orders of the two sides of the equation (1) of Proposition 4.1 are equal for $\sigma = \sigma_1$ and $H = T_1^{\psi}$ (cf. the criteria of Definition 5.1 and Proposition 4.6) and

$$\operatorname{ord} T_1^{\psi}(\sigma_1) = (-M_1)\lambda < 0.$$

Then it follows from $\delta_1 \geqq 0$ that

$$n\lambda + (-M_1)\lambda - 1 = -2 + \delta_1 \geqq -2.$$

We must have

$$-l - 1 \geqq -2$$

which implies

$$l = 1, \quad n\lambda \leqq -1$$

and

$$\delta_1 = n\lambda + (-M_1)\lambda - 1 + 2 \leqq (-M_1)\lambda < 0.$$

A contradiction. Q.E.D.

To deduce Proposition 5. 5 from Proposition 5. 6 we note that we only have to consider (cf. the first part of the proof of Proposition 5. 5)

$$\delta_2 = -1, \quad 0 < \delta_1 < 1.$$

Thus we have for suitable constants a and b the following

$$\sigma_1 = at^{-1} + b + \pi t^{\delta_1}.$$

And an automorphism of the following form

$$x \to x, \quad y \to y - ax - b$$

will change σ_1 to $\pi t^{-\delta_1}$. Proposition 5. 5 follows from Proposition 5. 6.

§ 6. The minor discs

In the last section we have discussed the construction of the towers of major discs (cf. Definition 5. 1 and Proposition 5. 3). Note that the minimal disc D_s which contains all roots of $g(y) \prod_{i=1}^{h} T_i^{\psi}(y)$ is a tower of one major disc (cf. Proposition 5. 1). Suppose that we have constructed a tower of major discs $D_s \supsetneqq D_{s-1} \supsetneqq \cdots \supsetneqq D_r$. The criterion (4) of Definition 5. 1 implies the conditions of Proposition 4. 6 are fulfilled. The conclusions of Proposition 4. 6 thus implies the existence of a certain polynomial $p(\pi)$. Let $(\pi - C_r)$ be a factor of the polynomial $p(\pi)$ with multiplicity V_r. If the multiplicity V_r satisfies

$$\deg p(\pi) = V_{r+1} \frac{d_r}{d_{r+1}} \geqq V_r > \frac{d_r}{n - M_r}$$

then we may extend the tower of major discs $D_s \supsetneqq \cdots \supsetneqq D_r \supsetneqq D_{r-1}$ according to the construction of D_{r-1} as described in Proposition 5. 3. On the other hand the multiplicity V_r may satisfy

$$\frac{d_n}{n - M_r} \geqq V_r.$$

We shall construct the corresponding disc D_{r-1}^* thus denoted a *minor disc* with the following characterizations.

Proposition 6. 1. *Suppose that $g(x, y)$ is monic in y with y-degree $n > 1$ and for $r \geqq 2$ a tower of major discs $D_s \supsetneqq \cdots \supsetneqq D_r$ is constructed. Let $\pi - c_r$ be a factor of $p(\pi)$ as in the conclusions of Proposition* 4. 6 *with multiplicity V_r satisfying*

$$\frac{d_r}{n - M_r} \geqq V_r \geqq 1.$$

Let τ *and* δ^*_{r-1} *be defined as*

$$\tau = \sum a_j t^j + c_r t^r \in D_r,$$

$$\delta^*_{r-1} = \min \Big\{ \operatorname{ord}(\tau_i - \tau_j) \colon \tau_i, \tau_j \text{ roots of } g(y) \prod_{i=1}^{r} T_i^{\psi}(y) \text{ with } \operatorname{ord}(\tau_i - \tau) > \delta_r \text{ and } \operatorname{ord}(\tau_j - \tau) > \delta_r \Big\}$$

$$= \text{the logarithmic radius of } D^*_{r-1} \text{ which is the minimal disc containing all roots } \tau_i \text{ of } g(y) \prod_{i=1}^{r} T_i^{\psi}(y) \text{ with } \operatorname{ord}(\tau_i - \tau) > \delta_r.$$

Then we have $\delta^*_{r-1} \geqq 1$. *Moreover let* $\sigma = \sum b_j t^j + \pi t^{\delta}$ *be any* π*-root of* $g(y) \prod_{i=1}^{r-1} T_i^{\psi}(y)$ *with*

$$\operatorname{ord}(\sigma - \tau) > \delta_r.$$

Then we have

(1) *the inequality* $\delta < 1$ *implies* $\operatorname{ord} g(\sigma) < 0$;

(2) *the inequality* $\operatorname{ord} g(\sigma) < 0$ *implies that* σ *is a distribution detector for* $g(y)$, $T_1^{\psi}(y), \ldots, T_{r-1}^{\psi}(y)$.

Proof. Suppose for a moment that $\delta^*_{r-1} < 1$. We shall deduce a contradiction as follows. Let σ^* be the unique π-root of $g(y) \prod_{i=1}^{r} T_i^{\psi}(y)$ in the disc D^*_{r-1}. Then it follows from Definition 5.1 and Proposition 1.3 that

$$\begin{aligned}
\operatorname{ord} g(\sigma^*) &= n\delta_s + \sum_{i=r+1}^{s} V_i\left(\frac{n}{d_i}\right)(\delta_{i-1} - \delta_i) + V_r\left(\frac{n}{d_r}\right)(\delta^*_{r-1} - \delta_r) \\
&= \frac{n}{d_r}\left[d_r\delta_s + \sum_{i=r+1}^{s} V_i\left(\frac{d_r}{d_i}\right)(\delta_{i-1} - \delta_i) \right] + V_r \frac{n}{d_r}(\delta^*_{r-1} - \delta_r) \\
&= \frac{n}{d_r}\lambda(\delta_r) + V_r \frac{n}{d_r}(\delta^*_{r-1} - \delta_r) \\
&= \frac{n}{n - M_r}(-1 + \delta_r) + V_r \frac{n}{d_r}(\delta^*_{r-1} - \delta_r).
\end{aligned}$$

For the definition of $\lambda(L)$ (or the special case of $L = \delta_r$) and the last equality the reader is referred to Lemma 5.2. Note that we have

$$\frac{d_r}{n - M_r} \geqq V_r, \quad 1 > \delta^*_{r-1} > \delta_r.$$

Thus we conclude trivially the following

$$(1) \qquad \operatorname{ord} g(\sigma^*) = n\left[\frac{1}{n-M_r}(-1+\delta_r)+\frac{V_r}{d_r}(\delta^*_{r-1}-\delta_r)\right] = n\lambda^* < 0$$

with λ^* defined by the above equation. Similarly we conclude

$$(2) \qquad \operatorname{ord} T_i^{\psi}(\sigma^*) = (-\mu_i)\lambda^* \quad \text{for} \quad i=1,\ldots,r-1.$$

We have thus verified the conditions (1) and (2) of Proposition 4.4. We shall establish the condition (3) of Proposition 4.4

$$\lambda^* < \frac{-1+\delta^*_{r-1}}{n-M_i} \qquad i=1,\ldots,r-1,$$

or equivalently the following inequalities

$$\frac{1}{n-M_r}(-1+\delta_r)+\frac{V_r}{d_r}(\delta^*_{r-1}-\delta_r) \leqq \frac{-1+\delta^*_{r-1}}{n-M_r} < \frac{-1+\delta^*_{r-1}}{n-M_i}$$

$$\Leftrightarrow (-1+\delta_r)+V_r\frac{n-M_r}{d_r}(\delta^*_{r-1}-\delta_r) \leqq -1+\delta^*_{r-1}$$

$$\Leftrightarrow V_r(\delta^*_{r-1}-\delta_r) \leqq \frac{d_r}{n-M_r}(\delta^*_{r-1}-\delta_r)$$

$$\Leftrightarrow V_r \leqq \frac{d_r}{n-M_r}.$$

For condition (4) and (5) (cf. Definition 5.1 and Proposition 4.6) we simply let $v=V_r$. We shall finally verify conditions (6) and (7) of Proposition 4.4. Note that

$$\operatorname{ord} T_r^{\psi}(\sigma^*) = (-\mu_r)\delta_s + \sum_{i=r+1}^{s} V_i\left(\frac{-\mu_r}{d_i}\right)(\delta_{i-1}-\delta_i) + \left[V_r\left(\frac{-\mu_r+M_r-n}{d_r}\right)+1\right](\delta^*_{r-1}-\delta_r)$$

$$= (-\mu_r)\left[\frac{1}{n-M_r}(-1+\delta_r)\right] + \left[V_r\left(\frac{-\mu_r+M_r-n}{d_r}\right)+1\right](\delta^*_{r-1}-\delta_r)$$

$$= (-\mu_r+M_r-n)\left[\frac{1}{n-M_r}(-1+\delta_r)+\frac{V_r}{d_r}(\delta^*_{r-1}-\delta_r)\right] + (-1+\delta_r)+(\delta^*_{r-1}-\delta_r)$$

$$= (-\mu_r+M_r-n)\lambda^* - 1 + \delta^*_{r-1} = \left\{\left[V_r\left(\frac{-\mu_r+M_r-n}{d_r}\right)+1\right]+\varepsilon\right\}\frac{d_r}{V_r}\lambda^*$$

where it follows from Definition 5.1 and Proposition 4.6

$$\deg T^{\psi}_{r,\sigma^*}(\pi) = V_r\left(\frac{-\mu_r+M_r-n}{d_r}\right)+1, \quad (1+\varepsilon)\frac{d_r}{V_r}\lambda^* = -1+\delta^*_{r-1}$$

which is equivalent to

$$\varepsilon = (-1+\delta^*_{r-1})\frac{V_r}{d_r\lambda^*} - 1.$$

Condition (7) follows from

$$\varepsilon > -1 \Leftrightarrow (-1+\delta^*_{r-1})\frac{V_r}{d_r\lambda'^*} > 0$$

$$\Leftrightarrow -1+\delta^*_{r-1} < 0 \Leftrightarrow \delta^*_{r-1} < 1.$$

Hence conditions (6) and (7) are satisfied. The conclusions of Proposition 4. 4 imply that the leading coefficient of $g(\sigma^*)\prod_{i=1}^{r} T_i^{\psi}(\sigma^*)$ has only one root. It thus contradicts the very definition of δ^*_{r-1} (cf. Proposition 1. 2). We have established $\delta^*_{r-1} \geqq 1$.

The second part of our proposition, i.e. the inequality $\delta < 1$ implies ord $g(\sigma) < 0$, follows almost verbatim from the second paragraph of this proof with σ^* replaced by σ and δ^*_{r-1} replaced by δ. The equations (1) and (2) become

$$\text{ord } g(\sigma) = n\left[\frac{1}{n-M_r}(-1+\delta_r)+\frac{V_r}{d_r}(\delta-\delta_r)\right] = n\lambda < 0 \tag{3}$$

with λ defined by the above equation and

$$\text{ord } T_i^{\psi}(\sigma) = (-\mu_i)\lambda \quad \text{for} \quad i = 1,\ldots,r-1. \tag{4}$$

For the last part of our proposition if $\delta < 1$ then the first part of our proposition shows that the leading coefficients of $g(\sigma)$, $T_1^{\psi}(\sigma),\ldots, T_{r-1}^{\psi}(\sigma)$ are powers of a common linear polynomial in π of degree $V_r\frac{n}{d_r}$, $V_r\left(\frac{-\mu_1}{d_r}\right),\ldots, V_r\left(\frac{-\mu_{r-1}}{d_r}\right)$ respectively. The second part of our proposition, i.e. the equation (3) and (4), shows

$$\text{ord } g(\sigma),\ \text{ord } T_1^{\psi}(\sigma),\ldots,\ \text{ord } T_{r-1}^{\psi}(\sigma)$$

are proportional to $n, -\mu_1,\ldots, -\mu_{r-1}$. Thus it follows from Definition 3. 1 that σ is a distribution detector for $g(y), T_1^{\psi}(y),\ldots, T_{r-1}^{\psi}(y)$. We are left with the case $\delta \geqq 1$. We shall apply Proposition 4. 2 inductively. Note that it suffices to check

$$\text{ord } g(\sigma) = n\lambda < 0, \tag{5}$$

$$\text{ord } T_i^{\psi}(\sigma) = (-\mu_i)\lambda \qquad i = 1,\ldots,r-1, \tag{6}$$

$$\lambda < \frac{-1+\delta}{n-M_i} \qquad i = 1,\ldots,r-1 \tag{7}$$

with equation (5) as the definition of λ.

Suppose that there is a σ which is not a distribution detector. Then we may choose one with the corresponding δ minimal. Proposition 1. 3 implies at once that equation (6) is satisfied. Equation (7) is trivial due to the fact $\delta \geqq 1$. Thus we conclude that σ is a distribution detector. A contradiction. Q.E.D.

The above proposition does not specify the logarithmic radius δ^*_{r-1} of D^*_{r-1}. We only get an estimate $\delta^*_{r-1} \geqq 1$. The significance of the minor disc D^*_{r-1} is that the roots of $g(y), T_1^{\psi}(y),\ldots, T_{r-1}^{\psi}(y)$ in it are all distributed proportionally according to the y-degrees up to the uncertainty of $2^0 = 1$. In other words if our estimate of the roots

is blurred as the order of $g, T_1^{\psi}, \ldots, T_{r-1}^{\psi}$ simultaneously reach 0, then all roots in the disc D_{r-1}^* of $g(y), T_1^{\psi}(y), \ldots, T_{r-1}^{\psi}(y)$ are indeed proportional to the y-degrees of $g(y), T_1^{\psi}(y), \ldots, T_{r-1}^{\psi}(y)$. This uncertainty is intrinsic due to two different reasons. First we may change $g(y)$ to $g(y)+c$. Let σ be a π-root of $g(y)$ such that $\operatorname{ord} g(\sigma)=0$. Then the leading coefficient of $g(\sigma)+c$ is with a variable constant c. There is no way to control the roots of it. Second, in all our arguments we may replace the approximate roots $T_i^{\psi}(y)$ by the corresponding quasi-approximate roots (cf. section 1). As the approximate roots $T_1^{\psi}(y), \ldots, T_{r-1}^{\psi}(y)$ simultaneously become units with y substituted by a π-root σ, the corresponding quasi-approximate roots will become units with different leading coefficients. This fact thus introduces some uncertainties in the conclusions.

We shall apply Proposition 6.1 to deduce some information about the Jacobian conjecture. As pointed out in Proposition 5.4 and Lemma 5.3 we shall only consider the case $M_s=n-2$ and the highest homogeneous form $g(x, y)$ equals $[(y-ax)^{v_s}(y-bx)^{u_s}]^{\frac{n}{d_s}}$ for $a \neq b$ where $u_s=d_s-v_s$. Note that Proposition 6.1 implies that the highest homogeneous forms in x and y of $T_1^{\psi}, \ldots, T_s^{\psi}$ are determined up to constant multiples. The major disc D_s is of logarithmic radius -1. It contains a major disc D_{s-1} and a minor disc D_{s-1}^* (cf. Propositions 5.3 and 6.1). The logarithmic radius δ_{s-1} of D_{s-1} is the following (cf. Definition 5.1)

$$\begin{aligned}\delta_{s-1} &= 1-\frac{(n-M_{s-1})\,[v_s(n-M_s)-d_s]}{(n-M_s-1)\,[v_s(n-M_{s-1})-d_s]} \\ &= 1-\frac{(n-M_{s-1})\,(v_s-u_s)}{v_s(n-M_{s-1})-d_s}=\frac{u_s(n-M_{s-1})-d_s}{v_s(n-M_{s-1})-d_s}.\end{aligned}$$

Recall that

$$d_s=\text{g.c.d.}\,(n, M_1, \ldots, M_{s-1}).$$

We have

$$\delta_{s-1} \geqq 0.$$

Lemma 6.1. *Suppose that $g(x, y)$ is monic in y with $\deg g(x, y)=\deg_y g(x, y)=n>1$. Let $D_s \supsetneqq D_{s-1}$ be a tower of major discs with logarithmic radius δ_s and δ_{s-1}. If $\delta_s=-1$ then $\delta_{s-1} \geqq 0$.*

Proof. Preceding discussions. Q.E.D.

It follows from Lemma 6.1 that all roots of $g(y)$ and $T_i^{\psi}(y)$ for $i=1, \ldots, s$ in the major disc D_{s-1} are of the form

$$y=at^{-1}+c_j+\text{higher terms in } t \tag{8}$$

where c_j's are constants. It follows from Proposition 6.1 that the logarithmic radius of the minor disc D_{s-1}^* is greater than or equal to 1. Thus all roots of $g(y)$ and $T_i^{\psi}(y)$ for $i=1, \ldots, s$ in the minor disc D_{s-1}^* are of the form

$$y=bt^{-1}+e+c_j^*t+\text{higher terms in } t \tag{9}$$

where e is common for all roots. We may choose y properly such that in equation (8) $a=0$. Let

$$z=y-bx-e. \tag{10}$$

Then we may trade the monic property with a reduction of y-deg (or z-deg) for the polynomials involved as follows.

Proposition 6. 2. *Suppose that* $g(x, y)$ *is monic in* y *with*

$$\deg g(x, y) = \deg_y g(x, y) = n > 1, \quad \delta_s = -1.$$

Furthermore suppose that in the equation (8) $a = 0$. *Let* z *be defined as in equation* (10). *Let* $g(x, y)$, $T_i^{\psi}(f(x, y), g(x, y))$ *for* $i = 1, \ldots, s-1$ *be expressed as polynomials* $\bar{g}$, $\bar{T}_i^{\psi}$ *in* y *and* z. *Then we have*

(1) $\deg_y \bar{g} = v_s \dfrac{n}{d_s}$,

(2) $\deg_z \bar{g} = u_s \dfrac{n}{d_s}$,

(3) $\deg_y \bar{T}_i^{\psi} = v_s\left(\dfrac{-\mu_i}{d_s}\right)$ *for* $i = 1, \ldots, s-1$,

(4) $\deg_z \bar{T}_i^{\psi} = u_s\left(\dfrac{-\mu_i}{d_s}\right)$ *for* $i = 1, \ldots, s-1$.

Proof. The proofs of the four equations are similar. We shall only prove the first one. Let the factorization of $g(y)$ in $k\langle\langle t\rangle\rangle$ be

$$g(y) = \prod_{i=1}^{n} (y - \tau_i)$$

where

$$\tau_i = \begin{cases} bt^{-1} + e + c_i^* t + \text{higher terms in } t & \text{for } i = 1, \ldots, u_s \dfrac{n}{d_s}, \\ c_i + \text{higher terms in } t & \text{for } i = u_s \dfrac{n}{d_s} + 1, \ldots, n. \end{cases}$$

Let

$$\sigma = bt^{-1} + e + \pi t.$$

Then we have

$$g(\sigma) = g_{\sigma}(\sigma) t^{(u_s - v_s)\frac{n}{d_s}} + \text{higher terms in } t.$$

Let

$$g = \bar{g} = \sum a_{ij} y^i z^j.$$

Then

$$\bar{g}(\sigma) = \sum a_{ij}(bt^{-1} + e + \pi t)^i (\pi t)^j = \sum a_{ij}(b^i \pi^j t^{j-i} + \cdots). \tag{11}$$

If we have in the above expression

$$i > v_s \frac{n}{d_s}$$

for some i, then we may take the largest one. In addition

$$i+j \leqq n.$$

Then we have

$$j-i < (u_s - v_s)\frac{n}{d_s}.$$

Note that the coefficient π^j in equation (11) guarantees that the terms with different j indices will not cancel out. Hence we conclude

$$\operatorname{ord} g(\sigma) < (u_s - v_s)\frac{n}{d_s}.$$

A contradiction. Q.E.D.

The following lemma will be used in Proposition 6. 3.

Lemma 6. 2 (Inversion lemma). *Let x and y be two elements in $k\langle\langle t\rangle\rangle$ with*

$$x = t^{-1}, \quad y = y(t) = \sum a_j t^j = \theta^{-1}.$$

Suppose that

$$\operatorname{ord}_t y(t) = -1 < 0.$$

Then we have $k\langle\langle t\rangle\rangle = k\langle\langle\theta\rangle\rangle$. Furthermore in the following expression

$$x = x(\theta)$$

we have for arbitrary rational q and positive integer p:

(1) $x(\theta) \bmod \theta^q$ *and* $y(t) \bmod t^q$ *uniquely determine each other,*

(2) $x(\theta) \bmod \theta^q \in k\big((\theta^{\frac{1}{p}})\big) \bmod \theta^q \Leftrightarrow y(t) \bmod t^q \in k\big((t^{\frac{1}{p}})\big) \bmod t^q.$

Proof. Since we have

$$\operatorname{ord}_t \theta = 1 > 0$$

then it follows from the definition of Puiseux fields that $k\langle\langle t\rangle\rangle = k\langle\langle\theta\rangle\rangle$. Suppose that

$$y(t) \in k\big((t^{\frac{1}{m}})\big)$$

for some positive integer m. Then let

$$T = t^{\frac{1}{m}}, \quad X = T^{-m}, \quad Y = bT^{-m} + \cdots.$$

Then the conclusions (1) and (2) follow from the classical inversion formula for plane meromorphic curves [M.3]. Q.E.D.

The following Proposition shows that in some cases the degrees of the polynomials involved can be reduced by some transformations while the Jacobian will be modified to a monomial.

Proposition 6.3. *Suppose that* $g(x, y)$ *is monic in* y *with*

$$\deg g(x, y) = \deg_y g(x, y) = n > 1,$$

$\delta_s = -1$ *and the logarithmic radius of* δ^*_{s-1} *of the minor disc* $D^*_{s-1} \geqq \frac{v_s}{u_s}$. *Let* z *be defined by equation* (10), $y^{-1} = \theta$ *and consider* $g, T_1^{\psi}, \ldots, T_{s-1}^{\psi}$ *as elements* $\bar{g}(z)\ \bar{T}_1^{\psi}(z), \ldots, \bar{T}_{s-1}^{\psi}(z)$ *in* $k\langle\langle\theta\rangle\rangle[z]$. *Then there is a* π*-root* σ *of the following form*

$$\sigma = \sum a_j \theta^j + \pi \theta^{\frac{v_s}{u_s}}$$

such that with

$$\gamma = \theta^{\frac{1}{u_s}}$$

we have

(1) $\bar{g}(\sigma), \bar{T}_1^{\psi}(\sigma), \ldots, \bar{T}_{s-1}^{\psi}(\sigma) \in k[\gamma, \pi]$,

(2) $\bar{g}(\sigma), \bar{T}_1^{\psi}(\sigma), \ldots, \bar{T}_{s-1}^{\psi}(\sigma)$ *are monic in* π *with* π*-degree*

$$u_s \frac{n}{d_s},\ u_s \frac{-\mu_1}{d_s}, \ldots, u_s \frac{-\mu_{s-1}}{d_s}$$

respectively,

(3) $J_{\gamma,\pi}\left(\bar{g}(\sigma), T_1^{-\psi}(\sigma)\right) = -\frac{u_s}{b} \gamma^{v_s - u_s - 1}$.

Proof. It follows from Proposition 6.2 that

$$\begin{aligned} \bar{g}(y, z) &= \left(y^{\frac{v_s n}{d_s}} + \cdots\right) z^{\frac{u_s n}{d_s}} + \text{lower terms in } z \\ &= \left(y^{\frac{v_s n}{d_s}} + \cdots\right) \prod_{i=1}^{\frac{u_s n}{d_s}} (z - \tau_i) \end{aligned}$$

where $\tau_i \in k\langle\langle\theta\rangle\rangle$. Let us consider all parametrizations of $g(x, y)$ in the following form

$$\begin{aligned} x &= t^{-1}, \\ y &= bt^{-1} + e + \prod_{i=1}^{<\frac{v_s}{u_s}} e_i t^i + \text{higher terms in } t. \end{aligned}$$

By the assumption on the logarithmic radius δ^*_{s-1} of the minor disc D^*_{s-1} we have all e_i's are common. Hence it follows from Lemma 6.2 that in the following inversion

$$x=\frac{1}{b}\left[\theta^{-1}+\sum_{i=1}^{<\frac{v_s}{u_s}} c_i\theta^i+\text{higher terms in }\theta\right],\quad y=\theta^{-1}$$

all c_i's are common. With z defined as

$$z=y-bx-e$$

we conclude that in all parametrizations of $\bar{g}(y,z)$ of the following form

$$y=\theta^{-1},\quad z=(c_0-e)-\sum_{i=1}^{<\frac{v_s}{u_s}} c_i\theta^i+\text{higher terms in }\theta$$

all c_i's are common. Thus if we let σ be defined as

$$\sigma=(c_0-e)+\sum_{i=0}^{<\frac{v_s}{u_s}} c_i\theta^i+\pi\theta^{\frac{v_s}{u_s}}$$

then we have

$$\begin{aligned}\bar{g}(\sigma)&=\left(\theta^{-\frac{v_sn}{d_s}}+\cdots\right)\left[\bar{g}_\sigma(\pi)\theta^{\frac{v_sn}{d_s}}+\cdots\right]\\&=\bar{g}_\sigma(\pi)+\text{terms of higher degree in }\theta\text{ and lower degree in }\pi\end{aligned}$$

where

$$\deg\bar{g}_\sigma(\pi)=\frac{u_sn}{d_s}.$$

The conclusions (1) and (2) for $\bar{g}(\sigma)$ are evident. Similar arguments prove the conclusions (1) and (2) for $\bar{T}_i^{\psi}(\sigma)$ for $i=1,\ldots,s-1$. To prove conclusion (3) we note that

$$\begin{aligned}J_{\gamma,\pi}\left(\bar{g}(\sigma),\bar{T}_1^{\psi}(\sigma)\right)&=J_{\gamma,\pi}(x,y)\,J_{x,y}\left(g(x,y),T_1^{\psi}\left(f(x,y),g(x,y)\right)\right)\\&=J_{\gamma,\pi}(x,y)\\&=\det\begin{bmatrix}* & -\dfrac{1}{b\gamma^{v_s}}\\ -u_s\gamma^{-u_s-1} & 0\end{bmatrix}\\&=-\frac{u_s}{b\gamma^{v_s-u_s-1}}.\end{aligned}$$

Q.E.D.

Proposition 6.4. *Suppose that $g(x,y)$ is monic in y with*

$$\deg g(x,y)=\deg_y g(x,y)=n>1,$$

$\delta_s=-1$ and $u_s=1$. Then the logarithmic radius δ^*_{s-1} of the minor disc $D^*_{s-1}\geqq\dfrac{v_s}{u_s}=v_s$.

Proof. Suppose for a moment that we have

$$\delta^*_{s-1} < v_s.$$

Let σ be the unique π-root in the minor disc D^*_{s-1} with

$$\delta^*_{s-1} = \sum a_j t^j + \pi t^{\delta^*_{s-1}}.$$

Then clearly we have

$$\operatorname{ord} g(\sigma) = -\frac{v_s n}{d_s} + \delta^*_{s-1}\frac{u_s n}{d_s} = -\frac{v_s n}{d_s} + \delta^*_{s-1}\frac{n}{d_s} < 0.$$

It follows from Proposition 6.1 that σ is a distribution detector for $g(y), T^{\psi}_1(y), \ldots, T^{\psi}_{s-1}(y)$. Let

$$g(\sigma) = g_\sigma(\pi) t^{n\lambda} + \cdots,$$
$$T^{\psi}_i(\sigma) = T^{\psi}_{i,\sigma}(\pi) t^{-\mu_i \lambda} + \cdots \qquad \text{for} \quad i = 1, \ldots, s-1$$

where

$$\lambda = \frac{1}{d_s}[-v_s + \delta^*_{s-1}] < 0$$

and

$$\deg_\pi g_\sigma(\pi) = \frac{n}{d_s}, \quad \deg T^{\psi}_{i,\sigma}(\pi) = \frac{-\mu_i}{d_s} \quad \text{for} \quad i = 1, \ldots, s-1.$$

Note that

$$\text{g.c.d.}\left(\frac{n}{d_s}, \frac{-\mu_1}{d_s}, \ldots, \frac{-\mu_{s-1}}{d_s}\right) = 1.$$

Thus the polynomials $g_\sigma(\pi), T^{\psi}_{1,\sigma}(\pi), \ldots, T^{\psi}_{s-1,\sigma}(\pi)$ are powers of a common linear polynomial. It contradicts the very definition of δ^*_{s-1}. Q.E.D.

Corollary 6.1. *Suppose that there is a pair of polynomials* $(f(x, y), g(x, y))$ *satisfying the Jacobian condition with*

(1) $f(x, y)$ *and* $g(x, y)$ *are monic in* y *with* y*-degrees* m *and* n *respectively,*

(2) *the associated characteristic data* $\{M_i, d_i\}$ *are with* $M_s = n-2$, $d_s = 3$,

(3) $\delta_s = -1$ (i.e., more than one point at infinite).

Then there exists a transformation χ *which send* $f(x, y)$, $g(x, y)$, $T^{\psi}_i(f(x, y), g(x, y))$ *for* $i = 1, \ldots, s-1$ *to polynomials* $\bar{f}(\gamma, \pi)$, $\bar{g}(\gamma, \pi)$, $\bar{T}^{\psi}_i$ *for* $i = 1, \ldots, s-1$ *in* $k[\gamma, \pi]$ *a polynomial ring in two indeterminates with*

(4) $\bar{f}(\gamma, \pi)$, $\bar{g}(\sigma, \pi)$ *are monic in* π *with* π*-degrees* $\dfrac{m}{d_s}$ *and* $\dfrac{n}{d_s}$ *respectively,*

(5) *the associated characteristic data are* $\left\{\dfrac{M_i}{d_s}, \dfrac{d_i}{d_s}\right\}$ *for* $i = 1, \ldots, s-1$,

(6) $J_{\gamma,\pi}(\bar{f}(\gamma, \pi), \bar{g}(\gamma, \pi)) =$ *a nonzero constant.*

Proof. Since we have (cf. Lemma 6. 3)

$$3 = d_s > v_s > \frac{d_s}{2}, \quad v_s + u_s = d_s = 3.$$

Then we must have

$$v_s = 2, \quad u_s = 1.$$

Our corollary follows from Propositions 6. 4 and 6. 3. Q.E.D.

The previous corollary shows that in looking for a possible counter-example for the Jacobian conjecture of the smallest possible degrees we must assume that $d_s > 3$.

We may summarize some of our results in the following theorem.

Theorem. *Let $f(x, y)$ and $g(x, y)$ be two polynomials satisfying the Jacobian condition and the following conditions*:

(1) $\deg f(x, y) = \deg_y f(x, y) = m$, $\deg g(x, y) = \deg_y g(x, y) = n$,

(2) *the associated characteristic data are $\{M_i, d_i\}$ for $i = 1, \ldots, h$,*

(3) *the number M_s is the largest one $\leqq n-2$.*

Then for any $r \geqq 2$ we may construct a tower of major discs $D_s \supsetneqq D_{s-1} \supsetneqq \cdots \supsetneqq D_r$ according to Definition 5. 1. *The roots of $\prod_{i=1}^{r} g(y)\, T_i^{\psi}(y)$ in D_r can be covered by a disjoint union of subdiscs $\bigcup E_j$ satisfying the requirements of Definition* 1. 1. *We have*

(4) *if the number of roots of $g(y)$ in E_i is $> \dfrac{n}{n - M_r}$ then the tower of major discs $D_s \supsetneqq D_{s-1} \supsetneqq \cdots \supsetneqq D_r$ can be extended to $D_s \supsetneqq D_{s-1} \supsetneqq \cdots \supsetneqq E_i$,*

(5) *if the number of roots of $g(y)$ in E_i is $\leqq \dfrac{n}{n - M_r}$, then E_i is a minor disc and any π-root σ of $g(y)$ with $\operatorname{ord} g(\sigma) < 0$ is a distribution detector for $g(y), T_1^{\psi}(y), \ldots, T_{r-1}^{\psi}(y)$ (cf. Definition* 3. 1).

(6) *The number of subdiscs E_i is bounded by*

$$(n - M_r) \cdot \frac{V_{r+1}}{d_{r+1}},$$

(7) *there must be at least one E_i satisfying the requirement in* (4).

Proof. Propositions 5. 4 and 6. 1.

One application of the theorem and the propositions of this article is to solve the Jacobian conjecture for all polynomials of degrees less than 100. For this purpose we consider a pair of polynomials f and g with the nonrestrictive assumption that $f = T_1^{\psi}$. We shall search for the sequences of integers $\{n, M_1, \ldots, M_s\}$ with

(1) $\deg f(x, y) = m = -M_1 < \deg g(x, y) = n \leqq 100$,

(2) $m \nmid n$, $M_s = n - 2$,

(3) $J_{x,y}(f(x, y), g(x, y)) = 1$ and the degrees of $f(x, y)$ and $g(x, y)$ can not be reduced simultaneously,

(4) $\{n, M_1, \ldots, M_s\}$ is the part of characteristic data of (f, g) which are less than $n-2$,

(5) $d_r = \text{g.c.d.}\{n, M_1, \ldots, M_{r-1}\}$.

Note that it follows from Corollary 6.1 that $d_s \geqq 4$. A simple computation shows that $s \leqq 5$. Combining this with Proposition 5.5 we conclude

(6) $3 \leqq s \leqq 5, \quad d_s \geqq 4$.

Furthermore we assign a sequence of integers $\{V_s, \ldots, V_2\}$ for a hypothetically existing tower of major discs $D_s \supset \cdots \supset D_1$ (cf. Theorem and Definition 5.1) with

(7) $$V_{r+1}\left(\frac{d_r}{d_{r+1}}\right) \geqq V_r > \frac{d_r}{n - M_r},$$

(8) the radii $\delta_s, \ldots, \delta_1$ of $D_s, \ldots, D_1$ can be computed from Definition 5.1. We shall consider the increment of the denominator A_{r-1} of δ_{r-1} with respect to $\delta_s, \ldots, \delta_r$. Namely let the l.c.m. of the reduced denominators of $\delta_s, \ldots, \delta_r$ be L. Then A_{r-1} is defined to be the reduced denominator of $L\delta_{r-1}$. Let us consider the following division algorithm equation

(9) $$V_r\left(\frac{d_{r-1}}{d_r}\right) = \triangle_{r-1} A_{r-1} + \square_{r-1}$$

Let $(\bar{t})^{LA_{r-1}} = t$. Due to the existence of the following automorphism of $k\langle\langle \bar{t} \rangle\rangle$ over $k\langle\langle \bar{t}^{A_{r-1}} \rangle\rangle$

$$\bar{t} \to \omega \bar{t}$$

where ω is a A_{r-1}-th root of unity, the value of V_{r-1} is further restricted by

(10) $$V_{r-1} \leqq \triangle_{r-1}$$

if the corresponding factor of $p(\pi)$ is of the form $\pi - a$ with $a \neq 0$ or for some j the following

(11) $$V_{r-1} = jA_{r-1} + \square_{r-1}$$

if the corresponding factor of $p(\pi)$ is of the form π. Finally, when $r=2$ we must have the following (cf. the proof of Proposition 5.5):

(12) $$A_1 \Big| \left(\frac{n}{d_2}\right) V_2, \quad A_1 \Big| \left(\frac{m}{d_2}\right) V_2 - 1$$

or

(13) $$A_1 \Big| \left(\frac{m}{d_2}\right) V_2, \quad A_1 \Big| \left(\frac{n}{d_2}\right) V_2 - 1.$$

Furthermore we shall note that the situation indicated by the equation (11) can not always happen as established by Proposition 5.6.

The above discussion can be easily put into a computer program. The computer program produces only the following exceptions which satisfy the necessary numerical restrictions.

n	$m=-M_1$	M_2	M_3	M_4	V_3	V_2	δ_2	δ_1
64	48	52	62	63	3	3	$\frac{1}{4}$	$\frac{9}{16}$
84	56	64 [72]	82	83	3	2 [5]	$\frac{2}{7}\left[\frac{1}{4}\right]$	$\frac{16}{21}\left[\frac{7}{12}\right]$
75	50	55	73		4	3 [2]	$\frac{1}{5}$	$\frac{1}{2}\left[\frac{1}{3}\right]$
99	66	77	97		8	8	$\frac{1}{3}$	$\frac{4}{9}$

It is easy to see that the above table is very restrictive. For instance given $\deg g=75$ $\deg f=50$ there are 25 possible values for M_2 and 2 possible values for V_3 etc. The above table determines only one possible value for M_2 and one for V_3. These data are very useful in excluding these four exceptional data from the possible counterexamples to the Jacobian conjecture. Without them a brute force computation will involve the four thousands coefficients of $g(x, y)$ and $f(x, y)$. The treatment of these four exceptional cases will be presented in Appendix II.

Appendix I. The polynomial solutions of some ordinary differential equations

Recall Definition 4.1 of the symbol $D(a, b, p(\pi), q(\pi))$ from Section 4. We shall further assume that $p(\pi)$ and $q(\pi)$ are nonconstants in this section.

Proposition A. 1. *Suppose that at least one of the constants a, b is not zero. Then the following equation*

$$D(a, b, p(\pi), q(\pi))=0$$

implies for some $e, c\in k$ the following:

(1) $a=e\deg_\pi p(\pi)$,

(2) $b=e\deg_\pi q(\pi)$,

(3) $p(\pi)^{\deg_\pi q(\pi)}=cq(\pi)^{\deg_\pi p(\pi)}$.

Proof. Let $p(\pi)$ and $q(\pi)$ be expressed as

$$p(\pi) = a_1 \pi^m + \text{lower terms in } \pi,$$
$$q(\pi) = a_2 \pi^n + \text{lower terms in } \pi.$$

Then we have

$$D(a, b, p(\pi), q(\pi)) = ap(\pi)\, q'(\pi) - bq(\pi)\, p'(\pi)$$
$$= (an - bm) a_1 a_2 \pi^{n+m-1} + \text{lower terms}.$$

Clearly we must have

$$an = bm.$$

Thus the conclusions (1) and (2) follow. Clearly we have

$$D(a, b, p(\pi), q(\pi)) = 0 \Leftrightarrow D(m, n, p(\pi), q(\pi)) = 0.$$

The following identity is elementary

$$\left(\frac{q(\pi)^m}{p(\pi)^n}\right)' = \frac{p(\pi)^{n-1}\, q(\pi)^{m-1}\, D(m, n, p(\pi), q(\pi))}{p(\pi)^{2n}} = 0.$$

Thus we conclude

$$\frac{q(\pi)^m}{p(\pi)^n} = c \in k$$

and (3) follows. Q.E.D.

Proposition A. 2. *Let* $m = \deg p(\pi)$ *and* $n = \deg q(\pi)$. *Suppose that* $n = (l-1)m + 1$ *and* $p(\pi)$, $q(\pi)$ *satisfy the following equation*

$$D(m, n + \varepsilon, p(\pi), q(\pi)) = cp(\pi)^l \neq 0$$

where $c \in k$ *and* ε *is a rational number* > -1. *Then* $p(\pi)$ *and* $q(\pi)$ *are powers of a common linear polynomial.*

Proof. Let us prove the case $l = 1$. Then we have $n = \deg q(\pi) = 1$. We may assume $q(\pi) = \pi$. The differential equation is reduced to the following form

$$-(1 + \varepsilon)\, \pi p'(\pi) = (c - m)\, p(\pi).$$

Note that the coefficients are not zero. Consider the multiplicities of roots of $p(\pi)$. We deduce that π is the only root of $p(\pi)$. Hence the proposition has been established for the case $l = 1$.

Inductively let us assume that $l > 1$ and the proposition has been proved for smaller l. We claim that

$$p(\pi) | q(\pi).$$

Let π (or in general $\pi - a$) be a factor of $p(\pi)$ with

$$p(\pi) = \pi^u p_1(\pi) \quad \text{with} \quad p_1(0) \neq 0,$$
$$q(\pi) = \pi^v q_1(\pi) \quad \text{with} \quad q_1(0) \neq 0.$$

To prove our claim it suffices to show that $u \leqq v$. Note that the differential equation is of the following form

$$mp(\pi)\, q'(\pi) - (n+\varepsilon)\, q(\pi)\, p'(\pi) = c\pi^{lu}(p_1(\pi))^l. \tag{1}$$

Let

$$p_1(\pi) = b_0 + b_1\pi + \cdots,$$
$$q_1(\pi) = c_0 + c_1\pi + \cdots.$$

Then the left hand side of the differential equation is of the following form

$$[mv - (n+\varepsilon)u] b_0 c_0 \pi^{u+v-1} + \text{higher terms in } \pi.$$

The restriction

$$\varepsilon > -1$$

implies

$$n + \varepsilon \geqq (l-1)m \geqq m.$$

Thus the inequality

$$u > v \tag{2}$$

implies

$$mv - (n+\varepsilon)u \neq 0.$$

After comparing π-orders on the two sides of the differential equation (1) we conclude

$$u + v - 1 = lu.$$

It contradicts the inequality (2). Thus we must have

$$u \leqq v.$$

It is clear that the above argument works for an arbitrary factor $(\pi - a)$ of $p(\pi)$. Thus we conclude

$$q(\pi) = p(\pi)\, q^*(\pi).$$

A simple computation shows

$$D(m, n+\varepsilon, p(\pi), q(\pi)) = cp(\pi)^l \Leftrightarrow D(m, (n-m)+\varepsilon, p(\pi), q^*(\pi)) = cp(\pi)^{l-1}.$$

Thus by the induction hypothesis we conclude that the polynomials $p(\pi)$, $q^*(\pi)$ and $q(\pi)$ are all powers of a common linear polynomial. Q.E.D.

Proposition A. 3. *Let* $m=\deg p(\pi)$, $n=\deg q(\pi)$. *Suppose that the polynomials* $p(\pi)$ *and* $q(\pi)$ *satisfy*

$$D(m, n, p(\pi), q(\pi))=cp(\pi)$$

where $c\neq 0\in k$. *Then we have*:

(1) *every root of* $p(\pi)$ *is a root of* $q(\pi)$,

(2) $q(\pi)$ *has no multiple root*,

(3) $p(\pi)$ *is not a power of* $q(\pi)$,

(4) *there is at least one root of* $p(\pi)$ *with multiplicity* $>\frac{m}{n}$.

Proof. The differential equation can be rewritten as

$$nq(\pi)\,p'(\pi)=(mq'(\pi)-c)\,p(\pi).$$

Let $(\pi-a)$ be a factor of $p(\pi)$. Comparing the multiplicities of $(\pi-a)$ on both sides of the above equation, we conclude that $(\pi-a)$ is a factor of $q(\pi)$. The conclusion (1) thus follows.

It is easy to see that every multiple root of $q(\pi)$ must be a root of $p(\pi)$. Let $(\pi-a)$, say $a=0$, be a multiple factor of $q(\pi)$ with

$$p(\pi)=\pi^u p_1(\pi) \qquad p_1(0)\neq 0,$$
$$q(\pi)=\pi^v q_1(\pi) \qquad q_1(0)\neq 0.$$

Then we have

$$\pi^{u+v-1}|D(m, n, p(\pi), q(\pi))=cp(\pi).$$

A contradiction. The conclusion (2) thus follows.

In case that $p(\pi)$ is a power of $q(\pi)$ it is elementary to verify

$$D(m, n, p(\pi), q(\pi))=0.$$

It contradicts our assumptions on c and $p(\pi)$.

Suppose that all roots of $p(\pi)$ are with multiplicities $\leqq\frac{m}{n}$. Then it follows from conclusions (1) and (2) that all multiplicities must be equal to $\frac{m}{n}$. Thus we must have

$$p(\pi)=c_1 q(\pi)^{\frac{m}{n}}.$$

It contradicts conclusion (3). Q.E.D.

Proposition A. 4. *Let* $m=\deg p(\pi)$ *and* $n=\deg q^*(\pi)$. *Suppose that we have*

$$D(m, n, p(\pi), q^*(\pi))=cp(\pi)^{l+1}$$

where $c\neq 0\in k$ *and* $l\geqq 0$. *Then we have*

(1) $q^*(\pi)=p(\pi)^l q(\pi)$,

(2) $D(m, n-lm, p(\pi), q(\pi))=cp(\pi)$.

Proof. For $l=0$ it is Proposition A. 3. Let us assume that $l\geqq 1$ and our proposition has been proved for all smaller l. Clearly we have

$$n>m.$$

Otherwise the right hand side of the differential equation has a larger π-degree. Let $\pi-a$, say $a=0$, be a factor of $p(\pi)$ with

$$p(\pi)=\pi^u p_1(\pi) \qquad p_1(0)\neq 0,$$
$$q^*(\pi)=\pi^v q_1(\pi) \qquad q_1(0)\neq 0.$$

Let

$$p_1(\pi)=b_0+b_1\pi+\cdots,$$
$$q_1(\pi)=c_0+c_1\pi+\cdots.$$

Then we have

$$D(m, n, p(\pi), q^*(\pi))=\pi^{u+v-1}[(mv-nu)b_0c_0+\cdots].$$

If we have

$$v<u$$

then the leading coefficient

$$(mv-nu)b_0c_0\neq 0.$$

After comparing the multiplicities of π on both sides of the differential equation we deduce a contradiction. Thus we conclude

$$v\geqq u.$$

It is clear that the above arguments work for any factor $(\pi-a)$ of $p(\pi)$. Thus we conclude

$$p(\pi)\,|\,q^*(\pi)=p(\pi)\,\bar{q}(\pi).$$

We clearly have

$$D(m, n, p(\pi), q^*(\pi))=cp(\pi)^l \Leftrightarrow D(m, n-m, p(\pi), \bar{q}(\pi))=cp(\pi)^{l-1}.$$

Our proposition has been established by mathematical induction. Q.E.D.

Proposition A. 5. *Let* $m=\deg p(\pi)$ *and* $n=\deg q(\pi)$. *Suppose that* $p(\pi)$ *and* $q(\pi)$ *satisfy*

$$D(m, n, p(\pi), q(\pi))=c$$

where $c\neq 0\in k$. *Then* $p(\pi)\cdot q(\pi)$ *has only simple roots.*

Proof. Routine.

Appendix II. Some special cases

A simple computer program shows that the only possible counter-examples to the Jacobian conjecture for polynomials of degrees less than 100 are of degrees (64, 68), (84, 56), (75, 50) and (99, 66) with some special data attached. A routine computation shows that the numbers of coefficients of the pairs of polynomials involved are 3370, 5308, 4352 and 7348 respectively. The task of direct verification of the Jacobian conjecture by computation for those pairs of polynomials with the said degrees is beyond the capacity of a large computer. In this Appendix we shall illustrate some methods to reduce the numbers of coefficients, and then settle the problems of computations.

We shall apply Proposition 6. 4 to the first three cases. Note that in these cases we always have

$$u_3=d_3-v_3=1.$$

Thus it follows from Proposition 6. 3 that our table can be transformed to the following table.

n	$m=-M_1$	M_2	V_2	δ_2	δ_1	Jacobian
16	12	13	3	-1	$\frac{1}{4}$	X
21	14	16 [18]	2 [5]	$-\frac{1}{2}[-1]$	$\frac{7}{6}\left[\frac{1}{3}\right]$	X
15	10	11	3 [2]	-1	$\frac{1}{2}\left[\frac{4}{3}\right]$	X^2

The numbers of coefficients are reduced to 244, 373 and 202 respectively.

We shall use the first case to illustrate the method of further reduction of the numbers of coefficients. We shall choose x, y properly so that the π-root in the major disc D_1 is of the following form

$$\sigma=\pi t^{\frac{1}{4}}$$

and the π-root σ^* in the minor disc D_1^* is of the following form

$$\sigma^*=t^{-1}+a_0+a_1 t+a_2 t^2+\pi t^3.$$

To truncate σ^* properly we define $\bar{\sigma}$ as

$$\bar{\sigma} = t^{-1} + a_0 + a_1 t + a_2 t^2 + \pi t^{\frac{11}{4}}.$$

Then it is easy to see that $\{\sigma, \bar{\sigma}\}$ is a coherent complete system of π-roots for $\bar{g}(y)$ (and $\bar{f}(y) = \bar{T}_1(y)$) with accuracy $-1\left(\text{resp. } -\frac{3}{4}\right)$. Then it follows from Theorem 1.2 that

$$\bar{g}(y) = h(y)^4 + \alpha_1(y)\, h(y)^3 + \alpha_2(y)\, h(y)^2 + \alpha_3(y)\, h(y) + \alpha_4(y),$$
$$\bar{f}(y) = h(y)^3 + \beta_2(y)\, h(y) + \beta_3(y)$$

with

$$h(y) = \text{the fourth approximate root of } \bar{f}(y)$$

$$\deg_y \alpha_i(y) \leqq 3, \quad \deg_y \beta_i(y) \leqq 3,$$

$$\operatorname{ord} \alpha_i(\sigma) \geqq i\left(-\frac{1}{4}\right), \quad \operatorname{ord} \alpha_i(\bar{\sigma}) \geqq i\left(-\frac{1}{4}\right),$$

$$\operatorname{ord} \beta_i(\sigma) \geqq i\left(-\frac{1}{4}\right), \quad \operatorname{ord} \beta_i(\bar{\sigma}) \geqq i\left(-\frac{1}{4}\right),$$

$$\operatorname{ord} h(\sigma) \geqq -\frac{1}{4}, \quad \operatorname{ord} h(\bar{\sigma}) \geqq -\frac{1}{4}.$$

It is easy to see that

(1) $h(x, y) = y^3(y - x) + b_1 y^3 + b_2 y^2 + b_3 y + b_4$
$= yA(x, y) + b_4$
$= y^2 B(x, y) + b_3 y + b_4$

with $A(x, y)$ and $B(x, y)$ defined by the above equations and

(2) α_1 is a constant,

(3) $\alpha_2 = c_1 A + c_2, \quad \beta_2 = c_3 A + c_4,$

(4) $\alpha_3 = c_5 A + c_5 B + c_7, \quad \beta_3 = c_8 A + c_9 B + c_{10},$

(5) $\alpha_4 = c_{11} A + c_{12} B + c_{13}(y - x).$

Thus the total number of coefficients is further reduced to 17.

Similar arguments can be applied to the second case to show the impossibility directly and to the third case to reduce the number of coefficients to 22 [15 or 13].

We shall use the following method to further simplify our computations. Let

$$f = \eta^{-12},$$
$$g = \eta^{-16} + a_1 \eta^{-12} + a_2 \eta^{-8} + a_3 \eta^{-4} + a_4 + \cdots.$$

Then we may replace (f, g) by $\left(f-\frac{3}{4}a_3,\ g-a_1 f-a_4\right)$ to reduce the above expression to

$$g=\eta^{-16}+a_2\eta^{-8}+\text{terms in } \eta \text{ with positive exponents.}$$

Note that

$$f=h^3+\beta_2 h+\beta_3=h^3(1+\beta_2 h^{-2}+\beta_3 h^{-3}).$$

It is easy to conclude that

$$g=h^4+\frac{4}{3}(\beta_2 h^2+\beta_3 h)+\frac{2}{9}(\beta_2^2+2\gamma)-\frac{4}{81}\delta+a_2 h^2$$

where γ and δ are defined by

$$\beta_2\beta_3=\gamma h+\gamma^*,\quad \beta_2^3=\delta h^2+\delta^*$$

with

$$\deg_y \gamma^*<\deg h=4,\quad \deg_y \delta^*<\deg h^2=8.$$

Thus the total number of coefficients are reduced to 10. This method can be used to reduce the numbers of coefficients of the third case to 12 [10 or 7].

In the fourth case there are complications. First, there are two possibilities according to the distribution of the roots of $g(y)$ in the minor disc D_2^*. Namely, there is an unique π-root σ of $g(y)$ of the following form

$$\sigma=a_{-1}t^{-1}+a_0+a_1 t+\pi t^2$$

with

$$g(\sigma)=g_\pi(\sigma)t^{-18}+\cdots.$$

The polynomial $g_\sigma(\pi)$ is either a power of a linear polynomial or the 9-th power of a cubic polynomial with precisely two roots. For the first possibility the data can be transformed to

n	$m=-M_1$	M_2	V_2	δ_2	δ_1	Jacobian
27	18	21	8	-1	0	X^4

The above mentioned method can be used to reduce the number of coefficients to 10. For the second possibility we can show that for suitable variables x and y there is a transformation Ω of the following form

$$x\to x^{-1},\quad y\to a_0+a_1 x+a_2 x^2+yx^3$$

such that

$$\begin{aligned}
\Omega(f(x,y)) &= x^6y^{18}+\cdots,\\
\Omega(g(x,y)) &= x^9y^{27}+\cdots,\\
\Omega(T_2^{\psi}(x,y)) &= x^5y^{15}+\cdots,\\
\Omega(T_3^{\psi}(x,y)) &= x^{15}y^{40}+\cdots,\\
J_{x,y}(\Omega f(x,y),\ \Omega g(x,y)) &= x.
\end{aligned}$$

A discussion of the solutions of $\Omega(g(x, y))=0$ in $k\langle\langle x^{-1}\rangle\rangle$ and in $k\langle\langle y^{-1}\rangle\rangle$ with a slightly modified version of the above method we may reduce the direct computation to involve only 11 variables.

Now we are in the position to settle these cases by direct computation. Let us handle the case $n=15$, $m=10$, $M_2=11$, $V_2=3$. After simplifying, we may assume

$$\begin{aligned} f &= h^2+2\beta = h^2(1+2\beta h^{-2}) = \eta^{-10}, \\ g &= \eta^{-15}+a\eta^5+b\eta^6+c(x^3+d)\eta^9+\cdots \\ &= f^{\frac{3}{2}}+af^{-\frac{1}{2}}+bf^{-\frac{6}{10}}+c(x^3+d)f^{-\frac{9}{10}}+\cdots \end{aligned}$$

where a, b, c, d are constants and $b\neq 0$, $c\neq 0$. Clearly we have

$$f^{\frac{3}{2}} = h^3(1+2\beta h^{-2})^{\frac{3}{2}} = h^3+3\beta h+\frac{3}{2}\frac{\beta^2}{h}-\frac{1}{2}\frac{\beta^3}{h^3}+\frac{3}{8}\frac{\beta^4}{h^5}+\cdots,$$

$$af^{-\frac{1}{2}} = a\left[\frac{1}{h}+\cdots\right].$$

Substituting the above expressions into the expression for g we have

$$(1)\qquad g = h^3+3\beta h+\frac{3}{2}\alpha+\frac{3}{2}\frac{\gamma}{h}-\frac{1}{2}\frac{\beta^3}{h^3}+\frac{a}{h}+bf^{-\frac{6}{10}}+c(x^3+d^*)f^{-\frac{9}{10}}+\cdots$$

where d^* is a constant and α, γ are defined by the following equations:

$$(2)\qquad \beta^2 = \alpha h+\gamma$$

with the restriction

$$\deg_y \gamma < \deg_y h = 5.$$

Since g is a polynomial in y, then the powerseries in y^{-1} in the expression (1) shall naturally cancel out. Thus we must have the following equations (3) and (4)

$$(3)\qquad \beta^3 = (3\gamma+a)h^2+\varepsilon h+\delta$$

where ε and δ are polynomials with y-degrees less than 4,

$$(4)\qquad \frac{1}{2}\varepsilon = bf^{\frac{4}{10}}+c(x^3+d^*)y+\text{a powerseries in } y^{-1}.$$

Let us consider the possible forms of the polynomial $h(x, y)$. Since we assume that the polynomials $f(x, y)$ and $g(x, y)$ are the transforms of some polynomials of degrees 50 and 75 respectively according to Proposition 6. 3, the inverse transformation implies that there is a minor disc. Thus in the major disc D_2 there are precisely three subdiscs which contains 2, 2, 6 of roots f respectively. The datum $\delta_1=\frac{1}{2}$ implies that the poly-

nomials $h(x, y)$ and $\beta(x, y)$ must be of the following form (cf. the beginning of this Appendix)

$$h=(y^2-x^2+a_1y+a_2x+a_3)y^3+(a_4x+a_5)y^2+(a_6x+a_7)y+a_8 \tag{5}$$
$$=Ay+a_8=By^2+(a_6x+a_7)y+a_8,$$

$$\beta=a_9A+a_{10}y+a_{11}x+a_{12} \tag{6}$$

with A, B thus defined. In fact the existence of the polynomials h and β of the above forms and satisfying the equations (2), (3), (4) will furnish a counterexample to the Jacobian conjecture.

From equation (3) we deduce

$$\deg_y\gamma\leqq 2. \tag{7}$$

Using the equations (5) and (6) we get

$$\beta^2=(a_9^2B+2a_9a_{10})h+[(a_9^2a_6+2a_9a_{11})x+(a_9^2a_7+2a_9a_{12})]A$$
$$-a_9^2a_{10}B+(a_{10}y+a_{11}x+a_{12})^2-2a_8a_9a_{10}.$$

In view of equation (2) we conclude

$$\alpha=a_9^2B+2a_9a_{10},$$
$$\gamma=[(a_9^2a_6+2a_9a_{11})x+(a_9^2a_7+2a_9a_{12})]A-a_9^2a_{10}B$$
$$+(a_{10}y+a_{11}x+a_{12})^2-2a_8a_9a_{10}.$$

In view of equation (7) we have the following equations

$$a_9^2a_6+2a_9a_{11}=0,$$
$$a_9^2a_7+2a_9a_{12}=0,$$
$$a_9^2a_{10}=0.$$

We have two possibilities $a_9=0$ or $a_9\neq 0$.

Case 1. $a_9=0$. We then have

$$\alpha=0,\quad \beta=a_{10}y+a_{11}x+a_{12},\quad \gamma=\beta^2.$$

Thus equation (3) can not be fulfilled.

Case 2. $a_9\neq 0$. We then have

$$a_{10}=0,\quad \gamma=(a_{11}x+a_{12})^2,\quad \deg_y\beta=4.$$

Thus equation (3) implies that the polynomial γ must be of y-degree 2. A contradiction. This establishes the impossibility for $n=15$, $m=10$, $M_2=11$ and $V_2=3$.

All other cases can be computed directly as above. There is no counter-example of polynomials of degrees less than or equal to 100 for the Jacobian conjecture.

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Correction to the paper

Moduli spaces of covers and the Hurwitz monodromy group

(this Journal **335** (1982), 87—121)

By *M. Fried* at Irvine and *R. Biggers* at Atlanta

The opening paragraph contains a serious misprint. The fourth sentence should read:

We give examples of Nielsen types for which the moduli space is *reducible*.

Further corrections:

p. 98, line 1 instead of $G(\sigma)$ read $G(\boldsymbol{\sigma})$,

p. 113, lines 14/15 instead of ... is a cyclic group whose generator ... read ... is a cyclic cover whose group generator ....

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